



T 2.2 Mapping the governance of the twin-transition

**Germany, Hungary, Ireland,
Italy, Portugal Spain,**



**Funded by
the European Union**

Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Executive Agency. Neither the European Union nor the granting authority can be held responsible for them.

PROJECT ACRONYM	FITTER-EU
Project ref. number	101132546
Document title	Mapping the Governance of the Twin Transition in the EU
Document type	Report
Due date of deliverable	31/03/2025
Submission date	31/03/2025
Status	Final
Dissemination level	Sensitive
Language	English
Organization responsible for deliverable	FGB
Author(s)	Silvia Sansonetti and Roberta Cupertino (FGB)
With contributions by	Alicja Bobek, Noirin MacNamara (TU Dublin); Olga Kolotouchkina, Paloma Piqueiras (UCM); Pia Wieser, Kiara Groneweg (WEFC); Kitti Kerekes, Zoltán Szekely (SFC); Fernando Ferrera, Joao Sarrapia (UNINOVA)

REVISION HISTORY			
Version	Date	Modified by	Comments
1	09.03. 2025	Silvia Sansonetti and Roberta Cupertino	First version sent to the Quality Reviewer
2	16.03.2025	Pia Wieser (WEFC)	Quality review. Main suggestions relate to the structure of the report. A reorganisation of chapter 1 recommended, as well as a shortening of chapter 2.2 and 2.3.
3	31.03.2025	Silvia Sansonetti	Revisions in line with feedback provided by the Quality Reviewer
4	31.03.2025	Sara Clavero	Final review
5	23.05.2025	Silvia Sansonetti	Executive Summary
6	29.05.2025	Sara Clavero	Final review for resubmission

Executive Summary

This report presents an in-depth examination of governance structures, mechanisms, and challenges in the context of the twin-transition. It explores regulatory frameworks, institutional responsibilities, and stakeholders' engagement strategies in the six EU Member States involved in FITTER-EU: Germany, Spain, Hungary, Ireland, Italy, Portugal.

The findings highlight the complexities of governance in navigating the twin transitions—digital and green—and the role of multilevel governance in policy coordination and stakeholders' participation. It reveals that governance models across different countries exhibit significant variations, shaped by historical, political, and economic influences.

The report identifies best practices while also highlighting key challenges, such as:

- Fragmented decision-making processes across governance levels.
- Limited coordination between national and EU policies, leading to inefficiencies.
- The need for greater inclusion of structurally vulnerable groups in governance structures.

Moreover, the study underscores the necessity of improving governance frameworks to ensure that the digital and green transitions are integrated effectively rather than being treated as separate initiatives.

A cross-cutting analysis explores key thematic areas that impact governance and provides suggestions for improvements:

1. Twin-transition or two transitions? A core debate in governance today is whether the digital and green transitions should be pursued as an integrated process or as separate policies. The report examines how different countries and governance structures manage these transitions, identifying barriers to coordination and synergies that can be leveraged for more effective policymaking.

Suggestion → Recognizing the need for a holistic approach that ensures both transitions are managed cohesively rather than in isolation. Strengthening policy coherence to ensure these transitions are mutually reinforcing. Encouraging cross-sectoral collaboration to align objectives.

2. National policymaking framed by the EU: EU policies provide a structural framework for governance at the national level, influencing how policies are developed and implemented. The report explores the extent to which national governance strategies align with EU objectives and how discrepancies between EU frameworks and national priorities impact policy coherence.

Suggestion → Highlighting the necessity for further alignment between EU and local strategies. Enhancing national policy alignment with EU objectives. Strengthening mechanisms for national-level adaptation of EU policies.

3. Multilevel governance and coordinating complexity: Governance is increasingly complex, involving interactions between local, national, and supranational levels. This section analyzes how governance structures can improve coordination mechanisms to enhance efficiency and avoid policy fragmentation.

Suggestion → Addressing the complexity of governance across different levels and advocating for streamlined coordination among national, regional, and local entities. Developing better mechanisms for intergovernmental coordination. Addressing the administrative burden of complex governance structures.

4. Structurally vulnerable groups: involvement and needs' addressing: Governance mechanisms must ensure that structurally vulnerable populations — such as low-income groups, marginalized communities, and those affected by climate policies — are actively involved in decision-making. The report assesses current gaps in engagement strategies of stakeholders and proposes ways to strengthen inclusivity.

Suggestion → Emphasizing the importance of inclusive policymaking that considers the unique challenges faced by marginalized populations. Ensuring inclusive policymaking and stakeholder engagement. Implementing targeted policies to support vulnerable communities in the twin transitions.

The report concludes by offering a forward-looking wider perspective on improving governance mechanisms in general and for each member state included in the analysis:

Overall conclusions:

The report presents the core insights from the analysis and suggests reinforcing the importance of dynamic governance models that adapt to evolving challenges. It concludes that governance effectiveness and efficiency as well as public trust are essential for the success of the twin transition.

Recommendations:

The report's conclusions suggest strategic recommendations on the type of governance models to adopt. These should foster cross-sectoral and inter-agency coordination, should support a further alignment of national and EU strategies, should promote robust transparency initiatives and active stakeholder participation. Implementing these measures can lead to increased goals' achievement, better resource allocation as well as stronger public confidence.

Specific recommendations for each member state:

The report provides also targeted, analysis-driven recommendations from project partners on how governance structures can be strengthened and ensure practical, actionable improvements in the analysed member states.

Overall, this report serves as a valuable resource for policymakers, institutional leaders, and governance stakeholders, providing data-driven insights and actionable strategies to drive meaningful improvements in governance structures and policy implementation.

Table of contents

1. Background and research question.....	7
2. Methodology	8
3. Literature review	9
3.1 Introduction.....	9
3.2 Emerging themes in the national literature on twin-transition governance	10
4. Country profiles.....	14
4.1 Introduction: funding the twin-transition	14
4.2 Governance Structures and Processes	17
4.3 Barriers	28
4.4 Factors for success	30
5. Cross-cutting analysis	34
5.1 Introduction.....	34
5.2 Twin-transition or two transitions?.....	34
5.3 National policymaking is framed by the EU	35
5.4 Multilevel governance, coordinating the complexity.....	37
5.5 Structurally vulnerable groups: involvement and needs' addressing	38
6. Concluding remarks.....	41
6.1 Improving the integration between the Digital and Green Transitions	41
6.2 EU Policy Framing in Member States of the Twin-transition	41
6.3. Improving Multilevel Governance and Coordination.....	42
6.4 Enhancing Structurally Vulnerable Groups' Involvement and Addressing their Needs.....	42
6.5 Overall conclusions.....	43
6. 6 Specific recommendations by partners.....	44
Literature.....	45
Literature by country.....	46
Annex I: Methodological templates	62
Literature review template	62
Country report template	63
Document analysis template.....	67
The Good practice report	70
Annex II: Document analysis table per country	73

Annex III: Good practices..... 103

Germany 103

Spain 119

Hungary 137

Ireland 153

Italy..... 170

Portugal 183

1. Background and research question

Governance is the “capacity of public organizations to provide public and other goods demanded by a country’s citizens or the representatives thereof in an effective, transparent, impartial, and accountable manner, subject to resource constraints” (Katsamunska, 2016: 133). It is an attribute of institutions as proper structures as far as it is related to steering and control (Torring, 2012) but it is also about the processes that develop among societal actors and therefore a dynamic concept connected to coordination among them to foster their cooperation (Peters and Pierre, 2006). Governance processes consist in “goal setting, resource mobilization, decision making, implementation and feedback and evaluation” (Peters and Pierre, 2020) and can develop across different institutional layers at local, regional, national and international levels (Peters and Pierre, 2001). Within this framework, robust governance fosters democracy by ensuring accountability, transparency, and representation in policy formation (Bingham, et al., 2005). In turn, democratic values require stakeholder involvement and meaningful engagement in decision making, elevating citizen trust and reinforcing institutional legitimacy. Sound governance thrives when diverse stakeholders engage in decision making processes, providing insights and shared accountability (Bingham, et al., 2005). Such collaborative participation enriches policy outcomes, nurtures trust and strengthens democratic legitimacy.

The European Union policy making is a good example of how multi-level governance develops across institutions and actors in a multilevel context. According to Piattoni (2006), this layered structure effectively combines supranational competencies with regional and municipal knowledge, granting civil society organizations, regional administrations, and other local actors a voice in shaping policies. By facilitating interaction among various governance levels, the EU promotes transparency, legitimacy, and cross-border coordination, reinforcing ideals and expanding opportunities for stakeholder participation throughout its policy cycle. Democracy, as a European Union value, fosters stakeholder involvement, from the Union’s main institutions down to local authorities, ensuring shared decisions.

The twin-transition is the most recent policy challenge of the European Union. In this context it was expected that cooperative policymaking (Hooghe & Marks, 2003) could facilitate coherent strategies and allow to address structural vulnerabilities effectively. This is especially needed as the twin-transition involves the alignment of diverse policy objectives and actors’ interests (Knill & Tosun, 2012) from two different policy areas: environment and digital. Scholars highlight how, by embedding transparency tools and accountability frameworks, the EU seeks to address concerns over bureaucratic complexity and democratic legitimacy (Jordan & Adelle, 2012). In parallel, “soft governance” instruments such as guidelines and voluntary commitments allow for policy experimentation and actors engagement (Radaelli and Franchino, 2003).

Civil society engagement is pivotal to democracy and governance in addressing vulnerabilities tied to the twin transition (FITTER D2.1: 63-74). By integrating diverse voices, institutions can better detect and mitigate hidden barriers across local, regional, national, and EU layers. According to the FITTER approach, recognizing different “subjects, domains, and objects of justice” ensures that socially and politically marginalized groups are not sidelined in policy cycles. This process involves “adopting a grounded approach,” in which authentic experiences and everyday challenges are

taken seriously, informing strategies that respond concretely to structural inequalities. Moreover, “addressing power relations” is fundamental to illuminate how certain actors maintain undue influence and others remain disenfranchised.

When civil society groups have a meaningful seat at the table—whether in formal consultations or as mobilized networks—they bolster institutional legitimacy, transparency, and accountability. The EU’s multilevel governance structure can serve as a catalyst here, facilitating ongoing exchange among local communities, regional bodies, member-state governments, and EU institutions. Through deliberative methods, these stakeholders co-create solutions aimed at environmental goals and digital expansion without deepening existing inequalities.

Ultimately, the FITTER framework underscores that coordinated governance is not solely about high-level policy alignment; it is about inclusive processes that empower those historically excluded. By weaving local knowledge, cross-border collaboration, and democratic oversight into twin transition efforts, civil society can help ensure that Europe’s pursuit of sustainability and digital innovation truly fosters social justice and protects those at greatest risk of being left behind.

Drawing from this literature and the theoretical framework elaborated in FITTER D 2.1 a research question emerges: how can a governance model ensuring broad civil society participation—across local, regional, national, and EU levels—effectively address structural vulnerabilities in the twin transition by fostering transparency, accountability, and stakeholder inclusion? Specifically, what mechanisms can encourage diverse voices, can strengthen democratic legitimacy, and can yield coherent policies bridging environmental and digital objectives in Europe? To address this question in FITTER an analysis of the twin-transition governance was conducted in six European Union countries (Germany, Spain, Hungary, Ireland, Italy and Portugal)¹ by experts of partner organisations. The methodology adopted is briefly sketched out in the next paragraph.

2. Methodology

Drawing on agreed concepts and definitions in FITTER, the six partners in charge of country level studies drafted their country reports on governance of the twin-transition, civil society participation and structural vulnerabilities with the aim of addressing the research question. They performed an in-depth desk review of policy documents, official government statements, and pre-existing literature on governance arrangements, stakeholder engagement strategies, and twin-transition initiatives in their respective country. This step helped identify formal regulations, institutional roles, and policy priorities. To this aim three templates were adopted as tools: Literature review template, Document analysis template, Country profile template.

In parallel, researchers were compiling a comprehensive map of civil society actors at country level (NGOs, community groups, labour unions, industry associations, etc.) for the Work Package 3 that integrated their assessment on how inclusively policies are designed and implemented in the country reports.

Finally, partners selected and analysed examples of governance practices exploring how governance structures and processes allow participation to decision making to civil society/social

¹ Whenever the logic of the analysis allows it, the countries are listed in the alphabetical order of their respective country acronyms: DE (Germany), ES (Spain), HU (Hungary), IE (Ireland), IT (Italy), PT (Portugal).

partner/other organisations/individuals to address needs of structural vulnerabilities. They documented both formal consultation mechanisms and less structured participation channels. An additional tool was designed to this aim: the Good practices template.

The methodology relied on partners' expertise for reporting on accountability measures, transparency practices, and pathways for bridging environmental and digital objectives. Findings from the six national case studies were compared to pinpoint recurring patterns of governance strengths and weaknesses, as well as innovative practices that foster broader societal inclusion and mitigate structural vulnerabilities.

This shared methodology ensured consistency across the six countries, thus enabling a robust comparative assessment of governance structures and practices in Europe's emerging twin transition. All mentioned tools are included in Annex I of this report.

3. Literature review

3.1 Introduction

The literature collected in the six countries covered by FITTER analysis reveals that a significant differentiation in the governance of the twin-transition—integrating both digital and green transitions—, exists in the selected countries reflecting different policy priorities and institutional capacities. Germany, is considered as treating the digital and the green transition separately, prioritizing the green transition while focusing on data protection, Artificial Intelligence and digital security. Spain scholars while acknowledging some synergies between the two transitions, raise concerns about widening digital and societal inequalities. The Hungarian debate is the least engaged with the twin-transition concept, treating digital and green policies as distinct topics with little academic or policy discourse linking them. Ireland's approach remains limited, with climate policies taking precedence over digital transformation. In Italy, the literature focuses on the role of the European Recovery Plan which is largely shaping national efforts towards a proper twin-transition, as national policies addressing both transitions in an integrated manner are scarce. Portugal scholars point out that the country has been able to actively merge these two transitions in a common governance framework.

The literature reviewed shows also a strongly differentiation in the attention to inclusivity in twin-transition policies. Germany has limited discourse on gender and social equality within the governance debate on the twin-transition. Spanish analysis places a strong focus on bridging the urban-rural divide, digital inclusion, and mitigating the negative socio-economic impacts of climate change. Ireland has begun exploring just transition governance, but discussions on gender and intersectionality remain largely absent. In Italy, inclusivity is discussed: experts point out that it is not properly addressed as a policy priority, though regional governance structures may hold potential for more inclusive approaches. Hungary does not appear to have a structured inclusivity framework in its discussions on the twin-transition, highlighting a gap in social equality considerations. Portuguese debate points out that the country demonstrates a strong commitment to address and work specifically towards gender and rural inclusion, recognizing the need to integrate intersectionality into the policy framework.

3.2 Emerging themes in the national literature on twin-transition governance

The governance of the twin-transition is not a priority in national scholars' debates, but it remains a topic often discussed in light of specific themes that are briefly introduced below.

1. Twin-transition Governance

Across the countries academic literature tends to treat the governance of the digital and green transitions separately rather than as a unified governance approach. This fragmented perspective focuses on either digitalization or sustainability policies individually, rather than exploring fully integrated twin-transition frameworks.

Multiple sources underscore the importance of multilevel governance—coordinated action across national, regional, and local authorities—to address both transitions effectively. Spanish and Italian sources highlight the need for stronger cooperation among all layers of government, for instance to reduce digital divides between rural and urban regions with tailored investments (FUNCAS SEFO / Bandrés, E., Gadea, L., Salas, V., & Sauras, Y, 2020). This emphasis on cohesive multilevel efforts is echoed by analyses in Portugal as well.

The literature highlights that across these contexts, national governments commonly set strategic objectives. In Italy, for instance the ministerial bodies formulate core NRRP measures (Midiri 2022), while Spain's central ministries define guidelines for digital and green initiatives (Bandrés et al. 2020). Portugal, similarly, drives its digital transition primarily from Lisbon (Open Access Government 2020). Despite top-down planning, bottom-up elements emerge: Ireland's Citizens' Assembly fosters citizen-led recommendations (Devaney et al. 2020), Spain's Just Transition Agreements require local input (The Just Transition Institute 2022), and Germany relies on private-sector momentum for digital innovation (Heyen 2019). Hungary's approach, though still centrally orchestrated, incorporates local-level mentoring programs in coal-transition zones (Jakab & Molnár 2022).

Although specific mechanisms differ, a similar challenge emerges across countries' literature: policy fragmentation. Portugal's structural issue of dispersed oversight is mirrored by other nations grappling with unclear mandates or insufficient coordination. Germany's distinct green versus digital policy tracks, and Ireland's slower climate governance structures, both illustrate how fragmented approaches can impede strategic alignment. In these contexts, overlapping responsibilities and lack of centralised direction often delay progress on twin-transition goals.

The role of local authorities emerges as pivotal in the literature. It commonly serves as implementers rather than policy architects. This dynamic is particularly visible in Italy and Spain, where provinces and municipalities struggle with capacity gaps (Giachetti Fantini 2022; Bandrés et al. 2020). Similarly, rural municipalities in Portugal need stronger digital infrastructure to reduce regional disparities (Durão et al. 2024). Ireland's councils face challenges in shaping climate policies due to central-level constraints (Dekker 2020), and Hungary's localities have limited autonomy to guide transition policies (Bojtor & Bozsó 2022). In Germany, regional authorities often coordinate energy and digital transitions but encounter fragmented mandates across federal, state, and municipal levels (Grossmann 2019).

2. National Recovery and Resilience Plans (RRPs)

There is agreement among scholars and experts about the role played by the EU governance as EU directives and funding strongly shape the transitions in Germany (Wurm & Wittmann 2023), Spain (European Commission 2023), Hungary (Bojtor & Bozsó 2022), Ireland (Slevin & Barry 2024), Italy (Polverari and Piattoni 2022) and Portugal (IPP-JCS 2022). According to the selected contributions for Spain, Italy, and Portugal, RRP play a central role in driving both the digital and green transitions (Fantini 2022, Travassos Rosário and Carmo Dias 2022). While these funding instruments are national in scope, their strategic focus—particularly in Spain and Portugal—leans heavily toward digitalization, reflecting a broader European push to modernize economies. Italy's RRP notably integrates green and digital dimensions and formally address equity but, like others, confronts challenges in coordinating across different administrative levels and across ministries and agencies. The digital focus emerges in the literature for Hungary as well, but the budget allocation for digital growth is in comparison to Spain and Portugal more limited.

3. Administrative Capacity

Limitations in administrative capacity to implement transition policies is identified in the literature review as a consistent barrier in several countries is the. Italy exemplifies this with the adequacy of the measures foreseen in the RRP to achieve its objectives, also in view of the short timeframe (Polverari and Piattoni 2022) and with concerns around insufficient local-level expertise and slow bureaucratic processes, especially in southern regions where public tenders may exceed the capacities of local authorities (Giachetti Fantini 2022). Hungary also reports ongoing efforts to strengthen the digital capabilities of public administrations (Bojtor and Bozsó 2022), reflecting a broader challenge: even if policies and funding are in place, delivering concrete outcomes can stall without strong institutional support.

4. Mission-Oriented Policies and Policy Learning

Germany's literature highlights the theme of needing more mission-oriented policies to bridge both digital and sustainability objectives. Wurm and Wittmann (2023), think that the transition in the EU faces transformational system failures due to a discrepancy between legitimate goals setting and implementation. They indicate international policy learning and collaborative governance involving civil society, academia, and public administration as strategies to overcome fragmentation. These recommendations parallel broader calls across Europe for tailored frameworks capable of directing innovation toward societal challenges, rather than letting each transition run on separate tracks.

5. Top-Down vs. Bottom-Up Dynamics

Common to multiple countries is the interplay between centralised directives and grassroots or private-sector initiatives. The German review typifies this split: top-down instruments drive sustainability, whereas digitalization is more bottom-up (Wurm and Wittmann 2023) and economic driven (Heyen 2019). Italian scholars indicate that Italy's RRP structure reveals a tension between national-level oversight and the role of regional authorities (Giachetti Fantini 2022). These patterns suggest that for both green and digital agendas, balancing top-down coordination with bottom-up innovation is a mutual challenge (Midiri 2022).

6. *Green vs. Digital Transitions*

A recurring theme in the collected contributions is the disparity in how governments and stakeholders prioritise each transition. In many cases—such as Germany, Spain, Hungary, Portugal—the digital dimension takes precedence, often driven by private-sector innovation and economic imperatives. Meanwhile, the green transition, emphasizing climate targets and sustainability, can be less dynamic or is framed more as a public-sector responsibility. This pattern is also observable in Ireland (O’Gorman, 2020; Torney and O’Gorman, 2019; Slevin and Barry, 2024), where protecting vulnerable groups through climate-related policies appears separate from, and sometimes lagging behind, digital transformation efforts.

7. *“Exnovation” and Ending Unsustainable Practices²*

In slower-moving, ecology-focused fields—like agriculture or mobility—policy proposals in the explored literature call for “exnovation,” or phasing out obsolete or harmful technologies and approaches. While emphasized in Germany’s context (Heyen 2019), the principle resonates with other countries aiming to shift from traditional, carbon-intensive sectors toward greener, digitally supported solutions. Ideological and political economy factors are argued to act as barriers to fulfilling Irelands’ climate obligations (Slevin and Barry, 2024). However, it has also been noted that significant change occurred in 2019 when the approach changed as new policies were introduced with a recognition that the old ones were not effective – an example here is the significant increase in carbon tax announced in the 2020 budget (O’Gorman, 2019). Even where digital transition dominates, as in Spain or Portugal, there is growing recognition that reducing unsustainable practices is integral to meeting climate goals.

Overall, these ten concepts reveal broad similarities across countries: a general struggle to integrate green and digital transitions into a cohesive governance framework, an uneven distribution of resources and administrative capacity, and a pressing need for coordinated, multilevel strategies that also address equity. Although each national context has unique specificities, the challenges of fragmentation, balancing top-down and bottom-up dynamics, and ensuring inclusive access to both digital and green benefits are widely shared.

8. *How inequalities are accounted for in the governance of the twin-transition*

According to the literature collected at country level all six countries— Spain, Germany, Hungary, Ireland, Italy, Portugal—demonstrate a growing, if uneven, emphasis on participatory governance mechanisms. Portugal’s digital inclusion programs (Estrutura de Missão Portugal Digital 2022) and Ireland’s Citizens’ Assembly on Climate Change (Devaney et al. 2020) exemplify structured consultation processes that involve citizens, civil society, and private-sector stakeholders. Spain’s Just Transition Agreements similarly engage multiple actors to reshape local economies amid coal closures (The Just Transition Institute 2022). Although Hungary’s twin-transition policies are judged less explicit, the Just Transition Fund encourages broader stakeholder input in affected regions (Jakab & Molnár 2022). Italy’s National Recovery and Resilience Plan (NRRP) references participatory ideals in principle, though its top-down structure has limited bottom-up engagement

² Exnovation is the termination of a practice, or the use of a technology or product, within an organization, community, or society. the concept has been increasingly adopted in sustainability and transition research to designate the phase-out of unsustainable technologies, products, and practices, in relation to energy transition.

(Giachetti Fantini 2022). Germany, for its part, engages stakeholders more robustly in certain policy areas (e.g., local energy transitions), but an integrated participatory framework for the entire twin-transition is still evolving (Wurm & Wittmann 2023).

In all countries scholars and experts acknowledge that the need to protect groups affected by structural vulnerabilities—women, rural communities, older workers—and mitigate risks posed by economic or digital disruptions. Spain’s Just Transition Agreements prioritize women’s employment and communities at risk of depopulation (The Just Transition Institute 2022), while Portugal’s inclusion programs target low-income and rural populations (Francescon 2021). Italy’s NRRP rhetoric addresses territorial and generational inequalities (Giachetti Fantini 2022). Ireland highlights just transition measures for impacted peat workers (Moore 2020). Hungary’s coal-phaseout strategies offer retraining, though systematic support for women and minority groups remains sparse (Jakab & Molnár 2022). Germany’s discourse often raises concerns about lower-income households facing higher energy costs or digital access barriers (Kelly et al. 2024).

To the aim of providing an adequate answer to individual and collective needs the literature points out that all six countries exhibit growing, though inconsistent, attention to intersectional issues. Portugal stresses women’s advancement in STEM fields (Francescon 2021), and Spain’s JIT framework explicitly references female workforce participation (The Just Transition Institute 2022). Ireland’s feminist climate justice reports highlight gender-blind spots in policy (Dukelow et al. 2024). Italy acknowledges gender inequities within its NRRP objectives but implementation lags (Menegus 2022). Hungary infrequently addresses intersectionality in official documentation (Bojtor & Bozsó 2022). Germany lacks explicit gender frameworks in twin-transition policies despite academic calls for mission-oriented approaches that incorporate equity at the outset (Wurm & Wittmann 2023).

4. Country profiles

To complete Country reports, partners relied mainly on desk research and their own expertise. Some of the most relevant documents they consulted are included in Annex II.

4.1 Introduction: funding the twin-transition

Twin-transition in the countries of analysis proceeds at different speeds, according to the national, political, social and economic national conditions. All the country's profiles show that the concept of twin-transition is not yet embedded in policy and the academic literature, while the green and digital transitions are present in all national policies, due to the financial support provided by EU funds.

The governance structures leading to policies and implementation of plans are also diverse, with a level of greater (Spain, Ireland, Germany, Portugal) or lesser involvement of civil society (Italy, Hungary). All governance structures have in common the key role of the government or single ministries. In the area of access to EU funds, diversified levels of national involvement can be reported: for example, both RRP loans and grants have been used by Italy and Spain, while only grants have been used by Germany and Ireland. The comparative analysis also highlights the existence of national policies and dedicated funds to support the green or digital transition in Spain and Ireland. Hungary is facing a specific situation linked to the fact that some funds are fully or partially frozen, making access to EU funds more difficult than in other countries studied.

As mentioned in Hungary, while several key EU instruments contribute to pursue the twin-transition objectives, including the European Regional Development Fund (ERDF), the Cohesion Fund, the European Social Fund Plus (ESF+), and the Just Transition Fund (JTF), the Country faces significant challenges in accessing EU financial support due to unmet conditionalities. In fact, the country is eligible for EUR5.8 billion under the EU's COVID-19 recovery package but these funds are frozen until Hungary fulfils 27 "super milestones" set by the European Commission, including judicial reforms and anti-corruption measures. Similarly, access to Cohesion Policy funds is restricted, as a percentage of them is frozen under the EU's rule of law mechanism due to concerns over corruption, judicial independence, and human rights violations, including restrictions on civil society, academic freedom, and discrimination against LGBTQ+ people, asylum seekers, and migrants. Key demands include strengthening judicial independence, increasing transparency in public procurement, and enhancing anti-corruption measures.

Hungary has established a multi-level coordination structure that governs the use of both EU and national funds. This governance framework involves various levels of government, including national, regional, and local authorities, working collaboratively to align their efforts and maximise the impact of twin-transition initiatives. At the national level, the central government, through ministries such as the Ministry of Innovation and Technology or the Ministry of Public Administration and Regional Development are responsible for overseeing the strategic direction and allocation of both EU and national funds. Regional authorities are tasked with implementing specific projects and initiatives, ensuring that the unique needs and priorities of each region are adequately addressed. Formally the promotion of equality is a key principle in the governance of the twin-transition.

In Italy, the RRP includes 150 investment streams and 66 reforms, with EUR 71.8 billion in grants and EUR 122.6 billion in loans. Of the total grants, 39% of the plan's resources will support climate objectives and 25.6% of the plan will support the digital transition. For the 2021-2027 programming cycle, for cohesion policies, Italy will rely on the amount of EUR 142,688.8 million for the seven-year period, while EUR 102,481.2 million are allocated to the Southern Regions (Open Coesione, retrieved in 2024).

On the one hand, the governance structures in place in Italy adapt to the compulsory European rules applicable to the funds; consequently, the governance of the Cohesion Fund can be considered as multi-level and pluralistic, given the significant involvement of stakeholders and local authorities in the planning phase (in addition to the implementation), in accordance with the European Code of Conduct on Partnership in the framework of the European Structural and Investment Funds. On the other hand, the National Recovery Plan, also due to its emergency nature, is affected by a strong centralisation: the whole policy making process is under the direct control of the central government, while the implementation is attributed to national, regional and local administrations, including small municipalities. Political power at national and local level is supported by technical/administrative bodies.

Both cohesion policies and Resilience and Recovery Plan are placed under the authority of the President of the Council, which for the present legislature, has delegated her powers to the Minister for European Affairs, the South, Cohesion Policy and the national Resilience and Recovery Plan (Ministro per gli Affari europei, il Sud, le politiche di coesione e per il national Resilience and Recovery Plan). The aim of the in-charge government is to assign to the Minister for European Affairs, the South, Cohesion Policy and the national Resilience and Recovery Plan (RRP) the role of policies coordinator, and in this view, also the RRP governance has been recently changed.

Germany's Recovery and Resilience Plan aims to support the green transition and the digital transformation of the country. The former includes measures for decarbonising the economy and industry, cleaner transport and energy-efficient building renovation. Digitalisation runs through almost all the measures in the recovery plan. In total, for the implementation of the plan, Germany plans investments for EUR 28.7 billion, of which EUR 28 million are grants and the remaining amount is financed from national funds. The value of the grants is equivalent to 0.8 % of the GDP. Germany has not requested loans to the EU for the implementation of the plan. More than EUR13 billion (47% of the funds) are invested in the green targets, while digitalisation represents 48% of the investment targets, almost EUR 4 billion for digitalising hospitals and local health administrations. Regarding the governance of the funds, Germany designated the Federal Ministry of Finance (BMF) as coordinator and interlocutor with the Commission for the overall implementation of the RRP. As such the BMF is responsible at national level for coordinating and monitoring the effective implementation of the measures envisaged and the achievement of the objectives set in the RRP

Also, in Spain the transitions are supported by Recovery and Resilience Plan approved by the EU for the period 2021-2026 amounts to EUR 163 bn (EUR 80 bn in grants and EUR 83 bn in loans) and structured through 142 investment streams and reforms. Green and digital transitions are supported by the 40% and 26% of the Plan respectively (European Commission, n.d.).

Spain has established a specific structure to manage and monitor the allocation and implementation of both EU and national funds for the twin-transition and has allocated national

funds to support the twin-transition through the General State Budgets, complementing EU financial resources with national funding, although NextGenerationEU contribution is more significant due to their scale and impact. The main areas these funds are used are renewable energy, infrastructure and innovation. The Spanish governance structure, led by the Ministry of Finance and coordinated with regional governments and civil society, ensures that these resources are effectively utilized to address local needs while aligning with national and EU priorities. The Recovery, Transformation, and Resilience Plan (RTRP) serves as the strategic framework in the country.

The Spanish government has established the National Fund for Energy Efficiency dedicated to improving energy efficiency across various sectors, particularly in residential and industrial contexts (IDAE, n.d.). Furthermore, the Digital Spain Agenda 2026 is supported by national investments aimed at expanding digital infrastructure, enhancing cybersecurity, and promoting digital skills across the population (Ministry of Economy, Trade and Entrepreneurship, n.d.). Finally, the Just Transition Strategy Fund supports regions and communities affected by the transition to low carbon use. This fund is aimed at mitigating the social and economic impacts of the transition, particularly in regions heavily dependent on fossil fuels, by promoting economic diversification, reskilling of workers, and investments in sustainable industries (Ministry of Finance, 2024b).

In Ireland, both the green and digital transformation are part of the Irish National Recovery and Resilience Plan, as part of two separate priorities: Priority 1 - Advancing the Green Transition and Priority 2 - Accelerating and Expanding Digital Reforms and Transformation. The plan allocates 42% of its budget to support climate investments and reforms and 34% to foster digital transition. The remainder of the plan is allocated to Social and Economic Recovery and Job Creation. With reference to Cohesion funds, European Regional Development Fund (ERDF), European Social Fund Plus (ESF+), European Maritime Fisheries and Aquaculture Fund (EMFAF), Just Transition Fund are benefited by the Country. The green and the digital transitions are also at the core of the Irish National Development Plan 2021-2030 (NDP), which is partially supported by the EU funds.

Government departments and Ministers have an agenda of business, which they are committed to develop and implement. Although the process varies, in general, there are three stages for national-level policy design – research and advisory meetings; cabinet decision making; and policy development.

Portugal heavily relies on EU funds, notably the Recovery and Resilience Plan (allocating EUR 13.9 billion in grants and EUR 2.7 billion in loans and a EUR 350 million state aid scheme), to finance decarbonization throughout renewables (large investments in wind and solar energy roll-out) and digital education. Cohesion Policy Funds also bolster regional sustainability and workforce reskilling. Centralised governance aligns EU and national budgets. National co-financing complements these mechanisms, bridging strategic gaps and ensuring timely, effective twin-transition investments.

4.2 Governance Structures and Processes

4.2.1 Introduction

Beyond the common framework of EU funding rules, it is important to consider that the green and digital transitions policies are established by countries with different structural governmental characteristics and ways of understanding the concept of multilevel governance. In principle, countries establish their dynamics of relationships across political/constitutional powers in parliamentary or presidential systems. Within these systems, in a concrete way, several variables are possible: by example more or less power can be provided to the executive power or to the parliament or various forms of decentralization of national power are possible, from the federal form to the administrative decentralization in a unitary State. Also, it should be considered that the multilevel governance approaches may vary state by state from the multilevel governance vertical approach (Hooghe and Marks) to the polycentric models, characterized by as “multiple governing authorities at different scale rather than a monocentric unit” (Moeko Saito-Jensen (2015), Ostrom (2010)).

In every country, green and digital strategies are being linked to broader objectives like equality, transparency, and social inclusiveness. The approaches range from specialized equality ministries (Spain, Italy, Portugal, Hungary) to active civil society participation (Germany) and citizens’ assemblies (Ireland). These evolving systems illustrate a balancing act between central government efficiency and the need for localized adaptation and democratic participation.

In our analysis all countries underline a vertical governance multi-level framework integrating national, regional, and local governance, with defined roles for various actors and mechanisms to ensure coordination and accountability.

The governance structures include, in some cases (Spain, Italy, Ireland, Germany, Portugal), administrative or political bodies dedicated to the management of the EU resources and their coordination or to guarantee the involvement of the civil society organizations (Spain, Germany, Ireland). With reference to equality and inclusion, mechanisms to ensure their enforcement are present in most examined countries, but there is no close link between twin-transition policies and the work of these institutions/administrations, which seem to act at the most general level. Anyway, as will be shown, differences emerge by country.

There is not a direct relation between governance complexity and the degree of centralisation of the decision-making process, on one hand and the opportunity to actively participate to decision making by stakeholders, on the other hand. Germany, Spain, and Ireland emphasize decentralisation in decision-making and favour effective multi-stakeholder participation. Portugal as well fosters multistakeholder participation but relies on centralised decision-making, the latter is also adopted by Hungary and Italy where civil society, however, does not enjoy a similar degree of active participation.

Despite these differences, all countries integrate national, regional, and local levels into policy implementation. Parliamentary oversight, civil society participation, and mechanisms for stakeholder inclusion are evolving, shaping how each country navigates the twin-transition's challenges and opportunities.

4.2.2. Commonalities and Differences Across the Six Countries

Multilevel governance between centralisation and decentralisation

Across the six FITTER-EU case study countries, governance for the twin-transition involves interplay among national, regional, and local levels, yet they vary significantly in how decision-making powers are distributed with a different degree of centralisation.

Spain, Italy, and Germany each feature constitutionally recognized forms of subnational autonomy -Autonomous Communities, regions, or federal states, respectively. Germany's constitutionally enshrined federalism besides the federal parliament (Bundestag) grants its states (Länder) direct legislative competencies and representation in the Bundesrat, providing genuine power-sharing and civil society participation. Spain also employs a notably decentralised model fostering genuine participation through its Autonomous Communities, each holding substantial authority. In Italy, regions possess legislative powers in specified sectors, although the national government retains core oversight (a mere consultative body between the Central State and the regions is in place but does not exercise formal power), under the control of the Parliament in addition the management of the RRP (the main funding tool for the Twin-transition in the country) is centralised with limited opportunities for civil society participation. Ireland operates as a unitary state but embeds significant consultative processes, particularly through local authorities that help shape policy implementation in a bottom-up manner. Portugal similarly centralizes finances and policy design at the national level, with ministries and agencies setting objectives, while municipalities and regions adapt measures to local contexts. In contrast, Hungary, formally has multiple tiers of administration but in practice concentrates authority at the centre; the strong role of the prime minister and legal instruments frequently bypass local bodies.

The overarching legal framework where parliaments enact legislation, ministries develop and implement policy, and local authorities manage on-the-ground initiatives fosters effective civil society engagement to a different degree, with most countries establishing consultation platforms or participatory processes- though the extent and impact of these vary.

Role of Parliament

All countries have parliamentary systems that enact laws and oversee the executive.

- Germany adds a Bundesrat for state representation, and Spain features a strong interplay between the national parliament and autonomous institutions. Both these countries ensure regular reporting and multi-level accountability through established committees and oversight mechanisms.
- Ireland, Portugal and Italy rely on parliamentary control for policy coherence and oversight. Italy's governance complexity that would expect the involvement of regional representative bodies is limited in the implementation of the RRP that is the main funding tool of the Italian twin-transition.

- Hungary's Parliament has limited de facto power, given the prime minister's dominance. Inputs from the local level are limited as monitoring and evaluation involve the State Audit Office and advisory committees, while the feedback loop from local civil society is less influential.

Engagement of Civil Society and Stakeholders

- Germany and Spain have institutionalised mechanisms (e.g., official committees, councils) and encourage civil society to influence policy.
- Hungary's centralised structure provides fewer practical avenues for local or civil society influence and its formal consultation infrastructure does not translate into substantial influence. In Italy civil society has been initially involved in the drafting of the initial RRP which is the main tool for funding the Twin-transition in the country. However, the participation of stakeholders has been reduced over time.
- Ireland makes broad use of citizens' assemblies, climate conversations, and stakeholder forums, while Italy involves social partners especially for cohesion funds (less so for RRP).
- Portugal integrates social partners through the Economic and Social Council and relies on participatory budgeting at local levels.

Equality and Non-Discrimination

Most countries incorporate EU principles for equality, diversity, and non-discrimination:

- Variable Capacity: Germany and Italy address these issues through advisory councils and specific working groups, whereas Hungary's approach depends largely on political will and available resources.
- Spain explicitly addresses gender equality in the twin-transitions via the Ministry of Equality. Italy has the Ministry of Equal Opportunities, and Portugal has the Commission for Citizenship and Gender Equality. These three Member States embed equality assessments in their policies, ensuring that structurally vulnerable groups are considered during policy development. Germany ensures civil society involvement and fosters open discourse, including special councils like the German Council for Sustainable Development, though it does not reference a dedicated ministry for equality which could coordinate gender mainstreaming in every ministry in the same manner.
- Ireland and Portugal incorporate the needs of vulnerable groups through consultations, dedicated advisory bodies, and climate or digital inclusion policies.

Use of EU Funds

- In all six countries, EU funding plays a significant role in advancing the twin-transition. Spain, Italy, and Portugal receive major EU support under frameworks such as NextGenerationEU or national recovery and resilience plans.
- Hungary and Germany manage EU funds within their structures (counties in Hungary and states in Germany), though the distribution of these funds reflects

differing degrees of local empowerment. Ireland channels EU funds through departmental and local authority mechanisms, with oversight from national bodies and parliamentary committees.

4.2.3 Country summaries

Hungary

Hungary is a unitary state with a centralised organizational structure and a parliamentary system.

The governance framework is divided into three primary levels:

- **National Level:**
 - The Parliament enacts laws and exercises oversight over the Government. In theory, it holds formal powers, such as legislative review through parliamentary committees.
 - The Government, led by the Prime Minister, executes laws, manages public administration, and has strong control over the national agenda.
 - Key ministries involved in the twin-transition include the Ministry of Innovation and Technology, the Ministry of Agriculture, the Ministry of Finance, and the Ministry of Public Administration and Regional Development. Ministries consult stakeholders, including industry experts and civil society, but consultation is limited in impact.
 - The Hungarian State Audit Office and advisory bodies such as the NCSD assess policy effectiveness.
- **Regional (County) Level:**
 - Hungary is divided into 19 counties, each governed by its regional authorities.
 - Regional authorities implement national policies, manage EU funds, and bridge national objectives with local needs. Regional Assessors ensure policy alignment and oversee the effective implementation of regional development initiatives.
- **Local Level:**
 - Municipalities (cities and towns) provide public services and implement policies that address community needs. They apply higher-level decisions directly to local conditions.

Despite these formal structures, governance is highly centralised, limiting the practical influence of Parliament and local entities. The government assumes the notable mechanism, “investment deemed of special interest for the national economy” which allows the central government to exclude projects from local or regional oversight, effectively weakening any checks or balances at subnational levels. Hungary’s governance of the twin-transition follows therefore a centralised approach with limited stakeholder influence. Ministries consult stakeholders, including industry experts and civil society, but consultation is limited in impact and civil society participation in evaluation is constrained. Policy development is led by the Prime Minister’s Office and relevant ministries, such as the Ministry of Innovation and Technology and the Ministry of Agriculture. These bodies set strategic priorities, which are then translated into policy frameworks.

Comparative Remarks:

Hungary's structure displays a clear parliamentary system with multi-tiered governance, but the central government exercises strong top-down control. Compared to some others, local and regional governance has a narrower scope because of legal instruments favouring the central government. The country aligns with EU principles on equality and inclusivity, requiring EU-funded projects to address structurally vulnerable groups. However, actual engagement varies, with political will and financial resources determining outcomes. Local municipalities interact more frequently with civil society, though smaller ones struggle with resource constraints.

Unlike Germany, Spain, and Ireland—where consultative bodies, citizens' assemblies, and formal dialogue mechanisms are prominent—Hungary's stakeholder engagement remains more advisory in nature, with a heavier reliance on top-down decision-making and less direct influence from civil society or local actors.

Italy

Italy is a parliamentary republic with multi-level governance structures, where both national and regional authorities have legislative and administrative powers.

- At national level the following actors are involved:
 - The Prime Minister directs the general policy of the government.
 - The Minister for European Affairs, the South, Cohesion Policy, and the RRP oversees EU fund programming and monitoring.
 - The Parliament exercises control powers over government actions, with permanent institutions like the Court of Auditors (Corte dei Conti) and the Constitutional Court intersecting with governance structures it exercises oversight, requiring annual reports on fund allocation and impact.
 - Legislative power is shared by the State and the regions, in compliance with the Constitution, EU law, and international obligations. Environmental protection is an exclusive State competence, while relations with the EU are a concurrent matter. Regions legislate in areas not explicitly reserved for the State. The Conferenza Permanente Stato-Regioni (Permanent Conference State-Regions) is the place where the central government and the regions openly discuss about governance issues and the resources allocation.

Italy follows dual governance approaches for Cohesion Funds and the Recovery and Resilience Plan (RRP). The Cohesion Funds governance is multilevel and participatory involving regional and local bodies and a broad set of stakeholders from the planning phase onward (in coherence with the European Code of Conduct on Partnership) while the RRP follows a more centralised, top-down model under the RRP Coordination Cabin (Cabina di Coordinamento RRP), led by prefects; it ensures cooperation across governance levels., though attempts to mitigate centralization have occurred (e.g., parliamentary hearings under the Conte and Draghi governments).

- Ministries and Departments involved in the Twin-transition are:
 - The Department for Digital Transformation coordinating national digital strategies within the Presidency of the Council of Ministers.

- The Ministry of the Environment and Ecological Transition leading on climate change, emissions reduction, and the promotion of renewable energy.
- Youth and citizenship gap initiatives are coordinated by the Ministry of Equal Opportunities under the Family Act and the National Strategy on Gender Equality, supported by two key bodies:
 1. Cabina di regia interistituzionale (interinstitutional control room): Connects national and local levels, promoting shared best practices.
 2. Osservatorio nazionale per l'integrazione delle politiche: Monitors and analyzes implementation of gender equality strategies.
- Role of the Ministry of Disability:
 - Requires central administrations to provide impact reports on reforms targeting persons with disabilities, ensuring accountability through the National Observatory on the Condition of People with Disabilities.

Comparative Remarks:

Like Spain and Germany, Italy has multi-level governance but also retains strong national oversight, especially regarding the RRP. Regions in Italy exercise certain autonomous powers, somewhat resembling Germany's federal states or Spain's Autonomous Communities, but the degree of actual regional control varies by policy area. Parliamentary oversight is present, though the specific "double track" approach (Cohesion Funds vs. RRP) distinguishes Italy from other contexts.

Italy faces challenges in fund absorption capacity, limiting the effective deployment of twin-transition initiatives. The participatory nature of the cohesion funds process contrasts with the more centralised, government-led RRP. Italy's system thus reflects an internal duality where some policies are co-created with diverse stakeholders, while others are directed primarily from above.

Germany

Germany is a federal state where powers are shared between the central government and the 16 federal states (Bundesländer). Governance of the twin-transition reflects this federalist principle:

- At the central level the following institutions are found:
 - The Bundestag (federal parliament) enacts laws, approves budgets, and supervises the executive. Parliamentary committees for Environment, Climate Protection and Energy, Transport, and the Digital Agenda shape legislation.
 - The Bundesrat represents state governments at the national level; it can veto legislation affecting state interests.
 - The Federal Chancellor leads the government, shaping and representing national policies. The Federal Cabinet comprises the chancellor and federal ministers, each heading a ministry with specific responsibilities.
- Key Federal Ministries for the Twin-transition:
 1. The Federal Ministry for Economic Affairs and Climate Action (BMWK), focusing on economic and energy policies, Industry 4.0, and hydrogen strategies.

2. The Federal Ministry for the Environment, Nature Conservation, Nuclear Safety, and Consumer Protection (BMUV), integrating environmental considerations and promoting digital solutions for environmental monitoring.
3. The Federal Ministry of Education and Research (BMBF), which drives innovation and funds research in digital and green technologies.

Other important actors include the Federal Ministry of Labour and Social Affairs (BMAS), since it monitors and manages the EU funds, the Federal ministry of Finance (BMF) which monitors and manages the EU funds, and the Federal Ministry of Digital and Transport (BMDV).

For a new policy the legislative process originates in ministries, undergo consultations, and require Bundestag and Bundesrat approval. The implementation is under the responsibility of federal, state, and municipal administrations. Each state sets its own digital and sustainability strategies, adapting national policies to local needs. No single institution oversees evaluations, but ministries conduct pre-implementation assessments. Civil society engagement is strong, and public consultations are frequent.

There is a coordination mechanism in place under the responsibility of the Staatssekretärsausschüsse (State Secretary Committees) allowing senior civil servants from different ministries to align strategies and resolve policy conflicts. The departmental circle process sees draft legislation negotiated at the specialist level, escalating unresolved items to higher political levels. The Federal Constitutional Court ensures laws comply with the Basic Law. It also hears constitutional complaints from citizens.

Shifting to the level of single federal state, the legislative authority is found in certain policy areas (e.g., education, culture, media). Single states also manage land use, building management, and sustainable energy production, sometimes setting stricter targets than the federal government. Many states have their own sustainability and/or digitalization strategies, reflecting diverse regional approaches within the broader federal system. Municipalities execute federal and state regulations and address local tasks like social welfare or school administration.

As far as civil society engagement is concerned Germany the over 660,000 registered organizations operating in sectors including environment, education, or democracy-building are involved via the participation of large umbrella organizations and alliances to parliamentary hearings. To take part organisations should register (according to the Lobby Register Act (effective from 2022)). Citizens' councils (Bürger*innenrat) are also periodically convened to gather direct public feedback on specific issues. Their recommendations are forwarded to the Bundestag, though parliamentarians have final decision-making power. Finally, there are Independent Councils: the National Hydrogen Council (NWR) which advises on implementing the National Hydrogen Strategy and the German Council for Sustainable Development (RNE) which advises the Federal Government on sustainability and fosters societal dialogue.

Comparative Remarks:

Germany stands out for its federalist structure and robust distribution of competences among federal, state, and local levels. This is reminiscent of Spain's Autonomous Communities, though Germany's states have constitutionally enshrined powers and representation in the Bundesrat.

Coordination across multiple layers can be complex, yet it allows states to adopt region-specific approaches. Civil society in Germany is notably active, enjoying both formal and informal avenues for participation.

Germany's Sustainable Development Strategy (DNS) provides a framework for policy alignment, involving advisory councils, parliamentary bodies, and public consultations. However, digital transition policies lack strong sustainability integration, leading to criticism about rebound effects.

Like Spain and Ireland, Germany emphasizes dialogue and public participation, ensuring that a broad range of voices—ranging from civil society to regional actors—can contribute to policy refinement.

Compared to countries with more centralised structures like Portugal, Ireland and Hungary, Germany's federalism allows each state to tailor policies to local conditions. However, while the green and digital strategies are discussed in multiple interministerial and consultative forums, there is also criticism that some initiatives (e.g., aspects of digital policy) do not fully integrate sustainability measures.

Spain

Spain follows a highly decentralised governance model, with significant regional and local government participation. The Sectoral Conference for the Recovery, Transformation, and Resilience Plan ensures coordination between national and regional levels. The governance is articulated as follows.

- At national level:
 - The Parliament legislates and oversees government action every three months according to the Parliamentary Oversight Mechanism, established by the Royal Decree-Law 36/2020.
 - The central Government with several ministries playing key roles in the twin-transition:
 - The Ministry for the Ecological Transition and the Demographic Challenge leads green initiatives (e.g., the National Integrated Energy and Climate Plan).
 - The Ministry for Digital Transformation and Public Function oversees digital policies (telecommunications, AI).
 - The Ministry of Economy, Trade and Entrepreneurship implements the "Spain Digital 2026" agenda.
 - The Inter-ministerial Commission for the Recovery, Transformation, and Resilience Plan coordinates EU NextGenerationEU funds.
 - The Ministry of Equality implements the Strategic Plan for the Effective Equality of Women and Men, ensuring gender considerations are embedded in environmental, digital, and labour market measures more in general by applying regular Equality Impact Assessments Spain integrates gender and social equity into twin-transition policies.
- At regional level the Autonomous Communities adapt national strategies to regional contexts. Their competences include managing digital infrastructure, promoting

eGovernment and environmental programs, and overseeing renewable energy, circular economy, and sustainable transport policies.

- At local level: the municipalities and provincial councils implement policies with direct local impact, especially in environmental competences (air pollution, water resources, waste management) and digital transformation (city-level technological innovation, digital administration).

Civil society participation to policies for the twin-transition is organised at the regional level (Autonomous Communities) through sectoral conferences (e.g., the Open Government Sectorial Commission organises meetings on mobility, energy, digital transformation). At the local and the national level citizens and social partners provide input through public hearings and specialized forums. At local level the Spanish Federation of Municipalities and Provinces (FEMP), plays a role in promoting public consultations. A Dialogue Table to foster collaboration between local, regional, and national authorities, was set up to ensure that local civil society voices can inform policy design and implementation.

Spain's system is characterized by constitutional recognition of strong regional autonomy. Regional actors not only implement policies but also participate actively in their design through sector-specific conferences and dialogue mechanisms. This creates a system where local needs and social equity measures (such as gender impact assessments) are integral to the process. Like Germany and Ireland, Spain's governance process is characterized by multi-level involvement and formal structures that promote transparency and dialogue. Furthermore, in terms of institutions compared to the Germany's federal structure, Spain involves multiple tiers, yet it maintains distinct committees dedicated to open government and dialogue with civil society.

Ireland

Ireland employs a system where the national government leads strategy, policy development, funding, and monitoring. The Irish Parliament (Oireachtas) has legislative oversight to ensure transparent and inclusive policy aligning with national objectives. Governance occurs in three main stages: research and advisory, cabinet decision-making, and policy formulation. As a result, Ireland's twin-transition governance blends central oversight with strong participatory mechanisms. Policy decisions follow structured consultation processes, incorporating advisory bodies and citizens' inputs. The institutions involved are:

- Research and Advice:
 - Government ministers receive input from departmental officials and various public policy institutes (e.g., National Economic and Social Council, Irish Fiscal Advisory Council).
 - Lobbying is regulated under the Regulation of Lobbying Act 2015, requiring lobbyists to register their activities.
- Cabinet Decisions:
 - Important policy proposals are prepared by departmental officials in the form of a Memorandum to Government. Once the cabinet agrees, it adheres to the principle of collective responsibility.
 - The Oireachtas comprises the President, Seanad Éireann, and Dáil Éireann. Committees of the Houses undertake detailed examinations of specific subjects.

- **Formulating Policy:**
 - A 'Green Paper' may be published to outline issues and propose options. Public consultations often follow.
 - A 'White Paper' (or Strategy Document) sets forth final policy directions, to be implemented via legislation or reforms.
- **Green Transition and Cross-Departmental Oversight:**
 - The Department of Environment, Climate and Communications (DECC) coordinates Ireland's annual Climate Action Plan under the Climate Action and Low Carbon Development (Amendment) Act 2021.
 - The Climate Change Advisory Council reviews performance annually, and ministers must explain or outline corrective measures if targets are not met.
 - Multiple cross-departmental taskforces coordinate sector-specific climate actions.
- **Regional Governance:**
 - Local Authorities and Municipal Districts: Elected councils make policy decisions, while municipal districts handle local matters. Four Climate Action Regional Offices (CARO) assist in creating Local Authority Climate Action Plans and assists in monitoring.
 - Regional Assemblies: Coordinate spatial, environmental, and economic planning, also managing EU funding.
- **Citizen Engagement:**
 - Citizens' Assemblies and Climate Dialogues (specific dialogues for Youth are also organised) solicit input at multiple governance levels.
 - Public consultations are common, and the National Climate Stakeholder Forum convenes diverse groups biannually.
 - The National Dialogue on Climate Action (NDCA) hosts workshops, focus groups, and surveys to foster awareness and behaviour change.

Comparative remarks:

Ireland's structured citizen participation differentiates it from countries with more top-down governance models like Hungary and Italy. Similar to Spain and Germany, Ireland relies on consultative processes to include civil society and stakeholder input in its policy formulation. While decision-making remains centralised (e.g., climate action planning), Ireland compensates for limited formal regional legislative powers by prioritising inclusive participation. The model is noted for its clear, staged approach that integrates public opinion directly into the policy cycle, which contrasts with Hungary's less influential consultation processes.

Portugal

Portugal's twin-transition governance is centralised in funding and policy design, but multi-level in implementation. The European Green Deal and Digital Europe goals guide national strategies, which are locally adapted.

- **National Level:**
 - Political level: Key ministries include the Ministry of Environment and Climate Action, the Ministry of Economy and Digital Transition, and the Ministry of Infrastructure. They lead policy formulation under the coordination of the Inter-

Ministerial Commission for Climate Action and the National Digital Transition Task Force. Parliamentary committees oversee legislation and incorporates possible stakeholders' inputs.

- Administrative level: National managing authorities like the Agency for Development and Cohesion (AD&C) manage the implementation of EU-funded programs. Specialized agencies (ADENE, ANI) oversee specific domains (energy efficiency, research funding).
- Portugal's Court of Auditors enforces accountability.
- **Regional Level:**
 - Political level: Regional governments adapt national policies to local contexts. Regional assessors oversee policy implementation in the region. Regional Development Councils coordinate across municipalities and align regional strategies with national priorities.
 - Administrative level: Regional Coordination and Development Commissions (CCDRs) manage regional funds and provide technical assistance.
 - Coordination:
- **Local Level:**
 - Political level: Municipal mayors and representatives tailor policies to local needs and implement them.
 - Administrative level: Local managing authorities handle sustainable infrastructure and digital transformation programs.
 - Coordination: Municipal Sustainability and Digitalization Councils engage stakeholders and plan local projects.
- **Commitments to Equality and Non-Discrimination:**
 - National-level bodies like the Commission for Citizenship and Gender Equality (CIG) and the High Commission for Migration (ACM) ensure policies address structurally vulnerable groups.
 - CCDRs assess gender and social impacts in regional planning, and municipalities often adopt participatory budgeting for inclusive community engagement.
- **Civil Society Engagement:**
 - At the national level, the Economic and Social Council (CES) brings together labour unions, employers' associations, and NGOs to influence policymaking.
 - Regional advisory councils integrate stakeholder input, and local participatory budgets allow citizens to propose and vote on sustainable and digital initiatives.
 - Consultation mechanisms go beyond EU requirements, covering both EU-funded and nationally funded programs.

Comparative remarks:

Portugal's governance shows a centralised financial management structure, yet it maintains multi-tiered coordination to adapt policies regionally and locally. Like Italy and Hungary, Portugal shows a tendency toward centralization in the creation of policy frameworks. Yet, similar to Germany and Spain, it employs consultative and coordination mechanisms that involve regional and local actors. However, Portugal's approach is distinct in its strong emphasis on financial control and monitoring, with a robust role for national oversight bodies. While the centralised design ensures

alignment with EU priorities, the multi-level execution allows for adaptation to local contexts—a balance that differs from Hungary’s more uniformly top-down process.

4.3 Barriers

4.3.1 Governance Processes and Challenges in the Twin-transition: Comparative Analysis

Effective governance for the twin-transition is a complex undertaking, involving multiple governance levels and stakeholders. Access to European Union funds is central, yet efficient utilization remains problematic across Hungary, Italy, Germany, Spain, Ireland, and Portugal. Despite common challenges, notable differences arise in each country's approach and difficulties.

4.3.2 Common Challenges Across Countries

All six countries struggle with similar overarching issues in implementing the twin-transition, prominently:

- Administrative capacity constraints: Limited financial resources, human resources, and technical expertise.
- Complexity in accessing and utilizing EU funds: Extensive paperwork, bureaucratic inefficiencies, and complicated procurement processes.
- Inconsistent coordination: Difficulty in aligning policies across national, regional, and local levels.
- Challenges related to equality and inclusion: Variable integration of gender and social inclusion measures in policies.
- Low public awareness and engagement: Limited public understanding of sustainability and digitalization benefits, leading to resistance.

In summary, each country exhibits distinctive governance challenges influenced by their political and administrative structures. Common threads—such as administrative constraints and the complexity of EU fund implementation—coexist with marked differences in degrees of decentralization, stakeholder involvement, and public participation levels. Addressing these shared challenges while respecting country-specific contexts will be crucial for an equitable and effective twin-transition.

4.3.3 Country-Specific Summaries and Comparisons

Germany

Germany’s federal structure promotes extensive regional autonomy but complicates governance coordination. The twin-transition processes encounter significant bureaucratic complexity and outdated administrative practices, especially digital administration. Limited integration of sustainability measures in digital policies leads to policy conflicts and departmental silos, complicating coherent implementation. Additionally, public resistance—particularly against renewable energy infrastructure—creates hurdles.

- **Commonalities:** Administrative complexity and public resistance issues resemble Hungary's and Ireland's challenges.
- **Differences:** Germany's federalism allows greater local adaptation and citizen participation, unlike Hungary's centralised model. Its notable lag in e-government activities contrasts sharply with other EU countries.

Spain

Spain employs a decentralised, regionally driven governance approach. However, significant administrative capacity disparities between regions hinder uniform policy implementation. Despite established coordination mechanisms (e.g., sectoral conferences), resource and capability gaps remain, exacerbating regional inequalities. The governance process integrates equality assessments actively, yet resource disparities among regions create uneven effectiveness.

- **Commonalities:** Administrative capacity limitations mirror those in Italy and Hungary.
- **Differences:** Spain's extensive decentralization and formal stakeholder engagement mechanisms contrast strongly with Hungary's limited participation.

Hungary

Hungary demonstrates a highly centralised governance process, where national ministries direct policy implementation despite nominally decentralised structures. Regulatory inconsistency with EU directives creates uncertainty, particularly regarding energy policy. Technical-managerial constraints—such as insufficient administrative capacity, limited human resources, and temporary employment linked to EU funding timelines—significantly impede implementation. Low public awareness and limited stakeholder influence further exacerbate these challenges.

- **Commonalities:** Shares administrative capacity constraints and difficulties with EU fund implementation, akin to Italy and Spain.
- **Differences:** Particularly marked by centralised control and low civil society influence, differing significantly from participatory approaches in Ireland and Germany.

Ireland

Ireland employs centralised governance but with strong participatory mechanisms through citizen engagement structures such as Citizens' Assemblies and climate dialogues. However, limited local autonomy, insufficient resources, and limited coordination between national and local governments hinder effectiveness. Challenges include inadequate public and private funding for adaptation measures, insufficient administrative capacity at local levels, and limited policy tools available to local authorities. Additionally, Ireland identifies substantial public resistance and disparities between required adaptation measures and current capacities.

- **Commonalities:** Ireland shares administrative and public participation challenges similar to Germany.
- **Differences:** Ireland uniquely emphasizes structured citizen engagement yet faces limitations due to centralised policy control.

Italy

Italy's governance features dual approaches—highly participatory for Cohesion Funds but strongly centralised and top-down for the Recovery and Resilience Plan (RRP). The primary barriers are inefficient resource management, complicated bureaucratic procedures, procurement difficulties, and limited local administrative capabilities, particularly in central and southern municipalities. Furthermore, gender equality and social inclusion measures, while formally integrated, frequently lack effective implementation mechanisms due to conditional exceptions within tenders.

- **Commonalities:** Like Hungary and Spain, Italy faces critical administrative capacity constraints, impacting resource management and timely implementation.
- **Differences:** Italy uniquely presents a dual-track governance model, creating internal contrasts between centralised and participatory structures.

Portugal

Portugal maintains centralised financial and policy control yet relies heavily on regional and local bodies for implementation. This governance model suffers from fragmented coordination, short-term political cycles, restrictive financing, and budget constraints. Limited digital literacy, inadequate local administrative resources, and restricted access to funding hinder local implementation, particularly impacting vulnerable groups. The complexity of EU regulations and conflicting priorities further complicate effective implementation.

- **Commonalities:** Administrative complexity and funding issues align closely with Spain and Italy.
- **Differences:** Portugal maintains a unique balance of centralised financial oversight and decentralised implementation. Its explicit commitment to addressing inequalities stands out compared to less integrated approaches in Hungary.

4.4 Factors for success

This comparative analysis identifies and evaluates the enabling factors for the twin-transition (digital and green) among six European countries—Germany, Spain, Hungary, Ireland, Italy, and Portugal—highlighting both commonalities and distinctive national differences.

4.4.1 Financial Resources: EU and National Funds

A central enabler across all analysed countries is access to financial resources, specifically European Union (EU) and national funding mechanisms. Despite acknowledged bureaucratic barriers, the availability of significant financial support from the EU and national budgets emerges as a common critical success factor.

- Germany explicitly emphasizes the strategic significance of EU and national funds, notably Cohesion and Just Transition Funds. The substantial financial commitments underline the political prioritization of twin-transition objectives. Furthermore, Germany notably

emphasizes substantial investment in research and innovation, which underpins both transitions.

- Hungary similarly acknowledges that EU resources, particularly cohesion and structural funds, significantly enable transition projects. Hungary explicitly mentions the importance of leveraging EU financial support despite facing administrative challenges in accessing funds.
- Italy highlights European funds, notably through its RRP, as fundamental, particularly due to targeted financial allocations. The effectiveness of these resources, however, heavily depends on overcoming administrative inefficiencies.

Overall, the consistent recognition across all countries illustrates EU funding as a fundamental pillar, though variations exist in administrative capacity to utilize these funds efficiently.

4.4.2 Institutional and Administrative Coordination

Institutional coordination emerges as another critical enabling factor, reflected variably across countries:

- Spain offers robust examples, including a specialized Technical Assistance Fund to directly support local administrations, especially those with fewer resources. Spain also highlights effective coordination and accountability through dedicated structures, such as the Monitoring Unit reporting directly to the Presidency. These mechanisms significantly support smaller municipalities, ensuring equitable regional participation.
- In Ireland, the Department of Environment, Climate and Communications (DECC) is a critical coordinating body, setting clear policy targets and implementing cross-departmental taskforces. This institutional clarity supports the twin-transition through strategic alignment across government departments.
- Italy emphasizes the importance of coordination through its institutional "control room" for managing RRP implementation. The new regulatory structure (Decree-Law 77/2021) strengthens coordination between national and local governance, creating clearer responsibilities and improving policy coherence. Despite criticism of overly centralised decision-making, these centralised mechanisms facilitate coherence and streamlined implementation.
- While Germany does not emphasize central institutional coordination as explicitly as Italy or Spain, its federal structure inherently provides regional flexibility, avoiding excessive centralization and enabling localized initiatives. This federal structure, combined with inter-ministerial cooperation, contributes positively to governance.

4.4.3 International Cooperation and Partnerships

International cooperation significantly facilitates the twin-transition, especially noted by Hungary and Italy:

- Hungary emphasizes international cooperation as an important mechanism to access best practices, technical expertise, and funding. Its engagement in European networks (EEAC, ESDN) provides valuable strategic alignment and resource sharing.

- Italy explicitly notes the advantage of adhering to the European Code of Conduct on Partnership, emphasizing that a pluralistic, multilevel governance approach mitigates policy inequalities and supports effective strategy design reflecting actual social needs.

Thus, while international partnerships emerge as an enabler across these countries, Hungary and Italy explicitly highlight the critical role of these networks in enhancing administrative and strategic capacities.

4.4.4 Democratic Structures and Checks and Balances

The effectiveness of constitutional and democratic structures, particularly regarding checks and balances, is emphasized notably by Germany and Italy:

- Germany stresses the benefits derived from its federalist governance model, notably preventing excessive centralization, promoting regional diversity, and ensuring effective representation of local needs. This governance model also facilitates accountability through clear separation of legislative, executive, and judicial powers. Germany specifically notes that its constitutional court and proportional parliamentary system significantly contribute to policy stability and long-term commitment to the green and digital transitions.
- Italy similarly underscores the role of Parliament as a significant enabler. The Parliament exercises critical control and evaluation during both the drafting and implementation phases of the RRP. Reports submitted regularly by ministries and monitored by the Corte dei Conti demonstrate rigorous oversight and accountability mechanisms. However, Italy's governance tends to be more centralised than Germany's federalist system, causing tensions regarding transparency and civil society participation.

Germany and Italy share a pronounced focus on democratic principles of checks and balances, but Germany's federalist structure contrasts with Italy's centralised governance style, impacting how effectively local needs are met.

4.4.5 Stakeholder Engagement and Social Dialogue

Countries differ significantly regarding stakeholder engagement, but the importance of inclusive governance and public participation emerges as a strong common theme:

- Germany highlights the importance of social dialogue, integrating perspectives of trade unions, employer associations, SMEs, and civil society organizations. Particularly, the role of SMEs is emphasized, suggesting they have considerable potential to influence transitions positively. Germany explicitly advocates differentiated governance structures tailored to the differing nature of green (top-down) and digital transitions (bottom-up), promoting stakeholder engagement as critical for transition legitimacy.
- Ireland strongly promotes public participation, notably through structured dialogues such as "Climate Conversations" and consultation processes embedded within policymaking cycles. This approach emphasizes engaging structurally vulnerable groups, making inclusion a cornerstone of successful governance.

- Spain similarly enhances stakeholder engagement through its Open RTRP platform, allowing direct public participation and transparent project tracking. These participatory structures help address regional disparities by ensuring equitable representation of less-resourced communities.

Italy, while generally adopting a centralised governance approach, acknowledged civil society contributions particularly at the planning stage, also through parliamentary hearings. Despite criticisms of limited transparency, CSOs have actively participated in presenting the needs of vulnerable groups.

Conversely, Hungary faces limitations regarding the depth of stakeholder engagement, highlighting the absence of transparent participatory mechanisms, which differentiates it from countries like Germany, Spain, and Ireland.

4.4.6 Integration of Social and Economic Dimensions

An overarching enabling factor emphasized particularly by Germany and Italy is the integration of social considerations alongside environmental and digital agendas:

- Germany specifically incorporates social considerations into its governance structure through dedicated ministries (e.g., BMAS and BMFSFJ). This ensures that policy development does not overlook impacts on workers, families, and structurally vulnerable populations. The presence of strong trade unions and active employer associations further enhances socially inclusive policy formulation.
- Italy, though acknowledging limitations in effectively implementing gender equality measures within its RRP, demonstrates legislative awareness and effort to integrate these social dimensions into funding mechanisms and policy assessments.

In contrast, countries such as Hungary and Portugal explicitly or implicitly face challenges in consistently integrating these social dimensions, partly due to administrative and political constraints, highlighting internal variation among European countries in effectively embedding social aspects into the twin-transition.

4.4.7 Public Acceptance and Awareness

Public acceptance and awareness of transition processes significantly influence governance outcomes:

- Germany acknowledges public resistance to some green initiatives, stressing the importance of engaging civil society and SMEs as credible sources of expertise. This engagement is seen as critical for increasing acceptance and overcoming barriers.
- Ireland similarly identifies public acceptance and enhanced community consultation as key to overcoming resistance, stressing the role of communication, transparency, and sustained dialogue.
- Hungary acknowledges challenges stemming from low public awareness and engagement, indicating the need for improved communication strategies and increased public awareness.

Thus, Germany and Ireland explicitly recognize public acceptance as crucial, suggesting active dialogue and knowledge dissemination as solutions, whereas Hungary still struggles with weaker public engagement strategies.

4.4.8 Conclusions

Across Germany, Spain, Hungary, Ireland, Italy, and Portugal, factors facilitating the twin-transition converge around the importance of financial resources, institutional coordination, democratic structures, stakeholder engagement, and social inclusivity. Differences are shaped by governance structures: Germany and Ireland excel in democratic participation and multi-level stakeholder engagement, while Hungary and Italy highlight centralization with varying degrees of success. Spain effectively bridges gaps in regional capacities through targeted administrative support. Overall, successful twin-transition governance depends critically on the integration of EU resources, robust institutional mechanisms, effective democratic practices, meaningful stakeholder participation, and comprehensive social integration.

5. Cross-cutting analysis

5.1 Introduction

The dual transition towards a green and digital economy is central to the future development of the European Union. How to manage this development at national level is the main challenge faced by countries, which are adopting differentiated strategies and instruments in their policy actions. As mentioned above, several internal and external factors influence the policy choices and the policy processes put in place by the countries, but the comparative analysis highlights common points for reflection.

5.2 Twin-transition or two transitions?

The twin-transition concept is not yet examined as a unitary concept in the countries of analysis: this data seems to be a general challenge for EU countries that is also due to a lack in the overall approach at EU level. Although several EU regulations address energy and IT independently, there is a notable legislative gap where these two systems intersect. While it is common for legislation to lag technological or societal advancements, this gap is particularly striking due to the absence of clear articulation of public values underpinning the twin-transition (Niet, 2023).

Across all six countries, the concept of a "twin-transition", integrating digital and green transformations, emerges more as two parallel processes rather than one fully unified strategy. Each country faces distinct governance and structural factors that influence the degree of integration.

A clear pattern is observed across all analysed countries: Portugal, Germany, Hungary, Italy, Ireland, and Spain all demonstrate difficulties in merging the green and digital transitions into one comprehensive, coherent policy framework. Despite the widespread acknowledgment of the relevance of integrating both dimensions, the actual operational and strategic approaches generally maintain clear separations:

- All countries report institutional fragmentation as a barrier to a fully integrated twin-transition. Ministries and administrative units responsible for environmental sustainability and digital transformation often operate independently, with separate strategic agendas and limited cross-sectoral collaboration.
- Despite overarching European frameworks and national strategic references to the "twin-transition", implementation remains segmented into distinct digital and green pathways. Ministries typically manage separate budgets, projects, and regulatory frameworks dedicated explicitly either to digital transformation or sustainability.
- Countries like Portugal, Spain, and Hungary explicitly highlight existing institutional mechanisms (e.g., inter-ministerial committees or commissions) aiming at cross-sectoral cooperation yet also acknowledge that the practical integration achieved remains minimal.

Despite these overarching similarities, countries display notable distinctions. Portugal and Hungary emphasize political fragmentation and the existence of ministerial silos as the main barriers preventing a cohesive twin-transition. Portugal explicitly points to the limited cross-sectoral cooperation even though institutional mechanisms to promote integration are formally in place. Germany highlights a structural difference in the nature of the transitions themselves, distinguishing between a green transition driven primarily by top-down governmental policies and a digital transition largely driven by bottom-up, private economic initiatives. This fundamental distinction between the actors and mechanisms behind each transition further accentuates the division. Italy describes its situation as particularly influenced by European Union policies, noting that although EU funding (such as through the RRP and Cohesion funds) promotes integration in principle, the actual governance and implementation structures at national and local levels remain separate and distinct, with RRP governance being significantly more centralised than that for Cohesion funds. Ireland clearly separates green and digital agendas within strategic documents (e.g., NRRP), reflecting an explicitly parallel approach without meaningful attempts at integration. Cross-references exist but are relatively superficial and do not form a coherent, integrated framework. Spain explicitly allocates distinct shares of financial resources (40% green, 26% digital) under its Recovery and Resilience Plan, reinforcing separate tracks.

Despite structured coordination mechanisms, practical execution favours separate approaches, suggesting administrative convenience or structural preference rather than fully integrative strategies. In conclusion, while the idea of a fully integrated "twin-transition" is conceptually acknowledged across all countries, institutional fragmentation, distinct operational frameworks, and differing governance approaches predominantly result in two separate transitions—green and digital—rather than one genuinely integrated twin-transition.

5.3 National policymaking is framed by the EU

The second point is the high relevance of EU policies, regulations and directives for the national policies, contents and strategies examined in the context of the comparative analysis on the governance of twin-transition. In fact, without claiming to be exhaustive, in the field of energy, the European Governance Regulation (EU) 2018/1999 ensures effective opportunities for the public to participate in the preparation of national plans and strategies as well as establishing rules underpinning the states' energy and climate plan. Also, the delegated regulation (EU) No 240/2014 establishes conduct on partnership in the framework of the European Structural and Investment Funds. It imposes clear governance guidelines on partners, identifying, inter alia,

relevant partners for programs (competent regional, local, urban and other public authorities, economic and social partners, bodies representing civil society, such as environmental partners, non-governmental organisations, and bodies responsible for promoting social inclusion, gender equality and non-discrimination) and providing key principles and good practices (consultation processes, information duties, monitoring and reporting practices. With reference to the Just transition fund, mentioned by many countries of analysis, the regulation (EU) 2021/1056 establishes that Member States shall prepare, together with the relevant local and regional authorities of the territories concerned, one or more territorial just transition plans, which should be prepared in social dialogue and cooperation with the relevant stakeholders, in accordance with the relevant provision of Regulation (EU) 2021/1060 on partnership.

This is reflected in the way countries understand and build Multi-Level Governance (MLG) systems. Despite the national differences highlighted, in general there seems to emerge a mutual recognition of the “advanced” multilevel governance, intended as collaboration, shared responsibility, and collective decision-making with active involvement and cooperation of a multiplicity of actors and stakeholders in the decision-making and implementation processes. This concept include non-governmental organisations and civil society more broadly, private sector entities, as well as intergovernmental cooperation structures at various levels ([CDDG(2023)12]. The embedding of this MLG model is mostly evident in Germany (e.g. German Digital Strategy), Ireland (e.g. National Adaptation Framework 2024 governance) and Spain (Renewable Energy Deployment and Integration governance) cases, where the NGOs and civil society organizations are broadly and regularly involved in the policy making processes.

More in details across all six analysed countries (Portugal, Germany, Hungary, Italy, Ireland, and Spain), it emerges clearly that national policymaking concerning the twin-transition is extensively influenced and shaped by European Union frameworks. EU funds, strategies, and directives exert significant influence on national policy agendas, particularly through the financial instruments provided by the EU, such as the Recovery and Resilience Facility (RRF), Just Transition Fund, Cohesion Funds, REACT-EU, and European Structural and Investment Funds (ESIF).

- Across all cases, the availability and conditionalities of EU funding substantially guide national policy directions. Countries such as Spain, Italy, and Portugal explicitly link their national policy frameworks to European funds and strategies, highlighting how these mechanisms influence domestic decisions on investment priorities, administrative structuring, and strategic planning.
- Strategic Alignment with EU Goals: Each country explicitly aligns national twin-transition objectives with EU strategic guidelines like the European Green Deal, Digital Europe Programme, and national implementation plans required by the EU (e.g., National Energy and Climate Plans - NECP). This alignment creates a common policy foundation across all analysed countries, fostering coherence in national-level strategies within a broader supranational framework.

However, despite these overarching commonalities, the degree, nature, and implications of EU influence differ in magnitude and details among the countries. Portugal and Hungary demonstrate a strong dependence on EU financial support and clearly identify it as essential for implementing transition policies. Portugal’s alignment is guided closely by European strategic frameworks, especially with regard to funding distribution and regulatory compliance. Germany also

acknowledges the substantial role of EU frameworks but highlights the coexistence of national resources alongside EU funding. Italy strongly emphasizes the influence of the EU framework, noting extensive reliance on EU guidelines and funds. Spain also recognises EU strategic and financial support as a primary driver for national initiatives and aligns strongly with EU strategies. Ireland similarly confirms significant alignment with EU frameworks through explicit strategic references in the National Recovery and Resilience Plan (NRRP), indicating heavy reliance on EU financial instruments for both green and digital transitions. However, Ireland particularly emphasizes sectoral and strategic documents (e.g., CAP24) referencing EU-mandated climate action targets, explicitly linking national goals with supranational directives and obligations.

5.4 Multilevel governance, coordinating the complexity

According to Tortola (2017) the ongoing process of multilevel governance (MLG) entails the issue of the capacity of coordination between regional and national levels, with an impact on structurally vulnerable groups and internal remote areas. This lies in the increased prominence of regions within the EU since the Maastricht Treaty of 1992, and even earlier with the revision of cohesion policies in 1988. This initiated a process of regionalization, transferring political authority from national governments to sub-national levels and enhancing regional participation in EU affairs both within and beyond member states. A key analytical focus of MLG is its implicit shift from formal government decision-making to the distribution of authority across different levels of government throughout the entire policy process, from formulation to implementation (Schakel 2020; Hooghe and Marks; Tortola, 2017).

Multilevel governance for the twin-transition across all six countries (Portugal, Germany, Hungary, Italy, Ireland, and Spain) involves interactions among national, regional, and local authorities, as well as structured mechanisms for policy coordination. Each country employs inter-institutional bodies, committees, and commissions to coordinate and implement transition policies effectively. Typically, higher-level institutions provide strategic direction and financial oversight, while regional and local actors focus on policy adaptation and direct implementation. In spite of the established institutional frameworks, all six countries identify significant coordination challenges, primarily linked to administrative complexity, resource disparities, political fragmentation, and limited local autonomy.

Even if several commonalities could be highlighted among the six countries, some specific characteristics emerge:

- Germany features a distinctly federal system where multilevel governance involves federal, state (Länder), and local governments. Its federal structure provides considerable autonomy and legislative competences to states, facilitating localized adaptation. However, Germany explicitly notes the complexity and difficulty of coordinating between these multiple governmental layers, citing political fragmentation, overlapping competences, and administrative silos as significant barriers. While federal-state mechanisms exist (e.g., GRW funds, inter-ministerial committees), practical intergovernmental coordination remains challenging and often fragmented.
- Spain has a notably elaborate and decentralised governance structure, with significant autonomy for Autonomous Communities. Coordination is institutionalized through sectoral conferences and inter-ministerial commissions. Nonetheless, effective

coordination faces major challenges due to significant regional disparities in administrative capacity and resources. While some regions are well-equipped to implement transition policies effectively, others face critical constraints, complicating efforts to achieve coherent multilevel governance.

- Hungary, in contrast, faces unique challenges related to its highly centralised governance model, where the central government exercises significant power over policy directions and funding. Despite formally having regional structures, real authority and autonomy at regional and local levels are limited, resulting in practical difficulties coordinating across governance layers. The centralised governance approach, driven politically, severely restricts genuine engagement and autonomy of local and regional authorities, weakening multilevel governance effectiveness and responsiveness to local needs.
- Ireland has established clear mechanisms for coordination across governance levels, including cross-departmental taskforces and climate-action regional offices (CAROs). However, in practice, the effectiveness of these coordination mechanisms is constrained by Ireland's highly centralised governance structure. National-level policymaking predominates, leaving local governments with limited autonomy and resources. Challenges persist due to the lack of robust coordination frameworks between national and subnational actors, thus limiting the overall efficacy of multilevel governance.
- Italy employs a structured multilevel governance framework, especially prominent in cohesion policies, involving regional authorities, local governments, and stakeholders in policy design and implementation. Despite established multilevel institutions such as the State-Regions Conference, the RRP governance tends to be more centralised and top-down. Italy highlights coordination difficulties due to limited administrative capacities and resources, notably in southern regions. Additionally, fragmented implementation mechanisms create significant barriers, limiting local effectiveness and complicating seamless inter-institutional cooperation.
- Portugal emphasizes multilevel governance through regional bodies like the Commissions for Coordination and Regional Development (CCDRs), responsible for adapting national policies into regional and local contexts. While Portugal has structured coordination bodies aimed at harmonizing regional initiatives with national objectives, coordination is hindered by disparities in administrative capacities, particularly affecting less-resourced municipalities and rural areas. Administrative fragmentation and the complexity of EU compliance further complicate effective multilevel coordination.

5.5 Structurally vulnerable groups: involvement and needs' addressing

The embedding of participatory processes does not directly result in fully inclusive policies, able to address the needs and wishes of civil society and in particular structurally vulnerable groups. This issue is relevant because, with regard to structurally vulnerable groups, some policies seem to be set up in a theoretical way, which can affect their social acceptance, especially by the most structurally vulnerable groups. In this sense, by example, according to the study by Segreto, Principe, Desormeaux and others (2020), which focuses on the social acceptance of renewable energies, while the global benefits of renewable energies are well known, there are still some concerns about their impact on the local environment. Even though non-renewable energy sources generally have an even worse environmental impact, the general uncertainty about the local impact of renewable energy installations has a negative impact on social acceptance and

might also negatively impact the living conditions of structurally vulnerable groups. In this context, social acceptance appears to be a significant barrier to achieving renewable energy targets.

Also, for this topic, the approaches adopted in the case study countries vary in accordance with national features and adopted government and governance systems.

- Germany features structured participatory mechanisms, such as the Forum Nachhaltigkeit, which engages stakeholders including labour unions and civil society. However, gatekeeping by federal authorities often limits access for structurally vulnerable groups. Recent policies show some legal and consultative advancements, such as public consultations on climate protection programs. An example is the good practice on Citizen participation in the Federal Ministry for the Environment, Nature Conservation, Nuclear Safety and Consumer Protection (BMUV in Dialogue) which engages stakeholders including labour unions and civil society. For Germany, key challenges include inadequate mainstreaming of an understanding of structural vulnerability and social inclusivity. Despite limitations this is a good example. It is analysed in-depth in Annex III a briefly described in Box 1 at the end of this section.
- Spain places considerable emphasis on structurally vulnerable groups in its twin-transition efforts. Specific initiatives, such as the Just Transition Agreements, address gender, employment, and digital access gaps in affected regions. These are tools implemented in Spain to respond to the socio-economic impacts of coal closure in the different regions. There exists a tool of co-governance that proposes support instruments for the process of just transition, based on public participation. In this framework, efforts like prioritizing women's employment and adapting infrastructure for older people and isolated communities stand out as examples of targeted support. In addition, Spain actively promotes inclusive governance structures, including mechanisms like the Open Government Sectorial Commission and Dialogue Tables, which facilitate civil society involvement. In Annex III of this report the good practice on Safe, Sustainable, and Connected Mobility Strategy 2030 points to the positive experiences about voicing structural vulnerable groups and women in planning mobility. A short summary of the good practice is included in Box 2 at the end of this section.
- In Hungary national governance often lacks a clear prioritization of social equality, partly due to centralised power dynamics that overshadow regional and local involvement. In fact, governance is centralised under the Prime Minister's office, with limited meaningful engagement from local governments or civil society.
- Ireland's approach integrates social justice considerations into its climate and digital policies. The Just Transition framework supports reskilling and energy poverty alleviation, while citizen assemblies have addressed equitable measures for climate action. Structurally vulnerable groups such as rural populations and workers in transition sectors are explicitly considered. Ireland showcases robust participatory governance through its Climate Citizens' Assembly and public consultations, ensuring inclusivity in policy deliberations. Two interesting examples are included in Annex III of this report on good practices, the analysis of Government of Ireland Energy Poverty Action Plan 2022 showcases how participation can effectively contribute to policy making. The good practice is summarised in Box 3 at the end of this section.
- Italy incorporates structurally vulnerable groups into its National Recovery and Resilience Plan (RRP) primarily through targeted policies such as gender equality certifications and

tender requirements promoting inclusion. However, actual results for these groups are limited. For instance, specific gender-focused measures show a meagre impact. In this sense, even if some measures have been adopted, the weak point seems to be represented to the fact that the measures introduced remain at the institutional/organizational level, without targeting policy content/impact and without proposing gender indicators for policy impact assessment. In this sense, as highlighted, although some stakeholder involvement is observed at institutional level, the inclusion of structurally vulnerable groups representatives in the governance of green/digital transition remains limited. Social organizations and local authorities participate mainly at the implementation level, with limited influence on decision-making. Outside the governance mechanisms, civil society organisations are active and engaged in monitoring government actions, but their demands do not always seem to be taken up in the institutional setting (see Data to count report).

- In Portugal, addressing the needs of structurally vulnerable groups within the twin-transition involves dedicated institutional mechanisms and targeted programs. The Portuguese High Commission for Migration specifically ensures that digital and green transition policies account for marginalized populations, such as low-income individuals, older people, and rural residents who often face significant digital divides and energy poverty. Additionally, initiatives like digital literacy training, green job training, and social impact assessments are conducted across governance levels. Civil society organizations and social partners actively represent structurally vulnerable groups through structured consultations, contributing to inclusive policymaking and ensuring equitable implementation. Nevertheless, administrative capacity disparities between regions remain challenging.

With reference to the six countries included in this analysis, the need for additional efforts in this direction clearly emerged with a particular focus on elements that can facilitate the involvement of citizens and also disadvantaged groups. Even in cases where the normative theories of public participation are applied, i.e. the principles of more dialogue, decision-making power, diversity and deliberation are met, policies, as emphasised by Perlaviciute (2021) with reference to the climate issue, may not be enough if they do not take into account if, when and how people want to participate.

Box 1: BMUV in Dialogue (Germany)

This is a cluster of initiatives that integrate diverse participants into policymaking and exchange. They involve citizens, researchers, civil society, and public/private stakeholders, both online and in-person, bridging multiple governance levels. However, the BMUV retains steering authority, limiting contributions. Participation usually covers only selected policy stages, and diversity criteria are not strategically embedded, leading to a focus on academic expertise over everyday experience. This can reproduce power imbalances. Despite some guidelines for reaching underrepresented groups, transparency issues persist, risking contradiction of the initiative's principles. Yet there is intent to bolster inclusivity for youth and less formally educated participants.

Box 2: Safe, Sustainable, and Connected Mobility Strategy 2030 (Spain)

This strategy demonstrates a firm commitment to gender equality and support for vulnerable groups in transport. Multi-level governance actions, including the OTLE and Inclusive Mobility Week, address inequalities in accessibility, gender-sensitive planning, and rural connectivity. Through digital tools, events, and direct stakeholder inputs, the late-2020 Open Mobility Dialogue engaged civil society, businesses, and advocacy groups, collecting diverse perspectives on mobility priorities for policy planning. However, further institutional learning on everyday transport barriers should be further developed to close the gap between policy and practice.

Box 3: Government of Ireland Energy Poverty Action Plan 2022

Within the Action Plan, stakeholders were actively engaged through broad consultations, especially with vulnerable groups, and a cross-departmental approach. The Plan's design and updates incorporate input from various representatives, demonstrating a commitment to inclusive policy making. A dedicated Steering Group and robust research efforts, notably with the Economy and Social Research Institute, reflect evidence-based engagement. Rapid market research during the communication campaign further acknowledges the importance of practical insights. While the Plan's emphasis on communications and partnerships with organizations like Money Advice and Budgeting Service and the non-governmental organisation ALONE is positive, additional collaboration with other underrepresented groups is advisable. Overall, the practice shows that an active engagement of different actors is possible, and that bottom-up involvement leads to improved policy making.

6. Concluding remarks

6.1 Improving the integration between the Digital and Green Transitions

The analysis reveals that all six countries— Germany, Spain, Hungary, Ireland, Italy, and Portugal— currently manage digital and green transitions as parallel processes, rather than as a fully integrated "twin-transition." To enhance integration, all countries would benefit from further dismantling institutional fragmentation and fostering cross-sectoral collaboration. However, the pathway to integration varies: Portugal, Hungary, and Ireland emphasize improving internal coordination mechanisms; Germany suggests harmonizing top-down policy initiatives with bottom-up innovation; Italy and Spain highlight the importance of integrating distinct policy and funding mechanisms explicitly.

6.2 EU Policy Framing in Member States of the Twin-transition

In summary, all six countries underscore significant EU influence on national twin-transition policymaking, primarily through EU financial mechanisms and strategic frameworks. However, differences arise regarding the degree of reliance and institutional embedding of EU frameworks. While Spain, Hungary, Italy, Portugal, and explicitly depict EU funding as indispensable, Germany portrays EU support as critical yet complemented by strong national resources. Italy uniquely highlights the constitutional alignment with EU sustainability obligations. Ireland acknowledges

EU frameworks in strategic documents but places relatively less emphasis on financial dependency compared to countries like Hungary and Portugal. Despite these nuances, the overarching conclusion remains that EU frameworks are vital, guiding national policy directions across all six countries studied.

6.3. Improving Multilevel Governance and Coordination

All countries face challenges in effectively implementing multilevel governance, yet their proposed recommendations for addressing these issues reveal interesting differences mirroring their specific governance models. Overall, while strengthening administrative capacities emerges as a common necessity, the paths toward better multilevel governance differ significantly: Portugal, Italy, and Spain recommend capacity-building measures targeted at local disparities; Germany highlights clarification of inter-level responsibilities; Ireland calls for greater local empowerment; Hungary emphasizes decentralization and increased local autonomy.

6.4 Enhancing Structurally Vulnerable Groups' Involvement and Addressing their Needs

The comparative analysis of structurally vulnerable groups' involvement in twin-transition policies reveals significant diversity among Italy, Germany, Spain, Ireland, Hungary, and Portugal, highlighting both shared barriers and distinct national approaches that suggest tailored recommendations to achieve inclusive, participatory policy structures for a socially accepted and fair transition, addressing concerns and conditions of the most excluded members of society.

Italy and Hungary share a need for substantial improvement in the genuine involvement of structurally vulnerable groups. Italy demonstrates formal inclusion through institutional policies and frameworks, such as consultations and gender equality certifications, yet these measures often fail to translate effectively into policy outcomes. Similarly, Hungary's centralised governance system severely limits regional and local representation, significantly reducing meaningful engagement with structurally vulnerable groups.

Germany exhibits structured participatory mechanisms, including forums involving civil society and labour representatives, which can be assumed as a good example. However, gatekeeping and institutional barriers limit meaningful involvement for structurally vulnerable groups. Germany would benefit from systematically mainstreaming structural vulnerability considerations and inclusivity principles across all participatory governance mechanisms.

Spain and Portugal present relatively advanced institutional mechanisms explicitly designed to address the needs of structurally vulnerable groups. While Spain's approach can serve as a model for countries facing similar structural transition impacts, Portugal's persistent regional administrative disparities require targeted capacity-building measures to ensure equitable policy implementation across all territories, addressing the specific needs of rural and economically disadvantaged areas. Spain's Just Transition Agreements, targeting gender equality, employment, and digital access can be considered as a good practice.

Ireland's experience highlights the importance of ensuring that deliberative processes (through Citizens' Assemblies and structured public consultations) are substantively reflected in

subsequent policy decisions, as demonstrated by parliamentary integration of citizen recommendations into Ireland's Climate Action Plan. The Irish example can be signalled as a good practice. However, to ensure sustained inclusivity, Ireland should further institutionalize ongoing dialogue mechanisms focused specifically on structurally disadvantaged and rural groups, ensuring continuous feedback loops between citizens' deliberations and policy implementation.

6.5 Overall conclusions

A cohesive governance model for the twin transition rests on three interlinked elements: broad stakeholder engagement, well-structured multi-level coordination, and explicit mechanisms to address structural vulnerabilities. First, consistent public participation opportunities—forums, assemblies, and consultations—are vital to capturing diverse perspectives and grounding policy in lived realities. When civil society, local communities, and socially marginalised groups help shape decisions, transparency and accountability flourish. Ireland's citizens' assemblies and Spain's Just Transition Agreements illustrate structured processes for inclusive input.

Second, robust coordination across local, regional, national, and EU tiers is needed to harmonise digital and environmental objectives. While each country integrates multiple layers of governance, fragmentation still creates inefficiencies. Setting up cross-departmental committees, as seen in Germany and Portugal, can clarify responsibilities and align top-down funding priorities with bottom-up innovation. Such mechanisms ensure synergy between climate action and digital development, even as they accommodate regional differences and resource disparities.

Finally, explicit support for structurally vulnerable groups must be hardwired into policy. Tailored training, inclusive budgeting, and targeted technical assistance help guarantee that digital expansion and green transitions do not replicate inequality. Emphasising gender, rural, or minority perspectives—such as Spain's focus on employment for women in coal-transition areas—bridges policy gaps. Evidence-based approaches, including regular impact monitoring, further maintain democratic legitimacy, since outcomes are measured against equity goals.

Altogether, by investing in inclusive decision-making, improving coordination among governance levels, and embedding social equity into every stage of the policy cycle, EU Member States can yield coherent strategies that marry environmental sustainability with digital progress. This integrative approach ultimately strengthens democracy, fosters accountability, and ensures that the twin transition serves all citizens—not just those already equipped to thrive in changing socio-economic landscapes.

6. 6 Specific recommendations by partners

Here below the specific recommendations formulated by partners

<i>Germany:</i>	<i>Spain:</i>	<i>Hungary:</i>	<i>Ireland:</i>	<i>Italy:</i>	<i>Portugal</i>
• Strengthen the cross-cutting role of social ministries in policymaking. • Establish advisory bodies for participatory governance. • Expand research on the social implications of the transition.	• Strengthen coordination mechanisms to reduce regional disparities. • Continue expanding digital literacy initiatives targeting rural populations.	• Reform governance structures to reduce centralization. • Address rule of law concerns unlock EU funding. • Enhance public engagement through targeted communication and education.	• Decentralize governance to enhance regional adaptability. • Focus on structurally vulnerable groups within the Just Transition framework. • Embed social equality into all levels of transition planning.	• Promote civic participation and access to monitoring tools. • Integrate gender-disaggregated data into planning. • Simplify administrative processes to ensure equitable regional access.	• Establish dedicated governance bodies to integrate digital and green transition strategies. • Increase local administrative capacity through financial and technical assistance. • Expand participatory governance mechanisms to enhance CSO involvement. • Implement measures to bridge the digital divide and alleviate energy poverty. • Strengthen transparency and accountability frameworks in policy implementation.

Literature

- Bingham, L. B., Nabatchi, T., and O'Leary, R. (2005). The new governance: Practices and processes for stakeholder and citizen participation in the work of government. *Public administration review*, 65(5), 547-558.
- Katsamunska, Polya. "The concept of governance and public governance theories." *Economic alternatives* 2.2 (2016): 133-141.
- Jordan, A., and Adelle, C. (2012). 20 EU environmental policy at 40. *Environmental Policy in the EU: Actors, institutions and processes*, (10), 369.
- Knill, C., and Tosun, J. (2020). *Public policy: A new introduction*. Bloomsbury Publishing.
- Marks, G., and Hooghe, L. (2003). Chapter two: contrasting visions of Multi-level Governance. *Multi-Level Governance: Interdisciplinary Perspectives*, edited by Ian Bache and Matthew Flinders Eds., Oxford Uni.
- Peters, B. G., and Pierre, J. (2006). Governance, accountability and democratic legitimacy. In *Governance and Democracy* (pp. 29-43). Routledge.
- Peters, BG, and Pierre, J. (2001). Sviluppo nelle relazioni intergovernative: verso una governance multilivello. *Policy and Politics*, 29(2), 131.
- Pierre, J., and Peters, BG (2020). *Governance, politics and the state*. Bloomsbury Publishing.
- Piattoni, S. (2006). Informal governance in structural policy. *Perspectives on European Politics and Society*, 7(1), 56-74.
- Radaelli, C., & Franchino, F. (2004). Analysing political change in Italy. *Journal of European Public Policy*, 11(6), 941-953.
- Schakel, A. H. (2020). Multi-level governance in a 'Europe with the regions'. *The British Journal of Politics and International Relations*, 22(4), 767-775.
- Segreto, M., Principe, L., Desormeaux, A., Torre, M., Tomassetti, L., Tratzi, P., ... & Petracchini, F. (2020). Trends in social acceptance of renewable energy across Europe—a literature review. *International journal of environmental research and public health*, 17(24), 9161.
- Torring, Jacob. *Interactive governance: Advancing the paradigm*. Oxford University Press, 2012.
- Tortola, P. D. (2017). Clarifying multilevel governance. *European Journal of Political Research*, 56(2), 234-250.

Literature by country

Germany

Außendorf, M. (2022, 08 31). Digitalstrategie lässt Luft nach oben: Zivilgesellschaft und Nachhaltigkeit müssen jetzt gestärkt werden. Retrieved 08 21, 2024, from <https://www.gruene-bundestag.de/presse/pressemitteilungen/digitalstrategie-laesst-luft-nach-oben-zivilgesellschaft-und-nachhaltigkeit-muessen-jetzt-gestaerkt-werden>

BDI. (2022). BDI-Unternehmensbefragung zu Genehmigungsverfahren nach Bundesimmissionsschutzgesetz (BImSchG). Berlin: BDI.

BMBF. (2019). Natürlich. Digital. Nachhaltig. Ein Aktionsplan des BMBF. Bonn: BMBF.

BMDV. (2022). Monitoring. Retrieved from <https://digitalstrategie-deutschland.de/monitoring/>

BMF. (2022, 07 25). Deut-scher Auf-bau- und Re-si-li-enz-plan (DARP). Retrieved from <https://www.bundesfinanzministerium.de/Content/DE/Standardartikel/Themen/Europa/DARP/deutscher-aufbau-und-resilienzplan.html>

BMU. (2020). Umweltpolitische Digitalagenda. BMU.

BMWK. (2023). Digitalpolitische Maßnahmen des BMWK für mehr Nachhaltigkeit. Berlin: BMWK.

Bundesregierung. (2016). Deutsche Nachhaltigkeitsstrategie. Neuauflage 2016. Retrieved from <file:///C:/Users/franz/Downloads/deutsche-nachhaltigkeitsstrategie-neuauflage-2016-download-bpa-data.pdf>

Bundesverfassungsgericht. (2021, 04 29). Verfassungsbeschwerden gegen das Klimaschutzgesetz teilweise erfolgreich. Retrieved 08 21, 2024, from <https://www.bundesverfassungsgericht.de/SharedDocs/Pressemitteilungen/DE/2021/bvg21-031.html>

Deutscher Bundestag. (2019, 11 15). Bundestag nimmt das Klimapaket der Koalition an. Retrieved 08 20, 2024, from <https://www.bundestag.de/dokumente/textarchiv/2019/kw46-de-klimaschutzgesetz-freitag-667244>

Deutscher Bundestag. (2019, 11 13). Regierung legt ihr Bundes-Klimaschutz-gesetz vor. Retrieved 08 21, 2024, from <https://www.bundestag.de/dokumente/textarchiv/2019/kw46-de-bundesklimaschutzgesetz-667190>

Deutscher Bundestag. (2022, 09 22). Kritik an schwammigen Zielen der Digitalstrategie. Retrieved 08 21, 2024, from <https://www.bundestag.de/dokumente/textarchiv/2022/kw38-de-digitalstrategie-909088>

Deutscher Bundestag. (2023). Haushalt 2024: Kürzungen im Digital-Etat. Retrieved from <https://www.bundestag.de/presse/hib/kurzmeldungen-963432>

Deutscher Bundestag. (2023, 11 08). Sachverständige kritisieren Bundes-Klimaschutzgesetz. Retrieved 08 20, 2024, from <https://www.bundestag.de/dokumente/textarchiv/2023/kw45-pa-klimaschutz-klimaschutzgesetz-974134>

Deutschlandfunk. (2021, 04 13). Deutsches Klimagesetz in Teilen verfassungswidrig. Retrieved 08 21, 2024, from <https://www.deutschlandfunk.de/erfolgreiche-klimaklagen-deutsches-klimagesetz-in-teilen-100.html>

European Commission. (2023). Shaping Europe's digital future. Retrieved 08 20, 2024, from DESI 2022 composite index: https://digital-decade-desi.digital-strategy.ec.europa.eu/datasets/desi-2022/charts/desi-composite?indicator=desi_sliders&breakdownGroup=desi&period=2022&unit=pc_desi_sliders

Falck, O., Czernich, N., Pfaffl, C., Ruthardt, F., & Wöfl, A. (2021). Benchmarking Digitalisierung in Deutschland. Munich: IFO.

German Federal Government. (2023). Steuerung der Deutschen Nachhaltigkeitsstrategie. Retrieved 08 21, 2024, from <https://www.bundesregierung.de/breg-de/themen/nachhaltigkeitspolitik/steuerung-nachhaltigkeitsstrategie-419776>

Grossmann, K. (2019). Using conflicts to uncover injustices in energy transitions: The case of social impacts of energy efficiency policies in the housing sector in Germany. *Global Transitions*, 1, 148-156.

Heyen, D. A. (2019). Governance-Ansätze für nachhaltige Transformationen auf dem Prüfstand dreier Praxisfelder. *GAIA* 28/S1, 198 – 203.

Hummel, S., Pfirter, L., & Strachwitz, R. G. (2022). *Civil Society in Germany: A Report on the General Conditions and Legal Framework*. Berlin: Maecenata Institut für Philanthropie und Zivilgesellschaft.

Kelly R., Mbah M. (2024). Regional energy transition governance for the co-transformation of sustainable energy infrastructure as a public service in rural areas. *RuR* 82(2), pp. 127-42. Available at: <https://doi.org/10.14512/rur.1729> .

Kemmerzell, J. (2023). Herausforderungen demokratischer Governance in der Energie- und Klimapolitik. *regierungsforschung.de*, pp. 3-9.

Knoefl, J., Sagebiel, J., Yildiz, Ö., Müller, J. R., Rommel, J. (2019). A consumer perspective on corporate governance in the energy transition: Evidence from a Discrete Choice Experiment in Germany. *Energy Economics* 75, pp. 440-448. Available at: <https://doi.org/10.1016/j.eneco.2018.08.025> .

Liebig, L. G. (2024). Subnational AI policy: shaping AI in a multi-level governance system. *AI & Soc* 39, 1477–1490.

Marschall, S. (2018). *Das politische System Deutschlands*. München: UVK Verlag.

Pulignano, V., Hauptmeier, M., & Frans, D. (2023). Determinants of union strategies towards the twin digital and green transitions in the German and Belgian automotive industry. *Transfer: European Review of Labour and Research*, 29(1), 121-138.

Schaller, M., Redlich, M.-C. & Fischer, F., 2023. Chapter 16 - Digital health policies and participation for digital transformation in German-speaking countries: A critical approach. In: P. Ordóñez de Pablos & X. Zhang, eds. *Information Technologies in Healthcare Industry, Accelerating Strategic Changes for Digital Transformation in the Healthcare Industry*. London: Academic Press, pp. 323-362. Available at: <https://doi.org/10.1016/B978-0-443-15299-3.00016> -

Grossmann, K. (2019). Using conflicts to uncover injustices in energy transitions: The case of social impacts of energy efficiency policies in the housing sector in Germany. *Global Transitions* (1), pp. 148-156. Available at: <https://doi.org/10.1016/j.glt.2019.10.003> .

Schubert, K., & Klein, M. (2020). *Das Politiklexikon*. Lizenzausgabe Bonn: Bundeszentrale für politische Bildung.: Dietz.

Schubert, P., Tahmaz, B., & Krimmer, H. (2023). *Erste Befunde des ZiviZ-Survey 2023 Zivilgesellschaft in Krisenzeiten: Politisch aktiv mit geschwächten Fundamenten*. Berlin: ZiviZ im Stifterverband.

SPD, B. G. (2021). *Mehr Fortschritt wagen – Bündnis für Freiheit, Gerechtigkeit und Nachhaltigkeit. Koalitionsvertrag 2021–2025 zwischen SPD, BÜNDNIS 90/DIE GRÜNEN und FDP*. Berlin. Retrieved 08 12, 2024, from <https://www.spd.de/koalitionsvertrag2021>

Spiegel. (2019, 10 09). Klimapakete und Klimaschutzgesetz vom Bundeskabinett angenommen. Retrieved 08 21, 2024, from <https://www.spiegel.de/politik/deutschland/klimapakete-und-klimaschutzgesetz-vom-bundeskabinett-angenommen-a-1290659.html>

Tagesschau. (2024, 04 26). Bundestag beschließt Reform des Klimaschutzgesetzes. Retrieved 08 20, 2024, from <https://www.tagesschau.de/inland/bundestag-klimaschutzgesetz-102.html>

Tagesschau. (2024). Deutsche Behörden verzetteln sich bei Digitalisierung. Retrieved 08 21, 2024, from <https://www.tagesschau.de/wirtschaft/digitales/digitalisierung-verwaltungen-deutschland-probleme-100.html>

tagesschau. (2024, 04 12). Warum Wissing Fahrverbote ins Spiel bringt. Retrieved 08 07, 2024, from <https://www.tagesschau.de/inland/wissing-fahrverbot-klimaschutzgesetz-100.html>

taz. (2024, 07 15). Bundesregierung bricht Klimagesetz. Retrieved 08 21, 2024, from <https://taz.de/Keine-Kurskorrektur-beim-CO2!/6020282/>

Tomor, Z., Meijer, A., Michels A., Geertman S. (2019). Smart Governance for Sustainable Cities: Findings from a Systematic Literature Review. *Journal of Urban Technology*, 26:4, pp. 3-27. Available at: <https://doi.org/10.1080/10630732.2019.1651178>

United Nations. (2023). *Global Sustainable Development Report: Times of Crisis, Times of Change*. Science for accelerating transformations to sustainable development.

Wurm, D., & Wittmann, F. (2023). Mission-oriented innovation policies for the twin-transition: a CO: DINA research report.

Spain

Bandrés, E., Gadea, L., Salas, V., & Sauras, Y. (2020). Spain and the European Recovery Plan. Funcas SEFO, 9(4), 5-14. <https://www.funcas.es/wp-content/uploads/2020/08/SEFO050art02.pdf>

European Commission (2023). Digital Decade Country Report 2023 Spain. <https://digital-strategy.ec.europa.eu/en/library/country-reports-digital-decade-report-2023>

European Commission. (n.d.). Spain's recovery and resilience plan. Commission.europa.eu. https://commission.europa.eu/business-economy-euro/economic-recovery/recovery-and-resilience-facility/country-pages/spains-recovery-and-resilience-plan_en

Gobierno de España. Portal de transparencia. (2018). Comisión Sectorial de Gobierno Abierto - Open Government - Portal de la Transparencia de la Administración del Estado. España - Inicio. Transparencia.gob.es. https://transparencia.gob.es/transparencia/en/transparencia_Home/index/Gobierno-abierto/comision-sectorialGA.html

IDAE. (n.d.). National Energy Efficiency Fund | Idae. www.idae.es. <https://www.idae.es/en/support-and-funding/national-energy-efficiency-fund>

La Moncloa.Gobierno de España. (2022). El Gobierno acuerda con los agentes sociales intensificar las reuniones de la Mesa del Diálogo Social del Plan de Recuperación e impulsar un plan específico de formación y recualificación de trabajadores. Lamoncloa.gob.es. <https://www.lamoncloa.gob.es/serviciosdeprensa/notasprensa/asuntos-economicos/Paginas/2022/100121-mesa-dialogo-prtr.aspx>

Madrid City Council. (2024). Madrid 360 * Madrid 360. Madrid 360. <https://www.madrid360.es/>

Ministry of Economy, Trade and Entrepreneurship. (n.d.). España Digital 2026. Portal.mineco.gob.es. https://portal.mineco.gob.es/en-us/ministerio/estrategias/Pages/00_Espana_Digital.aspx

Ministry of Economy, Trade and Entrepreneurship. (2024a). Plan de Digitalización de las Administraciones Públicas. Mineco.gob.es. https://portal.mineco.gob.es/en-us/ministerio/estrategias/Pages/Plan_Digitalizacion_AAPP.aspx

Ministry of Economy, Trade and Entrepreneurship. (2024b). Plan Nacional de Competencias Digitales. Mineco.gob.es. <https://portal.mineco.gob.es/en-us/digitalizacionIA/Pages/plan-nacional-competencias-digitales.aspx>

Ministry of Equality. (2023, July 5). El Consejo de Ministros aprueba hoy el nuevo Plan Estratégico para la Igualdad Efectiva de Mujeres y Hombres PEIEMH (2022-2025). Ministerio de Igualdad. <https://www.igualdad.gob.es/comunicacion/notasprensa/consejo-ministros-aprueba-hoy-el-plan-igualdad/>

Ministry of Finance. (2024a). DGFE:FONDO DE TRANSICIÓN JUSTA (FTJ). Hacienda.gob.es.
<https://www.fondoseuropeos.hacienda.gob.es/sitios/dgfc/es-ES/Paginas/FTJ.aspx>

Ministry of Finance. (2024b). Next Generation EU: Ministerio de Hacienda. Hacienda.gob.es.
<https://www.hacienda.gob.es/en-GB/CDI/Paginas/FondosEuropeos/Fondos-relacionados-COVID/Next-Generation.aspx>

Ministry of Finance. (2024c). Quiénes Somos: DGPMRR. Hacienda.gob.es.
<https://www.fondoseuropeos.hacienda.gob.es/sitios/dgpmrr/es-es/Paginas/quienes.aspx>

The Just Transition Institute (2022). Spain, towards a just energy transition.
https://www.transicionjusta.gob.es/Documents/Noticias/common/220707_Spain_JustTransition.pdf

Hungary

Jakab, N., & Molnár, B. (2022). A fenntartható foglalkoztatás kihívásai. (Challenges of sustainable employment) Pro Futuro, 12(2), 111-128. <https://doi.org/10.26521/profuturo/2022/2/12936>

András Bojtor, Gábor Bozsó (2022). Digitális forradalom a magyar közszoigálatban, Méresek és hatáselemzések tükrében VIII. Fejezet: A magyar közigazgatás-fejlesztés nemzetközi összehasonlítása (The digital revolution in the Hungarian civil service, Measuring and analysing impact Chapter VIII. International comparison of Hungarian public administration development).
<https://webshop.ludovika.hu/termek/konyvek/allam-es-jogtudomany/digitalis-forradalom-a-magyar-kozszoigalatban/>

Viktória Lilla Pató (2024). A digitális átállás nyersanyagszükséglete, Európa digitális szuverenitásának stratégiai kérdései a Brüsszel- és a Peking-hatás erőviszonyaiban / The raw material requirements for the digital transition, Strategic issues for Europe's digital sovereignty analysing the Brussels and Beijing effects in the balance of power.
<http://hdl.handle.net/20.500.12944/25278>

Ireland

Banerjee, A. and Schuitema, G. (2022). How just are just transition plans? Perceptions of decarbonisation and low-carbon energy transitions among peat workers in Ireland. Energy Research and Social Science, 88, 102616

CARO (2024a) Governance. Available at: <https://www.caro.ie/the-caros/governance>

CARO (2024b) Climate Action Regional Offices. Available at: <https://www.caro.ie/the-caros>

Citizen's Assembly (2018) Third Report and Recommendations of the Citizen's Assembly How the State can make Ireland a Leader in Tackling Climate Change. Available at:
<https://citizensassembly.ie/wp-content/uploads/2023/02/ThirdReportClimate-Change-Report-Final.pdf>

Citizen's Information (2021) Regulation of lobbying. Available at:

<https://www.citizensinformation.ie/en/government-in-ireland/how-government-works/standards-and-accountability/regulation-of-lobbying/>

Climate Change Advisory Council (2022) Memorandum of Understanding between the Irish Climate Change Advisory Council and all relevant Government Departments and Agencies.

Available at:

<https://www.climatecouncil.ie/media/climatechangeadvisorycouncil/contentassets/documents/memorandumsofunderstanding/2022%20Signed%20MoU%20between%20Government%20Departments%20and%20CCAC.pdf>

Climate Change Advisory Council (2023) Annual Review 2023. Available at:

<https://www.climatecouncil.ie/councilpublications/annualreviewandreport/CCAC-AR-2023-postfinal.pdf>

Climate Change Advisory Council (2024) The Climate Change Advisory Council. Available at:

<https://www.climatecouncil.ie/>

Commission on a Gender Equal Economy/Women's Budget Group (2020) Creating a caring economy, a call to action. <https://wbg.org.uk/wp-content/uploads/2020/10/WBGReport-v10.pdf>

Congress of Local and Regional Authorities (2023) Monitoring of the application of the European Charter of Local Self-Government in Ireland. CLRA 25 October. Available at:

[https://search.coe.int/cm#{%22CoEIdentifier%22:\[%220900001680acd809%22\],%22sort%22:\[%22CoEValidationDate%20Descending%22\]}](https://search.coe.int/cm#{%22CoEIdentifier%22:[%220900001680acd809%22],%22sort%22:[%22CoEValidationDate%20Descending%22]})

Department of Children, Equality, Disability, Integration and Youth (2023) National Youth Assembly on Climate 2023. Available at: <https://www.gov.ie/en/campaigns/3fd5d-national-youth-assembly-on-climate-2023/>

Department of Enterprise Trade and Employment (2021) AI - Here for Good: National Artificial Intelligence Strategy for Ireland. Available at: <https://enterprise.gov.ie/en/publications/national-ai-strategy.html>

Department of Enterprise Trade and Employment (2023) Digital Decade Policy Programme.

Available at: <https://www.gov.ie/pdf/?file=https://assets.gov.ie/292532/77a58105-f2ab-4c00-8a71-16a1c57042f1.pdf#page=null>

Department of Enterprise Trade and Employment (2024) Enterprise Digital Advisory Forum.

Available at: <https://enterprise.gov.ie/en/publications/enterprise-digital-advisory-forum.html>

Department of Environment Climate and Communications (2019) Climate Action Plan 2019.

Available at: <https://assets.gov.ie/25419/c97cdecddf8c49ab976e773d4e11e515.pdf>

Department of Environment Climate and Communications (2021) Climate Action and Low Carbon Development (Amendment) Bill 2021. Available at:

<https://www.gov.ie/en/publication/984d2-climate-action-and-low-carbon-development-amendment-bill-2020/>

Department of Environment Climate and Communications (2022) Digital Connectivity Strategy. Available at: <https://www.gov.ie/pdf/?file=https://assets.gov.ie/242271/96f1b6ad-b766-4ecb-95a1-bd3c9236f90b.pdf#page=null>

Department of Environment, Climate and Communications (2023a) National Dialogue on Climate Action (NDCA) Available at: <https://www.gov.ie/en/publication/4bf2c-national-dialogue-on-climate-action-ndca/>

Department of Environment, Climate and Communications (2023b) National Youth Assembly on Climate Summary Report 2023. Available at: <https://www.gov.ie/pdf/?file=https://assets.gov.ie/270981/1058697f-3bf4-4d2f-8b19-2d05646c872e.pdf#page=null>

Department of Environment, Climate and Communications (2023c) Climate Conversations Available at: <https://www.gov.ie/en/publication/bb1a3-climate-conversations/>

Department of Environment Climate and Communications (2024a) Climate Action Plan 2024. Available at: <https://www.gov.ie/pdf/?file=https://assets.gov.ie/296414/7a06bae1-4c1c-4cdc-ac36-978e3119362e.pdf#page=null>

Department of Environment Climate and Communications (2024b) Ireland's integrated National Energy and Climate Plan 2021-2030 July 2024. Available at: <https://www.gov.ie/pdf/?file=https://assets.gov.ie/299744/9a308db2-cbd5-46e8-8674-e939dca87263.pdf#page=null>

Department of Environment, Climate and Communications (2024c) National Adaptation Framework Planning for a Climate Resilient Ireland 2024. Available at: <https://www.gov.ie/pdf/?file=https://assets.gov.ie/298230/9af802e5-e601-488d-9ec1-db41279803cf.pdf#page=null>

Department of Environment, Climate and Communications (2024d) Ireland's Long-term Strategy on Greenhouse Gas Emissions Reduction 2024. Available at: <https://www.gov.ie/pdf/?file=https://assets.gov.ie/298534/3474b114-4d55-49f4-a31e-2cc74d403af8.pdf#page=null>

Department of Environment, Climate and Communications (2024e) Green Public Procurement Strategy and Action Plan 2024-2027. Available at: <https://www.gov.ie/pdf/?file=https://assets.gov.ie/288344/3b6eece7-7d30-47c5-895e-0512a0e9b3f8.pdf#page=null>

Department of Environment Climate and Communications (2024f) Just Transition Commission Terms of Reference 2024. Available at: <https://www.gov.ie/pdf/?file=https://assets.gov.ie/292130/50c502ee-2e58-4e8c-80dc-48a61af0b82b.pdf#page=null>

Department of Environment Climate and Communications (2024g) Statement of Strategy 2024-2025. Available at: <https://www.gov.ie/en/publication/6ed1b-statement-of-strategy-2024-2025/>

Department of Environment Climate and Communications (2024h) Long-term Strategy on Greenhouse Gas Emissions Reductions. Available at: <https://www.gov.ie/en/publication/e4e81-long-term-strategy-on-greenhouse-gas-emissions-reductions/>

Department of Environment Climate and Communications (2024i) National Climate Stakeholder Forum (NCSF). Available at: <https://www.gov.ie/en/publication/1aecf-national-climate-stakeholder-forum-ncsf/>

Department of Environment Climate and Communications (2024j) Climate Conversations 2023 Report from Individual Action to Collective Engagement. Available at: <https://www.gov.ie/pdf/?file=https://assets.gov.ie/297690/9395b1ab-7d7b-4ace-b111-ef77d699f7d5.pdf#page=null>

Department of Environment Climate and Communications (2024k) Climate Conversations 2024. Available at: <https://www.gov.ie/en/consultation/556db-climate-conversations-2024/>

Department of Housing, Local Government and Heritage (2024) Department of Housing Local Government and Heritage. Available at: <https://www.gov.ie/en/organisation/department-of-housing-local-government-and-heritage/>

Department of Public Expenditure NDP Delivery and Reform (2021) National Development Plan 2021-2030. Available at: <https://www.gov.ie/pdf/?file=https://assets.gov.ie/200358/a36dd274-736c-4d04-8879-b158e8b95029.pdf>

Department of Public Expenditure NDP Delivery and Reform (2022) Project Ireland 2024 Annual Report 2022. Available at: <https://www.gov.ie/pdf/?file=https://assets.gov.ie/272475/d6eea209-2404-490d-a6f6-0842ff6fe042.pdf#page=null>

Department of Public Expenditure NDP Delivery and Reform (2023) Digital for Good: Ireland's Digital Inclusion Roadmap. Available at: <https://www.gov.ie/pdf/?file=https://assets.gov.ie/267401/a898d78c-e234-465f-bedd-7ccd0655b7d2.pdf#page=null>

Department of Public Expenditure NDP Delivery and Reform (2024a) Project Ireland 2040. Available at: <https://www.gov.ie/en/campaigns/09022006-project-ireland-2040/>

Department of Public Expenditure NDP Delivery and Reform (2024b) About the Department of Public Expenditure, NDP Delivery and Reform. Available at: <https://www.gov.ie/en/organisation-information/919c7d-about-dper/>

Department of Public Expenditure NDP Delivery and Reform (2024c) Project Ireland 2040: Policy Documents. Available at: <https://www.gov.ie/en/collection/580a9d-project-2040-documents/>

Department of the Taoiseach (2022) Harnessing Digital - The Digital Ireland Framework 2022. Available at: <https://www.gov.ie/pdf/?file=https://assets.gov.ie/214584/fa3161da-aa9d-4b11-b160-9cac3a6f6148.pdf#page=null>

Department of the Taoiseach (2024a) Climate Action Delivery Board. Available at: <https://www.gov.ie/en/publication/24f0c-climate-action-delivery-board/>

Department of the Taoiseach (2024b) Harnessing Digital the Digital Ireland Framework 2023 Progress Report. Available at: <https://www.gov.ie/pdf/?file=https://assets.gov.ie/280518/e7f7f7c5-2af8-4209-849a-8c75caf5ceab.pdf#page=null>

Dekker, S., (2020). Responding to climate change: The role of local government in Ireland. In: D. Robbins, D. Torney, and P. Brereton: Ireland and the Climate Crisis. Palgrave, pp. 109-127.

Devaney, L, Torney, D, Brereton, P, and Coleman, M. (2020) Ireland's Citizens' Assembly on Climate Change: Lessons for Deliberative Public Engagement and Communication. Environmental Communication 14(2): 141-146. DOI: <https://doi.org/10.1080/17524032.2019.1708429>

Dukelow, F., Forde, C. and Busteed, E. (2024). Feminist Climate Justice Report. Feminist Communities for Climate Justice

Eastern and Midland Regional Assembly (2022a) EU Just Transition Fund Available at: <https://www.emra.ie/eu-jtf/>

Eastern and Midland Regional Assembly (2022b) European Commission adopts the EU Just Transition Fund for Ireland. Available at: <https://www.emra.ie/european-commission-adopts-the-eu-just-transition-fund-for-ireland/>

Environmental Protection Agency (2024a) Who we are. Available at: <https://www.epa.ie/who-we-are/>

Environmental Protection Agency (2024b) Latest Emission Data. Available at: <https://www.epa.ie/our-services/monitoring--assessment/climate-change/ghg/latest-emissions-data/>

European Commission (EC) (2022) EU Cohesion Policy: EUR1.4 billion for Ireland's economic and social development and green transition in 2021-2027. Available at: https://ireland.representation.ec.europa.eu/news-and-events/news/eu-cohesion-policy-eu14-billion-irelands-economic-and-social-development-and-green-transition-2021-2022-09-20_en

European Commission (EC) (2024) Ireland's recovery and resilience plan. Available at: https://commission.europa.eu/business-economy-euro/economic-recovery/recovery-and-resilience-facility/country-pages/irelands-recovery-and-resilience-plan_en

Ferris, T (2015) Reflections on the public policy process in Ireland. Administration 62 (4): 87–106

Friends of the Earth Ireland (2022) Friends of the Earth submission to Department of Environment Climate and Communications Call for Expert Evidence on the Climate Action Plan 2023. Available at: https://www.friendsoftheearth.ie/assets/files/pdf/friends_of_the_earth_submission_to_climate_action_plan_2023_call_for_expert_evidence.pdf

Goodbody, W (2024) Economic benefits at risk from data centre uncertainty – DII. RTE 17 May. Available at: <https://www.rte.ie/news/business/2024/0517/1449569-economic-benefits-at-risk-from-data-centre-uncertainty/>

Government of Ireland (2022) Territorial Just Transition Plan EU Just Transition Fund. Dublin: Stationary Office

Government of Ireland (2024a) Questionnaire for the Second Public Consultation on the draft updated National Energy and Climate Plan (NECP) 2021-2030. Available at: <https://www.gov.ie/pdf/?file=https://assets.gov.ie/295027/2761a5c4-fcac-4fb3-9322-a415b723dc24.pdf#page=null>

Government of Ireland (2024b) Climate Action Plan 2024 Public Consultation. Available at: <https://www.gov.ie/pdf/?file=https://assets.gov.ie/295601/55943119-474e-496d-a02b-b9ecd36a5955.pdf#page=null>

Heffron, R.J. and McCauley, D. (2017). The concept of energy justice across the disciplines. Energy Policy, 105, 658– 667.

Irish Fiscal Advisory Council (2024) About the Council. Available at: <https://www.fiscalcouncil.ie/about-the-council/>

Institute of International and European Affairs (2024) About the IIEA. Available at: <https://www.iiea.com/about/about-iiea/>

Irish Government Economic and Evaluation Service (2024) About the Irish Government Economic Evaluation Service (IGEES) Available at: <https://www.gov.ie/en/publication/b9e08-about-us/>

IHREC (2023). Policy Statement on a Just Transition. March 2023. Irish Human Rights and Equality Commission. Available at: <https://www.ihrec.ie/app/uploads/2023/04/Policy-Statement-on-a-Just-Transition-Final.pdf>

Irish Human Rights and Equality Council (2024) Who we are. Available at: <https://www.ihrec.ie/about/who-we-are/>

Local Government Management Authority (2024a) What we do. Available at: <https://www.lgma.ie/en/what-we-do/>

Local Government Management Authority (2024b) CCMA. Available at: <https://www.lgma.ie/en/ccma/>

Local Government Management Authority (2024c) Local Government Digital & ICT Strategy 2030. Available at: https://www.lgma.ie/en/what-we-do/programme-management-and-innovation/digital-ict-strategy_2024_2030_web_final-compressed.pdf

Mercier, S. (2020). Four Case Studies on Just Transition: Lessons for Ireland. NESC Research Series No. 15. Dublin: National Economic and Social Development Office NESDO.

Moore, J. (2020). Approaches to Just Transition. NESC Secretariat Paper No. 2020 (April), Dublin: National Economic and Social Development Office NESDO

Moore-Cherry, N., Calvin, A., Krawchenko, T. and Tomanye, J. (2022). Exploring Place-Based Opportunities and Practice in Transition. NESC Research Paper No. 24. Dublin: National Economic and Social Development Office NESDO.

Mullay, G. et al. (2022). A Roadmap for Local Deliberative Engagements on Transitions to Net Zero Carbon and Climate Resilience. EPA Research Report. Wexford: Environmental Protection Agency.

National Economic and Social Council (2024) Our Organisation. Available at: <https://www.nesc.ie/our-organisation/>

National Women's Council Ireland (2024) About us. Available at: https://www.nwci.ie/discover/about_us

NESC. (2021). Digital Inclusion in Ireland: Connectivity, Devices & Skills. Council Report No. 154. Dublin: National Economic and Social Development Office NESDO.

Nevin Economic Research Institute (2024) About the Nevin Economic Research Institute. Available at: <https://www.ner institute.net/about>

O'Gorman, R. (2020). Climate law in Ireland: EU and national dimensions. In: D. Robbins, D. Torney, and P. Brereton: Ireland and the Climate Crisis. Palgrave

Oireachtas (2021a) Committee on Environment and Climate Action. Available at: <https://www.oireachtas.ie/en/committees/33/environment-and-climate-action/>

Oireachtas (2021b) About Committees. Available at: <https://www.oireachtas.ie/en/committees/about-committees/>

Oireachtas (2024a) Joint Committee on Environment and Climate Action Work Programme 2024. Available at: https://data.oireachtas.ie/ie/oireachtas/committee/dail/33/joint_committee_on_environment_and_climate_action/reports/2024/2024-01-24_work-programme-2024_en.pdf

Oireachtas (2024b) Departmental Data Dáil Éireann Debate, Tuesday - 9 April 2024. Available at: <https://www.oireachtas.ie/en/debates/question/2024-04-09/68/>

Revez, A. et al. (2022). Mapping emergent public engagement in societal transitions: a scoping review. *Energy, Sustainability and Society* 12(2).

RPS Group (2024) Public Consultation Findings Report. Available at:

<https://www.gov.ie/pdf/?file=https://assets.gov.ie/294958/38d63e5e-4a44-47a9-8687-b1dab6bbd661.pdf#page=null>

Slevin, A. and Barry, J. (2024). Reconciling Ireland's climate ambitions with climate policy and practice: challenges, contradictions and barriers. *International Environmental Agreements*, 24, 29-48.

Sligo County Council (2023) Local Economic and Community Plan 2023-2030. Available at:

<https://www.sligococo.ie/lcdc/LocalEconomicandCommunityPlan/County%20Sligo%20Local%20Economic%20&%20Community%20Plan%202023%20-%202030.pdf>

Social Justice Ireland (2024) Delivering a Just Transition for Ireland. Policy Options. Available at:

<https://www.socialjustice.ie/system/files/file-uploads/2024-07/2024-07-11-Just%20Transition%20Briefing%20FINAL%20WEB.pdf>

Sustainable Energy Authority of Ireland (2024) About SEAI. Available at:

<https://www.seai.ie/about/>

TASC-FEPS (2022) The People's Transition. Community-led Development for Climate Justice.

Available at: https://www.tasc.ie/assets/files/pdf/feps-tasc_the_peoples_transition_-_2020f.pdf

TASC (2024) Our Work. Available at: <https://www.tasc.ie/work/>

Torney, D. and O'Gorman, R. (2019). A laggard in good times and bad? The limited impact of EU membership on Ireland's climate change and environmental policy. *Irish Political Studies*, 34(4), 575-594.

Italy

Bianchi L., RRP e Mezzogiorno: le potenzialità e il nodo irrisolto della governance locale, in *Il RRP alla prova del Sud* (a cura di) L. BIANCHI, B. CARAVITA, Napoli, Editoriale Scientifica, 2021.

Butera F., *Il PNRR per rigenerare le organizzazioni italiane nella transizione ecologica e digitale*

Costa, G., (2024), *Il sistema delle conferenze nelle vicende del RRP*, *Rivista di diritto Pubblico europeo* (rassegna online), 1/2014.

Davies, J. S. (2011). *Challenging governance theory: From networks to hegemony*. Policy Press.

Giachetti Fantini M., *La governance del Piano Nazionale di Ripresa e Resilienza: questioni critiche e nuove prospettive*, *Rivista italiana di public management*, Vol. 5 n. 1.

Gullo L., (2024), *Il ruolo del Parlamento nella fase di attuazione e controllo del RRP*, LUISS Working paper 2024.

Kersbergen, K. and Waarden F. (2004). "Politics and the transformation of governance. Issues of legitimacy, accountability, and governance in political science", *European Journal of Political Research*, Vol. 43 (2), pp. 143-171.

Jessop B. (2000). *The Dynamics of Partnership and Governance Failure* in G. Stoker, ed., *The New Politics of British Local Governance*, Basingstoke: pp. 11-32.

Lo, C. (2018). 'Between Government and Governance: Opening the Black Box of the Transformation Thesis' in *International Journal of Public Administration*, Vol. 41, No. 8, 650–656: <https://doi.org/10.1080/01900692.2017.1295261> L

Lupo N. (2022), *Il Piano Nazionale di Ripresa e Resilienza (PNRR) e alcune prospettive di ricerca per i costituzionalisti*.

Menegus G. (2022), *The Italian national recovery and resilience plan (NRRP) and the role of regions and local entities*, *Revista " Cuadernos Manuel Giménez Abad"*, 21-27.

Open Coesione (retrieved in 2024) https://opencoesione.gov.it/en/risorse_2021_2027/

Polacchini F., *I riflessi del Pnrr sulla forma di governo e sui processi di indirizzo politico*, in *Forum di Quaderni Costituzionali*, 4, 2022.

Polverari L., Piattoni S., (2022), *The Italian National Recovery and Resilience Plan and Administrative Capacity: A Real Game Changer?* *Rivista Italiana di Politiche Pubbliche*, Fascicolo 2, agosto 2022.

Polverari, *La capacità amministrativa in Italia: sfide, opportunità e prospettive - QUADERNI SVIMEZ*, 2020

Sciortino, A., (2022). *L'impatto del Pnrr sulle disuguaglianze territoriali: note introduttive*.

SVIMEZ (2023), *Report 2023.Citizenship, work, enterprises. Inclusion makes growing*, part. III.

Portugal

ADENE - Agência para a Energia (2023), *Renewable Energy and Efficiency Strategies in Portugal*. Available at: <https://www.adene.pt/>.

Alto Comissariado para as Migrações (2022), *Policies for Inclusive Digital and Green Transitions*. Available at: <https://www.acm.gov.pt/> .

Alves, I., Campos Pinto, P., & Pinto, T. J. (2020). *Developing inclusive education in Portugal: Evidence and challenges*. **Prospects*, 49*(3), 281–296. <https://doi.org/10.1007/s11125-020-09504-y>

ANI - Agência Nacional de Inovação (2023), *Innovation and Digital Transition in Portugal*. Available at: <https://www.ani.pt/> .

Assembleia da República (2021), Plano Nacional de Energia e Clima 2030 (PNEC 2030). Available at: <https://descarbonizar2050.apambiente.pt/> .

Autoconsumo e comunidade de energia renovável. Available at: https://www.adene.pt/wp-content/uploads/2022/11/Manual-Digital-Autoconsumo-e-Comunidade-de-Energia-Renovavel-Guia-Legislativo_vs2.pdf

CCDR-N - Comissão de Coordenação e Desenvolvimento Regional do Norte (2023), Regional Digitalization Strategies. Available at: <https://www.ccdr-n.pt/> .

CGTP - Confederação Geral dos Trabalhadores Portugueses (2023), Social Dialogue and Labor Rights in the Green Transition. Available at: <https://www.cgtp.pt/> .

CIP - Confederação Empresarial de Portugal (2023), Business Perspectives on the Twin-transition. Available at: <https://cip.org.pt/>.

CM Lisboa - Câmara Municipal de Lisboa (2023), Plano de Ação Climática de Lisboa. Available at: <https://www.cm-lisboa.pt/ficheiros/municipio/sustentabilidade/plano-de-acao-climatica-de-lisboa> .

Cidades pelo Clima. Available at: https://cidadespeloclima.pt/sobre_nos/

Commission for Coordination and Regional Development, Regulatory and Development Policies. Available at: <https://diariodarepublica.pt/dr/lexionario/termo/ccdr-comissao-coordenacao-desenvolvimento-regional>.

Confederation of Portuguese Business (CIP), Business Perspectives on Sustainability. Available at: <https://cip.org.pt/> .

DECO - Portuguese Association for Consumer Protection, Consumer Protection in Digital and Green Transition. Available at: <https://www.deco.proteste.pt/> .

Digital Transition in the Recovery and Resilience Plan: Challenges for Portugal. Available at: https://www.ipp-jcs.org/wp-content/uploads/2022/11/Policy-Paper_23.pdf .

Durão, N., Pereira, C. S., Lobo, C., & Moreira, F. (2024). Digitalization, sustainability, and internationalization nexus: Insights from Portuguese entrepreneurs. **Sustainability, 16*(11), 4786*. <https://doi.org/10.3390/su16114786>

Estrutura de Missão Portugal Digital. (2022). **Portugal, Nação Digital - 2 Anos de Transição Digital**.

EU Digital Scoreboard, Monitoring EU Digital Progress. Available at: <https://www.connecteurope.org/news/press-release-eu-digital-scoreboard> .

European Commission (2020), European Green Deal. Available at: <https://ec.europa.eu/green-deal> .

European Commission (2021), Digital Decade Policy Programme 2030. Available at: <https://digital-strategy.ec.europa.eu/en/policies/digital-decade> .

European Court of Auditors, EU Funding Oversight and Reports. Available at: <https://www.eca.europa.eu/en> .

Francescon, D. (2021, June 17). Portugal leads in gender equality in research: What can we learn from their leaders? Elsevier. Retrieved from <https://www.elsevier.com/connect/portugal-leads-in-gender-equality-in-research-what-can-we-learn-from-their-leaders>

GEOTA (2023), Opinion on the National Energy and Climate Plan (NECP) 2030. Available at: <https://geota.pt/blogs/o-geota-pede-rapida-aprovacao-do-plano-nacional-energia-e-clima-2030>

General Confederation of Portuguese Workers (CGTP), Labor Rights and Digital-Green Transition. Available at: <https://www.cgtp.pt/> .

Governo de Portugal (2021), Plano de Ação para a Transição Digital. Available at: <https://www.portugal.gov.pt/pt/gc21/comunicacao/documento?i=plano-de-acao-para-a-transicao-digital> .

Governo de Portugal (2023), Agendas Mobilizadoras para a Inovação Empresarial. Available at: <https://recuperarportugal.gov.pt/agendas-mobilizadoras/> .

IPP-JCS. (2022). Policy Paper 23: Digital Transition in the Recovery and Resilience Plan: Challenges for Portugal. Instituto de Políticas Públicas - José Crisóstomo de Sousa. Retrieved from https://www.ipp-jcs.org/wp-content/uploads/2022/11/Policy-Paper_23.pdf

National Energy Agency (ADENE), Energy Efficiency Strategies. Available at: <https://www.adene.pt/> .

National Innovation Agency (ANI), Innovation and Digital Transition Support. Available at: <https://www.ani.pt/> .

Open Access Government. (2020, September 23). Portugal: The development of a country requires knowledge and education. Open Access Government. Retrieved from <https://www.openaccessgovernment.org/portugal-the-development-of-a-country-requires-knowledge-and-education/94852/>

Participa Portugal (2023), Public Consultations on Climate and Digital Policies. Available at: <https://www.participa.pt/>

Plataforma ONGD (2022), Civil Society Contributions to the PRR. Available at: <https://plataformaong.pt/>

Portugal Energy Sector, Energy and Sustainability in Portugal. Available at: <https://www.portugalenergia.pt/setor-energetico/bloco-3/>

Portugal Inovação Social 2030. Available at: <https://inovacaosocial.portugal2020.pt/portugal-2030/>

Portuguese Parliament (Assembleia da República), Legislative Procedures and Policies. Available at: <https://www.parlamento.pt/>

Portugal, Nação Digital – 2 anos de Transição Digital. Available at: <https://portugaldigital.gov.pt/wp-content/uploads/2022/02/livro-portugal-nacao-digital-2-anos-de-transicao-digital.pdf>

Quercus (2021), Position on PRR and Green Transition. Available at: <https://quercus.pt/2021/07/quercus-emite-parecer-sobre-o-prr-plano-de-recuperacao-e-resiliencia/>

Recuperar Portugal (2023), Portugal's Recovery and Resilience Plan (PRR). Available at: <https://recuperarportugal.gov.pt/>

Rosário, A. T., & Dias, J. C. (2022). Sustainability and the digital transition: A literature review. *Sustainability, 14*(7), 4072. <https://doi.org/10.3390/su14074072>

Sustainability and the Digital Transition: A Literature Review. Available at: <https://www.mdpi.com/2071-1050/14/7/4072>

Trabalhos e competências verdes. Available at: <https://www.iefp.pt/programa-trabalhos-competencias-verdes-green-skills-jobs>

Tribunal de Contas (2022), Monitoring Public Expenditure on Twin-transition Policies. Available at: <https://www.tcontas.pt/>

ZERO (2023), National Energy and Climate Plan (NECP) 2030 - Analysis. Available at: <https://zero.org/noticias/plano-nacional-de-energia-e-clima-2030-aumenta-ambicao-mas-nao-traca-caminho-claro-para-a-neutralidade-carbonica-antecipada-para-2045/>.

Annex I: Methodological templates

Literature review template

1. Purpose of the review

The aim of this exercise is to collect recent literature at country level on the governance of the transition. You are firstly asked to fill-in maximum three significant pieces of literature describing them in depth and then to synthesise the overall debate on the governance of the twin transition that is taking place in your country.

2. Research literature cards

Please, identify up to 3 recent research literature (research reports, books) published in the last 5 years in your country focusing on the governance of the twin transition and on how the needs of women and diverse people are accounted for. If there are no contribution on the twin transition you can refer to publications focusing only on the digital or only the green transformation. Describe these sources by completing the table below. Next to this, give a summary of key points.

Table: Features of selected recent research literature

	Source 3
<i>Title in original language and in English</i>	
<i>Publication year</i>	
<i>Institution/Author(s)</i>	
<i>Scope (local, regional, national)</i>	
<i>Please provide original citation and internet-link, if available</i>	
<i>Summary of key points using bullet points (maximum of 200 words per book/article):</i> <i>What is the main evidence concerning how the needs of women and diverse people are accounted for in the governance of the Twin-transition?</i> <i>Any action suggested for improvement?</i> <i>What are the main reasons given for suggested actions (if any)</i>	
<i>Additional information</i>	

3. Additional literature inputs

If there are other contributions that could not be listed in Section 1.2 you can add them here.

As done above focus on the needs of women and diverse people: is there any discussion in the literature on governance?

If there are no contribution on the twin transition you can refer to publications focusing only on the digital or only the green transformation.

If you could not find any additional input, please state it clearly here.

4. Discussion

Please critically discuss the literature on the twin transition governance in your country.

Country report template

1. Introduction: the role of EU funds

This chapter should: briefly set out the context of the twin transition governance in the country. By clarifying the role of EU funds that are supporting the twin transition in Member States.

- *To what extent the twin transition is concretely supported by EU in your country? Completely, partly or not at all. Please detail your answer.*
- *Is your country using other national funds (beside co-financing of EU programmes) to fund the twin transition? If yes, to what extent national funding contribute to the twin transition (do not consider the co-financing of EU funds)? National contributions prevail, half and half, the twin transition is mainly supported by EU programmes and funding.*
- *Is there a coordination structure at higher level to coordinate the use of all available funding?*
- *Please summarise the different levels of governance involved in the twin transition in the jurisdiction and the key drivers for equality that operate at each level of governance. If applicable, please distinguish between the authorities responsible for national funding and for EU funding.*

Please note in the absence of a coherent strategy for a unified twin transition refer to the digital and green transition separately.

2. Governance structure

This chapter should provide summary information on the different levels of governance in the jurisdiction involved in governing the twin transition in the country by detailing:

- *the governance levels involved (national, regional local)*
- *the institutional actors including political positions (Ministries, Regional Assessors, local Assessors Deputy Ministers, Parliamentary Committees ...) and administrative actors (Managing authorities of national and EU funding, Director General in Ministries, Regional*

departments local departments) involved at each governance level and please clarify the competence of these public authorities

- *If they exist, please describe the composition and functions of the committees set up for governance coordination at each governance level.*
- *Consultation/co-decision arenas set up to allow the participation of civil society organisations (CSOs) to policy design, policy implementation, policy monitoring and evaluation.³ Please clarify if these arenas have been set up only to comply with EU requirements and are competent only for EU funding or if these arenas have their say more in general on the twin transition.*

<i>Governance level</i>	<i>Geographical area</i>	<i>Institutional actors</i>	<i>Committees</i>	<i>Consultation/co-decision CSOs</i>

The chapter should provide information about the key relevant institutional actors and committees in relation to:

- *The nature of the policy powers and functions of institutional actors (including committees set up for coordinating actions) at each level of governance and in particular their role in each step of the policy chain – planning for policy, drafting policy, implementing policy and monitoring/evaluating policy. Please clearly distinguish between the political and the administrative actors and roles. Please also clarify if these powers and functions are related to EU funds and so mirror the respective regulation requirements.*
- *The nature of the commitments made by the institutional actors (including committees set up for coordinating actions) at each level of governance to equality/diversity/non-discrimination and in accounting for the conditions of vulnerable groups.*
- *The capacity of these institutional actors at each level of governance (including committees set up for coordinating actions) to address issues of equality/diversity/non-discrimination and in accounting for the conditions of vulnerable groups.*
- *The manner in which civil society/social partner/other organisations engage with these public authorities at each level of governance in order to drive and support their focus on equality/diversity/non-discrimination and in accounting for the conditions of vulnerable groups.*

³ Civil Society Organisations – non-governmental groups such as trade unions, employers' associations and other social groups– allow citizens to take an active part in setting the political agenda.

Please note:

The detail required is only to a level sufficient to understand how the policymaking develops through the different levels of governance. Please also present the institutional actors responsible for issues of equality/diversity/non-discrimination and in accounting for the conditions of vulnerable groups. Please note in the absence of a coherent strategy for a unified twin transition refer to the digital and green transition separately.

3. Governance process

This chapter should provide information on the policy processes.

First of all, focus on the links between the different levels of governance and different institutional actors.

The needed level of detail should be sufficient to understand how the policy process governing the twin transition is linked through the different levels of governance, to appreciate in particular the nature and structure of the relationships established between the different levels of governance.

Please distinguish between the formal political steps and the administrative steps.

It should provide information on:

- *The nature and structure of the links between the different levels of governance established by the institutional actors for each step of the policy chain - planning for policy, drafting policy, implementing policy and monitoring/evaluating policy.*
- *The manner in which these links might operate to enhance performance on equality/diversity/non-discrimination and in accounting for the conditions of vulnerable groups at the different levels of governance.*
- *Please focus on the distinction we can operate among institutional actors between political actors and administrative actors. What are the different roles and responsibilities? What about the administrative capacity of administrative actors at different governance levels?*
- *The manner in which civil society/social partner/other organisations can contribute to these links between different levels of governance within the framework designed by institutional actors in order to drive and support a focus on equality/diversity/non-discrimination and in accounting for the conditions of vulnerable groups.*

Please note in the absence of a coherent strategy for a unified twin transition refer to the digital and green transition separately.

4. Barriers

This chapter should draw from the literature review conducted at the previous step of Taks 2.2 and from the paragraphs 1.1-1.3 of this report to identify and assess the barriers to effective governance of the twin transition. Please distinguish between the different levels of governance that enable these public authorities to address the needs of vulnerable group while implementing policies to best effect.

An **example of barrier** might be the lack of Institutional Administrative Capacity at local level.⁴

Please note in the absence of a coherent strategy for a unified twin transition refer to the digital and green transition separately.

5. Factors for success

This chapter should draw from the literature review conducted at the previous step of Taks 2.2 and from the paragraphs 1.1-1.3 of this report to identify and assess the factors that enable governance of the twin transition to best effect.

Please distinguish between the different levels of governance that enable public authorities to address the needs of vulnerable groups while implementing policies.

Example of a success factor is the availability of human capital in the institutions and of training opportunities for upskilling and re-skilling.

Please note in the absence of a coherent strategy for a unified twin transition refer to the digital and green transition separately.

6. Document list

While drafting this report you have probably encountered relevant documents on the twin transition governance. They might be laws, draft bills, etc., official strategic documents, or documents from the consultation processes (produced by different actors including both the CSOs.

All documents are relevant, when selecting the document to mention please ensure that the CSOs participation is also covered.

Please note in the absence of a coherent strategy for a unified twin transition refer to the digital and green transition separately.

Please list them in the table below.

Document titles in English and original language ↓	Governance level	Why relevant for governance of the twin transition	Link to document

⁴ According to the European Public administration Network this is the Institutional Administrative Capacity consists in the set of characteristics that public administrations and other public or private bodies involved in the delivery of public policies and services have to possess in order to be able to define 'good' policies, and to implement them effectively, thus achieving societal ambitions.

7. Good practices

While drafting this report you have probably encountered good practices showing the effectiveness of the governance of the twin transition. All practices showing how the governance work is interesting for us, but it would be good to include at least one good practice that considers the needs of vulnerable groups and that allow CSOs participation.

Please note it would be also very interesting for the research to report on one 'worst practice' as learning from it will be very interesting for the next steps of FITEER-EU.

Please note in the absence of a coherent strategy for a unified twin transition refer to the digital and green transition separately.

Please include the practices in the following list.

<i>Name of the practice in English and original language</i>	<i>Governance level</i>	<i>Objective</i>	<i>Link to documents</i>

8. Conclusion

This chapter should establish some priority areas for action in order to enhance of the governance structures and processes in the jurisdiction for effective representation of needs of vulnerable groups.

Please note in the absence of a coherent strategy for a unified twin transition refer to the digital and green transition separately.

References

Document analysis template

Documents list

Please choose three documents among those you listed in Chapter 6 of your Country Report. Choose the documents according to their relevance in setting the governance of the green-digital-twin transitions in your country and in the participation of stakeholders in decision making, ensuring that both government and stakeholders documents are included. Please provide a short title for each document and rank them (attributing score 1-3 in the last column) according to their relevance in providing information on the governance of these transitions.

Table 1: Documents list for <Country name>

<i>Document title in English and original language ↓</i>	<i>Author Who commissioned the document Year of publication</i>	<i>What is it about (Max 80 words)</i>	<i>Short title</i>	<i>Link to document</i>	<i>Ranking relevance 1-3</i>

Document 1: <please add here the short title of the document>

Repeat this section as many times as the number of documents you listed (minimum 3).

1.1 Roles and responsibilities

Please scan the document and indicate:

1. What are the main governing institutions (at the different governance levels) mentioned and what is the formal role/responsibility they are expected to play in the governance of the green-digital-twin transitions? What policy area (labour market participation, care, poverty...) each of the institutions mentioned is responsible for, and at which level of governance? Please specify the context in which this information is provided in the document and indicate the page and the paragraph number where this can be found in the text.
2. Which stakeholders different from public institutions are mentioned? What is the formal role/responsibility that is attributed to them? Specify the context in which this information is provided in the document and indicate the page and the paragraph number where this can be found in the text.

1.2 Relevant issues and stakeholders' involvement

Scan the document and please:

1. Detail if the document mentions/discusses the interests or the issues (gender equality, ethnicity, age, disability, poverty, workers, ..., etc.) that stakeholders represent, and the depth of analysis provided (i.e., if they are just mentioned, or entire paragraphs/sections are dedicated to these issues). Specify the context in which this information is provided and indicate the page and the paragraph number where this can be found in the text.
2. Briefly describe in relation to what policy area (labour market participation, care, poverty...) these interests or issues (gender equality, ethnicity, age, disability, poverty, workers, ..., etc.) are mentioned. Specify the context in which this information is provided in the document and indicate the page and the paragraph number where this can be found in the text. Clarify if any reference to stakeholders is made in those sections or in relation to the respective actions to be undertaken.

1.3 Policy-making process

According to the document, can stakeholders represent interests or issues (gender equality, ethnicity, age, disability, poverty, workers, ..., etc.) actively participate in the planning, implementation or evaluation of actions? Please provide details on their role in each of the policy-making steps:

1. Planning: provide details on their role and responsibility; specify the context in which this information is provided; indicate the page and the paragraph number where this information can be found in the text.

2. *Implementation: provide details on their role and responsibility; specify in what instance this is written (context); indicate the page and the paragraph number where the information is provided*
3. *Evaluation: provide details on their role and responsibility; specify in what instance this is written (context); indicate the page and the paragraph number where the information is provided*

1.4 *Reference to the European Union documentation*

Scan the document and please assess whether in the document any reference to the European Union policy documentation is made in relation to:

1. *the role and responsibilities of government and public authorities. - Specify the context in which this reference is made and indicate the page and the paragraph number where this can be found in the text.*
2. *the role and responsibilities of stakeholders. - Specify the context in which this reference is made and indicate the page and the paragraph number where this can be found in the text.*
3. *the interests or issues (gender equality, ethnicity, age, disability, poverty, workers, ..., etc.) that stakeholders may represent. Specify the context in which this reference is made and indicate the page and the paragraph number where this can be found in the text.*

1.5 *Overall judgment*

According to your expertise, do you think that the document content mirrors the real situation of the green-digital-twin transition governance in your country? Please explain the reasoning behind your judgment, referring to the literature identified and used in the first two steps of the country level analysis (Literature review and Country profile).

The Good practice report

1. Introduction

This report will provide information on the links between the different levels of governance of the Twin Transition in your country by the key relevant public authorities for each of the chosen sector.

2. What we expect from you

You are required to provide a level of detail which is sufficient to understand how the policy process is linked through the different levels of governance, to appreciate in particular the nature and structure of the relationships established between the different levels of governance by these public authorities

3. Focus

This report should provide data and information on two good practices, in particular on:

- The nature and structure of the links between the different levels of governance established by the public authorities for each step of the policy chain - planning for policy, drafting policy, implementing policy and monitoring/evaluating policy.
- The manner in which these links might operate to enhance performance on gender equality and/or support to vulnerable groups by these public authorities at the different levels of governance.
- The manner in which civil society/social partner/other organisations contribute to these links between different levels of governance established by these public authorities in order to drive a focus on gender equality and/or support to vulnerable groups.

4. Which good practice you should choose

The deciding factors on which good practice to choose are:

- The ability of that policy to illustrate the involvement of the key relevant public authorities at each level of governance.
- The ability of that policy to illustrate the manner in which the links established between the different levels of governance by these public authorities enhance the implementation and impact of the policy.
- Availability of information and evaluations.

5. The Good practice in the governance of the Twin Transition

5.1 Introduction

Please provide here the details on a good practice of governance on gender equality and/or support to vulnerable groups in the Twin Transition.

5.2 The aim

The aim, target, content, budget and impact of the policy and the levels of governance and key relevant public authorities involved in the Twin Transition. (20 lines)

5.3 Roles and responsibilities

Please describe roles and responsibilities in in the Twin Transition with regard to:

- The roles played by the key relevant public authorities at each level of governance in the policy chain - planning for the policy, drafting the policy, implementing the policy and monitoring/evaluating the policy.
- The roles played by civil society/social partner/other organisations at each level of governance in the policy chain - planning for the policy, drafting the policy, implementing the policy and monitoring/evaluating the policy.
- The commitment and capacity of the key relevant public authorities at each level of governance to address issues of gender equality and support to vulnerable groups and the contribution made by civil society/social partner/other organisations to this commitment and capacity.

5.4 Links between levels of governance

Please describe the links across different levels of governance of the Twin Transition that:

- enabled the effective implementation of each step of the policy chain - policy planning, policy drafting, policy implementation and policy monitoring/evaluation.
- enabled shared equality standards and objectives to inform all steps of the policy chain – planning, drafting, implementing and monitoring/evaluating.
- ensured a common capacity of the public authorities at each level of governance involved to promote equality and provide support to vulnerable groups.

Allowed learning on equality between public authorities at the different levels of governance to inform all steps of the policy chain.

5.5 Civil society

Please describe the role of civil society/social partner/other organisations in the governance (if any).

5.6 Barriers

Please identify and assess the barriers to effective links being established by the key relevant public authorities between the different levels of governance that enable these public authorities to implement gender equality and support to vulnerable groups policies in the Twin Transition to best effect.

5.7 Factors for success

Please assess the factors required for effective links to be established by key relevant public authorities between the different levels of governance that enable these public authorities to implement gender equality and support to vulnerable groups policies in the Twin Transition to best effect.

5.8 Conclusion

Please establish some priority areas for action in order to enhance links between the different levels of governance for effective action for gender equality and/or support to vulnerable groups in the Twin Transition.

6. Indications

This report is qualitative in nature with a view to providing learning on good practice in making links between the different levels of governance that enable public authorities to deliver effective gender equality and support to vulnerable groups in the Twin Transition.

The methodology is based on the information you can gather from desk research and contacts with informants (if any).

Annex II: Document analysis table per country

The Table below collects main findings emerged from the analysis of relevant documents at country level on governance.

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
Italy	5 th Progress Report on RRP Implementation	The report outlines RRP developments in early 2024, highlighting targets achieved, plan updates, and reforms to enhance synergies with cohesion policy. Key changes include the reorganization of the RRP Mission Structure (Law Decree No. 19/2024) and the introduction of local Coordination Cabins for project support. Additionally, the National Economic and Labor Council (CNEL) has been strengthened, and centralised governance mechanisms for	The report analyzes the progress of the 2030 Agenda Goals (SDGs) through Italy's Recovery and Resilience Plan (RRP) measures, particularly focusing on gender equality. It highlights the limited contribution of RRP measures to gender equality, inclusion, and poverty alleviation. A significant focus is on the gender equality certification system, introduced by Law No. 162/2021 and Budget Law 2022. This certification aims to improve gender equality in workplaces and is overseen by the Dipartimento delle Pari Opportunità. The system was developed through a working group that included institutional stakeholders, resulting in technical standards formalized by the Italian standardization body on 16 March 2022. Key milestones and incentives linked to the certification include:	The report highlights the reform for enhancing the implementation and efficiency of cohesion policy, introduced by Legislative Decree No. 60/2024, which was approved and integrated into the RRP in late 2023. According to the decree law: technical working group was formed to ensure institutional dialogue and cooperation at national level during the reform planning phase. This group included regional and national program managing authorities and representatives of the Directorate-General for Regional and Urban Policy (DG REGIO) of the European Commission. While regional authorities were involved, there is no evidence of participation by stakeholders representing	The document outlines the alignment of Italy's Recovery and Resilience Plan (RRP) with key EU policies and principles, focusing on environmental sustainability, strategic technologies, and anti-fraud measures: Do No Significant Harm (DNSH) Principle: The DNSH principle, outlined in Regulation (EU) 2021/241, requires RRP measures to conform to environmentally sustainable activity standards. Public authorities conducted self-assessments to ensure compliance, occasionally modifying investments to meet DNSH requirements. The Ministry of Economy, in consultation with the Ministry of the Environment and Energy Security, issued operational	This report reflects the current state of Italy's Recovery and Resilience Plan (RRP), focusing on recent governance measures aimed at accelerating implementation and supporting the green and digital transition. While local authorities are actively involved in planning and implementing cohesion policies (e.g., Decree No. 60/2024), civil society organizations are minimally engaged. Capacity building for local administrations is a key focus, with measures such as

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
		projects like housing and agriculture are established.	The certification system is now operational. Companies with the certification can gain advantages in public tenders, such as reduced guarantees (30% for service/supply contracts) and higher tender scores. Additional incentives under Law Decree No. 36/2022 include reduced surety guarantees (20%) and priority for companies adopting inclusive policies for gender equality and disadvantaged groups. The report notes that non-institutional stakeholders and the interests discussed during the certification's development are not disclosed. Overall, the certification system represents a structural step toward embedding gender equality into corporate practices through financial and procedural incentives in public procurement.	gender equality, ethnicity, age, disability, poverty, or workers. Concerning the implementation coordination, according to the document, The RRP control room played a crucial role. Control room meetings during the semester included only institutional stakeholders, such as the Conference of Regions, Autonomous Provinces, the National Association of Italian Municipalities (ANCI), and the Union of Italian Provinces (UPI). - No evidence was found of participation by previously engaged social stakeholders. 3. **Monitoring and Evaluation: - A total of 50 thematic tables, both technical and strategic, were coordinated by	guidelines and organized training events with ANCI-IFEL to support local authorities in adhering to the DNSH principle. Strategic Technologies for Europe Platform (STEP): STEP aims to increase investments in digital deep tech, biotechnology, and clean technology. Decree-Law 60/2024 mandates the Development and Cohesion Fund's Control Room to create national guidelines supporting investments and addressing labor shortages in STEP sectors. Funding includes the reprogramming of national and regional cohesion policy programs (2021–2027) and EUR300 million from the National Programme for the Green and Digital Transition for private company investments in these areas.	the Coordination Cabin introduced by Decree No. 19/2024 to address territorial challenges and support implementation. On social issues like gender equality, poverty, and inequalities, progress remains minimal, with achievement rates of 0%-1% for relevant SDGs. Despite good measures like gender equality certification reaching milestones, outcomes remain limited due to top-down policy-making and insufficient stakeholder engagement. A revision of indicators and broader stakeholder

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
Italy				the RRP mission structure. Participants included representatives of ministries, social partners, territorial authorities, and implementing institutions. - Monitoring and evaluation involved collaboration with the European Commission's technicians, but the report does not disclose the names of organizations involved. The document underscores a primarily institutional approach to stakeholder engagement, with limited transparency about broader social representation in key phases of planning, implementation, and evaluation.	Fraud Prevention and Combating: Decree-Law No. 19/2024 establishes a unified governance structure for anti-fraud controls in European funds, enhancing monitoring, information flow, and regulatory proposals. This approach aligns with the European Commission's 2022 recommendations in the 34th PIF report to strengthen fraud prevention and governance. The report emphasizes Italy's commitment to aligning with EU regulations by integrating environmental, technological, and anti-fraud measures into its national policies and operational frameworks.	representation is recommended to achieve tangible results. The findings align with existing literature and official documentation.
	Data to Count – Statistics and Indicators for Fair RRP	The report focuses on gender equality within Italy's RRP implementation and highlights gaps in governance and monitoring.	Although green and digital transitions are not directly addressed, the report highlights key topics: co-housing (reducing energy use), strengthening <i>community houses</i> for territorial healthcare,	Although recommendations and requests have been made to the relevant national authorities, effective joint actions between national public bodies and other stakeholders do not appear to have been put in place.	The document does not include references to the European Union policies and dedicated documentation	The report reflects Italy's governance of the green-digital transition and progress on gender equality. While reforms like the amended

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
		The campaign was aimed at evaluating the RRP, proposing suggestions and reducing gender inequalities. Stakeholders involved were representatives of local institutions, third sector organisations, health services and public administration, and universities. A list of additional gender indicators was elaborated by these stakeholders and proposed to the RRP control room. These suggestions seem to be not considered by political institutions.	and measures to ease women's care work, benefiting the labour market. Recommendations focus on integrated housing strategies, including improving data accessibility, promoting co-housing and intergenerational housing, strengthening the third sector, implementing targeted housing policies for women, migrants, and marginalized groups, and involving local communities in monitoring student housing allocation.			contractual code and gender certification system are positive steps, critical issues remain in implementation. The lack of concrete impact is evident in the 5th RRP Report, particularly regarding UN Agenda 2030 gender equality goals. Concerning data indicators, monitoring systems, and conditionality in RRP tenders are also confirmed by ASviS and Anac-Open Polis data analyses.

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
Germany	Deutsche Nachhaltigkeitsstrategie (German Sustainable Development Strategy)	The German Sustainable Development Strategy (DNS) operates primarily at the federal level but extends to EU and international policies. The Federal Chancellery oversees coordination, amendments, and updates, heading the State Secretaries' Committee for Sustainable Development, which manages content development and inter-ministerial implementation. Each ministry appoints a sustainability coordinator. Key bodies include: German Council for Sustainable	The German Sustainable Development Strategy (DNS) focuses on SDGs, highlighting issues like poverty, hunger, gender equality, employment, and global inequality, with dedicated chapters outlining priorities, indicators, and measures. However, the strategy lacks in-depth analysis of structural causes and intersectionality, such as gender-energy-nexus or overlapping vulnerabilities. Civil society stakeholders, including trade unions and youth councils, contribute through dialogue groups and annual forums like <i>Forum Nachhaltigkeit</i>, which involve 80-150 organizations. While stakeholders address slow implementation and financing issues, social vulnerability factors receive limited focus. Federal ministries, such as BMAS and BMFSFJ, implement related policies but are not directly involved in strategy formation, participating through sustainability coordinators.	The German Sustainable Development Strategy (DNS) involves civil society stakeholders and interest groups throughout planning, implementation, and evaluation. The Federation of German Trade Unions regularly contributes via the dialogue group supporting the State Secretaries' Committee, while additional CSOs are invited based on annual focus topics. Amendments occur every four years through open participatory processes, involving citizens, research institutions, and private sector actors. Federal Ministries play a central role in DNS implementation, ensuring new legislation aligns with its requisites. Monitoring and evaluation occur through biennial reports on sustainability indicators by the Federal Statistical Office. Each legislative period, departments report their DNS contributions, addressing conflicting goals, to the State Secretaries' Committee and	The German Sustainable Development Strategy (DNS) aligns with the UN SDGs and EU Agenda 2030, advocating for ambitious EU-level implementation. The Agenda 2030 serves as both the foundation and framework for DNS, aligning its indicators with the 17 SDGs and ensuring sustainability impact assessments for laws. The EU Green Deal defines key action areas like climate protection, clean energy, circular economy, and sustainable mobility. Germany's 2020 EU Council presidency advanced climate, energy, and mobility legislation. Key EU initiatives include the European Sustainable Development Network (ESDN) for country exchanges, the European Sustainable Development Week (ESDW) for outreach, and the Council Working Group, initiated by Germany in 2017, to coordinate trade policies among EU members.	The DNS is a key policy document for Germany's green transition, aligned with the SDGs and Agenda 2030, covering ecological and economic sustainability but underrepresenting social aspects and structurally vulnerable groups. Gender mainstreaming is inconsistently applied. Economic actors, such as agribusiness and industry groups, hold strong influence, sometimes contradicting ecological and social priorities. Conflicting interests are evident in government decisions, such as debates on reducing public transport subsidies

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
		Development (RNE): Consults and submits statements. Parliamentary Advisory Council: Assesses the sustainability impact of laws. Federal States and Municipalities: Implement strategies within their legislative scope. Civil society, economic, environmental, and social stakeholders collaborate via dialogue groups. Academia provides scientific input through the Science Platform Sustainability 2030, while economic actors and the financial sector contribute capital		the Parliamentary Advisory Council. These reports guide strategy updates and are submitted to the Bundestag.		while supporting car companies. While participatory processes exist, the federal chancellery and ministries act as gatekeepers, favouring interest groups with greater capacity to advocate for their positions. Sustainability remains closely tied to economic growth, reflecting trends at the EU level.

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
		for sustainable development.				
Germany	Bundes-Klimaschutzgesetz (KSG) (Federal Climate Protection Act)	The Federal Climate Protection Act (KSG), introduced on December 12, 2019, assigns implementation and monitoring responsibility to the federal government, including drafting remedial plans if sectoral emission targets are missed, with Bundestag approval. The Bundesrat has a limited role, mainly overseeing budget-related steps. Different ministries monitor sectoral targets, with the updated KSG highlighting the Ministry of Consumer Protection, Food,	The KSG aims to protect against global climate change by achieving national and European climate targets (§1), considering ecological, economic, and social impacts. However, it lacks detailed descriptions of the social consequences. The federal government conducts public consultations involving states, municipalities, business associations, and civil society (§9.3). Ministries must propose measures to meet annual emissions targets, including scientific assessments of social impacts, such as employment trends, economic structure, rural living conditions, and resource efficiency (§9.2). Despite this, the KSG provides no further detail on these social effects or links to other policy areas beyond emission reductions.	The KSG mandates civil society and interest group participation in climate protection planning through public consultations (§9.3), though the process details are not provided. Within 12 months of each legislative period, the federal government adopts a climate protection program outlining sector-specific and cross-sector measures (§3.1). An expert council, comprising members from various disciplines, operates independently and must include experts in climate science, economics, environment, and social issues, with gender balance (§11.1). The council evaluates the government's findings on social, cost, and climate effectiveness (§12.7) and provides statements on the	The KSG aligns with EU emission reduction targets and the UN Paris Agreement, with EU legislation serving as a reference and basis for its targets, which may change if EU laws are amended (§2.3). Key EU policies referenced include: European Governance Regulation (EU 2018/1999) European Climate Protection Resolution (EU 2018/842) European Climate Reporting Regulation (EU 2020/1208). The EU Commission oversees emission monitoring, receives sectoral reports, and must be notified of non-compliance. Beyond these references, no specific social issues are addressed.	The KSG offers limited participation opportunities beyond §9 and §12. A Frag den Staat study found that CSOs often face short deadlines, with less than 20 working days to contribute to most laws since 2018. Despite international law requiring public participation for environmental laws, three-quarters of cases have insufficient consultation periods. The 2019 KSG draft introduced binding national climate targets, strengthened in 2021 by raising

<i>Country</i>	<i>Document Name</i>	<i>Roles and responsibilities</i>	<i>Relevant issues and stakeholders' involvement</i>	<i>Policy-making process</i>	<i>Reference to the European Union documentation</i>	<i>Overall Judgment</i>
		and Agriculture due to new targets and the introduction of Carbon Capture and Storage (CCS) technologies. The Federal Environmental Agency oversees greenhouse gas emissions data, including sectoral monitoring and projections. An expert council, composed of interdisciplinary members, independently assesses findings from the government and the agency.		climate protection program (§9).		emission reduction goals to 65% by 2030. The 2024 amendment, however, removes sectoral targets and immediate action programs, focusing only on total emissions—a change heavily criticized by CSOs. CSOs can take legal action under the KSG. For example, Deutsche Umwelthilfe (DUH) successfully sued for stronger CO2 reduction measures, including limiting wood harvesting, re-irrigating marshlands, and ending wood burning in power plants. Other organizations like Greenpeace, Germanwatch, and BUND plan constitutional

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
Germany	Digitalstrategie Deutschland (German Digital Strategy)	The Digital Strategy for Germany (2022), led by the Federal Ministry for Digital and Transport (BMDV), sets digital priorities until 2025, aligned with the coalition contract. Updated in April 2023, a May 2024 progress report outlines its implementation. Key responsibilities lie with BMF, BMDV, BMWK, and the Federal Chancellery. An interministerial working group, headed by BMDV, oversees implementation, supported by a 17-member expert board from the economy,	The Digital Strategy for Germany focuses on three fields of action. Networked and Digitally Sovereign Society: Focuses on access to digital tools, high-speed internet, and participation. Vulnerable groups include women, youth, older people, people with disabilities, LGBTQI+ communities, ethnic minorities, and migrants. Measures include increasing digital literacy, family support (e.g., <i>Digital Family Assistant</i>), and improving digital accessibility (e.g., Accessibility Strengthening Act). Innovative Economy, Labour, Science, and Research: Targets women's underrepresentation in STEM, promoting initiatives like <i>GirlsDay</i> and <i>YouCodeGirls</i>. Addresses skilled worker shortages, focusing on women, migrants, and	The strategy's actions were planned through dialogue processes with "relevant" stakeholders, such as in health and youth digitalization strategies, though details on initiation and participants are unclear. The 2022 strategy, updated in 2023 without new actions, likely incorporated input from the advisory board (private sector, science, civil society), but this is not explicitly stated. Implementation is assigned to respective departments, as digitalization is a cross-cutting issue. While civil society, the private sector, and education are mentioned as target areas, their direct involvement is rarely specified. Exceptions include initiatives like <i>Civic Data Labs</i>, implemented with civil society organizations. Monitoring is overseen by the BMDV through the state	The strategy highlights key digital priorities: Data and AI Regulation: Supports the DSGVO as a global standard for safe data economies. Advocates for innovation-friendly EU regulations, including the EU Data Act for fair data use and the AI Act to balance innovation with security and rights. Media and Technology: Promotes media independence via the European Media Freedom Act.	complaints against these amendments. The strategy reflects Germany's challenges in embracing the digital age, fostering green tech innovation, streamlining bureaucracy, and leveraging AI to transform its economy. While emphasizing digitalization's role in economic growth, it highlights areas like autonomous vehicles—critical for Germany's car industry—while briefly mentioning public transport integration and predictive track maintenance (BMDV, 2023, pp. 20-21).

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
		academia, and civil society. Monitoring includes qualitative reviews for 19 lighthouse projects, quantitative tracking via a database, and impact evaluations for selected projects. Semestral state secretaries' meetings ensure coordination. Private sector, academia, and civil society primarily execute activities, such as <i>lighthouse projects</i> and initiatives like "<i>Civic Coding – AI for the common good.</i>" Local governance handles areas like smart cities and construction digitalization, with federal support for planning and	underrepresented groups. Companies are primarily responsible, with federal strategies as support. Learning, Digital State: Aims to make federal services more accessible, efficient, and citizen oriented. Highlights digital healthcare tools (e.g., e-prescriptions) to reduce health inequalities and ease caregiver burdens. Key groups mentioned include women, migrants, youth, older adults, people with disabilities, and LGBTQI+ communities, though some, like LGBTQI+ and socially disadvantaged groups, lack detailed strategies. Most initiatives are centred on labour policies, with limited focus on social and health policies.	secretary round on digitalization, with input from the advisory board. Horizontal and vertical monitoring involves federal, state, municipal, and EU levels, along with economic and societal actors, though details are limited.	Strengthens microelectronics through the European Chips Act and IPCEI initiatives. Platform Economy: Plans to safeguard platform innovation while improving employee conditions via EU legal frameworks. Funding Access: - Introduces the Z-EU-S IT funding portal for streamlined access to European Social Fund resources. Emerging Technologies: Promotes multilateral innovation through participation in NATO and EU initiatives like DIANA, NIF, and HEDI.	A "feminist digitalisation" approach is mentioned as a way to strengthen democracy and address power dynamics in digital spaces, but lacks detail, systematic critique, or measurable indicators (p.24). Despite acknowledging connections between digitalization, social aspects, and green technologies, the strategy fails to integrate these dimensions comprehensively. Implementation remains slow due to administrative complexities and budget cuts (Tagesschau, 2024a; Deutscher Bundestag, 2023).

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
Hungary		stakeholder inclusion.			Roles, responsibilities, and stakeholder involvement are not further detailed.	
	Action Plan HU	The Action Plan outlines four levels for climate policy: international, European, Carpathian Basin, and domestic, omitting sub-national governance. It emphasizes the Carpathian Basin as a unified ecological entity sustaining Hungary's wellbeing. Major polluters are targeted to fund climate neutrality, but Hungary's domestic responsibilities are overlooked. The Plan calls for protecting Cohesion Funds, supports nuclear energy in Europe,	The Action Plan addresses poverty and (implicitly) climate justice in Hungary's green transition. It highlights the issue in relation to transition costs, household energy expenses, and localized pollution. Four key criteria guide Hungary's climate-neutral goals, with three focusing on poverty and climate justice: Major polluters (countries and companies) should bear transition costs. Energy and food prices for Hungarian families must not increase. The EU's budget must not fund climate goals by reducing the Cohesion Fund for poorer countries. These issues appear in EU affairs, energy costs, and green economy transitions. Notably, the Plan does	The Action Plan mentions households and some market actors but does not address their formal or informal involvement in policy processes, leaving broader stakeholder engagement largely unaddressed.	The EU features in the Action Plan as one of four levels for climate policy implementation. The Plan asserts that the EU's budget must not reduce Cohesion Funds for poorer countries and stresses the need to support, not restrict, nuclear energy for climate neutrality. While aligned with the EU Green Deal's goal of achieving climate neutrality by 2050, the Plan offers little further detail on the EU's role.	The Action Plan is a strategic document for Hungary's green transition, focusing on carbon neutrality but omitting key areas like education, industry, and drought management. It selectively highlights government efforts, serving more as a political communication tool than a comprehensive plan. While some goals have been achieved—such as rapid solar power growth and the bottle deposit system—contradictions

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
		and remains largely strategic. National measures include creating a Waste Management Authority and a deposit return scheme led by state-owned MOL, central to Hungary's green transition. The Mátra Power Plant will phase out coal by 2025, transitioning to solar and waste-to-energy systems. Solar power incentives aim for 6,000 MW capacity by 2030, and a green bus program mandates electric buses in cities over 25,000 residents. Lastly, green bonds will finance climate-friendly public initiatives.	not mention any non-governmental stakeholders.			exist. The deposit system imposes costs on consumers, conflicting with the Plan's aim to avoid household expenses, though it supports climate justice by providing income opportunities for vulnerable groups. Overall, the Plan prioritizes broad visions over detailed actions, with uneven implementation results, reflecting political priorities rather than a holistic approach to the green-digital transition.

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
Hungary	Green Procurement	Hungary's Green Public Procurement Strategy identifies key institutions for implementation. The National Assembly ensures transparency, efficient fund use, and shapes legislation. The Public Procurement Authority leads strategy and promotes sustainable procurement, while local governments adapt national regulations to local practices. The Hungarian National Bank encourages green financial products in state-owned companies and financial institutions. Public procurement,	The document does not dedicate specific sections to social issues like gender equality, ethnicity, age, disability, or poverty but integrates social responsibility within green public procurement goals. It emphasizes balancing social well-being and environmental sustainability, particularly in areas like waste management and energy-efficient infrastructure, indirectly benefiting vulnerable groups (e.g., those in poverty) and creating jobs in green sectors. While disability and age are not explicitly addressed, universal design in public infrastructure implicitly supports accessibility for the elderly and disabled, aligning with the EU's Just Transition Fund. Energy-efficient public buildings also reduce costs, improving access for disadvantaged citizens (page 15). The strategy highlights businesses as key stakeholders, urging innovation in green products and services to align private interests with sustainability goals. Civil society, though briefly mentioned, contributes by shaping policies, raising awareness, and advocating	The strategy highlights stakeholders like local communities, NGOs, and businesses, but focuses more on environmental issues than social equity. Municipalities are expected to promote sustainable practices, businesses provide green technologies, and civil society monitors and influences policy. While the strategy emphasizes inclusive decision-making and broad consultations, it lacks specific mechanisms to ensure the active involvement of marginalized groups. This appears more rhetorical than practical, reflecting broader trends in the Hungarian government's approach.	Hungary's Green Public Procurement Strategy aligns with key EU policies, emphasizing sustainability and the green transition. It reflects the EU Green Deal goals of achieving climate neutrality by 2050, promoting energy efficiency, and using renewable energy to reduce emissions. The strategy incorporates the Circular Economy Action Plan (2015), focusing on resource efficiency, waste reduction, and sustainable solutions. It also aligns with the 2030 Climate Target Plan, which aims to cut greenhouse gas emissions by 55% by 2030, and supports the EU's Clean Vehicles Directive by promoting electric vehicles and infrastructure. Additionally, the strategy addresses the Sustainable Development Goals (SDGs), particularly those on sustainable cities, responsible consumption, and climate action. It integrates the Energy Efficiency Directive and Renewable Energy Directive to advance energy-efficient	The document outlines Hungary's framework for environmentally sustainable public procurement, aligning with EU goals like energy efficiency, renewable energy, and a circular economy. It positions Hungary within the EU's green agenda but reveals significant gaps between objectives and actions. Hungary's rule of law issues, corruption concerns, and judicial independence have led to EUR28.6 billion in frozen EU funds, undermining the strategy's focus on transparency and sustainability. Slow progress in reducing reliance

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
		accounting for 14% of GDP in the EU, can drive societal goals like the green transition. The strategy defines green public procurement as acquiring goods, services, or works with reduced environmental impact compared to alternatives, exceeding mandatory environmental requirements. Key policy areas include energy, transportation, infrastructure, and waste management. The strategy focuses on adopting energy-efficient technologies (e.g., solar energy), green vehicle procurement, sustainable building practices,	for marginalized groups. Local communities are entailing beneficiaries of green infrastructure improvements, such as better air quality and housing.		public buildings and renewable energy in services. The document highlights EU funding programs, like the Cohesion Fund and ERDF, as essential for achieving these goals. Overall, the strategy ties Hungary's procurement practices to EU climate and sustainability objectives.	on Russian fossil fuels and stalled wind energy development further highlight the gap between rhetoric and implementation. While the strategy appears ambitious, governance challenges and inconsistent actions weaken its credibility and effectiveness.

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
		and circular waste management solutions. Non-governmental stakeholders include businesses, civil society, and academia. Businesses are urged to provide eco-friendly solutions, civil society contributes to policy discussions, and academia offers expertise for developing and assessing green procurement strategies.				
Spain	Shock plan for sustainable, safe, and connected mobility in urban and metropolitan areas	The Shock Plan for Sustainable, Safe, and Connected Mobility (Component 1 of the RTRP) aims to decarbonize urban transport, improve air quality, and optimize urban	This component addresses social issues such as gender inclusion and universal accessibility, emphasizing the benefits of reliable public transport for vulnerable groups: Women: Studies highlight the gender perspective in electric mobility, noting women prefer	The policy-making process for Component 1 involves multiple stakeholders across planning, implementation, and evaluation phases: Planning:	The Sustainable Mobility Shock Plan aligns with key EU regulations and strategies: Directive (EU) 2018/2001: Promotes integrating renewable energy in transportation.	The Sustainable Mobility Shock Plan aligns with national and EU policies, emphasizing urban decarbonization and sustainable transport. It involves multiple stakeholders,

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
		infrastructure through targeted measures and investments. The Ministry of Transport and Sustainable Mobility (MITMA) leads the plan with cooperation from: MITECO: Focuses on environmental aspects and EV infrastructure deployment. General Administration: Oversees improvements to local train services (Cercanías) through ADIF, ADIF AV, and RENFE. Secretary of State for Transport: Drafts the sustainable mobility law, coordinated under	fewer polluting technologies but are less likely to purchase new vehicles short-term. Understanding these factors can improve support measures. Rural Households: Rural households spend a higher share (5.6%) on fuel compared to the national average (4.2%). The document stresses the need for electrification initiatives to address challenges in low-population areas, such as long distances and urban remoteness. People with Disabilities: Grants for adapted plug-in electric vehicles aim to improve accessibility and equity for this group. While the document highlights these issues and calls for tailored measures, it does not identify specific entities responsible for implementation or oversight.	MITMA sets national goals for sustainable urban mobility and collaborates with local governments to implement low-emission zones. PNIEC 2021-2030 provides direction for reducing greenhouse gas emissions. The National Air Pollution Control Program targets a 35% reduction in private vehicle use by 2030 (p.8). Implementation: Municipal authorities, guided by MITMA, implement low-emission zones and support public transport. Private-sector collaboration is encouraged for EV infrastructure and hydrogen-powered transport. Public funds aid municipalities in adopting sustainable technologies. Evaluation:	Regulation (EU) 2019/631: Sets CO₂ emission standards for cars and vans. European Green Deal: Supports climate neutrality by 2050 and urban decarbonization goals. Flagship 2: Recharge and Refuel: Aligns with European Commission recommendations and Spain's Strategic Energy and Climate Framework to reduce emissions and boost renewable energy use. PNIEC 2021-2030 and the Long-Term Decarbonization Strategy 2050: Focuses on greenhouse gas reduction and mobility transformation, aligning with EU climate targets.	including municipal authorities and private companies, to implement low-emission zones, digital transformation, and sustainable mobility initiatives. However, the plan lacks depth in addressing the needs of vulnerable groups, such as women, people with disabilities, low-income populations, and older individuals. Strengthening its focus on social equity could enhance inclusivity in Spain's mobility system.

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
		the Government's Delegate Commission for Economic Affairs. Governance and Stakeholders: Transport Sector Conference: Coordinates regional actions, allocating EUR900 M in investments and EUR400M for green transport programs. MITMA-FEMP Working Group: Manages relations with local councils. Annual Competitive Grants: Funds projects through applications from local councils. Cross-border Coordination: Ensures continuity		MITMA and local governments assess the impact of policies, monitoring emission reductions, EV adoption, and public transport usage, ensuring progress toward urban mobility goals.		

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
Spain		of EV charging corridors with neighbouring Member States. Private sector involvement is encouraged, especially in EV production and green hydrogen technologies, driving innovation and infrastructure development.				
	Renewable Energy Deployment and Integration (Component 7)	The MITECO oversees the renewable energy transition, ensuring the implementation of legislative frameworks in collaboration with ministries and local authorities. MITMA focuses on integrating renewable energy into transport and	The report highlights that the impacts of renewable energy deployment vary across population groups, emphasizing the need for gender, inclusion, and accessibility considerations, particularly in training and capacity-building activities. Gender Equality: Focuses on addressing employment and wage gaps, integrating gender perspectives in training, and aligning with the Long-Term Strategy (ELP 2050) to promote	The policy-making process for renewable energy reforms involves stakeholders at all phases: Planning: Government bodies (State and ACs) collaborate under laws like Royal Decree-Law 23/2020 and the Climate Change and Energy Transition Law, supported by environmental sensitivity maps.	The document aligns Spain's renewable energy initiatives with key EU policies and frameworks: Memorandum of Split: Signed in June 2020 with the EU and 23 Member States to support energy transitions on European islands. EU Flagship Projects: Power Up: Expands renewable technology adoption.	The document outlines Spain's renewable energy transition, aligning with EU goals on gender equality and energy access. However, it lacks clarity on stakeholder coordination at regional and local levels and provides limited focus on marginalized groups like migrants and people with

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
		distribution sectors. Key stakeholders at other governance levels: Energy Sector Conference: Coordinates regional projects, approves installations under 50 MW, and promotes self-consumption through working sessions with municipalities. Public Participation: A 2020 consultation for the "Self-consumption Strategy" gathered 91 contributions from civil society, businesses, and organizations. Local Governments:	gender equality in decarbonization efforts. Disability: Notes the need to reduce energy expenses for households with elderly women and people with disabilities through financial aid and energy efficiency measures. Social and Territorial Cohesion: Renewable energy projects aim to strengthen cohesion in rural, urban, and Just Transition Zones by improving services and infrastructure. Groups at Risk of Social Exclusion: Highlights training opportunities in energy communities to include people at risk of exclusion. Isolated Areas: Prioritizes renewable energy projects for rural and island regions to reduce energy costs and promote "Smart Islands" initiatives. However, there is little mention of migration backgrounds, the elderly, or other vulnerable groups, and the role of stakeholders remains	Public consultations: Key events like the National Self-Consumption Strategy and Energy Communities gathered input from civil society, businesses, NGOs, and universities. High-Level Forums addressed topics like biogas, offshore wind energy, and long-term strategies. EU Projects under Horizon 2020 focused on removing barriers for energy communities. Implementation: Supported by EU Regulation 2021/241 and a regulatory framework promoting renewables, including auctions, grid integration, and building integration. The National Self-Consumption Strategy established working groups, best practice guides, and technical outreach.	Recharge and Refuel: Integrates renewables into transport through electricity or renewable fuels. Modernize: Streamlines approval processes for renewable projects in line with Directive (EU) 2018/2001. Reskill and Upskill: Promotes professional training and education on energy communities and island energy transitions (p.8). Regulation (EU) 2020/852: Defines renewable energy activities as substantial contributors to climate change mitigation, including generation, storage, and grid reinforcements (p.52). Article 21 of Directive (EU) 2018/2001: Promotes self-consumption by removing barriers, protecting user rights, and encouraging consumer participation. Focuses on solar energy and identifies regulatory and coordination challenges,	disabilities. While public participation is noted, unclear stakeholder roles and insufficient attention to vulnerable populations could hinder effective implementation. Improved coordination and inclusivity are needed for the twin-transition.

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
		Develop good practice guides to support self-consumption initiatives at the local level.	unspecified beyond participation in citizen forums.	Evaluation: The PNIEC underwent Strategic Environmental Assessment with public and stakeholder consultations. Measures align with the “Do No Significant Harm” principle, ensuring water protection, circular economy goals, pollution control, and biodiversity preservation. Spatial zoning maps minimize environmental impacts, while sustainability criteria for biomass and biofuels meet EU Directive 2018/2001 standards.	with investments detailed under Investment C7.I1 (p.19).	
Spain	Digital Skills Plan (Component 19)	The Ministry of Economy, Trade, and Entrepreneurship leads the national digital skills component, focusing on key strategic actions:	This section examines stakeholder involvement and focus areas on the Plan: Gender equality: A key focus, with initiatives to promote women’s digital literacy, increase female STEM participation, and reduce the digital gender gap. Specific programs include collaborations with the Government Delegation	This section outlines stakeholder roles in the policy-making process across planning, implementation, and evaluation phases: Planning: A key challenge is stakeholder coordination. The Digital Skills Hub serves as a central body	The Digital Skills Plan aligns with key EU policies and frameworks: DigComp Framework: Guides the development of digital competencies, focusing on secure online skills, particularly for older adults, supported by volunteer networks (p. 11).	The National Digital Skills Plan (Component 19) addresses key aspects of the twin-transition, focusing on vulnerable groups such as women, unemployed youth, and older workers,

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
		National Digital Skills Plan (PNCD) Transversal digital skills Education digital transformation Digital skills for employment Training digital professionals Collaborative initiatives include: Digital training centres with the Ministry of Education (MEFD). Awareness campaigns led by the Sub Directorate of Talent and Digital Entrepreneurship and Red.es. Women's digital literacy in	for Gender Violence and the EOI Foundation to support unemployed women over 45 (p. 4; 5; 42). Age: Programs like Futur@ address youth (16-30) unemployment, while older workers receive support for transitioning into technological roles (p. 21). Child poverty: Initiatives aim to provide basic digital skills to children at risk of exclusion through tailored content and educational materials. Demographic inequalities: The Plan highlights Spain's digital divide, focusing on connectivity and socio-economic disparities in rural areas (p. 12). The document does not specifically address ethnicity, migration, or disability, opting instead for a general focus on digital inclusion (p. 1; 4; 5).	for collective decision-making among ministries and organizations (p. 10). Implementation: Public-private collaborations and competitive tenders focus on vulnerable groups, such as women, children in poverty, and workers. The Ministry of Economy and MEFD manage Digital Training Support Centres, while SEPE and FUNDAE drive upskilling and reskilling programs. Vocational Training (FP Digital) enhances workforce digital skills, and MEFD and INCIBE equip teachers with digital competencies to meet industry needs (p. 13; 23; 24). Evaluation: Specific evaluation methods are not detailed. Self-assessment tools help teachers identify training needs, while Horizon2020 and EU AI recommendations guide	Recognition of Digital Skills: Cites EU Recommendation 2006/962/EC, emphasizing digital competence as essential for living, learning, and working (p. 29). Code of Conduct for ICT: Ensures compliance with energy efficiency and material durability standards (Directives 2009/125/EC, 2012/19/EU, 2011/65/EU), promoting equipment reusability, recycling, and proper treatment at the end of its lifecycle (p. 35). EU Cost Estimation Methodology: Adopts EU guidelines to streamline cost calculations and ensure alignment with EU funding mechanisms (p. 35). These frameworks shape public authority actions and stakeholder roles in advancing digital literacy and sustainability.	with broad public-private stakeholder involvement. However, the plan lacks coordination between regional and local governments and fails to clarify specific stakeholder roles, which may hinder effective implementation. Additionally, limited attention to gender, ethnicity, and disability reduces its potential to ensure a fully inclusive transition. To achieve Spain's digital and green transition goals, stronger stakeholder engagement and clearer governance frameworks are necessary,

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
		partnership with ministries and the EOI Foundation Upskilling/Reskilling programs with MEFD, SEPE, and FUNDAE. Public workforce training in collaboration with Défense, Health, and Interior ministries. Social service digitalization by the Ministry of Inclusion and Migration. Regional and other stakeholders: Sectoral Conference for the Demographic Challenge: Implements training for digital citizenship.		program evaluation in technical, scientific, and social areas (p. 15; 27).		particularly at local and regional levels.

<i>Country</i>	<i>Document Name</i>	<i>Roles and responsibilities</i>	<i>Relevant issues and stakeholders' involvement</i>	<i>Policy-making process</i>	<i>Reference to the European Union documentation</i>	<i>Overall Judgment</i>
<i>Ireland</i>		State Foundation for Health and Welfare (CSAI): Develops digital skills to reduce child poverty. Red.es and INCIBE: Promote technological humanism and provide Open Educational Resources. These initiatives ensure broad collaboration across ministries, regions, and public organizations to advance Spain's digital transition.				
	National Adaptation Framework 2024 (NAF 2024)	The National Adaptation Framework (NAF), mandated under the Climate Act, guides Ireland's climate adaptation efforts at national,	The document broadly mentions vulnerabilities but lacks details on their assessment or the specific needs of marginalized communities. While NGOs are recognized as intermediaries between the government and	The National Adaptation Framework (NAF) outlines roles, processes, and reviews for Ireland's climate adaptation strategy: Planning:	The 2024 NAF lacks a dedicated section on EU documentation but references EU legislation throughout, highlighting the role of specific departments and bodies:	The document emphasizes a holistic governance approach involving government departments, civil society, and NGOs, with engagement

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
		sectoral, and local levels. Key actors and processes include: Government and Public Institutions: The Department of Taoiseach oversees coordination, supported by the Senior Officials Group and sub-groups on governance. The DECC leads NAF development and review, while Local Authorities (LAs) implement Local Climate Action Plans (LACAPs). Climate Action Regional Offices (CAROs) coordinate local climate action, and sector-specific ministries address	communities, the term "marginalized" is not defined. No clear policy areas are identified regarding vulnerable or marginalized groups. References to structural vulnerabilities are brief and general, with no specific stakeholders representing these groups mentioned in the text.	Led by DECC in collaboration with government departments, local authorities, and stakeholders, including EPA, Met Éireann, and OPW. Planning is evidence-based, informed by EPA research and data to ensure a climate-resilient society. Implementation: DECC leads implementation, fostering cross-sector cooperation and supporting local adaptation plans (LACAPs). Ministerial guidelines ensure sectoral adaptation plans (SAPs) align with the Climate Act, though vulnerable groups and stakeholder details are lacking. Evaluation: The NAF is reviewed every five years, with input from public consultations, sectoral stakeholders, and the Climate	The DECC oversees Ireland's reporting on climate adaptation and aligns with the EU Adaptation Policy, communicating developments to sectors and local authorities via NASC. The EPA is responsible for the National Climate Risk Assessment, linked to the European Climate Risk Assessment (EUCRA) published in March 2024. EU funding programs like Horizon Europe, LIFE, EU Innovation Fund, ERDF, and COST support climate research in Ireland. Stakeholder Roles: The Climate Research Coordination Group focuses on observation, climate modelling, risk assessment, and adaptation responses, aligning national research with EU developments. Issues and Interests:	mechanisms like Climate Conversations and the National Dialogue on Climate Action. However, it lacks recommendations for deeper democratic processes, particularly for evaluation and monitoring, and does not incorporate creative community engagement methods suggested by scholars like Devaney et al. (2020) and Mullay et al. (2022). While vulnerabilities and vulnerable communities are mentioned, they are addressed only in general terms, with no intersectional analysis or gender

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
		themes like natural resources, infrastructure, and health. The EPA supports adaptation with climate risk assessments, research synthesis, and policy advice, while the National Adaptation Steering Committee (NASC) ensures policy coordination. Stakeholders Beyond Government: Climate Ireland, managed by the EPA, provides tools, resources, and capacity-building for adaptation. Research institutions (e.g., Met Éireann, SEAI)		Change Advisory Council (CCAC). The CCAC assesses sectoral progress, provides independent advice, and emphasizes integrating just transition principles into adaptation policies. Public Engagement: Climate Conversations and the National Dialogue on Climate Action (NDCA) gather public input, engaging over 12,000 individuals since 2021. Platforms like the National Climate Stakeholder Forum and National Youth Assembly ensure wider participation in climate policy discussions. The document emphasizes governance and accountability but lacks clarity on specific stakeholders beyond government actors.	The document does not detail how stakeholders represent interests or address issues related to EU legislation.	focus. This aligns with critiques, such as those by Dukelov et al. (2024), that Irish climate policy includes marginalized voices minimally and remains gender blind. Additionally, the framework's focus on the 'green' transition overlooks the digital transition's complexity, as noted by Moore (2020), highlighting the need for greater scrutiny of its interconnected challenges. The document does, however, emphasize the critical role of research organizations in evaluating and advancing climate action.

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
---------	---------------	----------------------------	---	-----------------------	---	------------------

and the Climate Research Coordination Group advance climate data and solutions.

The private sector plays a key role in innovation and implementing resilient practices but requires enabling policies and further risk research.

Civil society and NGOs, such as An Taisce and Friends of the Earth, connect communities with government efforts and advocate for marginalized groups.

Mainstreaming Climate Adaptation: Government bodies integrate climate resilience

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
Ireland		into policies, infrastructure, and decision-making processes, supported by research institutions and local communities. Collaboration across sectors and stakeholders is emphasized as essential to tackling climate change effectively.				
	Harnessing Digital	The Digital Strategy for Ireland emphasizes strong governance under the Cabinet Committee on Economic Recovery and Investment, chaired by the Tánaiste, with cross-departmental cooperation and stakeholder engagement.	The Digital Strategy for Ireland addresses key issues related to inequality and stakeholder involvement but lacks clarity and depth in several areas: Key Interests and Issues: Women: Efforts to increase women-led startups and leadership roles are mentioned but lack detail or identified stakeholders (p.13). Vulnerable Groups: Older cohorts disadvantaged socio-economic groups, and workers at risk of job	The document lacks details on stakeholder participation in addressing structural inequalities during policy planning. While a “stakeholder-led approach” is mentioned, the engagement of groups like SMEs or start-ups remains unclear. Reports highlight the risks of digitalisation, noting 370,000 jobs at high risk of automation and further impacts on lower-educated workers, but the role of stakeholders in shaping	The document aligns Ireland’s Digital Strategy with EU policies, particularly the Europe Digital Decade 2030 and its four pillars: Skills, Digital Infrastructure, Business Transformation, and Digital Public Services (p.3). Key References to EU Policy: Ireland will contribute to EU targets through a Digital Decade Strategic Roadmap and annual reporting via the Digital	The debate on digital transformation in Ireland remains focused on jobs and skills, reflected in the strategy’s emphasis on enhancing digital skills and workforce upskilling. While the National Economic and Social Council (NESC) highlights digital inclusion through

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
		Four Dimensions of the Strategy: Digitalisation of Enterprise: Focuses on SME adoption of digital technologies, supported by Teagasc, Science Foundation Ireland, and local communities. Key actors include Enterprise Ireland (EI), the Enterprise Digital Forum, and Local Enterprise Offices (LEOs). Multinational Corporations (MCNs) and digital start-ups play a role, with initiatives like the National AI Cluster driving collaboration. The Strategy consists of four	loss due to digitalisation are noted, with emphasis on skills development and adaptability (p.30). Inclusivity: Basic digital skills and infrastructure are prioritized to ensure no one is left behind, with supports for assisted and non-digital options. Public Sector: Goals include 20% remote work, which could address caring needs indirectly.	policy is unspecified (p. 25, 41). Implementation of the strategy The strategy emphasizes an “ethical and rights-based” approach to digitalisation, ensuring no part of society is left behind. However, the role of regulators such as the Data Protection Commission, Media Commission, and Competition and Consumer Protection Commission in supporting vulnerable groups is not detailed. The Media Commission will regulate online safety through binding safety codes, while a Digital Services Coordinator will safeguard digital users’ rights under the Digital Services Act (p. 42). 2.3.3 Evaluation Oversight of the strategy involves the Cabinet Committee on Economic Recovery and Investment, supported by sub-groups led by the Taoiseach and Department of Enterprise. Ongoing stakeholder	Economy and Society Index (DESI) (p.44). EU legislation, including the Network and Information Service Directive (cybersecurity), Data Governance Act, and Open Data Directive, shapes Ireland’s policies on digital infrastructure and public services (p.39, p.43). A regulatory body for online service providers will be required to comply with future EU legislation on child sexual abuse (p.43). Stakeholder Roles: European Digital Innovation Hubs, established in 2022, aim to provide SMEs with access to digital technology. Interests and Issues: The document references the EU goal of a “digital society where no one is left behind” but lacks specifics on vulnerable groups or strategies to address their needs. Ireland's progress	connectivity, devices, and skills, the strategy addresses these only at a general level, with limited targeted examples for vulnerable groups like older citizens or disadvantaged students. The strategy lacks an intersectional approach to vulnerable groups. Gender is minimally addressed, with women mentioned only in relation to digital start-ups and leadership roles. There is no mention of ethnic minorities, Travellers, the Roma community, or disabled persons. Although stakeholder engagement and public participation are emphasized,

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
		dimensions: The Strategy consists of four dimensions: (1) Digital Transformation for Business; (2) Digital Infrastructure; (3) Skills; and (4) Digitalisation of Public Services. Other Stakeholders: National Cyber Security Centre (NCSC): Ensures infrastructure security. Technological Universities: Deliver digital skills aligned with industry needs. Public Involvement: Co-creation of services to meet societal needs.		engagement is mentioned at EU, national, and international levels, but no specifics are provided on including representatives of structurally vulnerable groups in monitoring or evaluation processes (p. 44).	will be benchmarked against EU leaders using DESI metrics (p.44).	the strategy provides no specifics or examples, omitting actors representing marginalized groups. Furthermore, local governance is limited, reflecting Ireland's centralised system, with minimal local-level involvement, particularly in the policy design phase.

Country	Document Name	Roles and responsibilities	Relevant issues and stakeholders' involvement	Policy-making process	Reference to the European Union documentation	Overall Judgment
		The strategy integrates green and digital transitions, emphasizing collaboration with businesses, public bodies, and research institutions to drive innovation and sustainable growth.				

Annex III: Good practices

Germany

Ariadne – Evidence-based Assessment for the Design of the German Energy Transition

1.1 Introduction

In July 2020, the Ariadne project was officially launched, driven by over a hundred scientists from twenty-seven research organisations/actors, such as research institutes, consulting companies, and universities taking part in it (Ariadne-Projekt, 2020). Ariadne is part of the 'Kopernikus project', and it is split into four subprojects conducted in three phases: Phase 1 (2016-2019) focusing on electricity and transport, Phase 2 (2019-2022) focusing on transport and heating, and Phase 3 (2022-2025) focusing on the effects of climate policy across society (ibid.; Ariadne-Projekt, n.d.-a). The project's core feature is that it itself is a policy monitoring process involving people from citizens to public authorities in governance to achieve energy and mobility transitions with social consensus. Scientists and selected citizens address situations and needs of low-income households through 'citizens' deliberation', research and analyses. These are delivered to stakeholders and authorities at regular dialogues as part of the 'learning process'.

1.2 The aim

Ariadne project is funded by the Federal Ministry of Education and Research (BMBF) with 30 million euro over three years, as part of the Kopernikus projects (Ariadne-Projekt, n.d.-a). It aims at incentivizing a collective learning process between science, society and decision-makers in politics and business about the systemic impacts of energy transition strategies and about policy instruments for a socially-just climate policy (Ariadne-Projekt, n.d.-b). The learning process is implemented through a combination of research publications, providing socially desirable policy options and diverse dialogue formats, inviting stakeholders across business, politics, NGOs, and civil society to address and debate about each topic (Ariadne-Projekt, n.d.-c). The dialogue formats consist of dialogue forums, stakeholder dialogues, parliamentary events, policy workshop, the Kopernikus Symposium and "citizen's deliberation" (Ariadne-Projekt, n.d.-d). The objective of the dialogues is to provide political decision-makers with diverse perspectives on each theme of the energy transition and to advise scientific policymaking (ibid.; Ariadne-Projekt, n.d.-c). The deliverables are mainly demonstrated in the form of published policy briefs, thematic dossiers, background papers, visualisations and interactive dialogue platforms (Ariadne-Projekt, 2020). Some of the publications and research address burdens on citizens, especially for low-income households, being caused by energy transition. For example, researchers analyse the rising CO₂ prices in the process of green energy transition and provide cost mitigating options for households, such as the withdrawal of the levy of the Renewable Energy Act (Ariadne-Projekt, 2022). In addition to stakeholder dialogue formats, the "citizens' deliberation" aims to include citizens in all stages of the Ariadne project to assess their perspectives and to strengthen the exchange between scientists and citizens (Ariadne-Projekt, n.d.-a; Ariadne-Projekt & PIK, 2023a). The results of the deliberations are incorporated in designing policy options and future solutions and they have been presented to stakeholders in politics, business, and civil society at the citizens' summit (Blum et al., 2021; Ariadne-Projekt, n.d.-a). By the process of citizen deliberation, hence by the process of including diverse needs and values, the social legitimacy of the project's research on policies is enhanced (Blum et al., 2021).

1.3 *Roles and responsibilities*

- **Role of scientists** – they are the drivers and coordinators of the Ariadne project as following: they design the project and its governance structure. A focus is on trying to generate more data and knowledge about impacts of the energy transition by detecting missing gaps (e.g., on the social perspective of energy and climate policy). The scientists invite multiple stakeholders and guide them to the gaps and solutions; and they provide a service with the Ariadne platform by bringing policy recommendations/instruments for reaching climate goals. Also, the scientists are mobilising citizens and increasing the acceptance of the energy transition in the public.
- **Role of public authorities** (e.g., Federal Ministry of Education and Research (BMBF) and Members of the Bundestag) – they are sharing windows of opportunity and current status quo of the climate and energy policy on national and international levels. The BMBF is also funder of the project, hence it is the receiver of the service – in consequence, the authorities need feedback on the impacts of their policy visions on society and industry, as well as more scientific data/information and future scenarios of the energy transition. Furthermore, they need the support of citizens for setting the political agenda and not perceiving any resistance on their policies' implementation. In response to the invitation of scientists, the public authorities join each parliamentary dialogue, workshops (or Politikwerkstatt), and events such as expert discussions, conferences, and symposia, where they listen to policy options from scientists and discuss them (Ariadne-Projekt, n.d.-c). In 2023, there was a cross party and ministry policy workshop, and the public authorities asked for advice of scientific policymaking to the scientists (ibid.). In 2024, the exchange between scientists and public authorities, as well as with civil society organisations was enabled through another workshop (Ariadne-Projekt, 2024).
- **Role of citizens** (randomly selected 150 citizens for the “citizens’ deliberation”) – they provide everyday knowledge, experiences, concerns and values for the research processes, allowing for alternative transformation approaches and diverse perspectives (close-to-the citizen-research/close-to-the-citizen politics) (Ariadne-Projekt, n.d.-a). They participate with self-efficacy and without intervention of stakeholders and develop their ideas throughout five steps: regional focus group discussions, co-creation workshops, citizens’ conferences, deliberative workshops, and citizens’ summit (ibid.). Their dialogue-driven participation in research enhances mutual understanding and communication between them and science/politics. This can lead to supporting a democratic culture in society and to more energy democracy, since the citizens are mobilised for energy transition projects and they get a saying, so that they experience a feeling of being seen and heard. Their selection process and representation are elaborated in chapter 2.5. Additionally, the citizens participate in joint discussions with stakeholders and their exchange is evaluated and included in Ariadne’s further work (ibid.).

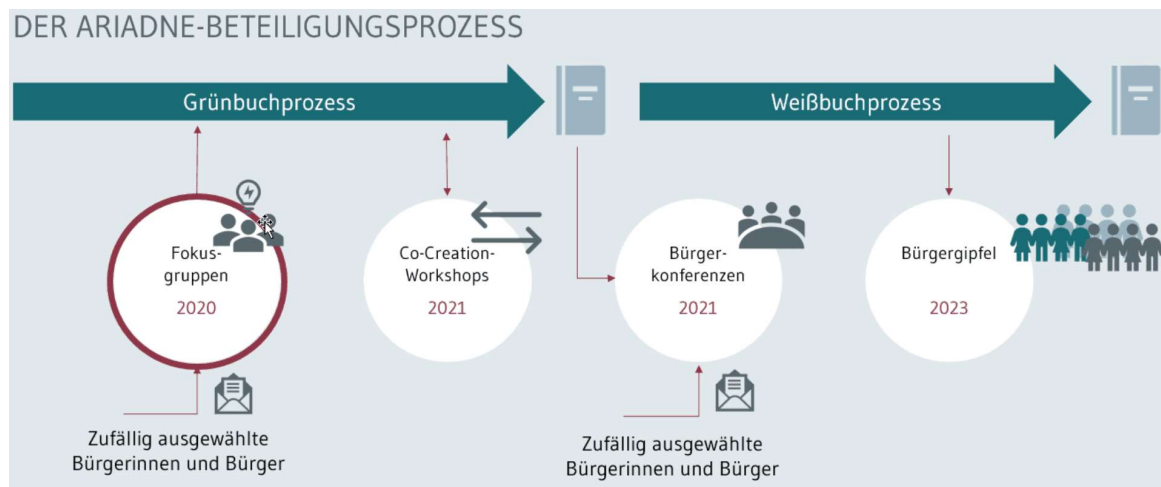


Image 1: The step-by-step process of citizens' participation (Blum et al., 2021).

- Role of civil society organisations and business – they join diverse dialogue formats as stakeholders; however, their selection criteria are uncertain, in terms of which interests and values they represent and how diverse they are. Their roles are also unclear in the Ariadne learning process.
- Commitment, capacity, and support of public authorities and citizens' contribution to gender equality and structurally vulnerable groups – the BMBF funds the Ariadne project to link energy and transport transition better to the society. Scientists invite the Members of Parliament (MPs) and ministry authorities regularly to present their research, develop their approaches together with citizens, advise the authorities, and give them a chance for discussion about each topic, regardless of their political stances. Nonetheless, gender equality is one of the barely discussed topics. Citizens addressed concerns of big industry players against citizen's energy initiatives and energy sufficiency throughout the energy transition, in terms of distributive justice (Blum et al., 2022). They also emphasised social justice of fair participation, social environment for growing ownership and responsibilities, satisfaction for all, energy sufficiency for all, and gap reduction between younger and older generations; cities and rural areas; tenants and homeowners (Blum et al., 2021; Blum et al., 2022). For instance, citizens addressed a potential NIMBY phenomenon regarding installation of energy system (Blum et al., 2021), meaning that citizens are only willing to accept energy transition measures if they do not have to cover the costs or bear the burdens (NIMBY is the abbreviation for "not in my backyard") (ibid.).

1.4 Links between levels of governance

- **Policy monitoring with diverse actors and stakeholders**

The Ariadne project aims at observing and influencing the policy cycle on a permanent basis by constantly monitoring the status quo and changes in energy and climate-policy making. Concretely, the project is providing publications, policy workshops and the participation in Kopernikus symposia. "Ariadne analyses the institutional design of climate policy planning and coordination processes between the political institutions, actors and levels involved [...]. One focus is on the design and adaptation of the climate policy instrument mix, the governance of

sector coupling and the handling of social conflicts arising from the energy transition” (Ariadne-Project n.d.-e). Some exemplary publications are:

- Duncan L. Edmondson, Oskar Krafft, Christian Flachsland, Christian van Ballegooy (2024): Governance capacities in reflexive climate policymaking: the scope, role, and institutional arrangements of policy mix evaluation in the German domestic buildings sector. Kopernikus project Ariadne, Potsdam. <https://doi.org/10.48485/pik.2023.011>
- Germany's foreign climate policy (2021-2025) focussing on the organisation within the German Federal Government and the responsible ministries:
- Ole Adolphsen, Marian Feist, Christian Flachsland, Oliver Geden, Benjamin Görlach, Michael Jakob (2024): Deutschlands Klimaaußenpolitik: Kontext – Rückschau – Weiterentwicklung. Kopernikus-Projekt Ariadne, Potsdam. <https://doi.org/10.48485/pik.2024.017>

1. **Sharing visions based on scientific models** - In October 2021, scientists launched a scenario model demonstrating the ways for achieving the climate neutral goal by 2045, within 25 years (Luderer et al., 2021). This model has incorporated data of industry, transport, building and energy sector to elaborate which are the most important goals for the private sector on the one hand and for policymakers on the other hand. This goal-sharing facilitates the entire policy chain by pointing out the most effective policy needs and options reflecting the cross-sectoral and industrial scenarios (ibid.). Nonetheless, this model lacks the societal perspectives, hence, it is not necessarily facilitating a holistic approach.
2. **Evaluation of the social desirability of policy planning** - Two citizen ‘s conferences were held in November 2021, each about the electricity transition and transport transition, to evaluate the social desirability of policy pathways (Blum et al., 2022). In each, 48 and 54 randomly selected citizens participated, driven by three scientists (ibid.). Citizens were selected with a consideration of gender, age, and origins; however, they were mostly formally well-educated (Blum et al., 2021; Blum et al., 2022). It was reflected by the scientists that immigration background and education levels should be further considered in the future (ibid.). This evaluation process enabled citizens ‘understanding of changes in the energy and mobility sector, which grew not only their ownership and motivation for the transitions (ibid.) but also contributed to grow their capacity to monitor the policymaking process. Additionally, some wishes that were expressed at the citizen conference on renewable energies have been addressed politically afterwards. These include fair wind energy distribution across Germany via the Wind-on-Land Act (2023), financial participation of local authorities in renewable energy projects, and using areas up to 500 meters wide along railways and motorways for solar energy under the Renewable Energy Sources Act (EEG, 2023) (Ariadne-Projekt, 2023).

1.5 Civil society

The framework for public participation in the Ariadne project has already been outlined in chapter 1.2 and 1.4. The following section focusses on the selection process of the citizens and the involvement of CSO’s.

The Ariadne project offers citizens and CSO's a variety of opportunities to engage throughout the learning process. Representatives of CSO's are invited to the diverse stakeholder formats, however, it is not transparent how many CSO's have been part of the project so far. For instance, representatives of The Nature and Biodiversity Conservation Union (Naturschutzbund, NABU), Friends of the Earth Germany, and World Wildlife Fund (WWF) had been invited as speakers during the Kopernikus symposium. In comparison to the rest of the speakers working either in politics, the private sector or research, the share of CSO representatives seemed low during that event.

However, CSO representatives had the chance to bring critical points in dialogue formats such as the third dialogue forum (Ariadne-Projekt, 2022b) – *Shaping the heating transition socially just* – or the fourth dialogue forum on *Challenges of the energy crisis for the heating transition*. Within the third forum, a critical analysis of ownership structures regarding heating systems was mentioned as a measure to enhance social justice. In the fourth dialogue forum social aspects, such as the financing of measures for low-income households, were seen as one of the main challenges for the heating transition and participants emphasised that social and intergenerational justice must be given greater consideration in political debates (Ariadne-Projekt & PIK, 2023b).

When looking at the background of citizens, the Ariadne project aimed at creating a representative mini-public: A two-stage random selection process was carried out targeting citizens from all over Germany with diverse backgrounds. In total, 177 citizens participated in the project, with a share of women ranging between 40% and 55% and a share of men between 45% and 59% (only binary gender data is available). The average age was between 40 and 53 years (Ariadne-Projekt, n.d.-c).

The public debate in the Ariadne project concerns both the transport transition and the energy transition.

- Regarding the citizen's consultation on the transport transition, 2 on-site events and 9 online events were conducted, with a total of 94 citizens. The average age of the participants was 48 years, six people had a migration background; 45% of the participants were women and 55% men.
- 83 citizens took part in 2 on-site events and 8 online events on the energy transition, of which 44% were women and 54% men, and four people had a migration background. The average age was 45 years (Ariadne-Projekt, 2023).

Involving citizens throughout the project creates an opportunity to incorporate an intersectional perspective into the discussions. The citizens' consultation emphasised the importance of addressing the needs of structurally vulnerable groups in both transport and electricity transitions. Participants highlighted the necessity of considering diverse mobility needs, such as those of low-income groups, families, and people with disabilities, to ensure fairness in transport systems. They also called for inclusive structures in the electricity transition, advocating for equitable decision-making, shared benefits, and accessible financial participation options, such as cooperatives (ibid.; Ariadne-Projekt, n.d.-c). However, in the citizen consultation, younger and older people are poorly represented, and the data collected only refers to binary genders. While social aspects are

mentioned in interactive discussions, they do not appear to be a significant focus compared to economic and environmental issues.

1.6 *Barriers*

Civil society organisations and associations do not have a clearly structured role in the governance of the project; their roles seem to be limited. The learning process can be characterised as a bottom-up process driven by scientists.

Lack of linkage between civil society organisations and selected citizens

There is no explicit governance structure for linking citizens to civil society organisations in the deliberation/learning process. This hinders citizens to be informed by advocates for the interests of different social actors and be empowered to address these through sound discussions. Out of the citizens' deliberation process, citizens highlighted the aspects of fair participation and its importance, in designing decentralised energy systems (e.g. citizens energy cooperatives that are highly encouraged on the national level). Nonetheless, the analyses of the aspects failed to be diverse in highlighting who can be marginalised in designing and managing the energy models. Only age and financial power were mentioned; lacking gender perspectives, considering people with health issues, education levels, migration biographies, care work, etc. Civil society organisations could provide citizens with knowledge on considering and fighting for all citizens.

Lack of direct links between citizens and public authorities, citizens and stakeholders

The political consultation is driven by the delivery of scientists' research result to political representatives at dialogues. Here, business and civil society organisations are invited, however, no citizen representatives are. This stands in contrast to citizens' huge role in the governance of the project, depriving them of the opportunity to directly exchange with the authorities and stakeholders. It prevents citizens from observing how the governance system works on all levels and learning about the consultation process. It also challenges them to track and monitor policymaking and whether their addressed values are both explicitly and implicitly well reflected. Furthermore, it worsens transparency, by lacking a regular top-down reporting.

1.7 *Factors for success*

An explicit citizen-inclusive governance framework

Citizens' participation is not an extra but plays a key role in the project. They do not deal with countless new topics but are rather deepening their own discussion further and further through each step (177 citizens discussed for 29 months) (Ariadne-Projekt, 2023b). This method clearly aims at a holistic approach for energy and transport transitions. It enables scientists to include perspectives from everyday knowledge and experience into their analyses. Also, scientists were concrete about involving their learning into developing policy options, assumably motivated by regular meetings with citizens. Despite the lack of top-down reporting, the results were scrutinised once by the authorities and stakeholders in the final phase (citizens' summit) and the need for barrier-free participation of low-income households and affordable electricity was urged (ibid.).

Policy Workshops (Ariadne-Politikwerkstatt)

Especially, the first policy workshop brought together political representatives, regardless of their own political stances, interests or goals, and they shared a common goal for the transition. This practice enables policymaking to be more scientific and fairer, which contributes to considering citizens' well-being. It fulfils a fundamental need to educate policy makers' attitudes towards transformative policies that should aim for social justice.

1.8 Conclusion

The citizen-inclusive governance framework of the Ariadne project was very promising in terms of creating a representative mini-public. However, governance structures need to clearly address every actors'/stakeholder's role and participation. When it comes to the participation of less influential actors, such as citizens, who are unfamiliar with policy consultation, their role should be more highlighted, their participation could be financially reimbursed, and they need to be supported by knowledge experts. Additionally, the Ariadne project lacks transparency regarding the selection of CSO's, especially in terms of CSO's working on gender or diversity topics. Selection criteria of stakeholders need to be clear and transparent; they should facilitate a representation of diverse interests and social backgrounds.

Further, vision sharing is important in the initial stage of a mutual learning or dialogue process. However, these processes must address societal visions as well and involve civil society organisations. Additionally, constructing a "check & balance" structure (e.g., regular joint meetings) between actors can further facilitate and motivate to reflect crucial values and ideas in research and policymaking.

Citizen participation in the Federal Ministry for the Environment, Nature Conservation, Nuclear Safety and Consumer Protection (BMUV in Dialogue)

2.1 Introduction

As the main responsible for government environmental policy, the Federal Ministry for the Environment, Nature Conservation, Nuclear Safety and Consumer Protection (BMUV) has put in place various formats for stakeholder and citizen participation in policy design, grouped in the so called "BMUV in Dialogue". Among the multiple mechanisms for participation, two initiatives are particularly interesting for the FITTER-EU project: the *Community on Sustainable Digitalisation* and the *Climate Adaptation Dialogue*.

Digitalisation is currently one of the main themes of the Ministry and therefore the BMUV has gathered experts from a range of sectors since 2021, including the private and public sector, research, and civil society for building a community on Sustainable Digitalisation. The community was created with the objective of developing joint strategies and gaining multi-level support for achieving a sustainable digital transformation (BMUV, 2024a). Any individual or entity engaged in political, economic, research activities, or other sectors can register to participate in this community by completing an online registration process. It has currently more than 1,400 members (ibid.). Moreover, it has organised multiple meetings on different topics related to (sustainable) digitalisation in climate change. This community enables to connect stakeholders across sectors and present and learn digital best practices (ibid.).

The Dialogue on Climate Adaptation, held mainly between 2023 and 2024, was created with the objective of involving stakeholders and citizens directly in the development of measurable objectives and actions for adaptation to climate change. The dialogue offered several participation formats, including on-site dialogue events, informal online consultations, regional dialogues with citizens, and online surveys.

2.2 The aim

In general, the Dialogue in BMUV processes aim to create dialogue spaces to incorporate the opinions of as many population groups as possible into political decisions, and to close the gap between the wider public, who has professional knowledge on the topic and the federal administration. There are different formats for participation, which are combined in different ways depending on the political programme or law in question. In the case of the Community on Sustainable Digitalisation, the first idea of initiating such a community was set with the *Digital Agenda for Environmental Policy* (BMUV, 2020). Activities include networking and knowledge exchange, participation in interactive events, access to knowledge and information about new trends, technologies, and policy initiatives and the sharing of expertise and contacts. For example, in the 2024 convention of the community, around 130 members of the community received information about projects from the funding programme of the Federal Ministry for Economy and Climate Protection (BMWK), including a project dedicated to developing an intelligent system for sustainable refurbishment measures ([naiS](#)) (BMUV, 2024b). Moreover, recently the BMUV Division for National and Fundamental Matters of Resource Efficiency presented the programme DigiRess (Digital Application for resource efficiency in the circular economy) and its contribution to the German Sustainable Development Strategy to the Community members. The community deals with a wide range of topics relating to both the ecological and social dimensions of digitalisation. For example, aspects such as sustainable consumption, resource efficiency and social justice in the context of digital transformation are discussed in workshops and publications. However, information on the results of such discussions is scarce.

The climate adaptation dialogue is a result of the entry into force of the federal law on adaptation to climate change in July 2024 and the process for the development of a precautionary climate adaptation strategy with measurable targets along 7 clusters:

- Health,
- Infrastructure,
- Land use,
- Urban development, spatial planning and civil protection
- Water
- Economy
- Cross-cutting cluster.

The BMUV led the target development process with the support of the Federal Environment Agency (UBA). In this context, associations, federal states, municipal umbrella organisations and the scientific community were involved to support the development of objectives in the clusters. The stakeholders and citizens participation process and recommendations were compiled in report by the Federal Environment Agency (UBA) (Hoffmann, et al., 2024).

2.3 Roles and responsibilities

The BMUV (2023) states that, considering that the ministry is taking decisions that affect citizens in their daily life, it is fundamental to include diverse perspectives by strengthening a participatory approach. For this reason, there are formats for citizen participation in the various programmes or laws, in which both civil society organisations and citizens play a primarily advisory role.

Regarding the *Community on Sustainable Digitalisation*, the BMUV acts as the main catalysator by incentivising the process, but also as a coordinator by managing the dialogue process. The BMUV wants to reach its goals for the implementation of its Digital Policy Agenda. The ministry is a platform provider – it provides the organisational structures, selects and invites stakeholders from business, science, civil society and administration to help shape the project. By guiding the process, the agenda setting power seems to remain in the hands of the BMUV, since it can select which themes are discussed in events or which areas are researched by scientific stakeholders. In consequence, this appears as a loose top-down approach. Loose in a way, that the BMUV offers space for critical discussion, and that it funds and strengthens an interdisciplinary collaboration between diverse stakeholders as well as raises awareness about innovations and best practices on sustainable digitalisation beyond the borders of the community.

Regarding addressing gender issues or groups in vulnerable situations, the BMUV has declared that it implements gender aspects as a cross-cutting issue, by referring to SDG5 and by the Parliamentary State Secretary Bettina Hoffmann and their minister Steffi Lemke announcing a feminist environmental policy (WELT am Sonntag, 2022; BMUV, 2022a; BMUV 2022b). However, these statements have not resulted in a profound dialogue process or clear guidelines for such a feminist environmental policy.

The stakeholders are suited to the spheres of the quadruple helix – politics, economy, academia and civil society. As members of the *Community on Sustainable Digitalisation*, they are invited as service providers to bring in their expertise and to deliver input presentations during events such as the Community Convention (Bilski, 2023). The selection of community members was initially based on existing stakeholder databases of the project initiators as well as on a thorough AI-assisted web search (ibid.). Experts for specific theme areas were consequently asked to provide information about potentially interested stakeholders. The websites of these persons were analysed, new contacts identified, and the data pool increased. By the method of “Scraping”, the project could identify 1000 organisations and persons and categorise them (ibid.). There is no information available on whether this process applied a gender-just or critical approach that would consider the diversity of the community members in terms of for example academic background, gender, age, or migration biography. It seems to have focussed on a specific expectation of expertise level, which means that people with a high academic level and in higher positions, were invited to the community in its first steps. However, now registration is open to everyone through the website.

In the Climate Adaptation dialogue, stakeholders – more specifically experts from municipalities, associations and academia – were invited to a two-days dialogue event in December 2023 to comment on a draft of the BMUV and other ministries on climate adaptation measures (UBA 2024). Stakeholders were thus involved in a second phase of the policy-making process as commentators and critics, but not by contributing their expertise before the policies were drafted.

Citizens in dialogue processes have a consultative role, meaning that they can improve the basis for a decision of policymakers by providing suggestions or information (BMUV, 2023). Before integrating citizens in a dialogue process, the BMUV will check if the participation is fruitful for the ongoing decision-making process and how to deliver the necessary information to the citizens (ibid.). More about the selection process, especially related to the Climate Adaptation Dialogue, is described in chapter 2.5.

2.4 *Links between levels of governance*

The BMUV's citizen participation processes are designed to enable multilevel engagement from actors at the local, state, and federal level. In general, participation processes in the policy chain start with a draft law or strategy on which stakeholders and citizens produce comments and recommendations in the dialogue events. This is the case of the dialogue on Climate Adaptation, which focused mainly on the drafting step of the Adaptation Strategy by inviting stakeholders to support the development of goals. In a first step, the BMUV, in consultation with other ministries and subordinated authorities, developed goals regarding each of the seven selected clusters. Representatives from public entities such as municipal umbrella organizations, civil society organizations such as environmental and social associations, the research and the private sectors were invited to participate in the stakeholder participation phase. This phase aimed to identify possible additions to the draft objectives and indicators, the proposed measures and instruments as well as recommendations for measures from other stakeholders. The citizen participation phase centred around individual perceptions of impacts, challenges and priorities for action (Hoffmann, et al., 2024). Regional face-to-face events with randomly selected individuals and online events were held in this phase. According to the report of the Federal Environmental Agency on the process, housing, supply and mobility, and livelihoods were the topics with the highest number of recommendations by the citizens. Citizens' recommendations focused mainly on measures for the local (municipal) and regional (states) levels, while stakeholders' recommendations included measures for other administrative levels (ibid.).

Regarding the housing sector, discussion was based on three main goals included in the strategy draft: 1) adaptation of buildings and properties to protect user groups, with a particular focus on people in vulnerable positions, 2) protection of buildings and properties (with a particular focus on existing buildings), and 3) reduction of financial risks. Citizens and stakeholders raised concerns regarding renovation measures that only address private buildings and therefore leave care facilities out and call for more financial support programmes and advisory centres (Hoffmann, et al., 2024).

Although the *Community on Sustainable Digitalisation* is not a consultation process on a specific policy or strategy but rather a platform for knowledge and information exchange, it allows communication among the different governance levels in order to strengthen existing initiatives and projects. In a broader sense, the community is closely related to the "Digital Agenda for Environmental Policy", which was developed in 2020 by the federal government with the support of around 200 experts. Since digital technologies are both a challenge and an opportunity for environmental policy, the agenda aims to create synergies between environmental and climate protection and digitalisation to ensure that opportunities are exploited and risks minimized. In this context, the Community offers spaces to showcase successful practices linking digitalisation to climate and environmental protection and strengthen accountability as participants can report

on progress, challenges, and outcomes, ensuring that initiatives and strategies are evaluated from multiple perspectives.

With regard to the promotion of equality, it is important to note that, with the exception of efforts to involve young people in participation processes, information on the participation of groups in vulnerable positions is very limited and there is some recognition of the need to create processes that facilitate the participation of people with lower levels of education. However, participation processes have become an important platform for stakeholders and citizens to demand more attention to social justice in climate adaptation and to the needs of people in vulnerable situations. In this matter, the Federal Environmental Agency has also implemented dialogue platforms for exploring the social dimension of climate adaptation in order to jointly develop ideas for policy design.

2.5 *Civil society*

The BMUV has created a separate department for citizen participation, and it offers dialogue processes for citizens on different theme areas that the ministry is working on. Citizens can participate through online dialogues on the website, live events (on-site or digital), and discussions with experts or ministry representatives. Recommendations from these engagements help shape political programmes. Usually, citizens older than 16 years old are invited to participate in these processes and can register online (with some exceptions). The term “citizen” is used in a broader way, including also residents who do not have German citizenship (BMUV, 2023). Furthermore, the BMUV states that it wants to address all population groups/milieus and prefers to apply a random selection approach for that purpose. The results of such dialogue processes are analysed by the BMUV itself and then presented on its webpage.

Regarding the dialogue process on Climate Adaption, between September and October 2023, an online participation process was opened for young people between 14 and 25 years old (UBA, 2024). Also, the Climate Adaptation dialogue process targeted some rural areas in Germany particularly threatened by climate change, by inviting randomly selected citizens to in-person regional workshops (ibid.).

There is a lack of data focusing on gender equality and/or support of structurally vulnerable groups in relation to these BMUV dialogue processes. The guidelines for a participative approach suggest to BMUV employees to implement targeted quotas based on socio-demographic characteristics (such age, gender and education) to ensure a diversity of citizens (Tran-Karcher & Füllekrug, 2024). More specifically, it is recommended to ‘overbook’ groups that are usually marginalised or had been non-participating in previous processes (ibid.).

Gender aspects are integrated in international forums that the BMUV is taking part in, but it does not seem that they have a detailed gender mainstreaming approach for its dialogue processes. However, social aspects are considered in some processes. To highlight one example:

Within the participatory research project “Strengthening, scaling and networking civil society sustainability initiatives on the way to sustainable policies”, which the BMUV has commissioned, neighbourhood initiatives were invited to a series of five innovation workshops, with a workshop on living costs & mobility in Berlin and one on energy & climate in Leipzig in 2022 (BMUV, n.d.). The results of these five workshops were evaluated and further elaborated by stakeholders from

science, politics, administration and civil society with regards to their feasibility and then handed over to policymakers (ibid.).

2.6 Barriers

As mentioned above, information on the integration of gender equality and support to structurally vulnerable groups in the BMUV participation platforms are very limited. Specific challenges related to these issues are not explicitly described on the ministry website. Therefore, the lack of a gender perspective (and potentially of gender expertise among the member of the community) in the *Community on Sustainable Digitalisation* and other BMUV initiatives constitute a significant barrier. There is some recognition of the linkages between gender equality and digitalisation at the federal level, as shown in the third report on gender equality published in 2021. The report pinpointed the need for gender equality in the context of digitalisation and specifically called for equal access to relevant resources and skills, gender equality-oriented use of digital technologies, and equality-oriented design of digital technologies. However, there is no evidence of the integration of those factors or the resulting recommendations in the digitalisation policy or agenda (BMFSFJ, 2021).

Additional challenges to establishing links between the different levels of governance are related to the acceptance of virtual formats for participation among citizens. Reports on participation on the BMUV dialogue initiatives show that response and attendance to online events is lower than in live events and the length of them is deemed insufficient for meaningful participation. Moreover, there are some issues of lack of trust from citizens and stakeholders, as they believe that their recommendations might not actually be integrated in the updated drafts of policies and strategies. Virtual formats for participation also face inclusivity challenges. In particular, the BMUV recognises that in processes such as the Climate Adaptation Dialogue, the participation of people with lower education levels requires more attention and the development of specific strategies to ensure that these groups have access to digital resources and the necessary skills to use them (Hoffmann, et al., 2024). Similarly, participation of young people is below expectations. In this regard, the *Community on Sustainable Digitalisation* held a workshop aiming to co-create recommendations and guidelines for making young people's everyday digital lives more sustainable (Community Nachhaltige Digitalisierung, 2023).

2.7 *Factors for success*

Improving coordination between policy areas

Although the creation of the digital environmental policy agenda and the various thematic dialogues within the BMUV are an important step towards creating synergies between policy areas and between various stakeholders, the coordination and cohesion of strategies remains a significant challenge in the German context. Therefore, coordination mechanisms that enable the alignment of priorities, sharing of best practices, and resolution of conflicts across governance levels is an essential factor for success.

Specific approach in the selection of target groups and engagement strategies

Dialogue processes in the BMUV thematic areas involve a wide range of target groups from companies, the public sector, research and civil society. In the involvement of civil society, the consideration of aspects such as age (aiming to engage young people) or location (focusing on rural areas particularly affected by climate change) demonstrates a more targeted approach. The ministry has developed comprehensive guidelines for the responsible of carrying out participation processes that aim to ensure the quality of the processes (IFOK, Deutsches Forschungsinstitut für öffentliche Verwaltung, 2023).

Strategies to ensure inclusive citizens and stakeholders' involvement

Considering existing barriers to participation for certain population groups (including people in vulnerable positions, it is essential that participation formats are developed in a way that facilitates access and encourages engagement of these groups. For instance, registration for the *Community on Sustainable Digitalisation* is simple, which facilitates the process for those interested in the topics of the community, regardless of their level of expertise. Further barriers for groups in vulnerable positions such as access to digital technologies or experiences of discrimination in participation spaces need to be assessed and addressed.

2.8 *Conclusion*

In conclusion, it can be said that the initiatives of the BMUV clustered under “BMUV in Dialogue” establish mechanisms and platforms with the potential of integrating diverse members of society into policy making and information exchange. The different dialogue formats include individual citizens as well as stakeholders from research, civil society, and the public and private sector. Taking place as in-person-events as well as online, the initiatives aim to bring different levels of governance together, enabling dialogue between them.

However, there are some remarks on issues of inclusivity within the initiatives. In both initiatives presented, the steering function remains with the BMUV throughout the entire process, which could be a limiting factor for the success of diverse contributions. Additionally, participation mechanisms usually only address specific stages of policy formulation, further narrowing the sphere of influence. Diversity criteria and corresponding approaches are not a strategic part of the initiative, although some guidelines in the BMUV exist, such as the above-mentioned guidelines for a participative approach which aim at targeting underrepresented groups. Furthermore, there is a strong focus on academic expertise rather than everyday life experience.

These limitations could potentially result in the reproduction of structural power imbalances. There are, however, acknowledgements to strengthen the inclusivity especially regarding people with lower levels of formal education as well as young people. Finally, there seems to be a lack of transparency in several stages of the participation process, contradicting its own principles.

References

Ariadne-Projekt (2020). Start des Kopernikus-Forschungsprojekts Ariadne. In: [Start des Kopernikus-Forschungsprojekts Ariadne | Ariadne](#), 03.07.2020, (last accessed on 02.01.2025).

Ariadne-Projekt (2022a). Wie ein Klimageld einfach und spürbar bei den Menschen ankommen kann. In: <https://ariadneprojekt.de/news-de/wie-ein-klimageld-einfach-und-spuerbar-bei-den-menschen-ankommen-kann/>, 02.06.2022, (last accessed on 02.01.2025).

Ariadne-Projekt (2022b). Dokumentation zum Ariadne-Dialogforum: Die Wärmewende sozial gerecht gestalten: Diskussionen jüngster Forschungserkenntnisse und einer Haushaltsbefragung. Drittes Dialogforum zur Wärmewende im Rahmen des Forschungsprojekts Ariadne am 17. November 2021, https://ariadneprojekt.de/media/2022/05/Ariadne-Dokumentation_Dialogforum-Waermewende_November2021.pdf, (last accessed on 02.01.2025).

Ariadne-Projekt (2023a). Ergebnisse der Ariadne Bürgerdeliberation: BÜRGERSICHTEN AUF DIE STROMWENDE. Teilhabe und Gerechtigkeit für eine klimawirksame Energiewende, https://ariadneprojekt.de/media/2023/03/20230315_Faltposter_Strom_Final.pdf, (last accessed on 02.01.2025).

Ariadne-Projekt (2023b). Bürgerinnen und Bürger stellen mit der Wissenschaft entwickelte Zukunftspfade für die Energie- und Verkehrswende vor. In: <https://ariadneprojekt.de/news-de/ariadne-buergergipfel-buergerinnen-und-buerger-stellen-mit-der-wissenschaft-entwickelte-zukunftspfade-fuer-die-energie-und-verkehrswende-vor/>, 31.03.2023, (last accessed on 02.05.2025).

Ariadne-Projekt (2024). Schnittstelle zwischen Forschung und Klimapolitik stärken. In: <https://ariadneprojekt.de/news-de/zweite-ariadne-politikwerkstatt-schnittstelle-zwischen-forschung-und-klimapolitik-staerken/> (last accessed on 02.01.2025).

Ariadne-Projekt (n.d.-a). Citizens' Deliberation (n.d.). In: <https://ariadneprojekt.de/en/citizen-deliberation/> (last accessed on 02.01.2025).

Ariadne-Projekt (n.d.-b). Aspirations of Ariadne: The read thread through energy transition. In: [Aspirations of Ariadne: The read thread through energy transition](#) (last accessed on 02.01.2025).

Ariadne-Projekt (n.d.-c). Der Ariadne-Lernprozess. In: <https://ariadneprojekt.de/lernprozess/> (last accessed on 02.01.2025).

Ariadne-Projekt (n.d.-d). Energy transition in dialogue. In: <https://ariadneprojekt.de/en/energy-transition-in-dialogue/> (last accessed on 02.01.2025).

Ariadne-Projekt (n.d.-e). Institutions for effectively governing the energy transition. Coordinated and target-oriented climate policy governance. In: [Governance and institutions of climate policy](#) (last accessed on 27.01.2025).

Ariadne-Projekt & Potsdamer Institut für Klimafolgenforschung (PIK) (2023a). Ergebnisse der Ariadne Bürgerdeliberation: BÜRGERSICHTEN AUF DIE VERKEHRSWENDE. Stadt-Land-Gerechtigkeit, Mobilitätsalternativen vor Preiserhöhungen und Berücksichtigung unterschiedlicher Mobilitätsbedürfnisse – zentrale Bedingungen für das Gelingen der Verkehrswende aus Sicht von Bürgerinnen und Bürgern, ariadneprojekt.de/media/2023/03/20230315_Faltposter_Verkehr_Final.pdf (last accessed on 02.01.2025).

Ariadne-Projekt & Potsdamer Institut für Klimafolgenforschung (PIK) (2023b): Herausforderungen der Energiekrise für die Wärmewende: Eine Einordnung auf Grundlage jüngster Forschungserkenntnisse, [Herausforderungen der Energiekrise für die Wärmewende: Eine Einordnung auf Grundlage jüngster Forschungserkenntnisse](#) (last accessed on 02.01.2025).

Bilski, Noah (2023). BMUV Community Nachhaltige Digitalisierung. In: public, 01/2023, [BMUV Community „Nachhaltige Digitalisierung“ | Public Magazin](#) (last accessed on 07.01.2025).

Blum, M., Colell, A., Hoffmann, J., Karohs, K., Kowarsch, M., et al. (2021). *Ariadne-Report: Was ist uns wichtig bei Verkehrs- und Stromwende? Bürgerinnen und Bürger sprechen über Herausforderungen und Ziele*. Retrieved from [210120 Synthesebericht Layout V1.0](#).

Blum, M., Treichel-Grass, K., Kowarsch, M. (2022, July). *Report: Views of Citizens on Four Future Pathways of the Transport Transition – Results of the Ariadne Citizens' Conference on the Transport Transition*, Kopernikus-Projekt Ariadne. Retrieved from [Report: Views of Citizens on Four Future Pathways of the Transport Transition - Results of the Ariadne Citizens' Conference on the Transport Transition | Ariadne](#)

BMFSFJ (2021). *Dritter Gleichstellungsbericht: Digitalisierung geschlechtergerecht gestalten*, Berlin: BMFSFJ. Retrieved from: <https://www.bmfsfj.de/bmfsfj/service/publikationen/dritter-gleichstellungsbericht-184546>

BMUV (2020). Die Umweltpolitische Digitalagenda: Wie ein Problem zur Lösung wird, Berlin: BMUV. Retrieved from: https://www.bmuv.de/fileadmin/Daten_BMU/Download_PDF/Digitalisierung/digitalagenda_bf.pdf

BMUV (2022a). Umweltpolitik und Geschlechtergerechtigkeit gehören zusammen - BMUV wirbt bei den Vereinten Nationen für feministische Umweltpolitik. In: [BMUV: Umweltpolitik und Geschlechtergerechtigkeit gehören zusammen - BMUV wirbt bei den Vereinten Nationen für feministische Umweltpolitik | Pressemitteilung](#) (last accessed on 07.01.2025).

BMUV (2022b). Speech by Steffi Lemke at the information event for the Permanent Representations (StäV) of the UN member states on the appointment of a UN special representative for water, 15.06.2022. In: [BMUV: Speech by Steffi Lemke at the information event](#)

[for the Permanent Representations \(StäV\) of the UN member states on the appointment of a UN special representative for water | Rede](#) (last accessed on 07.01.2025).

BMUV (2023). Gute Bürgerbeteiligung. Leitlinien für Mitarbeiterinnen und Mitarbeiter des Bundesministerium für Umwelt, Naturschutz, nukleare Sicherheit und Verbraucherschutz, [Gute Bürgerbeteiligung: Leitlinien für Mitarbeiterinnen und Mitarbeiter des Bundesministeriums für Umwelt, Naturschutz, nukleare Sicherheit und Verbraucherschutz](#) (last accessed on 07.01.2024).

BMUV (2024a). Community Nachhaltige Digitalisierung. [Online] Available at: <https://www.bmuv.de/themen/digitalisierung/community-nachhaltige-digitalisierung> (last accessed on 15.12.2024).

BMUV (2024b). Everything CO₂units – Jahreskonferenz Nachhaltige Digitalisierung, Available at: <https://www.bmuv.de/meldung/everything-counts-jahreskonferenz-nachhaltige-digitalisierung> (last accessed 10.12.2024).

BMUV (n.d.). Forschungsprojekt: „Nachhaltigkeitsinitiativen stärken und vernetzen!“. In: [BMUV: Forschungsprojekt "Nachhaltigkeitsinitiativen stärken und vernetzen!"](#) (last accessed on 07.01.2025).

Community Nachhaltige Digitalisierung (2023). Nachhaltiger digital unterwegs, Berlin: BMUV. Retrieved from: https://www.bmuv.de/fileadmin/Daten_BMU/Download_PDF/Digitalisierung/guide_zum_mitm_achen_bf.pdf

Hoffmann, E., Rupp, J., Schönthaler, K. & von Andrian-Werburg, S. (2024). *Empfehlungen aus dem Dialog KlimaAnpassung*, Dessau-Roßlau: UBA. Retrieved from: <https://www.umweltbundesamt.de/publikationen/empfehlungen-aus-dem-dialog-klimaanpassung>

IFOK, Deutsches Forschungsinstitut für öffentliche Verwaltung (2023). Gute Bürgerbeteiligung. Berlin: BMUV. Retrieved from: https://www.bmuv.de/fileadmin/Daten_BMU/Download_PDF/Buergerbeteiligung/leitlinien_buergerbeteiligung_bmuv_bf.pdf

Luderer, G., Kost, C., Sörgel, D., Günther, C., Benke, F., et al. (2021). *Report: Deutschland auf dem Weg zur Klimaneutralität 2045 – Szenarien und Pfade im Modellvergleich*, Kopernikus-Projekt Ariadne. <https://ariadneprojekt.de/publikation/deutschland-auf-dem-weg-zur-klimaneutralitaet-2045-szenarienreport/>.

Tran-Karcher, Huy & Füllekrug, Alina (2024). Kriterien für gute Online- und hybride Beteiligung. Hinweise und Empfehlungen für die Mitarbeiter*innen des Bundesministeriums für Umwelt, Naturschutz, nukleare Sicherheit und Verbraucherschutz. In: BMUV, [Kriterien für gute Online- und hybride Beteiligung](#) (last accessed on 07.01.2025).

Treichel-Grass, K., Blum, M., Kowarsch, M. (2022). *Report: Citizen's Views of Future Energy Worlds – Results of the Ariadne Citizen's Conference on Electricity Transition*, Kopernikus-Projekt Ariadne.

Retrieved from <https://ariadneprojekt.de/en/publication/report-results-citizensconference-electricitytransition/>

Umweltbundesamt (UBA) (2024). Dialog KlimaAnpassung. In: [Dialog KlimaAnpassung | Umweltbundesamt](#) (last accessed on 07.01.2025).

WELT am Sonntag (2022). Was ist “feministische Umweltpolitik”, Frau Lemke?, In: [Umweltministerin: Was ist „feministische Umweltpolitik“, Frau Lemke? - WELT](#) (last accessed on 07.01.2025).

Spain

National Strategy Against Energy Poverty (ENPE) 2019–2024

2.1 Introduction

The equitable distribution of costs and benefits during the twin transition is essential to preventing the deepening of energy poverty, particularly among vulnerable groups. In Spain, the [National Strategy Against Energy Poverty \(ENPE\) 2019–2024](#) (Ministry for the Ecological Transition and the Demographic Challenge [MITECO], 2019), represents a good practice by integrating multi-level governance mechanisms that involve central, regional, and local governments. The strategy ensures that different levels of public administration coordinate through specific governance structures, such as intergovernmental working groups and regional implementation plans, to develop targeted energy poverty mitigation measures.

In addition to public authorities, the private sector and civil society organizations play a pivotal role in the strategy’s implementation and evaluation. The ENPE establishes structured stakeholder engagement mechanisms, including formal consultations with social organizations, consumer advocacy groups, and industry representatives, ensuring that policies reflect the realities of those most affected by energy poverty.

Moreover, the ENPE is aligned with Spain’s ["The Long-Term Decarbonization Strategy 2050: Long-term strategy for a modern, competitive and climate-neutral Spanish economy in 2050"](#) (LTS), to guarantee that the twin transition is socially inclusive. This involves a targeted subsidy framework for low-income households, regulatory mechanisms that ensure fair access to renewable energy, and investments in social energy infrastructure to prevent regional disparities in energy access. By integrating these elements, the ENPE not only addresses vulnerable groups but also creates long-term resilience for disadvantaged communities during the energy transition.

2.2 The aim

The ENPE aims to reduce energy poverty ensuring universal and equitable access to affordable, sustainable, and reliable energy. It integrates social, economic, and environmental policies into a coordinated governance framework to address structural inequalities (MITECO, 2019, p. 6). The strategy sets specific goals including a 25% reduction in energy poverty by 2025 and the renovation of 120,000 low-income households (MITECO, 2019, p. 28). To achieve these objectives, the ENPE employs:

1. Short-term financial relief, including energy subsidies and emergency aid during extreme weather.
2. Long-term structural reforms, such as energy-efficient housing renovations and alignment with EU climate adaptation policies.

Multi-level governance structure:

- Ministry for the Ecological Transition and the Demographic (MITECO) setting regulatory frameworks, allocating funds, and ensuring policy coherence with EU directives.
- Regional governments adapting national policies to territorial needs and overseeing implementation.
- Local governments managing direct aid programs and community-based solutions.
- Private sector and civil society enhancing execution through public-private partnerships and advocacy efforts.

The budget is primarily funded through national public funds and EU recovery mechanisms, with €900 million allocated to direct aid and €2.5 billion for energy efficiency improvements (MITECO, 2019, p. 171). It aligns with Royal Decree-Law 15/2018, the UN Sustainable Development Goals (SDG Goal 7), and Spain's National Energy and Climate Plan (PNIEC).

The strategy prioritizes vulnerable groups, including older adults, individuals with disabilities, migrants, and single-parent households. Recognizing the gendered dimension of energy poverty, targeted financial support for women-led households is integrated (MITECO, 2019, p. 28).

2.3 *Roles and responsibilities*

The ENPE 2019–2024 employs a multi-level governance framework to combat energy poverty, integrating social, environmental, and economic policies (MITECO, 2019, p. 6). The governance structure ensures vertical coordination between national, regional, and local authorities, while incorporating civil society and private sector actors to enhance execution and oversight.

Public authorities' roles across the policy chain

- **Policy planning & drafting:**

The **MITECO** defines the national framework, setting regulatory standards and financial priorities in alignment with EU directives. It coordinates with regional governments and international bodies to integrate best practices. Civil society and private stakeholders contribute through public consultations to refine policy scope and objectives (MITECO, 2019, p. 162–163).

- **Implementation:**

Regional and local governments translate national policies into targeted programs, overseeing direct financial aid, energy efficiency programs, and housing renovations. The Institute for Energy Diversification and Saving (IDAE), a state-owned entity, ensures

technical and financial assistance reaches municipalities while maintaining policy coherence (MITECO, 2019, p. 171).

- **Monitoring & evaluation:**

The monitoring body for the strategy is the IDEA, a state-owned enterprise under the Secretariat of State for Energy which leads performance tracking through **annual reports and gender-disaggregated indicators**, ensuring compliance with national and EU energy poverty benchmarks. The **Interministerial Working Group on Energy Poverty** provides policy oversight, while the **Social Roundtable on Energy Poverty** institutionalizes civil society's role in monitoring impact and advocating for adjustments (MITECO, 2019, pp. 173–175).

To date, specific monitoring and evaluation reports for the **National Strategy Against Energy Poverty (ENPE) 2019–2024** have not been published. However, in December 2022 the MITECO published the Update Report on Energy Poverty Indicators (MITECO, 2022).

Civil society and social partners' contribution

The coordination between **public administration and civil society** is of great importance. **ACs**, together with **Local Entities**, **Private entities**, and **civil society organizations**, play a crucial role in managing consumer protection mechanisms.

- The **Cruz Roja**, a humanitarian organization operating as a **private entity**, collaborates in tackling **energy poverty** through **advisory and support programs** for vulnerable households. Its initiatives include **energy efficiency workshops** that help families optimize electricity consumption and access social assistance. Additionally, in 2018, it distributed **more than 6,400 energy efficiency kits**, containing low-energy light bulbs, insulation strips for doors and windows, and power strips, helping families improve their household energy use (MITECO, 2019, p. 145)
- The **Third Sector Platform** and other advocacy groups influence policy adjustments, particularly in **gender-responsive financial aid mechanisms**.
- **Local entities and organizations** are focused on the **early detection of vulnerable individuals**. Measures to facilitate effective communication include **information channels**, **extrajudicial conflict resolution procedures**, such as **voluntary mediation**, among others. Also, in **preventing supply suspension**. Social services act as an **information channel** to provide **advice and support** throughout the process and to identify the mechanisms available to users to prevent disconnection of their household energy supply (MITECO, 2019, p. 109).

Commitment to gender equality & vulnerable groups

The ENPE integrates gender-sensitive measures, recognizing the disproportionate impact of energy poverty on women-led households. Policy commitments include targeted subsidies, prioritization of female-headed low-income households in housing renovations, and gender-responsive financial tracking tools (p.28). The Sectoral Conference facilitates coordination with regional governments to ensure localized gender equality measures are effectively implemented (MITECO, 2019, p. 173).

2.4 *Links between levels of governance*

The governance framework integrates national, regional, and local authorities, civil society, and the private sector.

Effective implementation of the policy chain

- **Policy Planning & Drafting**

The **MITECO** led an **inclusive public consultation process** to incorporate diverse stakeholder perspectives. Civil society organizations, **particularly NGOs and advocacy groups**, played a critical role in highlighting **social and gender-based vulnerabilities**. However, **local and regional governments** showed lower engagement levels, requiring additional efforts to strengthen their involvement (**MITECO, 2019, p. 163–168**).

Policy implementation

Operational plans, developed by the **MITECO** and executed by **regional and local governments**, guided implementation efforts. The **IDAE** played a key role in **technical and financial support**, ensuring localized adaptation of national strategies. Coordination mechanisms, such as the **Interministerial Working Group on Energy Poverty** and the **Social Roundtable on Energy Poverty**, facilitated cross-sectoral collaboration (**MITECO, 2019, p. 171–173**).

Policy monitoring & evaluation

The National Statistics Institute (INE) and IDAE conducted annual gender-disaggregated assessments of energy poverty, tracking financial allocation, execution rates, and policy effectiveness. Civil society and research institutions contributed independent evaluations, fostering adaptive policy responses (**MITECO, 2019, p. 173–175**).

Shared equality standards & objectives

The ENPE aligns with key international and national equality frameworks

- EU & UN frameworks: SDG 7, the EU Winter Package, and the European Pillar of Social Rights.
- National strategies: The PNIEC, the Just Energy Transition Strategy, and Royal Decree-Law 15/2018 provided a regulatory foundation for equality-driven policies.

These frameworks integrated gender and social inclusion considerations across all governance levels, ensuring that policy planning, drafting, implementation, and monitoring systematically addressed vulnerabilities (**MITECO, 2019, p. 14–23**).

Capacity to promote equality & support vulnerable groups

To strengthen institutional capacity, the strategy introduced mechanisms to enhance public authorities' ability to address energy poverty (**MITECO, 2019, p. 80–88**):

- Gender-disaggregated data collection by INE and IDAE, informing targeted interventions for women-led households and marginalized groups.
- Transparency and dissemination data initiatives, publishing annual reports to facilitate policy comparisons and adjustments at different governance levels.
- Identification of climate zones and vulnerable consumers, leveraging academic research (e.g., studies by the Pontifical University of Comillas-ICAI) to tailor localized responses.
- Collaboration between the Directorate-General for Consumer Affairs, ACs, and consumer associations, as well as the Municipal Consumer Information Offices (OMIC), to expand the reach of awareness campaigns. Electronic media and printed materials are used to ensure information accessibility (MITECO, 2019, p. 150).
- “Consumópolis”: An initiative launched in 2005-2006 by the Directorate-General for Consumer Affairs, aimed at raising awareness among schoolchildren about responsible consumption and its connection to quality of life. The project has had an average participation of 10,000 students per edition, promoting justice, equity, and solidarity in consumption (MITECO, 2019, p. 151).

Learning on equality across governance levels

The **ENPE institutionalized learning mechanisms to mainstream equality considerations** into governance processes (MITECO, 2019, p. 139–141):

- **Training for frontline professionals**, to identify vulnerable households through medical consultations. A screening protocol was proposed to integrate energy poverty assessments into routine healthcare services.
- **Capacity-building programs for public officials**, improving understanding of gender-based disparities in energy access.
- **Knowledge-sharing through advisory bodies**, particularly the **Social Roundtable on Energy Poverty**, ensuring continuous exchange of best practices between public authorities and civil society.

2.5 Civil society

The consultation process was in line with Article 1.3 of Royal Decree-Law 15/2018, which mandates the involvement of regional and local governments, social actors, and affected communities in the strategy’s development. This framework ensured that the perspectives of vulnerable groups most affected by energy poverty were actively incorporated into policy formulation (MITECO, 2019, p. 162).

To enhance participation, two public consultations were conducted via the MITECO website, facilitating transparent and accessible engagement for stakeholders, including NGOs, social service organizations, consumer associations, and research institutions.

Civil Society’s Role in Advancing Equality and Support for Vulnerable Groups

The consultations allowed civil society organizations to advocate for structural improvements, ensuring more targeted and inclusive interventions:

- Redefining energy poverty criteria: NGOs pushed for income-based indicators adjusted to household size, while also advocating for the inclusion of social and geographic vulnerabilities to capture hidden energy poverty.
- Strengthening policy accessibility: Civil society urged for simplified administrative procedures and automatic subsidy allocation to reduce bureaucratic barriers for vulnerable households. They also advocated for territorial standardization to minimize regional disparities in aid distribution.
- Enhancing monitoring mechanisms: Organizations lobbied for the establishment of a National Observatory on Energy Poverty, a permanent body responsible for tracking regional disparities, measuring policy impact, and suggesting continuous improvements based on real-time data.
- Expanding energy efficiency programs: Recommendations included increased financial aid for home energy rehabilitation, with a focus on low-income households, tenants, and single-parent families. Proposals also called for subsidies for energy-efficient appliances and regulatory adjustments to support energy efficiency upgrades in rented homes.
- Gender-sensitive policy considerations: While gender was not explicitly prioritized, civil society emphasized the need for social protection mechanisms that indirectly benefit women-led households, particularly single mothers, who are disproportionately affected by energy poverty. Their recommendations aligned with efforts to bridge gender-based economic disparities through stronger coordination between energy policies and social services.
- Ensuring the Precautionary Principle: Establishing a Minimum Vital Supply, which will remain in place for a period of four months in cases of non-payment, preventing total disconnection with the support of local administrations. This proposal was put forward by various stakeholders during the preliminary public consultation and the drafting phase.

Through these contributions, civil society played a crucial role in shaping policy implementation to be more equitable and responsive to the needs of vulnerable groups.

2.6 Barriers

The strategy identifies several barriers that hinder effective coordination among public authorities across different governance levels, affecting the implementation of policies on gender equality and support for vulnerable groups.

- **Fragmentation and lack of coordination** remain a significant challenge. Governance involves multiple actors at national, regional, and local levels, often leading to overlapping responsibilities and inefficiencies. While mechanisms such as the Sectoral Conference on Energy aim to facilitate collaboration, their impact is limited by bureaucratic inefficiencies and the lack of a centralized coordination body. Additionally, ACs and municipalities face implementation challenges due to inconsistent procedures and uncoordinated databases (MITECO, 2019, p. 172–175).
- **Regulatory and administrative complexity** further complicates policy execution. The regulatory framework addressing energy poverty is fragmented across various policies, making policy coherence difficult. Inefficient bureaucratic processes delay aid implementation, with automated verification of energy subsidy eligibility hampered by

poor inter-agency data-sharing (MITECO, 2019, p. 103). Moreover, regional disparities in policy execution led to unequal access to energy poverty relief programs (MITECO, 2019, p. 142).

- **Limited integration of a gender perspective** weakens the effectiveness of the strategy in addressing social inequalities. Despite acknowledging gender disparities, the strategy does not sufficiently prioritize gender-sensitive policies. Gender-disaggregated data remains insufficient, making it difficult to assess the differentiated impact of energy poverty. Single-parent households, particularly those headed by women, are disproportionately affected, yet specific measures targeting their needs remain underdeveloped (MITECO, 2019, p. 81).
- **Challenges in data collection, public consultation, and monitoring** reduce the capacity to assess and improve policy outcomes. The strategy lacks statistical tools to measure energy poverty with a gender-sensitive approach, failing to capture intra-household inequalities (MITECO, 2019, p. 81). The absence of a national database consolidating income, energy use, and social vulnerability data results in inefficiencies in policy targeting and monitoring (MITECO, 2019, p. 142). Furthermore, **public consultations often have limited timeframes** (sometimes under a month), restricting participation from affected groups and civil society organizations (MITECO, 2019, p. 110).
- **Financial and structural constraints** undermine the long-term sustainability of policy interventions. Many energy poverty relief programs depend on European funding, leading to implementation delays and uncertainty in long-term planning (MITECO, 2019, p. 119). Additionally, funding mechanisms prioritize short-term measures, such as bill subsidies, rather than structural solutions like housing rehabilitation or renewable energy adoption (MITECO, 2019, p. 121). Insufficient investment in public awareness campaigns and advisory services further reduces the effectiveness of support programs (MITECO, 2019, p. 145).
- The **landlord/tenant dilemma in rental housing**: Is a major obstacle to improving energy efficiency, particularly for vulnerable consumers. The misalignment of incentives prevents necessary renovations, as **property owners are reluctant to invest in energy efficiency improvements while tenants lack the authority to make changes**. Additionally, **an aging housing stock, administrative fragmentation, and a lack of awareness and data on housing conditions** further hinder the effectiveness of current policies (MITECO, 2019, p. 123).

2.7 Factors for success

To deliver lasting social impact two key enablers must be reinforced, **institutional coordination and evidence-based policy learning**.

1. Strengthening multi-level institutional coordination

Effective governance of energy poverty reduction requires alignment across public administrations to ensure that vulnerable groups receive consistent and timely support. The ENPE has established vertical and horizontal coordination mechanisms, yet **fragmentation and unequal implementation across regions remain challenges**. Some plausible changes could be:

- **Reinforce intergovernmental coordination mechanisms** such as the Sectoral Conference on Energy, ensuring that national, regional, and local policies are fully aligned.

- **Improve local implementation capacities**, particularly in municipalities with limited resources, by expanding technical and financial support through IDAE and regional governments.
- **Strengthen civil society participation** in decision-making processes through bodies like the Social Roundtable on Energy Poverty, ensuring that policies reflect the realities of affected communities.

2. Enhancing Data Collection, Policy Evaluation, and Training

A major barrier to effective policy implementation is the limited availability of detailed, gender-sensitive, and regionally disaggregated data on energy poverty. Strengthening policy evaluation and institutional learning will require:

- **Expanding the role of INE and IDAE** in monitoring energy poverty indicators, integrating geographical, socio-economic, and gender-based disparities.
- **Creating a centralized database** that consolidates real-time energy consumption, income, and social vulnerability data, ensuring that policy interventions are targeted and responsive.
- **Improving training for public officials and service providers**, particularly in externalized assistance programs, to ensure that frontline workers are equipped to support vulnerable households effectively.

Reinforcing these enablers will be crucial to making the **twin transition socially inclusive, protecting vulnerable groups, and moderate energy poverty gaps**.

2.8 Conclusion

The governance framework of the twin transition in the energy sector, as reflected in the ENPE 2019–2024, has established a structured multi-level approach to addressing energy poverty and social inequalities.

While the strategy integrates various governance levels and aligns with EU social and climate policies, some challenges persist in ensuring **effective coordination, consistent implementation across regions, and the full incorporation of gender-responsive policies**.

One of the main obstacles is **the complexity of intergovernmental coordination**, where overlapping responsibilities between national, regional, and local authorities create inefficiencies. Existing mechanisms, such as the Sectoral Conference on Energy, provide a space for dialogue, but **a more structured and binding coordination framework could enhance policy coherence** and reduce disparities in implementation. Ensuring that policies are adaptable to regional and local contexts remains a priority.

The **integration of a gender perspective in energy policies requires further strengthening**. While gender considerations are acknowledged in the ENPE, **more systematic data collection and targeted support measures with gender indicators** would improve the effectiveness of policies in addressing the specific needs of women, particularly single mothers and female-headed households.

Financial sustainability is another key area for consideration. **Balancing short-term relief mechanisms, such as energy subsidies, with long-term structural solutions**, including energy-efficient housing renovations and decentralized renewable energy projects, remains a challenge. Greater investment in sustainable solutions could enhance resilience and reduce long-term dependency on temporary assistance.

Perhaps, the ENPE represents significant progress in Spain's twin transition in addressing energy poverty and ensuring social justice objectives.

Safe, Sustainable, and Connected Mobility Strategy 2030

2.1 Introduction

The transition towards sustainable and digital mobility in Spain is guided by the [Safe, Sustainable, and Connected Mobility Strategy 2030](#) (Ministry of Transport, Mobility, and Urban Agenda [MITMA], 2021), which ensures that transport policies align with both environmental objectives and social equity. A key governance practice within this framework is the enhancement of **public transport, rail connectivity, low-emission vehicles, and smart mobility solutions**, aimed at improving accessibility and reducing transport-related inequalities.

Axis 1, **"Mobility for All,"** prioritizes universal accessibility by investing in **public transport expansion, electrification of fleets, and the development of intelligent mobility systems**. These initiatives particularly benefit **low-income individuals, persons with disabilities, and rural communities**, ensuring that transport decarbonization efforts do not exacerbate existing social disparities. The strategy is integrated into the **PNIEC 2023–2030** and the **Recovery, Transformation, and Resilience Plan**, requiring regional and local governments to develop **Sustainable Urban Mobility Plans (PMUS)** that address territorial mobility needs.

Governance mechanisms foster **multi-level coordination** between national, regional, and local authorities, alongside structured **public consultations** that engage civil society organizations and vulnerable communities.

2.2 The aim

The strategy redefines Spanish transport policy, shifting from infrastructure development to integrated mobility governance. Led by the Ministry of Transport, Mobility, and Urban Agenda (MITMA), the strategy aligns with EU climate and digitalization policies to ensure an inclusive, decarbonized, and technologically advanced transport system. It prioritizes universal accessibility, emission reduction, and digital innovation to prevent transport inequalities during the twin transition. Financial backing from Spain's Recovery, Transformation, and Resilience Plan, implementation is guided by nine strategic axes addressing infrastructure, governance, social equity, and environmental sustainability (MITMA, 2021, p. 8–11).

Governance is structured through multi-level coordination among the General State Administration (AGE), Autonomous Communities (ACs), and local governments.

- The MITMA leads national and regional mobility policies, supported by the Spanish Railway Infrastructure Administrator (Adif), the Spanish Airports and Air Navigation (Aena), the

National Network of Spanish Railways (Renfe), the State Ports Authority (Puertos del Estado), and the Spanish Air Traffic Management Entity (Enaire), ensuring implementation across all transport modes. Public-private collaboration is reinforced through the Open Mobility Dialogue, which integrates civil society, academia, and industry stakeholders into policy development.

The strategy is structured around **three pillars**:

- **Safe Mobility**: Strengthening transport security and resilience.
- **Sustainable Mobility**: Promoting low-emission transport and climate adaptation.
- **Connected Mobility**: Advancing digitalization and multimodal integration.

2.3 *Roles and responsibilities*

Policy Planning and Drafting

- **The Inter-ministerial Working Group on Mobility** is responsible for drafting the **Sustainable Mobility Law**, a key legislative instrument integrating sustainability, equity, and digitalization.
- **The Sectoral Transport Conference** is a forum for coordination between the MITMA and regional transport ministries.
- **The Local Government Coordination Mechanism**, in collaboration with the Spanish Federation of Municipalities and Provinces (**FEMP**), facilitates the implementation of **PMUS** (MITMA, 2021, p.104–105).

Public participation was ensured through the **Open Mobility Dialogue**, a 100-day consultation gathering **746 citizen contributions** and hosting **100 events** with over **10,000 participants** from local governments, NGOs, businesses, and advocacy groups. Civil society input shaped priorities, emphasizing **universal accessibility, rural mobility, and gender-sensitive transport policies** (MITMA, 2021, p. 79).

Policy Implementation

The MITMA oversees execution through **state-owned transport entities**, including **Adif** (rail infrastructure), **Aena** (airport operations), **Renfe** (passenger rail services), **Puertos del Estado** (ports and maritime transport) and **Enaire** (air traffic management).

Implementation is guided by:

- **Sustainable Urban Mobility Plans (PMUS)**, mandatory for cities over **50,000 inhabitants**, ensuring inclusive transport policies (MITMA, 2021, p. 105).
- **NextGenerationEU-funded Low Emission Zones (ZBE)**, promoting urban decarbonization and public transport expansion (MITMA, 2021, p. 111).
- **The Strategic Plan for Infrastructure and Transport (PEIT 2023–2030)**, aligning mobility policies with environmental and social objectives (MITMA, 2021, p. 107).
- **The “Mobility for All” program, financing accessible transport for people with disabilities and older adults** (MITMA, 2021, p. 108).

- **The Rural Mobility Strategy**, improving transport access in sparsely populated areas through **demand-responsive services** (MITMA, 2021, p. 112).

Policy Monitoring and Evaluation

The MITMA, in collaboration with the **Transport and Logistics Observatory of Spain (OTLE)**, conducts ongoing policy assessments using:

- General Mobility Indicators, tracking public transport use, emissions, and accessibility (MITMA, 2021, p. 106).
- Specific Policy Indicators, measuring the impact on gender equality and vulnerable groups. The MITMA also organizes territorial workshops on daily mobility, evaluating policy effectiveness with local governments and citizens (MITMA, 2021, p. 86, 87).

To date, no specific monitoring and evaluation reports on the **Safe, Sustainable, and Connected Mobility Strategy 2030** have been published.

Roles of Civil Society and Social Partners

Beyond consultations, institutional mechanisms ensure civil society participation in governance:

- The Higher Council for Sustainable Mobility and Transport, a permanent advisory body integrating public institutions, businesses, unions, and NGOs.
- Local Mobility Forums, engaging municipalities in the execution of PMUS (MITMA, 2021, p. 104–105).

Commitment to Gender Equality and Vulnerable Groups

To embed social inclusion in transport policy, the **MITMA launched thematic plans addressing equity gaps**:

- Inclusive Mobility Week, focusing on accessible transport for people with disabilities, older adults, and vulnerable groups.
- Training and Labor Aspects Week, promoting women's employment in the transport sector.
- Rural and Low-Density Mobility Week, addressing transport access in remote areas (MITMA, 2021, p. 83).

2.4 *Links between levels of governance*

Governance links that enable effective policy implementation

The MITMA leads the policy planning, drafting, implementation, and monitoring process with strong institutional, regulatory, and financial mechanisms to ensure a coordinated approach (MITMA, 2021, p. 90).

- **Policy planning and drafting:** The MITMA works closely with 21 other ministries, regional governments, and municipalities through working groups, coordination councils, and

sectoral conferences. The upcoming Sustainable Mobility Law is designed to align mobility governance at all levels.

- **Implementation mechanisms:** The MITMA oversees public agencies such as Adif, Renfe, Aena, and Puertos del Estado, which are responsible for transport infrastructure and service delivery. The MITMA Strategy Office ensures that policies align with national and EU frameworks while adapting to local and regional needs.
- **Monitoring and evaluation:** The MITMA-led Rural Mobility Table fosters collaboration among national, regional, and local governments, ensuring that policies remain adaptable to changing mobility needs (MITMA, 2021, p. 132–133).

These mechanisms facilitate coordination, ensure accountability, and strengthen intergovernmental cooperation across all phases of the policy cycle.

Links that enable shared equality standards and objectives

The harmonization of equality standards is ensured through alignment with national and EU frameworks, integrating principles of gender equality, territorial cohesion, and accessibility.

- **National strategies:** The National Demographic Challenge Strategy (2019) and the Plan for Territorial Cohesion (2021) establish a framework for reducing disparities in transport access, particularly in rural areas (MITMA, 2021, p. 132–133).
- **European funding and policy alignment:** The EU Recovery and Resilience Plan (Next Generation EU) provides €70 billion in direct funding, facilitating investments in inclusive and sustainable transport infrastructure (MITMA, 2021, p. 99).
- **Legal commitments to accessibility:** Spain comply with the UN Convention on the Rights of Persons with Disabilities, ensuring universal design principles in transport systems (MITMA, 2021, p. 145).
- **Urban-rural connectivity measures:** The National Urban Transport Financing Framework, under development, aims to improve access to public transport for marginalized communities (MITMA, 2021, p. 120).

By embedding these equality standards into policy planning, drafting, implementation, and monitoring, Spain ensures a cohesive and inclusive mobility strategy.

Public authorities' capacity to promote equality and support vulnerable groups

To enhance the capacity of public authorities at all governance levels, Spain has implemented training, financial, and policy coordination mechanisms to support gender equality and vulnerable groups. These governance structures reinforce institutional capacity, ensuring that equality measures are consistently implemented across all levels of government.

- **Rural and depopulated areas:** The MITMA Rural Mobility Table works with regional and local governments to develop alternative mobility solutions for low-density areas, such as on-demand mobility services and car-sharing initiatives (MITMA, 2021, p. 132–133).
- **Women in transport initiatives:** The MITMA collaborates with Women in Motion (Mujeres en Movimiento in Spanish), and other networks to promote gender equality in the mobility sector (MITMA, 2021, p. 364).

- **Social and territorial cohesion:** The EU TRAN Committee (European Parliament Transport Committee) provides technical support to Spanish policymakers, ensuring that mobility policies address social inequalities and enhance equal access to transport (MITMA, 2021, p. 133).

Links that enable learning on equality by public authorities

To enhance knowledge-sharing and adaptation to equality challenges, Spain has developed capacity-building initiatives and international collaborations.

- The MITMA's Transport Sector Training Program promotes knowledge exchange on gender equality and workforce diversity (MITMA, 2021, p. 357).
- EU Pact for Skills in Transport facilitates vocational training programs, integrating underrepresented groups into the transport workforce (MITMA, 2021, p. 357–358).
- The MITMA-led Working Group on Transport Education collaborates with universities, training centres, and professional associations to develop gender-sensitive mobility curricula (MITMA, 2021, p. 358).
- Awards and recognition programs: The Matilde Ucelay Award honors companies and organizations that advance gender equality in transport (MITMA, 2021, p. 364).

2.5 Civil society

The **Open Mobility Dialogue**, conducted in late 2020 engaged civil society, businesses, and advocacy groups through a combination of digital participation tools, public events, and direct stakeholder contributions, ensuring that diverse perspectives were integrated into policy planning (MITMA, 2021, p. 86). The **structure of the Consultation Process** was organized through three main channels:

1. **Digital Platform:** esmovilidad.mitma.es Hosted the Document for Debate, alongside surveys, idea labs, and open feedback forms. The platform received 80,000 visits and collected 746 public contributions, providing insights into mobility priorities (MITMA, 2021, p. 86).
2. **Events and Workshops:** Over 100 events, with more than 10,000 participants, were held to discuss key mobility issues. Organizers submitted official reports, which the MITMA reviewed to inform the strategy (MITMA, 2021, p. 86).
3. **Direct Stakeholder Submissions:** Organizations, companies, and advocacy groups formally contributed recommendations, including input from disability rights advocates, transport industry representatives, urban planners, and environmental organizations (MITMA, 2021, p. 379–388).

Several **key collaborating entities** submitted proposals aligned with their sectoral interests. Among the participants, the most notable contributors included **business associations, civil society organizations, and private entities**:

Business associations:

- **SERNAUTO** (automotive suppliers): Proposed incentives for vehicle purchases and fleet renewal.

- **CEOE** (Spanish employers' confederation): Submitted a comprehensive report covering all aspects of the Strategy.
- **AOP** (petroleum product operators): Emphasized the need for technological neutrality and fiscal measures for new fuels.
- **CETM** (freight transport): Proposed measures for more sustainable freight transport.
- **CNAE** (driving schools): Advocated for an active role in public awareness and road safety education.

Civil society and environmental organizations

- **Fundación ONCE**: Contributed insights on universal accessibility in mobility.
- **Zeroport and Ecologistas en Acción**: Criticized the strategy's treatment of air and maritime traffic.
- **Valencia en Bici – AE.A_**: Advocated for a more inclusive and sustainable mobility approach, integrating urban planning and active mobility.

Rail sector:

- **MAFEX** (railway industry): Highlighted the railway's role in achieving the Strategy's objectives.
- **AEFP** (private railway companies): Proposed incentives to shift freight transport to rail.

Private companies in technology and mobility:

- **KAPSCH**: Provider of technological solutions for mobility.
- **QOOB**: Specializes in bicycle and personal mobility vehicle parking infrastructure.
- **Dedocar**: Shared mobility platform.

This consultation process allowed for the integration of diverse perspectives, ranging from the business sector to environmental and accessibility concerns.

Civil Society's Influence on Gender Equality and Vulnerable Groups

- Universal accessibility was a key concern, with the ONCE Foundation and other disability organizations advocating for the inclusion of universal design and accessibility principles in mobility planning.
- Gender and mobility issues were also addressed, as groups such as Women in Motion Spain highlighted safety concerns, infrastructure gaps, and the gender disparity in the transport workforce, pushing for targeted policies (MITMA, 2021, p. 379–388). Public surveys revealed a strong preference for investments in public transport and active mobility over traditional large-scale infrastructure projects, influencing investment priorities.
- Environmental considerations were also a significant theme, with Zeroport and Ecologistas en Acción advocating for stricter environmental regulations, particularly concerning aviation and maritime transport emissions (MITMA, p. 379–388).

Transparency and Public Engagement

The Open Data Initiative facilitate public access to transport-related data, enabling stakeholders to provide evidence-based recommendations and strengthening government accountability and transparency is an initiative to highlight (MITMA, 2021, p. 260–261).

2.6 Barriers

One of the **major barriers** in Spain is the **fragmentation of competences** across different levels of governance. The governance of urban transport falls primarily under the jurisdiction of **local administrations**, making it difficult to ensure coherence in policies at a national level. The role of the **AGE is limited to that of a facilitator or promoter**, which can slow down the harmonization of public transport policies across regions and cities (MITMA, 2021, p. 120–121).

Infrastructure and service gaps

The **low investment in certain transport services** has resulted in infrastructure and service gaps that disproportionately affect vulnerable groups. For example:

- Renfe operates **Cercanías**, Spain's commuter rail network, which connects suburban areas with major cities. **These trains carry 90% of railway passengers**, yet investment remains insufficient, limiting service efficiency and accessibility (MITMA, 2021, p. 34).
- **Electric vehicle (EV) charging infrastructure is underdeveloped**, despite Spain's extensive **11,400 service station network**. Only **11,800 public charging points** exist, and most lack fast-charging capacity, slowing adoption of electric mobility (MITMA, 2021, p. 227).
- The **uncertain financing model for metropolitan transport** poses risks to **service continuity and affordability**, particularly for lower-income populations (MITMA, p. 120–121).

Mobility disparities between urban and rural areas

One of Spain's most persistent challenges is the rural-urban mobility divide, which disproportionately affects older adults, low-income individuals, and those without private vehicles.

- By 2050, 88% of Spain's population will reside in urban areas, with Madrid and Barcelona expected to concentrate 30% of the national population. This urbanization trend risks further exacerbating transport inequalities (MITMA, 2021, p. 120–121).
- Rural areas with low population density struggle to maintain viable public transport services, limiting access to jobs, healthcare, and education (MITMA, 2021, p. 132–133).
- Expanding infrastructure does not guarantee improved connectivity in rural areas. Tailored solutions such as on-demand bus services and ridesharing initiatives are needed (MITMA, 2021, p. 133–134).

Challenges in integrating public service obligations (OSP)

The **OSP framework**, designed to guarantee **essential transport services**, faces governance and coordination challenges:

- **Divergent regional OSP models** result in **inefficiencies and service gaps**, affecting accessibility for **remote and underserved areas** (MITMA, 2021, p. 137–138).
- **Rail OSP services**, including *Cercanías* and **Mid-distance trains (*Media Distancia*)**, remain underfunded despite their **key role in mobility for vulnerable populations** (MITMA, 2021, p. 138–139).
- **Bus routes in the national concession system** are largely based on **historical demand patterns**, which no longer reflect modern mobility needs. A **revised concession framework** is required to optimize service coverage (MITMA, 2021, p. 139).

Dependence on private vehicles and environmental concerns

The **high level of private vehicle ownership** in urban and metropolitan areas presents another **significant barrier** to sustainable mobility:

- Cities like Madrid, Cáceres, Zaragoza, Mallorca, and Valladolid have motorization rates exceeding 535 vehicles per 1,000 inhabitants, contributing to congestion, pollution, and road safety risks (MITMA, 2021, p. 120–121).
- Urban areas remain major sources of air pollution, affecting public health and quality of life, particularly for children, older adults, and individuals with respiratory conditions (MITMA, 2021, p. 121).
- Gender and income disparities limit access to affordable transport options, disproportionately impacting women and lower-income groups, who rely more on public transit (MITMA, 2021, p. 363–364).

Technological and regulatory challenges

The rapid evolution of transport technologies presents regulatory and governance challenges:

- Regulations often lag behind technological advancements, creating uncertainty for integrating autonomous vehicles, electric mobility, and data-driven transport planning (MITMA, 2021, p. 91).
- Administrative barriers remain for international transport, highlighting the need for harmonized driver certification and streamlined digital procedures (MITMA, 2021, p. 400).
- Big data and open data offer significant potential for transport optimization, but public agencies lack technical expertise to leverage these tools effectively (MITMA, p. 260–261).

2.7 *Factors for success*

The successful integration of gender equality and support for vulnerable groups in mobility policies relies on three key enablers: data-driven decision-making, institutionalized gender mainstreaming, and inclusive governance mechanisms.

Data-driven decision-making and institutional learning

Effective governance in transport requires robust data collection and cross-institutional learning to align mobility policies with real needs. The Independent Authority for Fiscal Responsibility (AIReF) published the AIReF's Transport Infrastructure Report revealed that investment

inefficiencies persist, particularly in suburban rail services *Cercanías*, which serve 90% of rail passengers but remain underfunded. A standardized metropolitan transport financing framework is needed to prevent fragmented PMUS and enhance transport equity (MITMA, 2021, p. 34).

Additionally, the *Universal Accessibility Policy* recognizes that inclusive mobility is a prerequisite for social participation. It calls for integrating accessibility standards across infrastructure, vehicles, and information systems. However, implementation remains uneven, requiring mandatory enforcement of accessibility standards in public transport and urban mobility planning (MITMA, 2021, p. 148).

Strengthening gender equality in transport policies

Gender disparities in mobility persist, particularly in workforce participation (women represent only 22% of transport employees, falling below 20% in land and maritime transport, p.363) and in access to safe, inclusive mobility. The *National Strategy* acknowledges that women's travel patterns differ from men's—women take shorter, more frequent trips, often using public transport due to caregiving responsibilities. Yet transport networks remain designed around male commuting patterns, limiting accessibility for women, caregivers, and low-income workers.

To address these disparities, the strategy proposes:

- **Gender-disaggregated data collection** to inform inclusive transport policies (*Measure 1.5.5*).
- **Best-practice guidelines** to ensure gender-sensitive transport planning (e.g., adequate service frequency, intermodal coordination).
- **Safety enhancements**, including improved lighting, flexible bus stops, and gender-sensitive urban design.
- **Stronger gender balance policies in transport employment**, aligned with EU initiatives (Women in Transport, Women in Motion).

Despite these initiatives, gender equality remains aspirational without binding measures such as mandatory gender impact assessments and enforceable employment quotas.

Institutional recognition and governance for vulnerable groups

The *Accessibility Universal Policy* outlines five essential factors for equitable transport: infrastructure, vehicles, embarkation systems, information accessibility, and service provision. While progress has been made—61.8% of suburban trains and 100% of high-speed trains are accessible —implementation gaps persist. A fully integrated **universal design approach is required to ensure accessibility across all modes of transport** (MITMA, 2021, p. 148–150).

Key governance initiatives include:

- **The OTLE Transport Observatory**, which consolidates mobility data for vulnerable groups.
- **The Adif Acerca Program (formerly Atendo)**, a nationwide assistance service for passengers with disabilities, providing support in 141 stations and handling nearly 700,000 assistances annually. Despite its reach, accessibility remains uneven across different transport modes.

- **Collaborations with disability organizations**, between the **Spanish Committee of Representatives of People with Disabilities (CERMI)** and **Renfe (Cermi-Renfe agreements)** to enhance accessibility services and ensure inclusive mobility policies.
- **The Plan *Cercanías* 2025**, which integrates accessibility measures such as tactile paving, real-time train accessibility alerts, and station redesigns.
- **The PlayRenfe digital accessibility initiative**, which provides free Wi-Fi and entertainment onboard, designed to accommodate passengers with disabilities by offering adapted content and user-friendly digital interfaces.

Spain's alignment with EU gender and accessibility frameworks is a positive step, but sustained impact requires **institutional enforcement, financial incentives, and clear accountability mechanisms**.

2.8 Conclusion

Spain's commitment to integrating gender equality and support for vulnerable groups into transport policy through the Safe, Sustainable, and Connected Mobility Strategy 2030 represents a significant step toward ensuring a fair and inclusive Twin Transition. The fact that the multi-level governances are actively working on accessibility, gender-sensitive planning, and rural connectivity demonstrates a structured approach to tackling inequalities in mobility. Remarkable initiatives as the **OTLE** and the **Inclusive Mobility Week provide valuable frameworks for assessing progress**. However, to ensure that these policies translate into real, enforceable improvements, three priority areas for action must be reinforced.

Enhancing cooperation between AGE, ACs, and local governments is essential to align policies and prevent fragmentation. Mechanisms such as the **Sectoral Transport Conference** and the **Local Government Coordination Mechanism** must be strengthened to ensure that PMUS systematically integrate universal accessibility, gender-sensitive planning, and rural connectivity.

Institutional learning and evidence-based policymaking must be strengthened through gender- and accessibility-disaggregated data. While the OTLE provides a foundation for mobility analysis, further **integration of academic research, qualitative studies, and more users experience surveys would enhance understanding of how transport inequalities manifest in everyday life**. This includes insights into how safe, comfortable, and accessible people feel when using public transport.

Additionally, public authorities must reinforce oversight of private and outsourced service providers, such as those operating *Acerca*, to guarantee that accessibility measures are effectively implemented and that personnel receive **specialized training**. Ensuring that all transport workers—whether in public or private services—receive comprehensive training in accessibility, gender-sensitive mobility, and inclusive customer service is fundamental to closing the gap between policy and real-world impact.

References

Ministry of Transport, Mobility, and Urban Agenda (MITMA). (2021). *Safe, Sustainable, and Connected Mobility Strategy 2030*. Madrid: MITMA.

Ministry for the Ecological Transition and the Demographic Challenge (MITECO). (2019). *National Strategy Against Energy Poverty (ENPE) 2019–2024*. Madrid: MITECO.

Ministry for the Ecological Transition and the Demographic Challenge. (2022). *Update report on energy poverty indicators*. Madrid: MITECO.

Hungary

Energy: Enhancing Solar Capacity

2.1 Introduction

Hungary has seen the solar photovoltaic (PV) capacity grow at a quick pace, representing an important transformation towards a greener energy mix. It is not a mere endogenous transformation but a multifaceted interaction of state strategies and supranational pressures, predominantly originating from the climate and energy policy framework of the European Union (EU). Solar power deployment in Hungary has also been spurred forward through the Renewable Energy Directive and EU financial instruments particularly the Modernisation Fund and the Recovery, and Resilience Facility.

Hungary has hit its 2030 National Energy and Climate Plan (NECP) solar capacity targets way ahead of schedule, and projections are that the country will meet them as early as 2024. This shows that targeted policy interventions and financial support work effectively in promoting renewable energy. If looked at more critically, however, the governance structures and implementation mechanisms have several pertinent challenges.

Although the EU framework gives general guidance, its translation into effective national policies is challenging and regulatory tools need strong governance. This encompasses the simplification of permitting processes and grid (electric network) connection procedures, and the creation of ancillary services to provide grid stability with more intermittent energy sources as renewable energies are. In addition, the efficiency of the Hungarian Energy and Public Utility Regulatory Authority (MEKH) in regulating the transition and maintaining market competitiveness is important.

The infrastructural constraints, especially related to grid capacity and storage options, present formidable challenges to the long-term incorporation of solar power. The current grid infrastructure, intended for centralised fossil fuel generation, needs to undergo sweeping upgrades to support the distributed nature of solar PV. Spending on smart grid equipment, energy storage systems (such as battery storage, pumped hydro), and grid reinforcement is critical to avoid curtailment and maintain system reliability.

Critical necessity does exist, however, in the justification for access to solar energy benefits on an equitable basis. While large solar farms make up most of the added total capacity, distributed solar PV systems in the form of rooftop solar installations within homes and SMEs would be essential for building energy democracy and alleviating energy poverty. Policy instruments like feed-in tariffs and net metering plans with subsidies focused on particular groups must be drafted with sensitivity to cover those who are frequently displaced and made unaffordable.

Environmental impacts of solar PV manufacturing and decommissioning, regarding the management of end-of-life solar panels and the sourcing of critical raw materials in favour of long-term sustainability, will determine the sustainability of the solar energy transition. Whole-of-life assessments of solar energy systems must drive circular economy principles.

2.2 The aim

Reacting to the EU's commitment to clean energy transition and climate protection, Hungary had made a legally binding long-term commitment to achieve decarbonisation and reaching climate neutrality by 2050. This include shifting away from fossil energy sources towards sustainable ones. One guiding document that supports this action is the NECP, that outlines this road up to 2030, and it was just recently revised reacting to the current trends and evolving needs. The updated NECP addressed the necessity of strengthening the energy independence of Hungary and the securing of energy supply chains. There are aspirations towards a diversified energy mix to avoid the possibility of leaning on heavily on one energy source while neglecting the others. Meaning the utilisation of the "unpopular" wind energy (tripling the country's capacity to 1 GW by 2030). This also supported by modifying the regulatory framework that surrounded the deployment of wind energy power plants. The NECP also favours the exploration of biomass, geothermal energy, biogas, hydrogen and biomethane.

The plan also recognises the importance of a stable, sustainable and independent energy system to support the civilian's and economy's need. Despite these commitments, there is no real political commitment to achieve this sovereignty, due to the government having years-long contract with the Russian Gazprom to purchase fossil energy (natural gas) through the TurkStream despite the geopolitical tension that is present in the continent.

According to the current NECP, Hungary's key targets to achieve green transition are the following:

- **Emission reduction** by at least 50% from the levels recorded in 1990 by 2030. (Previously it was 40%, due to reaching this goal in 2023.)
- The gross final **energy consumption up to be 30% sourced from renewable energy** by 2030. (Previously it was 21%.) Due to the rapid expansion of solar energy farming, the initial target 6 000 MW of total **solar capacity** was achieved in 2024⁵, six years ahead of the deadline. The Ministry of Energy ambitiously set a new target for Hungary to reach by 2030: 12 000 MW. Also, to support the greening of the sector, the nation aims for having 90% of their **electricity generation coming from low-carbon sources** by 2030 – mixing renewable energy and nuclear power.
- Improving **energy efficiency**. The cap for final energy consumption should be at 740 PJ by 2030. (Previously it was 785 PJ). This action includes various, if not all, sectors: households, industry, transport.
- Ensuring a **robust energy storage** system to utilise and stabilise renewable energy sources like wind and solar, that are fluctuating in their intensity throughout the year. (Investment in battery technologies, pumped hydro storage.) Upgrading the grid infrastructure is also

⁵ The "[Napenergia Plusz Program](#)" is the main instrument for the country's ambition to come to fruition. It is providing the essential and much needed financial support to both residential and commercial entities for solar PV application.

unavoidable step to take to accommodate the rapid growth of weather-dependant renewable energy sources, and the transmission and distribution of clean energy.

- Promoting and supporting **electromobility** (see more in the next chapter).
- Paks II, which is a crucial instrument in Hungary's plan to achieve independency in energy sourcing while adhering to climate goals, is a **nuclear energy** project launched in 2014, to add two new reactors to Paks nuclear power plant and extend the lifetime the already operating four.
- Phasing out coal by 2030, while launching a new gas power plant as interim during the transition period.

In accordance with the NECP goals, Hungary had created a grant system for supporting residential PV installation⁶. In 2024, around 288.000 households had PV installed, which had 2.63 GW of solar energy generated. The government also favours the planting of large-scale solar parks⁷, like the 7MWp solar farm in Balatonberény, or the 138 MWp in Szolnok, to reach the newly set goal of 12 GW energy "farmed from the Sun" by 2030.

Among all the above, as good practice that fuses green transition with social equity considerations, we have mention the "[Felzárkózó Települések](#)" program. It is an ESF+ project, also supported by the Hungarian government. Their goal is to help and uplift the most disadvantaged settlements (villages, isolated and integrated farmsteads).

The program helps along the transition from highly polluting (trash, wood) energy sources to renewable alternatives like installing PVs in these areas. By investing in these measures, their goal is to lower financial burden on these disadvantaged communities. They also encourage the use of biomass heating systems – utilising the locally available resources for warmth. They support the establishment of local energy communities, to empower them and collectively invest in green energy projects. The programs also consider the education of these disadvantaged groups to raise awareness about energy conservation and teach about the benefits of adopting sustainable habits. This initiative supports and adheres to the "just transition" principles advocated by the EU.

2.3 Roles and responsibilities

The implementation of Hungary's solar energy expansion strategy involves a complex network of stakeholders with distinct roles and responsibilities.

National Government:

Primarily through the Ministry of Energy the government plays a central role in forming the policy and legislative landscape for green energy, that means doing the following tasks:

⁶ The "[Napenergia Plusz Program](#)" is the main instrument for the country's ambition to come to fruition. It is providing the essential and much needed financial support to both residential and commercial entities for solar PV application.

⁷ METÁR (Mandatory Renewable Energy Support Scheme) is a Hungarian tender system. The purpose of the system is to upscale the share of renewable energy sources in the nation's energy mix. Channels financial support for energy producers, encourages investment in projects and market competition.

- *Strategic planning and policy development*, which means formulating and implementing national energy strategies, including the NECP, to align with EU directives and other national objectives.
- *Regulatory framework and permitting* to facilitate the deployment of solar energy, including streamlined bureaucratic processes, grid connection protocols hand in hand with the national energy providers, and taking part in market mechanisms such as the METÁR tender system.
- *Ensuring EU compliance and reporting*
- *Financial management*, means directly overseeing the allocation of national and EU funds for renewable energy projects, including the management of programs such as the Napenergia Plusz Program.

Municipalities:

Local governments have an important role in the practical implementation of the country's solar energy initiatives, focusing on the community level. They oversee the designated *land-use planning processes and zoning* to facilitate the development of solar installations. They are also – for public uses the main *collaborators with utility companies and private stakeholders* to integrate distributed solar generation into local grids. Municipalities *engage with the communities to raise local and public awareness* of the possibilities for e.g. implementing PVs. They also raise awareness about the benefits of solar energy and address the potential concerns of civilians. Local governments keep a close eye on the disadvantaged areas in their jurisdiction thus are able to implement programs with targeted aid for this communities, such as: Felzárkózó Települések program.

European Union Institutions:

The EU institutions are naturally playing a crucial role in financial and strategic support for Hungary's green – or in this case more like solar - energy transition. Through different frameworks, they provide funding: European Union's Recovery and Resilience Facility (RRF), the Modernisation Fund, and other relevant programs. They set the overarching policy objectives and targets through directives such as the RED and the European Green Deal for all the members of the EU, thus forming the landscape for Hungary too. Their "last" role in helping the green transition for Hungary is monitoring all processes and evaluate the results.

Private Sector:

The private sector is the one that basically "prefinances" the green transition of Hungary through keeping up with technological trends and if necessary, develop solutions for both residential and utility-scale e.g. solar installations. Although, we have to note, that implementer of these projects is not necessarily 100% part of the private sector, because most of the energy provider companies are more or less state-owned companies, e.g. MVM. They often time use subcontractors to do the installations. Utility companies play a vital role in grid integration, distribution, and the provision of ancillary services.

2.4 *Links between levels of governance*

In the recent years, with the rapid expansion of PV capacity across country encouraged and fuelled by EU funding and policy directives, consequently demanding a complex, cross-level cooperation between national, regional, and local governances. As established, the driving force for Hungary's solar initiatives stems from the EU's RED and associated financial instruments (RRF, Modernisation Fund). The Mandatory Renewable Energy Support Scheme (METÁR) tender system, for instance, serves as a link between investment opportunities and regulatory policies, aligning market-based incentives with broader sustainability objectives.

This approach ensures a coherence between national policy frameworks and local implementation -- vertical integration of policy objectives and the centralised coordination of activities across agencies and authorities.

The regional development agencies and the municipal governments localise renewable energy targets by analysing the needs of the region; thus, able to tailor projects to the exact needs of the locals. This collaboration between different levels propels the adaptation of national policies to local contexts and ensures the effective utilisation of resources, creates and inherent need within the community for green transition by creating achievable opportunities for them.

Despite this aspiration for a successful, smooth established coordination mechanism, gaps remain within the overall system for implementation. There are neglected areas, such as grid modernisation and equitable access to subsidies. Bureaucratic slowness and complex, hard-to-follow administrative procedures pose significant challenges for the simple- everyday people, hinder the timely implementation of solar energy projects (some experience annual delays).

These oftentimes administrative and resource deficiencies crave the streamlining of permitting processes and the reduction of bureaucratic burdens. As mentioned, grid limitations, including insufficient capacity and inadequate infrastructure, pushback the efficient and quick integration of solar energy into the existing electricity grid. All of the infrastructure-related delays and problems could use a substantial investment to achieve the needed and necessary grid modernisation, solve the existing issues regarding energy storage shortage, and lack of applied smart grid technologies.

We must also mention that the distribution of subsidies and incentives may not always be equitable, potentially exacerbating existing socio-economic disparities – only those can successfully apply for these tenders who have the necessary fiscal background, digital literacy and overall infrastructural stability in the neighbourhood. Until this day, the government fails in successfully addressing these issues, they lack targeted interventions and the successful implementation of inclusive policy mechanisms.

2.5 *Civil society*

NGOs actively engage in policy advocacy, work on pressing policymakers to enhance the efficiency, transparency, and inclusivity of green transition policies. Similarly to other European societies, they advocate for realistic and just renewable energy targets, simpler and achievable permitting processes, and equitable distribution of these solutions.

In theory, CSOs/NGOs are granted the possibility to present and advocate (lobby) for their views and be part of the governance itself, according to Act XCIII of 2011 about National Economic and Social Council of Hungary. This Council serves as consultative platform for proposal making, or if necessary and advisory board for the government. This body is independent from the Parliament, has members from across the sectors. Inherently, the NESC should be effective in facilitating dialogue between the state and the civil society, but its effectiveness is largely debated by experts and critics. We can also state, that according to research⁸ conducted in 2022 the effectiveness of these type of institutions are hugely dependent on the political environment, their independence from the state and commitment of the governments to consider and adopt their inputs. Knowing this, and all-around political climate of Hungary⁹, we can safely say, that the consultations – if they happen – between the government and the NGOs/CSOs are mostly symbolic.

In response to advocating climate and green energy NGOs, the Hungarian Ministry of Energy launched a collaborative consultation on climate and energy policy planning as part of the NECP review. This public consultation¹⁰ addressed the new draft NECP by the end of June 2023. Around this time, the participating NGOs emphasised the need for a transparent drafting process, incorporating both scientific and expert opinions, and aiming for energy efficiency, renewable energy diversification, and support for energy communities.¹¹ It had mostly fallen on deaf ears.

NGOs/CSOs, notably NSC-FoE and the international 1point5 closely followed the implementation of solar energy policies to try to hold policymakers accountable for their commitments, or to draw attention to any discrepancy. They conducted research, analysed data, and published reports, articles to assess the effectiveness of government initiatives. They also raised concerns about the equal distribution of subsidies and the potential for marginalised communities¹² to be excluded from the benefits of the solar energy transition. They have called for improved subsidy distribution mechanisms that prioritise the needs of low-income households and disadvantaged communities.

Table 1 - Notable NGOs/CSOs regarding green energy transition of Hungary

Name of the organisation	Focus Area	Activities
Friends of the Earth Hungary (NSC-FOE)	Broad environmental issues (climate, energy, agriculture)	Advocacy, campaigning, policy participation, local projects
Greenpeace Hungary	Climate change, energy, forests, oceans, waste	Activism, campaigning, public awareness
WWF Hungary	Protection of rivers, forests, sustainable agriculture, climate change	Conservation projects, advocacy, partnerships
Göncöl Foundation	Environmental and social issues, sustainability	Education, research, community engagement
Energiaklub	Energy efficiency, renewable energy, transparency	Research, analysis, advocacy, public education
Hungarian Geothermal Association	Geothermal energy development	Promotion, networking

⁸ von Zeschau, Juan & Sanchez-Mosquera, Marcial. (2022). A literature review on the Economic and Social Council's effectiveness and evaluation. *Revista de Administracao Publica*. 56. 465-487. 10.1590/0034-761220220033x.

⁹ <https://bti-project.org/en/reports/country-report/HUN>

¹⁰ <https://1point5.caneurope.org/speed-energy-transition-necps-review-process-hungary-and-ngo-messages/>

¹¹ <https://mtvsz.hu/en/news/2023/06/green-ngos-call-for-a-complete-energy-planning-redesign-in-hungary-june2023>

¹² Vadovics, E., Szöllőssy, A. (2025). Energy Citizenship in Hungary: Diversity, Actors, Focus and System-Contestation. In: Fahy, F., Vadovics, E. (eds) *Energy Citizenship Across Europe*. Palgrave Macmillan, Cham. https://doi.org/10.1007/978-3-031-70157-3_4

Hungarian Association of NGOs for Development and Humanitarian Aid (HAND)	Sustainable development, global citizenship education	Umbrella organization, advocacy, capacity building
Hungarian Civil Liberties Union (HCLU)	Human rights, rule of law	Advocacy, legal aid, monitoring
Transparency International Hungary	Anti-corruption, transparency	Monitoring, advocacy, research

In the recent years, CSOs have become louder in addressing energy inequality and advocated for vulnerable groups in Hungary. The cross-country Women for Solidarity in Energy (WISE) project¹³ aims to raise awareness about energy inequality and empower low-income women, including single mothers and retirees, to participate in the related policy development. The project disseminates information on energy policy and efficiency, focusing on women's unique challenges in the time of an energy crisis.

There are also several think-tank organisations that are focusing on energy-related issues specific to Hungary – like Energiaklub. They mostly work on promoting energy efficiency, green energy and advocate for transparency in all processes and decisions regarding the energy sector in the country. They also conduct research and in-depth technical analysis to help decision makers the general public to understand the most critical energy issues that endanger the population, and to offer possible solutions that are either good practices proven by other, similar countries or a niche way to solve existing problems.

A good addition to also mention, is the organisation called HAND that acts as an umbrella organisation for numerous CSOs, promotes cooperation and advocates for local and global sustainable development goals. Hungary also has several human rights and anti-corruption NGO, e.g. HCLU and Transparency International Hungary. Although they are not exclusively deal with energy issues, they play a crucial role in promoting the rule of law, transparency, and accountability, and monitoring all processes which are – ideally - fundamental for good governance. Their work in monitoring government actions and speaking out for greater transparency will hopefully help to create a more enabling environment for environmental and energy policy advocacy in the near future.

Keeping in mind the importance and necessity of that the two lastly mentioned organisation, that although the formal structures exist for civil society engagement as it is their right to exist and participate in the workings of a society, unfortunately their influence on shaping policy, particularly regarding gender equality and support for vulnerable groups¹⁴, are intentionally limited by the government¹⁵. The proactive efforts of NGOs e.g. WISE highlight a bottom-up approach for achieving fair green transition goals, but the integration of their perspectives into national policies remains constrained or ignored.

2.6 Barriers

Technical and infrastructural barriers:

¹³ <https://www.rug.nl/rechten/news/archief/2024/dr-marlies-hesselman-and-colleagues-awarded-funding-for-women-in-solidarity-for-energy-wise-pr?lang=en>

¹⁴ <https://apnews.com/article/hungary-orban-crackdown-media-ngo-38776560a2edf5948482dd4839461411>

¹⁵ <https://www.amnesty.org/en/latest/press-release/2017/06/hungary-ngo-law-a-vicious-and-calculated-assault-on-civil-society>

As mentioned in previous documents of FITTER: HUN case study, with Hungary being historically – and politically -reliant on fossil energy sources, the transition to green energy is inherently hard on the country. Also, with the rapid expansion of PV capacity and the growing popularity of acquiring it, all sectors of Hungary face serious barriers that makes the process impossible to be smooth and the transition to be equitable for all. The infrastructure existing now is designed for fossil fuel generation, therefore ill-fitted to handle the fluctuating and distributed nature of solar energy within the system. This often times – during when the daylight lasts longer, and the Sun is stronger – leads to grid congestion, instability and power outage. Investing in monumental grid modernisation efforts, including smart grid technologies, energy storage solutions, and transmission infrastructure, are essential and inevitable.

The permitting processes are largely delayed by bureaucracy and inconsistent regulatory approaches. The lack of clear communication also sets back the timely implementation of e.g. solar energy projects. In the end, the administrative hurdles create an uncertain environment for investors (civilian and market) and slows down the pace of solar deployment. Also, despite the availability of subsidies and financial incentives, the prefinancing the costs of solar PV installations makes it deterrent for many households and businesses, mainly in the rural and economically disadvantaged areas – peripheral regions of Hungary. While the existence of “Napenergia Plusz Program” is a good practice in Hungary, it also hinders the closing of socio-economic inequalities and limit the widespread adoption of solar energy. Without ironed out and simplified just frameworks, addressing the specific needs of vulnerable populations remains unanswered and challenging.

Legislative, political and policy barriers:

Several years in a row, Hungary ranked very low among the EU member states in the Transparency International’s Corruption Perceptions Index¹⁶, in 2024 the country reached all time low: dropped down on the global ranking too. The energy sector also not exempt from the corruption that rage havoc in Hungary, especially when we talk about public procurement and tax administration.¹⁷ NGOs highlighted that government’s habit of launching centralised, capital-intensive projects are the breeding ground for corruption, especially when one begins to investigate the Paks II project (which is an intensively debated topic inside and outside of Hungary).

NGOs and politicians also voiced their concerns regarding the government push for increasing their ownership in strategically important sectors like this one, thus creating disadvantageous market field for foreign investors, companies resulting in a distorted market competition and slowing the adoption of green technologies. There is no enforcing power in place to mitigate these risks created by the systematic corruption that is present in the country, which slowly but clearly visibly undermines the fair and successful transition to green energy. That means the allocation of resources are questionable, the prioritisation of politically connected companies and contractors are easily detectable.

The opacity of the policy and decision making, including the procurement processes and the sponsoring of energy projects makes it especially challenging for NGOs, CSOs – the public – and the investors to conduct investigations, scrutinise the situation present in the energy sector, and

¹⁶ <https://transparency.hu/en/adatok-a-korrupcirol/korrupcio-erzekelesi-index/cpi-2024/>

¹⁷ <https://www.state.gov/reports/2024-investment-climate-statements/hungary/>

if needed hold relevant actors accountable for their corruption or wrongdoing. E.g. Paks II nuclear power plant project has been under scrutiny since the launching of it, they voiced their concerns regarding the lack of transparency, they questioned the reason behind the large costs, the potential illegality of the state aid, and the lack of competition among contractors, purveyors.¹⁸

All above points to this: the lack of publicly available, open information on the ongoing processes can cover the corruption that surrounds the energy sector. This also hinders every try for a successful public debate on what is necessary for a fair green transition, or just to understand the nature and reason of state-backed projects. The lack of fairness in competition also impairs the effectivity of potential stakeholders and investors in shaping and helping the country's energy future. As remarked by the Transparency International Hungary, the current political situation and their methods in using special websites to publish procurement data also hinders and destroys transparency by fragmenting the landscape of information.

Organising impactful public consultations and the actively involving CSOs in the policymaking processes should be natural step to be taking for a government, especially if they want to ensure that any executive decision they make will long-termly accepted and effective. However, in Hungary – like most things – this is different too, and not in a good way. These good practices are mostly powerless and symbolic across all policy areas.

With the changes happened under the regime of the ruling Fidesz-KDNP, the legal and regulatory environment for NGOs and CSOs are became highly restrictive, impeded their operation and hindered their ability participate in policy processes. The debated and highly criticised Law on the Transparency of Organisations Receiving Foreign Funding of 2017¹⁹ stated that any NGO/CSO receiving more than a certain amount of fiscal support from abroad must disclose their donors publicly. In 2020 the European Court of Justice ruled²⁰²¹ that this law violated the EU law and the fundamental laws by stigmatising and intentionally silenced critically important voices within the civil society.²²²³ Despite this, the Hungarian government was slow to adhere to decision, and introduced a new law called Protection of National Sovereignty Act in 2023.²⁴ This norm raised even more questions about the intention of the government, and pointed to further restrict the NGOs, media and the general opposition of Fidesz-KDNP that are receiving support from foreign sources. These measures introduced a more complicated and burdensome administrative environment for NGOs/CSOs, particularly for those who rely on international funding for their activities. The law introduced a system of fines or in drastic cases imprisonment discouraging these groups in engaging in critical debates, conversation or activities regarding government policies.

¹⁸ https://www.sgi-network.org/docs/2024/country/SGI2024_Hungary.pdf

¹⁹ <https://www.opensocietyfoundations.org/newsroom/open-society-welcomes-court-of-justice-of-eu-ruling-on-hungary-anti-ngo-law>

²⁰ <https://abouthungary.hu/blog/hungary-on-ecj-ruling-we-will-continue-to-insist-on-the-transparency-of-ngos>

²¹ <https://rm.coe.int/168070bfbb>

²² <https://www.amnesty.org/en/latest/news/2019/02/global-assault-on-ngos-reaches-crisis-point/>

²³ <https://www.civicus.org/index.php/re-imagining-democracy/stories-from-the-frontlines/3257-democratic-backsliding-and-civil-society-response-in-hungary>

²⁴ <https://www.theparliamentmagazine.eu/news/article/hungarys-ngos-feel-chilling-effect-of-foreign-funding-law>

The government also doesn't shy away from voicing a hostile rhetoric against NGOs, portraying them as "foreign agents" serving outsiders' interests (e.g. George Soros, "Brussels")²⁵ rather than the nation. This narrative, amplified through the mainly state-controlled media, aims to slowly destroy public trust in NGOs and delegitimize their role as stakeholders in policy debates²⁶. High-ranking government officials (politicians) have also made vilifying statements against NGOs, and some of them have faced smear campaigns and accusations of illegal activities in the name of foreign powers. This negative environment hinders NGOs, CSOs from participating in policy discussions, fearing negative repercussions. The government has also been withdrawing funding from certain NGOs and denied them representation on national bodies, further limiting their ability to influence policy decisions.

Research indicated the steady decline in the *quality* of public participation from draft to the final versions of NECP, followed by above mentioned draft update of 2023.²⁷ The opportunities for consultation are becoming more and more limited, resulting in policies being out of touch with reality.²⁸ They do not really address public concerns – e.g. battery factories –, or leverage the readily available civilian expertise of CSOs that are actively engage in battling energy and environmental issues. There is no real mechanism or system in place to incorporate the feedback from relevant NGOs, CSOs or public into the formulating policies, which further alienates the citizens to embrace green transition.

Table 2 – Legal impacts on the activities of NGOs/CSOs

Year	Development	Impact on NGOs
2017	Law on the Transparency of Organisations Receiving Foreign Funding	<i>Imposed registration and disclosure requirements on foreign-funded NGOs, leading to criticism and legal challenges</i>
2018	"Stop Soros Package"	<i>Further restricted NGOs working on migration issues, criminalizing certain activities and imposing taxes on related funding</i>
2020	European Court of Justice Ruling on Foreign Funding Law	<i>Ruled the 2017 law was in breach of EU law and fundamental rights</i>
2023	"Protection of National Sovereignty Act"	<i>Introduced broad powers to investigate organisations suspected of foreign influence, raising concerns for NGOs, media, and opposition parties</i>

NGOs, like NSC-FoE Hungary and the Clean Air Action Group, poses as strong opposition to the government's current, not so green energy policies. They rightly argue that the political fascination with relaying on fossil fuel-driven re-industrialization hinders a just and fair transition to sustainable or green energy. Their opinion on the matter is discarded or muted by restrictive measures introduced by "Lex NGO" law, and the Sovereignty Protection Office's crusade against NGOs like Transparency International Hungary further deepen the problem. Experiences, like the one of The Clean Air Action Group's: exclusion from policy discussions and funding cuts, represents the government's zero-tolerance of critical voices.

²⁵ <https://www.transportenvironment.org/articles/hungarys-ominous-rhetoric-ngos-condemned>

²⁶ https://www.sgi-network.org/docs/2024/country/SGI2024_Hungary.pdf

²⁷ <https://www.tandfonline.com/doi/full/10.1080/07036337.2025.2460194>

²⁸ <https://mtvsz.hu/en/news/2023/06/green-ngos-call-for-a-complete-energy-planning-redesign-in-hungary-june2023>

2.7 *Factors for success*

While the above-mentioned laws are in effect, there is no hope for successful incorporation of civilian voices within the policy and decision making regarding green transition in the energy sector. Until retraction of these norms, there is no full compliance with EU law and international human rights – regarding freedom of association and expression. Even if the law is in effect, the government, as an example for all, should omit using derogatory rhetoric and smear campaigns that erode public trust in CSOs. The government should recognise their role as independent watchdogs and partners in creating a just and fair environment for the citizens for green transition.

The Hungarian government should prioritise the enhancement of transparency across the sector especially their own involvement in the processes. They should ensure the access to relevant information are released in a timely and open manner, including the details of procurement, contracts and the environmental and societal impact assessments for all energy projects. They should have an independent oversight mechanism in place that targets suspected corruption. The ones that currently exist in Hungary is the [Integrity Authority](#) (recently established due to EU demand) and the [National Protective Service](#), which are mostly symbolic in their role. If Hungary would have one or more institutions that serve as watchdogs, they could have effectively monitored all processes concerning public funding, they would be made more transparent through an established objective and clear criteria-system created by independent bodies, open bidding procedure would become a reality.

Having a clear, non-restrictive legal framework that adheres to meaningful public consultation and serve the active participation of NGOs and CSOs in energy – or any - policy development and implementation is non-negotiable for the successful adaptation of EU values and goals regarding green energy. Regular engagement between government agencies responsible for energy and environmental policies and relevant NGOs and CSOs should be facilitated: consultations with them should be initiated in advance early in the policy cycle, even before the drafting to be inclusive of diverse perspectives. Same way, they should ensure the consideration of the feedback from civil society and is incorporated into policy decisions. Resources should be also allocated to support public awareness / government campaigns regarding energy and environmental issues, thus fostering a more informed and engaged society.

Regarding our example of “Napenergia Plusz Program”, the successful implementation of Hungary's solar energy ambitions largely depends on (without talking about speculations) a combination of enabling factors that address the identified barriers and promote a just and sustainable transition. The government-backed grants, subsidies, and EU funding play an important role in reducing the financial burden of e.g. solar PV installations and encouraging investment. Although we must note, the effectiveness of these incentives depends on their accessibility, transparency, and equitable distribution. Strategic partnerships between government agencies, private investors, and research institutions could enhance the development and deployment of solar infrastructure.

2.8 *Conclusion*

The current situation in Hungary is far from optimal, but not the worst regarding achieving progress regarding green energy transition. Thank to the EU funding and directives in place and incorporated within Hungary's framework the progression to reaching climate and green goals is

timely successful. However, substantial challenges remain concerning lack of transparency, equitable access and the overall ineffective participation of CSOs/NGOs within policy making.

Although the policy implementation currently overseen by the Ministry of Energy involves a complex web of stakeholders, there are hindering gaps in coordination and implementation – i.e. administrative slowness, grid limitations. The source of weaknesses is most like originate from within the overcentralised and untransparent governance structure.

Regarding CSOs and NGOs, their power in shaping and helping the transition within the energy sector is hugely limited and restricted. Former consultation, in a limited form, exist, however their effectiveness and usefulness are debated and often questioned. Consultation is mostly symbolic or pre-determined, the government is overall reluctant to incorporate any feedback or recommendations coming from these organisations. The unethical laws surrounding these institutions also hinders any activities taken by them, and the rhetoric targeting them also create an alienation between NGOs/CSOs and the society.

Considering these limitations Hungary's energy policy cannot be considered as a strong or any kind of example of good governance, especially considering the involvement (or lack of it) of civilian or non-governmental organisations. The challenges discussed during this chapter raise all kind of concern about Hungary's policy's long-term effectiveness, its impact and their overall intentions with their programs in place. Notable progress aside especially talking about solar capacity expansion, the sustainability, inclusivity and equability of this progress is questionable and so far, unproven by the government.

Decarbonising transports

2.1 Introduction

Similarly to the energy sector, Hungary has a clear vision on how to create a more green, more sustainable mobility and transportation sector. The currently running programs put strong emphasis on promoting the general use of electric vehicles (EVs), and the overall electrification of public transportation systems. One of the most recent approaches in adhering to the Green Deal agreement is the government's 5-point Transport Action Plan, that focuses on the infrastructural and technological modernisation of the national rail system, including the improvement of passenger experiences.

In the previous chapter the already mentioned NECP also sets a goal to reach by 2030 about achieving 14% renewable energy usage within the transportation sector, to further support the government's commitment to build a greener and sustainable future for all. There is also a law in place to encourage the use of EVs while also instructing the relevant authorities to develop the charging infrastructure to an optimal level.

Despite the steps already taken towards green transition, Hungary has several challenges ahead of them yet to be successfully solved or planned for: **infrastructure gap**, the **number of electrified railway lines** (according to 2022 data, it was around 40.7% in 2022, which is way lower compared to other European countries). As briefly discussed in the previous chapter, with the continuously increasing electricity demand from fuelling EVs and electrified public transport with green energy sources an old-new **financial challenge** of grid modernisation shows up.

The **affordability** of purchasing electric cars has also remains a major challenge for the government to eliminate. Even with the tenders in place to lessen the burden on the households and companies, the upfront cost of the EVs remains to be fiscal barrier. Also, the greening of public transport does not necessarily lead to the lowering ticket prices or related fares – rather the opposite -, which puts unproportionate strain on low-income households.

The **risk of social and territorial disparities** arising from the green transition becomes a more and more pressing issue, especially in the light of the current economic situation within the country (an all-time high inflation rate). The benefits originating from green transition are mostly concentrated in the more mid-to-high income urban areas, where the municipalities and locals have a generally better access to infrastructure and resources. This leads to leaving behind rural, peripheral communities – socioeconomically vulnerable groups. With the unintentional exclusion of low-income groups, who simply cannot afford or prefinance EVs, or lack access to sustainable public transport options are critical concern that is largely unaddressed by the government despite of the available statistical and research data about this pressing issue.

2.2 *The aim*

Hungary is set in its way to achieve a carbon-neutral economy by 2050. To reach this, the government also put an emphasis on decarbonisation target on the mobility and transportation sector. This goal aligns with EUs, and beyond mitigating the effects of the climate change, Hungary also wants to improve the heavily effected areas' air quality thus reducing the development of related public health issues. The governments also promote zero-emission and low-emission vehicles and tries to (in policy) encourage electric and hydrogen-powered options for businesses. Although initially the transition aligns with the EU level energy security goals of achieving energy independence through relying more on green energy sources than fossil fuels, the government has years-long contract with the Russian Gazprom to purchase fossil energy (natural gas) through the TurkStream.

A policy instrument that has a great effect on the green transition of mobility & transportation sector is Hungary's National Recovery and Resilience Plan (NRRP)²⁹, that was introduced in response to negative effects on the country's economy and the well-being of the society caused by the COVID-19 pandemic. Besides mitigating these, the NRRP also tries to be source of bolster for building up resilience, ensuring sustainability and making the country's economy ready for tackling any challenge posed by the green transition.

Within NRRP there is a dedicated green transition section, that targets transportation as a strategically important area for future investment and reform: modernising the current infrastructure, encouraging consumers to use greener methods for private and public transport. Modernising in this case means e.g. the insurance of having technologically advanced railway lines, including their management and decision-making systems. (Currently, the situation is dire, the railway system infrastructure is from the previous century, some lines are operated manually.) It also means the procurement of zero-emission buses for public transport, improvement of tram and trolleybus infrastructure within the capital city. The government also intend to offer support for the purchase of at least 12 500 EVs, while also securing the necessary amount of charging

²⁹ [https://www.europarl.europa.eu/RegData/etudes/BRIE/2023/747098/EPRS_BRI\(2023\)747098_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/BRIE/2023/747098/EPRS_BRI(2023)747098_EN.pdf)

station for them. *The overall budget of this program (NRRP) is around 6,512 billion euro (that stems from grants) and there is an additional 3,918 billion euro acquired from loans.*

Hungary's strategy to expedite the green transition in transportation relies heavily on targeted financial incentives and infrastructure development. This approach primarily aims to promote the adoption of EVs and the electrification of public transportation fleets.

Existing financial incentives for EV adoption:

Corporate subsidies: The dedicated programs offer grants ranging from HUF 2,8 million to HUF 4 million per vehicle (this ranges between roughly €7,035 to €10,050 on current average exchange rate). A budget of €79 million (around HUF 30 billion) has been allocated to support companies to purchase battery EVs for business use. By early 2025, over 5 000 businesses had applied for these programs, to support the procurement of 6 400 vehicles – that means the distribution of HUF 21 billion financial aid.

Civilian subsidies: The government offers some low-level incentives for private persons, such as: tax investment incentive for the purchase of zero-emission cars (0-15% of the price), different depreciation rules, exemption from pollution tax. These cars receive special license plates (green), which means they do not have to pay registration tax, vehicle tax, property transfer tax. Beside these, many municipalities support this with offering free parking in public spaces for cars with green plates.

Expansion of the charging infrastructure: The government offers non-refundable grants and loans for planting EV charging stations. The preferred stations by the government should prioritise energy storage capabilities, power generation. Funding is a combination of national budget and EU fiscal support – RRF. Over a hundred new chargers will be financed from REPowerEU, and the EU will help the installation of more than 200 EV charging stations in 60 different locations, to the current (as of 2023) more than two thousand EV charging stations.

Electrification of public transportation: The "[Green Bus Programme](#)" (2019-2029) is an important initiative to modernise Hungary's public transportation fleet. This program supports the procurement of new electric buses and the necessary infrastructure. Currently they financed 139 electric buses (beside the 100 that the Volánbusz public transportation company had procured), 4 self-propelled trolleybuses, and 148 charging stations. The initial calculations prognosticated that around with 70 thousand tonnes the CO₂ emissions could be reduced with the improvement of public transportation fleet. Using e-buses lowers the operating costs, amounting to around HUF 3,5 million savings per year (electricity is HUF 60 cheaper than diesel fuel). The total budget of the program is HUF 35,9 billion.

2.3 Roles and responsibilities

The role of the EU level authorities is clear, they set the guidelines, objectives and recommendations form the national strategies, through frameworks like Green Deal and RRF. They also mandate the minimum amount of investment that is required to achieve climate objectives. The EU is not directly involved in the drafting of national plans, but they offer feedback on whether the document is in alignment with the overarching objectives of the EU or not. The European Commission, as auditor, execute rigorous assessment following a well-defined criteria

system. In terms of the actual implementation, the EU is a supervisor, monitors progress, the moving of funds and the achievements of initial targets and milestones.

Naturally, the government is in central position in the “command chain of” green mobility transition. They have direct responsibility over the whole continuum: setting national priorities, establishing frameworks (Jedlik Ányos Plan 2.0), drafting plans (National Recovery and Resilience Plan [NRRP]), implementation, establishing tender system, managing projects, evaluating and monitoring progress, coordinating between authorities. For example, sector specific organisations like the recently founded waste management authority (under MOL) is a key implementer of the transport related components of the circular economy. The government is also the primary investor in infrastructural developments (e.g. management system for railways).³⁰

The municipalities of Hungary have a circumstantial role in contributing. While they take part in the planning with offering insight, identifying localised needs, their input in policy formulation is historically limited. Their main role is to implement the projects – like Green Bus programme – and engage with civilians to assess impact.

Independent NGOs and CSOs (with other type of stakeholders) have some kind of influence over policy making. They – if asked – contribute through consultations, advocacy and bring attention to social justice considerations. The Hungarian Lean & Green Platform is dedicated to facilitating cross-sectoral collaboration in order to achieve climate and transition goals. Their direct role is also heavily restricted, (see previous chapter), but they try to hold the implementers accountable, as well as the public authorities who are handling the flow of finances.

2.4 Links between levels of governance

The implementation of Hungary's transport electrification strategy - similarly to its green energy plans – involves a complex cooperation across public authorities and other type of organisation, that emphasises collaboration across all levels.³¹

Level of Governance	Key Responsibilities related to Green Mobility
National	Setting national strategies and targets (NECS, NECP), implementing key programs (Green Bus Program, EV subsidies), financing infrastructure development (railways, charging), coordinating national efforts (e.g. Ministry of Energy, HUMDA), long-term strategic planning (NTS)
Regional	management and maintenance of some regional transport infrastructure (roads, railways), determining the modes, frequency, and quality of regional public transport, rate of fare subsidies and revenue support
Local	development and maintenance of local public roads, provision of local public transport services, implementation of green parking initiatives (under national framework), applying for and implementing national programs (e.g., Green Bus Program)

Table 3 – Links between actors

³⁰ <https://www.rvo.nl/sites/default/files/2021/04/Smart-sustainable-mobility-market-Hungary.pdf>

³¹ https://www.sgi-network.org/docs/2024/country/SGI2024_Hungary.pdf

2.5 Civil society

The situation concerning independent NGOs and CSOs are quite similar in this sector to the one in the energy sector.

Regarding NRRP, the government held a public consultation³² that was under scrutiny by the European Parliament. They were concerned about the lack of meaningful consultation with relevant stakeholders, CSOs and municipalities.³³ CSOs themselves spoke up about that their input were not considered or even slightly incorporated into the final form of the plan. The consultation was also distorted by not localising its content, thus not providing real picture about the different situations coexisting in the different parts of the country. The government restricted themselves by not including those kinds of organisations that are in day-to-day contact with local communities and vulnerable groups.

The integration of CSOs into transport policy development has been exemplified by initiatives like the [SUNRISE project in Budapest's Törökőr neighbourhood](#)³⁴. Although this project is not closely linked to mobility sector, but it uses a co-creation approach (that should be more common in Hungary), involving local civilians and local stakeholders in engaging them in the sustainable mobility planning and implementation.

Consultation process structure:

- Problem identification: The process commenced with a comprehensive SWOT analysis, facilitated through diverse engagement formats such as workshops, neighbourhood festivals, online platforms, walking tours, and thematic walks. These activities support the locals voice their mobility challenges and needs, ensuring that the perspectives of vulnerable groups are considered.
- Solution design: Building upon the identified issues, collaborative efforts led to the development of strategic directions: enhancing infrastructure with considerations for vulnerable groups, including those with mobility- and visual impairments.
- Measurement and implementation: The community participated in selecting and implementing measures, such as improving accessibility for mobility- and visually impaired individuals and introducing low-scale measures supporting cycling.

The active involvement of CSOs and residents ensured that the mobility measures addressed the specific needs of vulnerable populations in this area. The Institute of Blind People in Zugló emphasises the critical need for making the public infrastructure more accommodating visually impaired individuals. The co-creation process created a stronger community spirit and empowered residents to shape their environment, making it safer, sustainable, and liveable.

³²

<https://www.hungarianconservative.com/articles/current/hungarian-government-green-energy-consultation-launch/>

³³ <https://www.theparliamentmagazine.eu/news/article/hungarian-mep-raises-questions-over-his-countrys-national-recovery-and-resilience-plan>

³⁴ https://urban-mobility-observatory.transport.ec.europa.eu/resources/case-studies/co-development-and-co-implementation-sustainable-mobility-measures-torokor-neighbourhood-budapest_en

2.6 *Barriers*

Due to the specific situation currently present in Hungary, some of the barriers are already identified and described in the previous chapter. The centralised governance structure limits the autonomy and effectiveness of the lower-level authorities, leaving them out of policy planning, thus lowering the efficiency of the implementing period. The independent organisations' and stakeholders' influence is also meagre, which makes it especially challenging to incorporate the needs of the disadvantaged or vulnerable groups.

2.7 *Factors for success*

Similarly to the previous chapter, the situation is similar.

Until Hungary does not nurture a genuine and working multi-level governance structure with focus on stakeholder engagement, thus having mechanisms in place for collaboration, shared decision making, the root of the problem will remain paralysing for NGOs and CSOs in having their voice heard and respected.

2.8 *Conclusion*

Conclusion is the same as in the previous chapter.

Ireland

Government of Ireland Energy Poverty Action Plan 2022

1.1 *Introduction*

The [Energy Poverty Action Plan 2022](#) was published in December 2022 as a response to the unprecedented situation of the rising energy costs between November 2021 and November 2022. It adopts a whole of Government approach with two overarching categories of action: near term (supporting people with energy costs through the winter) and medium term (supporting energy efficiency upgrades to help to tackle the roots of energy poverty, with the commitment to inclusive and just transition to a carbon neutral society). It contains a mixture of measures, including financial supports for vulnerable groups, financial measures focused on redistribution of revenue surplus, consumer protection enhancement, retrofit programme supports, as well as increased investment in research on energy poverty in Ireland, improved communication campaigns, and ensuring the whole of government approach at all levels of the policy development and implementation. It also recognises that while the energy poverty definition and measurement framework has been put in place, more information is needed to provide enhanced and targeted supports the vulnerable groups, particularly low-income households and elderly citizens. It builds on earlier versions of the 2016 Strategy to Combat Energy Poverty and follows a consultation campaign carried out in August 2022.

1.2 The aim

The Energy Poverty Action Plan 2022-2023 sets out actions in four main areas, aiming at alleviating energy poverty in Ireland. Energy poverty for the purpose of the Plan is defined as a function of three elements: household income, the cost of energy, and the level of energy efficiency of the home. It is closely related to basic material deprivation. It was estimated that 29% of Irish households were in energy poverty by 2022, projecting that the proportion can reach 43% in a scenario where there is another 25% increase in energy prices. In this context, the main areas of action set by the plan are as follows:

1. **Meeting the Cost of Energy**, with the main focus on supporting consumers through the winter 2022/2023. Actions are closely linked to the Communications section and the market research, which is led by the Department of Environment, Climate and Communication (DECC), and aimed at measuring consumers' sentiment and experiences, and to ensure that these inform ongoing policy action.
2. **Energy Efficiency**, with measures funded by the Government and delivered by the Sustainable Energy Authority Ireland (SEAI) as well as the Local Authorities. The main focus of this area of action is the improvement of the energy rating of residential buildings, with particular focus on vulnerable persons such as **older people** and **children** living in energy poor households, as both cohorts are especially vulnerable to negative health implications of living in houses with low energy efficiency.
3. **Research**, with a new work programme lead by the Economic and Social Research Institute (ESRI) with the aim to propose how to improve measuring of energy poverty in Ireland. The research recognises shortcomings of the current measuring methodology, with the aim to improve this, to allow for a better understanding of the intensity of energy poverty in Ireland and its correlations with energy efficiency level of homes.
4. **Governance and Communications**, with actions focused on the Reduce Your Use and Stay Warm and Well campaigns lead by the DECC in cooperation with Money Advice and Budgeting Service (MABS) and ALONE to ensure that the campaign reaches people and communities that are most in need.

While the Plan is part of wider government strategy and complements the whole of Government 'Roadmap to Inclusion 2020-2025', specific budget figures have been provided in the Plan document with regards to the near-term actions. These include:

- **Income supports** with €1.57 billion allocated to the two Electricity Costs Emergency Benefit Schemes
- **Targeted social protection** with €1.2 billion of social protection lump sum payments.

Additional €10 million is also provided through a fund established by the DECC to deliver extra safeguard for people struggling with their energy bills.

The Action Plan emphasises the need for appropriate communication between the Government, the stakeholders and the consumers for the effective implementation of the measures. The Plan thus pledges to ensure that there is a "coherent and consumer centred approach to monitoring the impacts of these measures, which will inform and underpin the development of the next phase of the policy response" (DECC, 2022, p.4). Finally, the Plan acknowledges that all these action areas need to be effectively communicated to enhance public knowledge and understanding of what

supports are available. In this regard, the most vulnerable groups will also be additionally targeted through the involvement of the MABS and ALONE which are engaging with those in financial difficulties (MABS) and senior citizens (ALONE), two groups identified throughout the plan as particularly vulnerable to energy poverty.

1.3 *Roles and responsibilities*

The Energy Poverty Action Plan adapts a whole of Government approach and was developed with the participation of: Department of Environment, Climate and Communication (DECC), Department of Social Protection (DSP), Department of Housing, Local Government and Heritage (DHLGH), Department of Rural and Community Development (DRCD), Department of Health, as well as the Commission for Regulation of Utilities (CRU), Sustainable Energy Authority of Ireland (SEAI) and the Health Service Executive (HSE).

In order to ensure the whole of Government approach, the **Energy Poverty Steering Group** was established to develop, implement and oversee policies and measures aimed at alleviating energy poverty. The following are the members of the Group: DECC, DSP, DHLGH, Department of Health, SEAI, CRU, HSE, Department of Public Expenditure and Reform (DPENDR), Department of Finance and the DRCD. The Steering Group is led by the **DECC**.

DECC has a central role in the development and the delivery of this policy. While the document consistently refers to the whole of Government approach, the DECC is among the most frequently mentioned actors who are responsible for establishing some of the income supports, as well as supports for the community and volunteering sector, and the retrofit programme. They also lead the Communications Campaign and initiated the market research on energy poverty. Other government departments who have important roles in this Action Plan include: the **Department of Social Protection (DSP)** who is responsible for allocating social welfare payments, the **Department of Rural and Community Development (DRCD)** who is involved in the community and voluntary sector supports, the **Department of Housing (DHLGH)** who provides **Local Authorities** with funds for retrofitting social housing stock and is also currently collaborating with the ESRI on the issue of 'split-incentive' scheme research to enhance private rented properties retrofit.

Other actors who are playing a core role in the Action Plan include:

Commission for Regulation of Utilities (CRU): Independent body responsible for overseeing the regulation of electricity and gas sectors in Ireland, under the Electricity Regulations Act 1999 (Cassidy et al., 2021). It has a legislative obligation to protect the interest of customers and promote competition in retail energy markets, with a responsibility for implementing customer protection measures for electricity and gas. This is done by placing requirements on electricity and gas suppliers, and electricity and gas network companies; these are set out in the Electricity and Gas Suppliers Handbook (CRU, 2022). CRU is also responsible for monitoring disconnections and arrears, extending non-disconnection moratoriums, as well as setting out enhanced consumer protection measures during specific periods, such as winter 2022/23.

Sustainable Energy Association Ireland: This is the Irish national energy agency of Ireland established under the Sustainable Energy Act 2002. The main role of the SEAI in the actions set out by the Energy Poverty Action Plan relate to the retrofit schemes, which SEAI delivers. This

includes schemes for privately owned homes, as well as the Warmers Homes Scheme and Social Housing Retrofit Programme. The former is designed to provide energy upgrades to eligible homeowners who are at most risk of energy poverty with funds secured from the European Regional Development Fund (2022-2027). The latter programme is funded by the Government with work done by DHLGH and Local Authorities. SEAI also has retrofit schemes available for landlords and the Approved Housing Bodies. All the SEAI schemes are aimed at improving energy efficiency of buildings across Ireland.

Economic and Social Research Institute (ESRI): ESRI is a leading research body in Ireland, partially funded by the Department of Public Expenditure and Reform. It produces independent, high-quality research that informs policy making and promotes social progress (ESRI, 2025). As part of the Action Plan on energy poverty, the Government tasked the ESRI to lead the research component, with the focus on energy poverty and inequality. The methodology for studying energy poverty is under scrutiny under this task, as the ESRI research to date highlights that current measures generally target those on lower incomes, however the approach does not ensure that all households at risk of poverty are adequately targeted (Barret et al., 2022). As laid out in the Plan, these are to be improved through the ESRI led research.

Money and Budgeting Advice Services (MABS): MABS provides supports with budgeting and debt, all free of charge (MABS, 2022). Within this Action Plan, MABS is among the main partners of the Government in terms of delivery and implementation, with particular focus on vulnerable communities. It plays an important role in the Communication Campaign led by the DECC as it provides detailed information on the consumer experience on the ground. It also provides assistance to those who are facing difficulties in paying their energy bills, and helps them to engage with the energy supplier, while also working with the electricity and gas sector.

Local Authorities: While the role of Local Authorities in the twin transition is limited due to the centralised style of governance, Local Authorities are an important actor for the implementation of the Retrofit area action of the Plan. In this case, the Local Authorities are responsible for the social housing stock in their areas, and these include competing stock condition survey, subsequent development of strategic work programmes (together with the DHLGH). Local Authorities are also legally obliged to manage and maintain their housing stock, including pre-letting repairs to vacant properties, the implementation of planned maintenance programme and carrying out responsive repairs.

ALONE: National organisation that enables older people to age at home as well as providing them with services, advocacy and supports. In the context of the Energy Poverty Action Plan, ALONE is partnering with the government to ensure the appropriate outreach to those most in need, as older citizens are recognised as a group at risk of energy poverty.

1.4 Links between levels of governance

The current plan was developed with a cross-departmental approach, with the main actors listed in the point 2.3 of this document. Following the adoption of the plan, the **Energy Poverty Steering Group** is responsible for the implementation of the plan and for drafting further updated plans. The Group has been meeting since August 2022 and held eight meetings before publishing their Annual Report in 2024 (Energy Poverty Steering Group, 2024). Several organisations were present at these meetings, including ALONE, Amarach Research and the Electricity Association of Ireland.

At the core of the work of the Steering group is providing a structured, whole of Government stakeholders engagement. This includes engagement with relevant stakeholders (such as NGOs) at events and during consultation process, to ensure that their voices are part of the annual review process. This follows the previous significant engagement with stakeholders since the initial consultation process conducted during the development of the current Action Plan.

The plan also puts an emphasis on communications and research, which constitute two of the main areas of actions and are laid out in 2 chapters of the current Plan.

The Research work plan is led by the ESRI and involves other actors such as DECC, DSP, SEAI and the Central Statistics Office, which together are working on the methodology allowing for a better understanding of the energy poverty in Ireland. The work of the ESRI has already been taken into account while developing improvements for the current plan, and the institute also measures the impact of current Government measures on inequalities. The analysis is also complemented by the ongoing work of the **Irish Government Economic Evaluation Services (IGEES)**, which is an integrated cross-Government service to support the Government in progressing cross-cutting policies³⁵. Furthermore, as explained in the Action Plan, the Government is funding EIRI to work with a research network and (among other issues) to quantify the characteristics behind dwelling-related fuel poverty, as well as to quantify which socio-economic characteristics drive income-related fuel poverty (e.g., homeowners vs. renters), and to prepare a map of deprivation and energy efficiency distribution in Ireland.

The Communication Strategy, which is included in the Governance and Communications chapter of the document, builds on the work from the Reduce Your Use Campaign which began in Spring 2022 and was developed as a whole of Government message. However, it has been acknowledged that the reduction in energy usage is not possible for all, especially those who are at risk of energy poverty as they don't use enough energy in the first place. Thus, the improved campaign also aimed to provide consumers with 'entry point' information on income, social protection and energy efficiency supports available. The campaign, which was led by the DECC, prompted the department to initiate a consumer sentiment research initiative using marketing research approaches to better understand consumers experiences of the impact of the rising energy costs (Energy Poverty Steering Group, 2024). The research was conducted by the Amarach Research nationally, with representative groups surveyed to test response to government messaging and consumer sentiment. In addition to the details laid out in the main document, the Energy Action Plan Steering Committee Annual Report (2024) emphasized that the research part of the Communications Strategy allowed the Government to "listen to people actively" and ensure that their needs are understood in order to determine supports and advice to be provided. Other key partners involved in the campaign included CRU, MABS, ALONE and SEAI.

In relation to the near-term supports aim (Meeting the Cost of Energy), a range of supports has been coordinated by various departments, mainly through the cooperation between the DECC and the DSP. Furthermore, so-called **Additional Needs Payments** are available through the **Community Welfare Centres**. These are designed to help those on low incomes to meet their essential costs and can be accessed through **51 local Intreo Centres** – public service employment centres, available nationwide. In this context, the role of **CRU**, as well as **ESB Networks**, and the

³⁵ <https://www.gov.ie/en/organisation-information/8f949-irish-government-economic-and-evaluation-service-igees/>

energy service providers is also explained. CRU, as the main regulator, has been monitoring disconnections and arrears as critical markers of the state of the retail electricity and gas markets, and this allows them to monitor concrete situations ‘on the ground’. Under the **Energy Engage Code**, customers in difficulties who engage with the suppliers are not disconnected. In this respect, the **MABS** work with the electricity and gas sector to provide financial advice to customers (the document notes that it is often difficult for the vulnerable groups to engage with the supplier), while the **ESB Networks** maintains a **Vulnerable Customer Register** based on the information provided to them by the supplier. Further to that, **CRU** oversees the regulation of electricity and gas in Ireland. Following engagement with customers representative groups, NGOs, energy suppliers and network operators, the consumer protection was enhanced for the winter 2022/2023 with special emphasis on vulnerable customers who were experiencing financial difficulties. The Meeting Costs of Energy area of action also contains supports for community and voluntary sector groups operating in disadvantaged areas – **Community and Volunteer Energy Support Scheme (CVESS)**. The fund is allocated by the DRCD (with €10 million Euros listed in the document) and administrated locally by the **Local Community Development Committees (LCDCs)**.

In relation to mid-term supports and the second area of action (retrofit), most of the supports are funded by the **central government** and delivered through the **SEAI**. Some deviation is observed in terms of the social housing stock, which is the responsibility of the **Local Authorities** who are in charge of the management and the maintenance of their housing stock. They also have a legal obligation to ensure that all their tenanted properties are in compliance with the Housing Regulations of 2019. In this regards, **Local Authorities** work together with the **DHLGH**, who is the main provider of the funding.

1.5 *Civil society*

The Energy Poverty Action Plan document, as well as other government publications related to the plan (such as report with stakeholder’s forum or the Steering Group annual report) emphasise the importance of the stakeholder engagement throughout the report. These include NGOs and representatives of the Civil Society Organisations. In fact, some of the NGOs, such as ALONE, have been involved in the information campaign led by the DECC (described in detail in point 2.4). Other organisations, such as MABS were involved in the initial development of the Plan. The role of the Energy Poverty Steering Group also includes a provision of a structured, whole of Government stakeholders’ engagement, with arrangements for consultation with relevant stakeholders such as NGOs, and events (e.g. annual plenary session with stakeholders) to ensure that their voices are heard as part of the annual report process.

The initial consultation, which was published in August 2022, was set up to review the previous ‘Strategy to Combat Energy Poverty’ (2016) and involved (along other activities) a public consultation that ran for 4 weeks in August 2022, inviting participants to answer 5 questions. 70 submissions were received, including 1 group response representing 42 NGOs and 28 individual NGOs submissions. Following the publication of the Energy Action Plan 2022-2023, a new consultation process was launched in May 2024 with potential participants asked to answer several questions related to the areas of actions as laid out in the Plan, including Research, Communications and Governance, and Policy Design (DECC, 2024a). Those who responded to the consultation also received an invitation to the Energy Poverty Stakeholder Forum (DECC, 2024b). The Forum took place in July 2024 and allowed industry representatives, NGOs and advocacy

groups to provide feedback on existing measures and put forward recommendations for future supports (Energy Poverty Steering Group, 2024). It was attended by representatives of government departments and agencies, business organisations, and a number of civil society organisations, representing such vulnerable groups as: senior citizens, vulnerable children, persons with disabilities, members of the Traveller community, low income families and tenants in private rented accommodation (DECC, 2023). As will be discussed in point 2.6 and 2.7 below, various stages of the consultation process with civil society organisations allowed the sector to express their opinions regarding the existing measures and the development of future Energy Poverty Action Plans. However, the exact mechanisms by which the recommendations were considered are not clear from the documents publicly available from the DECC.

1.6 *Barriers*

As noted in the previous section, the Action Plan was a subject of a consultation process, which took place at different stages of the Plan, including the development of the current plan, as well as its subsequent updating. Most of the submissions identified barriers related to policies and actions themselves, however several weaknesses regarding the current governance of the Plan were also highlighted. The way in which the policy-making process is informed by the evidence was identified as in a need of improvement. It was noted that the government is currently lacking disaggregated data to represent in detail the disproportionate impact of energy poverty experienced by marginalised communities in Ireland, in addition to an overreliance on household surveys, which may have negative effects on the Governments' ability to ensure that no one is left behind (NCWI, 2024; Winston et al., 2024). The lack of data sharing practices among key stakeholders was also identified as a barrier to collective knowledge on drivers to energy poverty (Energy Poverty Steering Group, 2024). Broadening the definition of energy poverty was also proposed by several organisations, with some also calling for inclusion of a 'subjective method' allowing for self-reporting of energy poverty (Irish Rural Link, 2022; Pavee Point, 2024). There were also some voices calling for inclusive transport in the definition of energy poverty, particularly in the context of 'forced car ownership' (Winston et al., 2024). Enhanced engagement with communities, with a shift from 'communicating to' to 'working with' to allow for greater public participation and inclusion of those experiencing energy poverty during all stages of the design and the delivery of the new Plan was also proposed (Irish Rural Link, 2022; SVP, 2024; Pavee Point, 2022). In terms of the delivery of the Action Plan, some also identified the lack of inclusion of clear and specific targets in the plan as a barrier to effective monitoring and evaluation (Irish Rural Link, 2022; Pavee Point, 2022).

1.7 *Factors for success*

The main Energy Poverty Action Plan provides little details on the effectiveness of links between different levels of governance in relation to supporting vulnerable groups. However, several points can be made in this context. First, the document repeatedly mentions 'whole of government approach' to the design, delivery and implementation of the plan, which is a positive message, and it is also reflected in the composition of the Steering Group which has several government departments representatives as members. Second, the Steering Group also reflects the need for a wider engagement, with membership also drawn on inter-agency basis, with SEAI, HSE and CRU included. Third, it is also worth noting that the stakeholders' engagement is emphasised in the document as being at the core of the Action Plan, with consultation process in place. The need for

ongoing engagement with the representatives of NGOs was also voiced during the consultation process, and this is taken to consideration in the future plans. Fourth, communication strategy led by the DECC also should be noted as positive, particularly in relation to engagement with vulnerable groups, which was recognised as a challenge, with subsequent further measures (such as involvement of other partners) undertaken by the government. Nevertheless, more details would be needed to fully assess factors for success.

1.8 Conclusion

Overall, in relation specifically to governance, the Action Plan appears as a positive step forward in terms of the increased wholistic approach to policy making, as well as engagement with representatives of vulnerable groups. As mentioned in point 2.7, the plan has been designed and updated with consultation mechanisms in place and has received meaningful engagement from various stakeholders. The whole of Government approach is emphasized as being at the core of the Plan, however, this would require further scrutiny. Although the term ‘whole of Government’ is mentioned 11 times in the document, it only features in the introduction and the governance section, without providing any further information of how the approach is realized in practice. The way in which the Action Plan was drafted (cross-departmental approach), as well as the composition of the Steering Group suggests that the approach is being adopted, yet more details are needed for a full assessment.

In terms of engagement with vulnerable groups, in addition to consultation processes, there are also several positives, particularly in relation to adopting evidence-based approaches to policy making. First, the research component has a significant place in the Plan with a dedicated chapter and concrete actions taken in the form of assigning the research programme to the ESRI. Second, more rapid market research was also conducted during the communication campaign, which signals that the need for a better understanding of the situation on the ground was recognised.

Finally, the Plan puts a lot of emphasis on communications, and the need to be able to reach groups which are at most difficulty. In this case, engagement with such organisations as MABS or ALONE is also a positive sign, however it could be argued that more organisations should have been brought on board as partners to enable effective communications with other vulnerable groups currently not mentioned in the document (e.g. migrants, Travelers). Overall, the value of consistent messaging has been adequately recognised, with more improvements as a possibility for the updated plan.

National Dialogue on Women in Agriculture – Event and Action Plan

2.1 Introduction

The agri-food sector - comprising primary agriculture, food and drink processing/manufacturing, fisheries, aquaculture, fish processing, forestry, forestry processing and the equine sector - is Ireland’s oldest and largest indigenous industry. In 2020, it accounted for almost 7% of modified Gross National Income (GNI) and 10% of exports in value terms (DAFM, 2021: 11). Its continued viability is dependent on ensuring the economic, environmental and social sustainability of the sector for all, and particularly for primary producers. To support social sustainability, issues such as low levels of generational renewal, gender inequality, the poor health and wellbeing of a

significant proportion of primary producers and insufficient rural development need to be addressed.

One of the actions specified as part of Food Vision 2030 was a National Dialogue on Women in Agriculture under goal of building the social sustainability of the sector.

2.2 *The aim*

There are two primary relevant missions to the twin transition in Food Vision 2030 - Mission 2: Viable and resilient primary producers with enhanced well-being; - Mission 4 - An innovative, competitive and resilient agri-food sector, driven by technology.

Under Mission 2, Goal 4 actions include publishing gender disaggregated datasets and reports; women only knowledge transfer groups for primary producers; **a National Dialogue on Women in agriculture**; inclusion of the agri-food sector in the National Women and Girl's Strategy; promoting LGBTI+ inclusion; combatting isolation; and rapid roll out of the National Broadband Plan.

The **aim** of the **National Dialogue on Women in Agriculture** was to 'explore gender issues encountered by women within the sector and to draw conclusions, identifying barriers and possible solutions, and making recommendations to inform future policy on gender equality in farming' (DAFM, 2024a: 7). As for the **content**, issues were discussed as part of 2 panels: The Role of Women in the Agri-Food Sector; Women in Farming – Challenges and Opportunities, - and 4 workshops: Leadership and Visibility; Education and Awareness; Financial and Legal Matters; and Sustainability.

No information on budget is available. In terms of **impact**, a report was produced and its recommendations were translated into a [12-point Action Plan](#) by a dedicated working group ([DAFM 2024c](#)). This working group will also oversee the implementation of these recommendations.

2.3 *Roles and responsibilities*

Public Authorities

The **Department of Agriculture Food and the Marine (DAFM)** has primary responsibility for implementing actions arising from the **National Dialogue on Women in Agriculture**. Policy planning, drafting, and implementation is thus the responsibility of DAFM.

Teagasc – the Agriculture and Food Development Authority – also has responsibility for policy implementation (actions 3 and 4). Teagasc is the Irish national body providing integrated research, advisory and training services to the agriculture and food industry and rural communities.

Some women farmers are disadvantaged by the combined impacts of agriculture policy, tax policy and social policies. Multiple reports have recommended changes to agriculture tax policy and social policy as they impact on the 'active farmer's' spouse (most of whom are women). Greater integration of policy planning and implementation by **Revenue**, the **Department of Agriculture Food and the Marine** and **Department of Social Protection** is part of actions 5 and 6.

The **Department of Education** is responsible for implementation of action 2.

Civil Society Organisations

The working group overseeing the implementation of the action plan has representatives from the Women in Agriculture Stakeholders Group (WASG), Dairy Women Ireland (DWI), Young Female Farmer Representatives, Teagasc, Agri-food Diversity and Inclusion Forum (AgDIF), Irish Forest Owners, Waterford Oysters (Aquaculture), Education and Research representative, and Macra na Feirme ([DAFM, 2024b](#)).

The Women in Agriculture Stakeholders Group (WASG) is an independent, voluntary body bringing together representatives from most of the Irish farming organisations, including the Irish Creamery and Milk Suppliers Association, Irish Cattle and Sheep Association, Irish Grain Growers Group, Irish Co-Operative Organisation Society, and the Irish Natura and Hill Sheep Farmers Association.

WASG have taken a leadership role in advocating for gender equality in agriculture in Ireland. WASG representatives are very active in advocating for the implementation of the actions. However, decision making and implementation capacity rests primarily with DAFM.

Dairy Women Ireland are engaged in awareness raising activities on the role of women in dairy farming.

2.4 *Links between levels of governance*

The National Dialogue on Women in Agriculture and subsequent action plan is linked to Ireland's obligations under CAP 2023-2027.

The Common Agricultural Policy (CAP) 2023 - 2027 specifically highlighted the need to support women in farming. There are 10 key objectives including the objective on:

- Rural Areas - promote employment, growth, gender equality, including the participation of women in farming, social inclusion and local development in rural areas, as well as the circular bio-economy and sustainable forestry.

The EU advised that under CAP 2023-2027 Members States should 'be required to assess the situation of women in farming and address challenges in their strategic plans. Gender equality should be an integral part of the preparation, implementation and evaluation of CAP interventions. Member States should also strengthen their capacity in gender mainstreaming and in the collection of sex disaggregated data.' ([European Union, 2021: L435/6](#))

DAFM were thus obliged to account for gender equality as part of CAP intervention preparation, implementation and evaluation.

[Ireland's SWOT analysis](#) in preparation for CAP 2023-2027 highlighted the low numbers of women farmers and women in leadership roles in the agriculture sector. The SWOT analysis cited 5 of the 6 actions (DAFM, 2020: 235) assigned to DAFM as part of the [National Strategy for Women and](#)

[Girls 2017-2020](#) (Department of Justice and Equality, 2017) as **opportunities** to address gender inequality in the sector.

1. Provide support for rural female entrepreneurs at start-up stage through the 'ACORNS' programme.
2. Improve statistics and reporting on women's involvement in the agri-food sector.
3. Clarify any perceived taxation barriers to registering farms in joint ownership and publish the outcomes.
4. Ensure that gender equality and the empowerment of women and girls is prioritised in Ireland's overseas development assistance programme, particularly in relation to agriculture and nutrition.
5. Encourage female involvement in decision making and leadership in all parts of the agri-food sector, especially through mentoring and positive case studies

In August 2021 the WASG proposed several [Common Agriculture Policy \(CAP\) interventions](#) to advance gender equality in the agriculture sector. Much of the evidence base for the proposed interventions was obtained via Freedom of Information (FOI) requests.

DAFM are responsible for proposing and implementing CAP interventions.

Ireland's CAP Strategic Plan did include measures to support greater gender equality.

- an increased rate of grant aid of 60% for women aged 18-67 years,
- women-only knowledge transfer groups,
- and a call under the European Innovation Partnerships (EIP) initiative for proposals to examine women's participation in agriculture.

These measures are not designed to address the needs of women who do not already hold legal status as an active farmer and having no legal status is one of the main issues for women farmers.

Outside of the CAP interventions, DAFM also

- commissioned a research project to establish a baseline position on women in agriculture and provide evidence for future policy in the area (HER-SELF project)
- hosted a National Dialogue on Women in Agriculture
- agreed a 12-point National Action Plan to advance gender equality in agriculture

The 12-point action plan reiterates many previous recommendations on advancing gender equality but there is a lack of firm commitments to address key issues such as the negative impact of agriculture, tax and social welfare policies on the status of the active farmer's spouse (most of whom are women).

Governance and reports on gender equality more broadly in Ireland have yet to prioritise the needs of women in rural areas and particularly women farmers. In February 2022 the [WASG made a submission](#) to the Joint Committee on Gender Equality. This Committee was established to consider the recommendations contained in the [Report of the Citizens' Assembly on Gender Equality](#). Neither the initial report of the Citizens' Assembly on Gender Equality nor the

subsequent [report from the Joint Committee](#) accorded priority to the needs of women in agriculture and/or rural women.

2.5 Civil society

As part of the policy planning and drafting process all government departments invite submissions from civil society groups. Submissions are taken into consideration as part of policy design, but final decision-making powers remain with government.

Responsibility for agriculture policy planning, drafting, implementation and monitoring rests primarily with the Department of Agriculture Food and the Marine (DAFM). Civil society groups do engage regularly with DAFM. In relation to gender equality this is now primarily through the working group to oversee the implementation of the action plan. Final decision-making powers rest with DAFM.

2.6 Barriers

Agriculture policy making is highly centralised and has historically benefited more profitable farms. There is poor understanding of gendered norms and the ‘farm family-farm’ relationship by policy makers. Policy makers tend to prioritise economic levers only to encourage behaviour change and not account for the social norms and values embedded in the agriculture sector.

Key dominant norms such as the ‘son as heir’ norm, the ‘woman as helper’ norm and the sole trader model of farm ownership makes invisible the contributions of women to the agriculture sector.

At a structural level, agriculture law and policy establish ‘the land’ as the primary unit of analysis. The people who work on farms, and particularly the people within a family unit other than the farm holder, are generally not the primary concern of law or policy (Shortall and Marangudakis, 2022). Irish law allows for a holding to be passed to one child intact without having to compensate other children (Shortall 2015). This, combined with how different types of work are gendered, supports the continuation of a **patrilineal impartible succession model** whereby one son is designated as the successor and socialised from a young age to take over the farm holding (Cassidy, 2019).

At an institutional level, a range of criteria - particularly related to tax policy, agriculture policy and social welfare policy - shape the legal status of family members on family farms. These have particularly detrimental impacts on the spouse of the farm holder (usually a woman). They also contribute to low levels of generational renewal (older farmers retiring and younger farmers entering the sector). A recent report on the *Sustainable Transition of the Rural Economy through Generational Renewal* found that:

The Irish system, by not having mandatory registration for spouses/civil partners and “prescribed relatives” working on farms, and by having a means testing system which imputes notional income to non-cash generating assets and attributes 50% of farm asset ownership to spouses/partners regardless of the cash reality, has created significant gaps in social welfare pension coverage within the farming community. This has consequent potential effects on poverty for farmers in old age

and creates a reluctance to retire and pass on farm businesses to the younger generation. [...] Ireland lags in terms of its social security pension system, as it applies to farmers [...] reform is needed to ensure that a basic State pension is available for all engaged in farming to protect their financial welfare and the financial welfare of their spouses/partners in retirement thereby facilitating full retirement and supporting generational renewal in agriculture ([Hayden et al, 2024: 13](#))

Supporting the **social sustainability** of farming would require co-ordinated change across a range of policy areas – agriculture, taxation and social welfare. Although multiple reports have called for such a systems approach to policy design little progress has been made.

Advancing gender equality is also accorded relatively low priority in practice. In relation to SO8 - Promote employment, growth, gender equality, including the participation of women in farming, social inclusion and local development in rural areas, including the circular bio-economy and sustainable forestry – Ireland’s CAP Strategic Plan places gender equality last in the list of priorities.

SO8 – Identification of Needs and Prioritisation at the level of the CAP Strategic Plan

1. SO8 N1 Maximise the opportunities presented by the circular and bio-economy in rural areas
2. SO8 N2 Improve infrastructure and access to services in rural areas including broadband
3. SO8 N3 Support sustainable community development in rural areas, in particular to address E+C challenges
4. SO8 N4 Support the further development of the tourism industry in rural Ireland
5. SO8 N5 Improve on farm safety and increase awareness of the importance of farmer well-being
6. SO8 N6 Increase opportunities for women in agriculture and business development ([DAFM, 2024d: 148](#))

Anecdotally there are also resistances to questioning gendered norms or considering the injustices of those norms on the part of key actors in the sector including those in leadership positions.

2.7 *Factors for success*

As decision making and policy implementation is highly centralised it is important that DAFM take a real leadership role in supporting the **social sustainability** of the agriculture sector. Meaningful advancement of gender equality in agriculture, and indeed meaningful progress on related social challenges such as generational renewal, improved work/life balance and the improved health and wellbeing of primary producers, will require a systems approach to policy design, implementation and evaluation.

The combined impact on farmers of policy measures across key departments – Department of Education; Department of Social Protection; Revenue; Department of Agriculture Food and the Marine; Department of Rural and Community Developments – needs to be assessed. Integrated policies which support the sustainability of farm families need to be designed.

In a recent report *Just Transition in Agriculture and Land Use* (2023) the National Economic and Social Council (NESC) identified four areas of action to support an effective, fair, and inclusive transition in the agriculture and land-use system in Ireland.

1. **Socially and Farmer-Inclusive Processes** - participatory processes of dialogue, which clearly communicate the scale of change needed, could develop a shared national narrative and strengthen consensus and decision-making. The active participation and engagement of farmers in innovating and changing practice is identified as critical.
2. **Enabling People to Benefit from Opportunities of Transition** - scale up and more fully align farm advisory services with environmental objectives; public and private funding sources to reward farmers for protecting and enhancing ecosystem services should be significantly increased; efforts must be made to better understand the social and cultural barriers to action, to align all policies with the transition, and reduce risks for farmers by creating more market certainty for income diversification.
3. **Sharing and Mitigating the Costs of Transition** – the costs of transition need to be accounted for, shared equitably, addressed and/or mitigated
4. **Co-Ordinating Action** – the transition must be co-ordinated and governed so that it can deliver real change in a balanced, inclusive and just manner. Within this area of action the Council further recommends
 - establishment of an **implementation group** for climate transition in agriculture in 2023 to consider the report recommendations.
 - that NESC work on transition should inform and shape the next stages in the development of the **Land Use Review**.
 - the **agriculture and land-use system** should be a **priority focus** for Just Transition institutions, processes and resourcing in Ireland, notably in the work of the Just Transition Commission.
 - a **Just Transition in agriculture and land use fund** should be established, consolidating available carbon tax revenues and other public resources.
 - a **Just Transition lens** should be applied to climate adaptation in agriculture, forestry and other land use.
 - monitoring transition should include economic, **social** and environmental **aspects and data** at the local level in support of place-based transition (NESC, 2023: viii-x)

A decision was taken by the Minister for Agriculture in March 2024 that implementation of the NESC recommendations would be monitored through the Food Vision 2030 Environmental Sub-Group (Oireachtas, 2024a)

2.8 Conclusion

All government departments have a responsibility to eliminate discrimination, promote equality of opportunity, and protect the human rights of those to whom they provide services. Currently 90% of CAP payments go to male recipients (DAFM, 2024a: 35). This demonstrates the need for measures to address the reproduction of inequalities. Such measures could include:

A full Equality Impact Assessment (EIA) of agriculture policy and practices that shape the status of 'active farmer' and the wider legal status of farm family members, farm business owners and farm employees. To include gender disaggregated data on accessibility and participation of all schemes

and measures. To include the impacts of the intersections of agriculture policy with tax rules and social welfare entitlement criteria. Any discriminatory impacts, particularly on the basis of the intersections of gender, civil status, family status, age and ethnicity need to be considered. To progress this action the establishment of a Working Group, led by DAFM, with representatives with relevant knowledge and expertise is recommended (to include Revenue, Dept of Social Protection, Teagasc, WASG)

Just Transition initiatives are an opportunity for eco-social policy development which takes a holistic approach to solving economic, environmental and social challenges concurrently. Funds allocated to addressing social and environmental challenges in agriculture (access to land for new entrants, sustainable farm practices, succession, supporting farmer's health and wellbeing) should have advancing gender equality as a core objective in addressing each challenge. Addressing social challenges, with advancing gender equality as a key priority, should be allocated a percentage of any just transition in agriculture and land use funds.

Qualitative research suggests that the process for issuing or amending the herd number (to include additional owners for example) is very straightforward in some Regional Veterinary Office (RVOs) but unduly complicated in others (DAFM, 2024a). The current process for issuing and amending herd number should be reviewed to ensure consistency across all Regional Veterinary Offices.

Qualitative research suggest poor governance practices in some agriculture sector organisations are leading to the reproduction of inequalities. Organisations under the remit of public authorities should report on their governance practices; identify any practices which may inhibit gender equality (e.g. no term limits); and produce and implement recommendations to ensure equality of opportunity and due regard for the value of diversity and inclusion.

References

Barrett, M., Farrell, N. and Roantree, B. (2022). *Energy Poverty in Ireland*. ESRI Research Series No. 144, June 2022. URL: <https://www.esri.ie/system/files/publications/RS144.pdf>

Cassidy, A. (2019) Female successors in Irish family farming: four pathways to farm transfer *Canadian Journal of Development Studies / Revue canadienne d'études du développement*, 40 (2), pp. 238-253. Available at : <https://www.tandfonline.com/doi/abs/10.1080/02255189.2018.1517643>

Cassidy, E. McLay, P. and Carmody, W. (2021) *Ireland*. In: J. Dewar (ed). *Electricity Regulation 2021*. Lexology: Getting The Deal Through, Law Business Research LTD London.

Citizen's Assembly (2021). *Report of the Citizen's Assembly on Gender Equality*. Available at: <https://citizensassembly.ie/wp-content/uploads/2023/02/report-of-the-citizens-assembly-on-gender-equality.pdf>

CRU (2022). *Electricity and Gas Suppliers' Handbook*. Dublin: Commission for Regulation of Utilities.

DECC (2022). *Strategy to Combat Energy Poverty – Progress Review and Public Consultation*. July 2022. Government of Ireland.

DECC (2023). *Energy Poverty Action Plan: Stakeholder Forum Report*. Government of Ireland.

DECC (2024a). *Revised Energy Poverty Action Plan 2024 Consultation*. Government of Ireland.

DECC (2024b). *Consultation on the Revised Energy Poverty Action Plan*. Government of Ireland: <https://www.gov.ie/en/consultation/c07a4-consultation-on-the-revised-energy-poverty-action-plan/> (accessed 30.01.2025).

Department of Agriculture Food and the Marine (DAFM) (2020). *Ireland's SWOT Analysis Preparations for Ireland's CAP Strategic Plan 2023-2027*. Dublin: Stationary Office. Available at: <https://assets.gov.ie/137582/1640bd23-22ce-45d4-ae32-09e7ed6d7a0a.pdf>

Department of Agriculture Food and the Marine (DAFM) (2021). *Food Vision 2030*. Dublin: Stationary Office. Available at: <https://www.gov.ie/en/publication/c73a3-food-vision-2030-a-world-leader-in-sustainable-food-systems/>

Department of Agriculture Food and the Marine (DAFM) (2024a). *National Dialogue on Women in Agriculture Report*. Dublin: Stationary Office. Available at: <https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://assets.gov.ie/282362/66655b42-18ab-4552-8c36-7941b8b8f686.pdf&ved=2ahUKewi46YX0n9GIAxVxRkEAHaIXMcwQFnoECBMQAQ&usg=AOvVaw16m2QDnAovR5bjC0Dylb9R>

Department of Agriculture Food and the Marine (DAFM) (2024b). *Food Vision 2030 - A World Leader in Sustainable Food Systems – Women in Agriculture*. Dublin: Stationary Office. Available at: <https://www.gov.ie/en/policy/b2a3c-food-vision-2030-a-world-leader-in-sustainable-food-systems/>

Department of Agriculture Food and the Marine (DAFM) (2024c). *Action Plan arising from the National Dialogue on Women in Agriculture*. Dublin: Stationary Office. Available at: <https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://assets.gov.ie/282363/b3f2cb5c-9daf-42d8-a514-953471b3d0da.pdf>

Department of Agriculture Food and the Marine (DAFM) (2024d). *CAP Strategic Plan Third Amendments*. Available at: <https://www.gov.ie/pdf/?file=https://assets.gov.ie/301467/ce6681d0-a519-4724-ba04-54593d2c1c41.pdf#page=null>

Department of Justice and Equality (2017). *National Strategy for Women and Girls 2017-2020*. Dublin: Stationary Office. Available at: <https://assets.gov.ie/6427/88cc57e206a34e94aeff2310f2b100d5.pdf>

Energy Poverty Steering Group (2024). *Energy Poverty Action Plan Steering Group. 2022-2023 Annual Report*. Government of Ireland.

European Union (2021). *Regulation (EU) 2021/2115 of the European Parliament and of the Council*. Available at: <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32021R2115>

Hayden, M., McNally, B., and Kinsella, A., Wakamatsu Shin, M. Hlochova, H (2024). *Sustainable Transition of the Rural Economy through Generational Renewal*. Dublin: Department of Agriculture Food and the Marine. Available at: <https://www.maynoothuniversity.ie/sites/default/files/assets/document/Sustainable%20Transition%20of%20the%20Rural%20Economy%20Through%20Generational%20Renewal%20Project%20Report.pdf>

Irish Rural Link (2022). *Submission to Energy Poverty Action Plan Consultation*. Irish Rural Link: Nasc Tuaithe na hEireann, September 2022.

NESC (2023). *Just Transition in Agriculture and Land Use Council Report*. No 162 June 2023. Dublin: National Economic and Social Council. Available at: <https://www.nesc.ie/publications/exploring-a-just-transition-in-agriculture-and-land-use/>

NCWI (2024). *Submission to the Consultation on the Revised Poverty Action Plan*. Feminist Communities for Climate Justice.

Oireachtas (2022). *Unfinished Democracy: Achieving Gender Equality Final Report*. December 2022. Joint Committee on Gender Equality. Available at: https://data.oireachtas.ie/ie/oireachtas/committee/dail/33/joint_committee_on_gender_equality/reports/2022/2022-12-15_final-report-on-unfinished-democracy-achieving-gender-equality_en.pdf

Oireachtas (2024a) 'Just Transition' *Oireachtas*, 05 March. Available at: <https://www.oireachtas.ie/en/debates/question/2024-03-05/507/>

Pavee Point (2022). *Submission to the Department of the Environment, Climate and Communications. Energy Poverty Action Plan Consultation*. Pavee Point Traveller and Roma Centre.

Shortall, S (2015) Gender mainstreaming and the Common Agricultural Policy. *Gender, Place & Culture*, 22 (5), pp. 717-730. Available at: <https://doi.org/10.1080/0966369X.2014.939147>

Shortall, S and Marangudakis, V. (2022). Is agriculture an occupation or a sector? Gender inequalities in a European context. *Sociologia Ruralis*, 62, pp. 746–762. Available at: <https://doi.org/10.1111/soru.12400>

SVP (2024). *Submission on the Revised Energy Poverty Action Plan*. Society of St Vincent de Paul.

Winston, N., Dingley, O. and de Silva Pedroso, M. *Submission on the Revised Energy Poverty Action Plan*. Submission to the Department of Energy, Climate and Communications. UCD School of Social Policy, Social Work and Social Justice.

Women in Agriculture Stakeholders Group (WASG) (2021) *Draft Interventions for CAP Strategic Plan*. Available at: <https://wasg.ie/wp-content/uploads/2024/08/Draft-interventions-for-CAP-strategic-plan.pdf>

Women in Agriculture Stakeholders Group (WASG) (2022) Submission to Joint Committee on Gender Equality. Available at: <https://wasg.ie/wp-content/uploads/2024/08/Submission-to-Joint-Committee-on-Gender-Equality.pdf>

Italy

Emilia-Romagna Region Pact for Work and Climate

1.1 Introduction

On 14 December 2020, the Emilia-Romagna Region, together with local authorities, trade union, business, professional and third sector representatives, the regional school office, universities and research institutes, chambers of commerce and banks, signed the Pact for Work and Climate. The Pact outlines a shared relaunch and development project aimed at generating new quality jobs, accompanying Emilia-Romagna in its ecological and digital transition. The document considers the Agenda 2030 for Sustainable Development as a reference, with a focus on sustainability concept, as declined in its three inseparable components, environmental, social and economic.

1.2 The aim

The pact is aimed at safeguarding health and employment, fighting inequalities, as well as achieving the climate neutrality by 2050 and tackling the transition towards a new, more environmentally and socially sustainable economy.

The pact assumes that the work should be **quality work**, for both employees and the self-employed; stable, adequately paid and secure work.

Quality relationships between institutions and economic and social stakeholders, mutual recognition and accountability of different social players, as well as sharing strategic objectives, are key principles for the governance of the pact.

Strategic objectives include the environmental transition with five dedicated lines of action: 1) accompanying the ecological transition of all size companies by directing and incentivizing investments in renewable energy and processes with a lower environmental impact, through targeted aid, regulatory simplifications and support for the transition to sustainable production and consumption models; 2) developing new green supply chains with a focus on both the climate/energy and material recovery industrial supply chains, 3) investing in research and innovation with a focus on the use of materials with high strategic potential such as hydrogen, electricity and green chemistry, 4) building a research team for sustainable and social impact finance consistent with the Pact objectives, 5) increasing the production and use of renewable energy and storage, also in a widespread form, through a regional law on energy communities.

These actions are implemented with regional and European financial resources, including RRP funds, assigned to Emilia Romagna.

Several stakeholders belonging to trade unions, employers' organization, universities, municipalities are involved in the pact policy making and implementation³⁶.

1.3 *Roles and responsibilities*

As mentioned above, the Pact for Work and Climate is based on the quality of the relationships between the parties, the mutual recognition of their social role, the sharing of strategic objectives and the transversal assumption of responsibility by all the actors involved in the process, such as regional institutions, universities, municipalities (also belonging to other regions), economic and social representatives. Meetings between stakeholders (all the ones which signed the pact) are held every six months. Thus, we can assume that planning, drafting, implementing and monitoring/evaluating of regional policy processes are equally shared among all the pact components.

Civil society is not represented in the list of institutions adhering to the pact. However, the document mentions the intention to capture the contribution of civil society and the possibility to identify the most effective way to ensure information, participation and reporting to citizens, to strengthen transparency, accountability and shared responsibility for the change to be pursued.

With reference to the tasks assigned to the stakeholders involved, the pact outlines the strategic framework and guidelines for the projects up to 2030. They are implemented through operational agreements and operational strategies, defined on the basis of the same participative method.

Meetings among stakeholders involved in the pact are held every six months³⁷ to:

- monitor the progress of the actions undertaken and to evaluate their impact, based on a common and uniform database for the whole territory,
- evaluate possible additions or modifications based on new scenarios, new criticalities and new opportunities,
- promote the implementation of the strategies identified in the territorial pacts and programmes.

Monitoring is based on indicators defined by the signatories of the Pact. However, the list of detailed indicators is not available; the only [document](#) available seems to deal with the objectives of the Pact and the results achieved.

The governance and strategic control of regional programming is ensured by a dedicated control room. Among the functions and purposes of the control room, there is the monitoring of the

³⁶ For the list of all stakeholders involved please, refer to the dedicated link <https://www.regione.emilia-romagna.it/pattolavoroeclima/gli-aderenti-al-patto-per-il-lavoro-e-per-il-clima>.

³⁷ Topics discussed during the meetings held since the document was signed on 14 December 2020, to mention the most important, were the 2021-2027 RSP, the 2021-2027 Smart Specialisation Strategy (S3), the 2021/2027 ESF+ and ERDF programmes, the 2023-27 Rural Development Programming Complement to the CAP Strategic Plan, the main sectoral plans, some regional laws, such as no. 2 /2023 'Attraction, retention and development of highly specialised talent' and several operational agreements.

strategic control of regional programming: it aims to verify the full coherence between the regional programming, the strategic objectives defined by different sources, such as the Employment and Climate Pact, the Regional Strategy for Sustainable Development - Agenda 2030, the Emilia-Romagna Digital Agenda 2020-2025.

With specific reference to the governance of the **work and climate pact**, the control room deals with

- secretariat and organisational coordination of the table work,
- monitoring the progress of the actions undertaken,
- definition of indicators and data sets, common and uniform for the entire regional territory, for the preparation of the impact assessment - (integrated in the Regional Strategy for Sustainable Development - Agenda 2030),
- drafting the impact assessment,
- definition of integrations or modifications to the document, based on new scenarios, new critical issues and new opportunities,
- drafting and implementation of the strategies, together with the relevant regional+ Departments,
- application of the regional strategies at local level,
- organisation of dissemination initiatives, including seminars, addressed to the members of the table³⁸.

In addition, in order to guarantee the necessary transversality and implementation of the regional pathway to carbon neutrality before 2050, a regional **technical coordination for Climate Transition**, composed of experts in the topic of interest, has been set up to act as a control room and to monitor the effectiveness of regional sectoral policies on the subject. The coordination, also through the support of agencies, universities, research centres and the Chamber of Commerce system, is intended as a reference point for the definition of common and uniform methods and measurement systems and monitoring the effectiveness of regional sectoral policies on the topic.

On the basis of the data collected, it can be assumed that the planning, the drafting and the monitoring/evaluation of the policy are shared between all the institutions involved in the Pact and between the public and private sectors. Coordination powers, as mentioned, are assigned to the control room for coordination and governance of regional programming, which is composed of internal regional members: it is coordinated by the Undersecretary and established within the Cabinet of the Presidency of the Regional Government under the responsibility of the Head of the Cabinet. In order to perform the assigned tasks, the control room relies on the responsible for the coordination of the Jobs and Climate Pact, the Agenda 2030, the Digital Agenda and International Relations, operating under the Cabinet Office of the Presidency of the Regional Council.

With reference to vulnerable groups, despite Emilia Romagna region wellbeing level seems to be higher than other Italian regional ones, some challenges are highlighted by the document: as mentioned by the pact document, for example, higher unemployment among women, a marked

³⁸ <https://bur.regione.emilia-romagna.it/area-bollettini/bollettini-in-lavorazione/maggio-periodico-parte-seconda-1a-quindicina.2021-05-11.1864155375/istituzione-cabina-di-regia-per-la-governance-e-il-controllo-strategico-della-programmazione-regionale/allegato>.

gender pay gap, youth unemployment, accompanied by high insecurity and low incomes, a stagnation in disposable income and wages, a failure to recover per capita levels pre-crisis 2008 in real terms and an increase in people at risk of poverty or social exclusion (percentage share of at risk of poverty or social exclusion (percentage share increased to 17.8 per cent in 2013, now reduced to 14.2 per cent). The labour and business system is experiencing an unprecedented polarisation between highly specialised and highly paid occupations and those with low qualifications, poorly protected and poorly paid, which increasingly become poor jobs, whether employed or self-employed, and which do not allow employed people to cross the poverty line. In this framework, it is very relevant the contribution to the pact provided to the **Permanent Regional Table for Gender Policies** as it made it possible to give centrality to women's issues within the regional strategies for the sustainable future.

The Permanent Regional Table for Gender Policies is a consultative body of the Region, with the task of providing a unified framework of the gender dimension within regional policies. It carries out activities of sharing, reflection and comparison on territorial positive actions and on the issues impacting on the gender dimension at a regional level; it coordinates the activities carried out directly by the Region and the ones carried out by the Local Authorities. It promotes the implementation an effective and coordinated equal opportunities policy in the whole regional territory.

The Table composition mirrors the pluralism of the civil society active in the policy area at regional level reflecting the aim of establishing a permanent social dialogue involving all stakeholders. Members by right are Councillors in charge of equal opportunities of Local Authorities, Representatives of ANCI (Associazione nazionale comuni italiani) and UPI (Unione Province italiane) and of the metropolitan city of Bologna; Regional representatives of confederal trade unions; Regional representatives of employers organisations, Representatives of the Coordination of regional Anti-Violence Centres; Representatives of regional Universities and research institutes; Regional Equality Advisors; Representatives of the regional Single Guarantee Committee for Equal Opportunities; The Regional Guarantor of persons subject to measures restricting personal freedom; The Regional Ombudsman for Children and Adolescents; The President of the regional Corecom (Comitato regionale per le comunicazioni); The Regional Ombudsman. With reference to civil society organisations, they are mainly women's associations such as women's documentation centres, migrant women's associations, disabled women's associations, women's health associations, and other associations that promote gender equality, equal opportunities, combating stereotypes and women's rights in the region. Regional representatives of recreational, sports and cultural associations even if not specifically active in the policy area are also involved. To join the table, it is requested to fill in the module available on the institutional [website](#).

In collaboration with the **Permanent Regional Table for Gender Policies** and the widespread associationism active on civil, social and gender rights issues, the plan intends to design innovative policies that promote: the quality and stability of women's work; the elimination of gender pay gaps; career paths training in all disciplines; female entrepreneurship; reconciliation between life and work times, also by strengthening the network of services; better distribution of the burden of care; interventions on city timetables and times; combating all forms of discrimination and violence against gender and homophobic violence.

Specifically, among others, regional strategic guidelines on the subject were outlined as:

- 1) Consolidation of the network of guidance services and contrast of gender stereotypes in training and professional choices, and promotion and enhancement of all vocational and technical training paths, also through the dissemination in schools of structural and permanent actions to bring young people closer to technical-scientific subjects.
- 2) Strengthening the Regional Employment Agency and the integrated public-private system for active policies for young people, women, and those who have lost or are at risk of losing their jobs.
- 3) Within the framework of collective bargaining, encouragement of flexibility and reconciliation tools - such as parental leave - that make it possible to meet both the needs of companies and those of workers.
- 4) in cooperation with the Permanent Regional Table for Gender Policies and the widespread associationism active on civil, social and gender rights issues, design of innovative policies that promote the quality and stability of women's work; the elimination of gender pay gaps; career paths; training in all disciplines; women's entrepreneurship; the reconciliation of work and life times also through a strengthening of the network of services; the better distribution of the burden of care; interventions on city schedules and times; the fight against all discrimination and gender and homo-bi-transphobic violence.
- 5) Creation and strengthening of new enterprises and new professional activities, particularly young people and women, with a special focus on innovative start-ups, defining a regional hub with the role of research, support and codification of innovative entrepreneurship projects, also making use of the regional observatory.

1.4 *Links between levels of governance*

The Pact for Work and Climate was signed, after a long process of elaboration and discussion, by the Emilia-Romagna Region and several, diversified economic and social actors, in line with the Mandate Programme, on 14 December 2020³⁹.

At regional/institutional level, policy implementation is facilitated by the *control room for the management and strategic control of the regional planning contribution*. This body is responsible for systematic and collaborative coordination with the regional directorates, monitoring, processing and providing the Presidency with "data" in the form of reliable, complete and relevant information, enabling it to strengthen its steering and decision-making role⁴⁰. In a context where several institutions, bodies and organisations work under a single strategy, the control room plays a key role for ensuring the effectiveness of the action.

With reference to the attention to vulnerability and inequalities, the pact aims to fight territorial, economic, social and gender inequalities and pursue fair and sustainable development in the green transition. Inter alia, actions mentioned by the pact, out of the specific field of the green transition, concern health care rights, proximity territorial services, community welfare, birth and

³⁹ Source <https://www.pandorarivista.it/articoli/che-cose-il-patto-per-il-lavoro-e-per-il-clima/>

⁴⁰ <https://bur.regione.emilia-romagna.it/area-bollettini/bollettini-in-lavorazione/maggio-periodico-parte-seconda-1a-quindicina.2021-05-11.1864155375/istituzione-cabina-di-regia-per-la-governance-e-il-controllo-strategico-della-programmazione-regionale/allegato>.

parenthood, work life balance and active labour policies. The interaction between the pact governance structure, regional offices, permanent table (such as the regional table for gender policy) and directorate allows the region to implement actions for equality in an integrated manner, through internal and external instruments and to combine them with those of the strategy. The contribution of the permanent regional table for gender policies is also relevant to avoid the top-down approach of the mentioned policies: thanks to its multilevel and multisectoral composition, the table seems to be the place of connection of governance levels and instances.

Given the integrated approach of regional policy, it is relevant to mention also the regional engagement for gender equality, shown by the Law for Equality and Against Gender Discrimination (Law No. 6/2014) and the gender balance release: the latter makes it possible to ensure common standards of equality by providing adequate information on the state of play. For the 2022 edition, the latest one, the analysis has been extended to regional policies on employment, work-life balance and gender-based violence, which were the focus of the last editions, as well as to other dimensions such as the gender dimension in the health sector, with a particular focus on prevention, made possible thanks to an exceptional collaboration with the Emilia-Romagna Regional Health Authorities. The gender equality balance sheet is an instrument for monitoring regional equal opportunities policies as part of the overall assessment of regional public policies. The tool analyses the differential impact of policies on the status of men and women in different areas of public intervention. It enables learning about gender equality by the region and other actors in the policy chain at all levels.

1.5 *Civil society*

The Climate and Labour Pact involved actors from the social, economic and third sectors. The Pact was the result of a continued tripartite social dialogue (Adapt, 2021) involving unions, institutions and employers' organisations, as well umbrella organisations of civil society organisations. In addition, it focused also on gender equality policies with a dedicated regional table: the Permanent Regional Table for Gender Policies. Participation in the table is open to interested women's associations such as women's documentation centres, migrant women's associations, disabled women's associations, women's health associations, and other associations that promote gender equality, equal opportunities, combating stereotypes and women's rights in the region, as well as regional representatives of recreational, sports and cultural associations. To join the table, it is requested to fill in the module available on the institutional [website](#).

It is very relevant that the members of this table have been involved in the pact planning and drafting phases and thus, with reference to the gender issues, we can assume that civil society organizations were actively involved.

1.6 *Barriers*

As mentioned above, the regional and local levels seem to be able to implement synergetic actions, thanks to the preparatory work done in the planning phase and the active institutional coordination of the **Control room for Governance and Strategic Control of Regional Planning** (Cabina di regia per la governance e il controllo strategico della programmazione regionale).

The governance in place is not always able to ensure full consistency between the Pact's objectives and regional actions, and that the divergent views of associations and stakeholders are not always

considered. In this, sense, after the floods of 2023, Legambiente left the pact membership complaining discrepancies between the pact objectives and the concrete actions carried out ([Bologna today](#), 2023). An example of the mentioned discrepancies is that the regional endorsement to develop a regassification facility in Ravenna, in agreement with national governments but in disagreement with environmental organisations.

In addition, quantitative targets and specific indicators (also about the equality) are not clearly expressed by the pact⁴¹. This can contribute to the misalignment above mentioned.

1.7 Factors for success

The region seems to adopt the European approach to policy making based on the bottom up and pluralistic method, as shown by the wide number and the variety of the organizations involved in the planning and monitoring/evaluation phase.

The governance structure, with the **Control room for Governance and Strategic Control of Regional Planning** (Cabina di regia per la governance e il controllo strategico della programmazione regionale), seems to be a success factor to reach the qualitative/quantitative objectives set out in the regional strategies (Mandate Programme, Jobs and Climate Pact, Regional Strategy for Sustainable Development - Agenda 2030, Emilia-Romagna Digital Agenda 2020-2025), thanks to its ability to provide reliable, timely and complete information to the Regional Presidency, thus strengthening the regional governance and decision-making capacity.

Another strength is the *all-government approach* which implies that the public authority, integrating interventions on human capital, innovation, territory welfare overcomes the consolidated organisational administrative bodies segregation. This approach enhances the balance between sustainability objectives, economic development, work qualification, welfare, democracy in which the collaboration between the administration and local actors.

The involvement of pact members in the monitoring/evaluation phase can be seen as a success factor, as the ongoing monitoring enables them to discuss emerging needs based on new criticalities and/or opportunities.

With specific reference to gender equality and issues concerning vulnerable groups, the contribution to the permanent regional table and bodies, such as the **Permanent regional table for gender policies**, which is open to different stakeholders and associations wishing to contribute to the regional action, is a key strength factor. The civil association involvement is facilitated by the regional widespread active associationism on civil, social and gender rights issues.

⁴¹ Data on main indicators are available at the following link <https://notizie.regione.emilia-romagna.it/comunicati/2023/marzo/patto-per-il-lavoro-e-per-il-clima-dall2019andamento-dell2019economia-al-contrasto-alle-disuguaglianze-in-regione-il-punto-sulle-cose-fatte-le-prossime-sfide-da-affrontare-e-le-leve-finanziarie-per-raggiungere-gli-obiettivi-condivisi-bonaccini-confermata/dati-sui-principali-indicatori-del-patto.pdf/@@download/file/Dati%20sui%20principali%20indicatori%20del%20Patto.pdf>.

1.8 Conclusion

The participatory and inclusive working method, with the broad involvement of organisations, economic and social actors and regional bodies and services, is a priority area for action.

Another priority area is the whole-of-government approach, which covers all areas of public intervention, such as human capital, innovation and territorial well-being. This approach makes it possible to overcome the fragmentation of the consolidated organisational administrative bodies and, at the same time, to promote coherent and integrated policies for vulnerable groups directly or indirectly linked to the twin transition area (e.g. social support for families, territorial care support, retraining and upskilling of workers, role of local employment services, reconciliation of work and family life).

In addition, the region seems to be advanced in terms of monitoring and evaluation, thanks to the diversified tools adopted, such as a regional integrated monitoring of policies, the monitoring/evaluation work of the members of the Pact. It is important to note that, depending on the results of the monitoring, the members of the Pact may modify or integrate the Pact. Finally, from the perspective of the regional integrated approach, we can highlight the appropriateness of drawing up a regional gender balance sheet.

Despite the virtuous governance approach, the objectives are not always in line with the regional actions actually implemented (see the Legambiente case) and there is no information on the effective involvement of civil society organisations.

Marche Region Quality Agrinido Model

2.1 Introduction

Social agriculture can be considered as a relevant action for social innovation, thanks to its capacity to integrate rural development with the social and care needs of disadvantaged groups (Genova, 2022). Since 2010, the Marche region has promoted farm diversification and services for the elderly and children, with initiatives such as the “Marche Region Quality Agrinido Model” (DGR 722/2011, coordinated by the Montessori Foundation. In 2016, the region also established the Active Longevity Laboratory Model (DGR no. 336/2016), in collaboration with the Istituto Nazionale ricovero e cura anziani IRCSS INRCA, which focuses on the care of the elderly in rural areas. The regional initiative, which anticipates national legislation, is relevant to the green and just transition, as it has a positive social impact on the territories, while helping to increase farmers' resilience and facilitate their transition to the new agricultural model designed by the new CAP.

The region's efforts over the years involve collaboration between agricultural services, private organizations (Montessori Foundation, INRCA), and agricultural entrepreneurs.

2.2 The aim

The regional action focuses on two strands of social agriculture: support for elderly people longevity and providing agri-nursery services according to the Montessori approach.

For the elderly people, the 2016 Active Longevity Laboratory Model (DGR no. 336) was created in partnership with Istituto nazionale ricovero e cura anziani IRCSS INRCA, envisioning farms as spaces providing social and community support through adaptable activities in a non-medicalized setting.

For children, the Agrinido model is the result of the partnership between the Marche Region and the Municipality of Chiaravalle, founder and promoter of the Chiaravalle-Montessori Foundation⁴². The format was developed by a technical-scientific committee composed of professionals with diversified skills and knowledge and included regional bodies and organisations active in the agricultural sector and experts in psycho-pedagogy and social vulnerability⁴³.

Under this framework, over the years the region dedicated resources of the rural development plan 2007-2014 and 2014-2020 and 2023-2027 (RDP). As there are not yet updated information on the most recent RDP, we will consider the project developed under the RDP 2014-2020.

2.3 *Roles and responsibilities*

A central role in the development of the social innovation policy is played by the regional head of the Department of Agriculture, with the support of highly specialised organisations, the Chiaravalle-Montessori Foundation for the agri-nursery and INRCA for the longevity policy; they were involved in the policy-making process from the design phase (Genova, 2018). When the action was launched, the farms registered on the regional list of social agricultural operators (established by DGR 345 of 18/04/2016) were only involved in the implementation phase: they supervised the implementation of the agri-nursery and active longevity activities, in exchange for a small financial contribution from the region.

As mentioned above, the action launched in 2010 has evolved over the years. During the 2014-2020 period, the governance structure changed. The regional level was responsible for planning, funding and monitoring the actions funded by the RDP. In detail, the region managed the RDP with these bodies:

- the Managing Authority in person of the manager of the Environment and Agriculture Service of the Marche Region,
- the Paying Agency is the Agenzia per le erogazioni in agricoltura (AGEA), which is also certifying body for the region.

Measure No. 16.01 of RDP included two actions:

- 1) Setting up operational groups - consisting of an initial phase in which at least two project partners (agricultural and agri-food enterprises, research organisations, other organisations) submit an action plan. This phase was aimed at 1) supporting the companies' aggregation in the legal forms envisaged by the Rural Development Plan; 2)

⁴² They signed a cooperation agreement in accordance with DGR 1107 of 12 July 2010.

⁴³ In detail the committee was composed of two representatives of professional agricultural organisations, the director of the Social Services and Agriculture Service of the Marche Region, the Director of the Montessori Foundation, as well as experts in psycho-pedagogy, pedagogy and problems of social fragility.

- creating connection with the knowledge system; 3) prepare an action plan that may lead to the subsequent presentation of a project in phase 2.
- 2) Funding operational groups - in this phase, potential OGs presented their innovative projects to be funded by the single measure 16.1 or together with other additional PDP measures.

In this framework, the operational group Agricoltura sociale Marche was responsible for the development of the operational plan and its implementation, thanks to the funding of the RDP, measure No. 16.01.

The group seems to be adequately composed to foster the networking development: lead partner role was filled by the farm La Fattoria del Borgo. The financial and technical/scientific coordination is performed by INRCA (Istituto di Ricovero e Cura a carattere Scientifico), the dissemination and transfer of results and knowledge to the outside world by AMAP (Agency for Innovation in the Agri-food and Fisheries 'Marche Agricoltura Pesca') and AiCare (Agenzia Italiana per la Campagna e l'Agricoltura Responsabile e Etica-Roma - Italian Agency for the Countryside and Responsible and Ethical Agriculture). Chiaravalle-Montessori Foundation is in charge to provide advice for the activation of the Montessori courses and the support for technical/administrative coordination.

Five other farms were in charge to implement the action (Azienda Agricola Fattorini Paola - Fattoria Didattica L'asino che ride; La Quercia della Memoria - Società Agricola - Agrinido della Natura; Azienda Agrituristiche La Castelletta; Azienda Agricola Lupini - Agrinido l'Orto dei Pulcini; Azienda Agricola Delmanowicz Monika - Agrinido La Fornace degli Gnomi; Azienda Agricola Bagalini Roberto - Agrinido Arca di Noè), the cooperative society H.O.R.T.; INRCA – Istituto Nazionale di Ricovero e Cura per Anziani; Chiaravalle Montessori foundation, and the AMAP - Agenzia per l'innovazione nel settore agroalimentare e della pesca "Marche Agricoltura Pesca"; and AiCare (Agenzia Italiana per la Campagna e l'Agricoltura Responsabile ed Etica).

The association Ai Care, specialized on social innovation, was in charge of the construction of a policy making analysis pathway and, in this field, has launched several activities (including workshops, interviews, dissemination events) to favour the creation of new relations in the territories between multiple actors and the definition of strategies to interconnect public actors (different levels of government) and private actors (companies, civil society, economic and representative organisations) in multi-level governance.

In the domain of the Ai care action, several tools have been used to expand the social farming network: 1) preliminary meetings to present the project and the related criticality to be resolved; 2) dissemination seminars with farms; meetings between politicians and other actors; 3) peer-to-peer comparisons and debates; 4) presenting the solutions identified to overcome obstacles in starting up or carrying out activities; 5) presentation of new opportunities, 6) provision of sustainability report and set up of a community the new network participants may have direct expertise and direct experience; 6) - meetings between heterogeneous groups aimed at getting to know the community service perspective.

The analysis carried out by Ai care identified the suitable key actors in a territorial social innovation network: the broadening action seems to have been aimed at administrators, agricultural professional organisations, social territorial ambits, labour inspectorates and related offices, agro-

food companies, social and trade representatives, cooperatives and voluntary associations. And where possible or where there is active research and study, universities and research bodies.

Organisations representing civil society or committed to gender equality and gender equality were not mentioned nor included in the possible new allies list nor the funded operational group Agricoltura sociale Marche.

With reference to the public capacity to address vulnerable groups, according to the monitoring report shared by INRCA⁴⁴, the operational group Agricoltura sociale Marche generated a good impact on vulnerable groups (elderly people and families with children between 1-6 years of age). In terms of governance, the involvement of highly specialized organizations, such as INRCA and Fondazione Montessori, which has started to work on these topics since the launching of the initiatives, seems to be relevant and helpful to reach the action objectives.

The action produced a positive impact also on farmers, which, as mentioned, were involved in the group implementing activities. Very relevant are the results gained by the operational project about farmers involved in the activities: with reference to the farms multifunctionality, the project has focused on the promotion of a stable partnership and a network by the participating farms, the acquisition of new skills through the exchange of knowledge with actors from other sectors - social, agricultural, third sector, local authorities). It allowed to the farms to be more likely to confrontation and dialogue, more able to find complex solutions.

2.4 *Links between levels of governance*

About the governance structure and the links between all actors involved in the social agriculture project the governance structure is not linear, given the changes occurred over the years. As mentioned, the cooperation between the region and Fondazione Montessori and INRCA seem to have enabled all the phases of the policy as well as shared equality standards and common objectives⁴⁵. The links between the regional agriculture service and the two organisations are sketched below followed by the description of the interactions among the governance levels.

Since 2010 the Chiaravalle Montessori Foundation started a cooperation to create a network of agri-nido (agri-nursery) with the Marche region. Since 2014, the cooperation was gradually renewed each year through a special agreement. For this activity the Foundation, provided significant and fundamental training, coordination and advisory activities to support the various structures of the agri-nido Network. In this framework, highly professional pedagogues were engaged in the project. The FCM followed the activities from the planning phase to the start of the service and then continued as Pedagogical Coordination and followed the training of the educational groups with periodic meetings on different topics.

Following up on the path started by the regional law on multifunctionality and farm diversification (Regional Law no. 21 of 14 November 2011), in 2012 the Marche Region's, Agriculture

⁴⁴ For further information and details please consult the report at the link https://www.agricolturasocialemarche.it/index.php/pubblicazioni/addon/addon_26?addon_task=attachment.download&addon_data_id=35_6dbcb7870c06f6aa166cbfb5d8401bc7&article_id=83.

⁴⁵ For detailed information please refer to the document Il percorso di Agricoltura sociale Marche, URL <https://www.agricolturasocialemarche.it/index.php/notizie/45-manuale-esperienze-di-policy-making-in-agricoltura-sociale-il-percorso-del-progetto-agricoltura-sociale-marche>.

Department, approved a collaboration agreement with the IRCSS INRCA (Institute for Research and care with a scientific character on ageing). With this agreement an experimentation in rural areas of longevity services was started. The INRCA, as an expert partner in the theme of ageing, collaborated with the regional technical-scientific committee in the definition of the characteristics and needs of the elderly for whom these services were intended.

This collaboration led in 2014 to the launch of the call for ideas with regional resources 'Active Longevity in Rural Areas', aimed at bringing out, consolidating and facilitating the development of initiatives to support active longevity in rural areas. The project stemmed from the region's objective to test innovative initiatives and services for the frail elderly population, to prevent social isolation and non-self-sufficiency in peri-urban and disadvantaged areas of the territory lacking in services for the elderly. The INRCA was also entrusted with the task of coordinating and monitoring the experimentation with the aim of assessing the quality of the projects launched, supporting their implementation and evaluating their impact on the end users (the elderly over 65), in terms of improvement of health conditions, maintenance and/or recovery of mental abilities improvement of the sense of inclusion and learning of new knowledge.

As mentioned, since 2019, INRCA and Fondazione Montessori are members of the operational group "Agricoltura sociale Marche", within the framework of the European Partnerships for Innovation (EIP-AGRI). The group work is aimed at introducing and testing in the farm context innovative social farming practices to foster the multifunctionality of the farm and promote entrepreneurial opportunities for enterprises in the sector. The operational group is managed by private organisations, and it is implemented by farms. The regional contribution in this governance structure is limited to the activities linked to the funding, managing and monitoring the RDP 2014-2020 European and regional resources.

2.5 *Civil society*

With regard to the consultation process, as mentioned above, it is necessary to distinguish between the initial policies implemented by the Region and those implemented by the Operational Group Agricoltura Sociale Marche. In the first case, the consultation process seems to be top-down, as the regional initiative was favoured and supported by Fondazione Montessori and INRCA, which have provided their qualified know-how and experience to the region.

In the second case, the partnership between the private organisations and the farmers seems to be inclusive and structured to consider the farmers' instances in action from the drafting phase of the plan.

In both cases, civil society seems to be excluded from policy making processes. It is possible that civil society is involved in the dissemination activities of some operational groups, but we cannot report cases of structured involvement due to lack of information.

2.6 *Barriers*

In line with Genova's (2018) governance analysis, local, social and health services, as well as third-sector organisations and trade union representatives are not included in the governance structure.

In our opinion, there are also other barriers: the governance structure seems to reflect the lack of a long-term strategy, as it changes over the years, involving new stakeholders, from time to time, in a non-linear way. In this sense, we remark the absence of effective involvement of civil society organisations in the governance structure and thus the scarce representation of the community needs (out of the Ai care activities, addressed to user services).

At the level of regional policy strategy, moreover, the fragmentation of the welfare sector runs the risk of stifling the innovation potential found. It is precisely on the capacity of the different welfare areas to integrate that the development of social agriculture seems to be at stake in its great potential for innovation (Genova, 2018).

In addition, the desk analysis did not reveal any results regarding gender equality policies adopted or the involvement of organizations representing gender equality issues in the governance structure. This is despite the Common Agricultural Policy (CAP) 2023-2027, which identifies gender mainstreaming as a fundamental principle of the Union and urges Member States to prioritize the participation and promotion of women's roles in agriculture.

2.7 *Factors for success*

The governance experience of the Marche Region in designing and implementing social farming initiatives highlights the capacity of the Regional Agricultural Service to promote innovative actions by involving actors with extensive expertise in the themes of the proposed experiments. In doing so, the Regional Agricultural Service demonstrates its ability to detect social needs. It has been able to address these needs by adopting the perspective of a social innovator capable of building networks with private actors (companies), foundations, and other public entities with significant expertise in the sector targeted by the policy.

2.8 *Conclusion*

The practice examined demonstrates that public-private cooperation and, most importantly, the involvement of the policy's target group (e.g., farmers) can enhance the policy's effectiveness. However, the fragmentation of the regional strategy and the resulting lack of integration of other social services into regional policymaking may limit the policy's potential positive impact, particularly concerning the vulnerable groups it seeks to support.

Some efforts have already been made in the Marche Region to integrate social innovation policies with other sectors of public action, creating opportunities for broader stakeholder engagement. For instance, references to social agricultural policy have been incorporated into the Pesaro Social Area Plan, a tool connected with the regional planning, which develops specific objectives and innovation activities to answer, at local level, to the citizen's needs.

Nonetheless, the network can be further expanded to include civil society organizations, trade unions, and entities promoting equality and gender equality. This broader inclusion would support the structural implementation of the new CAP strategy and further improve its alignment with its overarching objectives.

References

Adapt (2021). “I patti per il lavoro: il caso della regione Emilia-Romagna” in VII Rapporto ADAPT sulla contrattazione collettiva in Italia: parte II capitolo 2. 2020.

Butera F. (2022). Il PNRR per rigenerare le organizzazioni italiane nella transizione ecologica e digitale.

Gagliardi C., Piccinini F., Contiero D., Fabbietti P. (2023). *Progetto Agricoltura Sociale Marche. Report finale INRCA*

Genova, A. (2018). *L'innovazione nel welfare regionale: la governance dell'agricoltura sociale nel caso studio delle Marche*. Argomenti 11.

Genova A., Nothdurfter U., Viganò F. (2020). Agricoltura sociale tra politiche di sviluppo rurale e politiche di welfare, Bolzano University press.

Martellini C. and Durasanti F. (Agenzia Italiana per la Campagna e l'Agricoltura Responsabile ed Etica) (2022), *Esperienze di Policy Making in agricoltura sociale. Il percorso del progetto Agricoltura Sociale Marche*

Portugal

Portuguese Twin Transition Strategy

1.1 Introduction

Effective governance is essential for navigating complex transitions, such as the shifts toward sustainability and digitalization, ensuring that policies are well-coordinated, inclusive, and capable of addressing challenges across multiple sectors of the society. The Portuguese strategy for the twin transition, as outlined in the National Energy and Climate Plan 2030 (PNEC 2030), demonstrates its commitment to carbon neutrality and renewable energy, underlying the commitment to follow the Paris Agreement's goal of limiting global warming to 1.5°C. The plan includes reducing greenhouse gas emissions by 43% in European Emissions Trading System (ETS) sectors and 30% in non-ETS sectors by 2030, with a target of 47% renewable energy use in final consumption. Measures are based on several key decision, such as phasing out coal, adopting hydrogen-based energy systems, and enhancing energy efficiency with a 25% reduction target. The strategy emphasizes decentralized energy production to reduce grid losses, hybrid energy systems for efficiency, and interconnections to integrate the European energy market. In the transport sector, it is aimed to take measures such as shared mobility, incentive adoption of electric vehicles, and railroad transportation.

Concerning the just transition, Portugal integrates financial instruments like EU funds and state budgets to mitigate social and economic impacts by promoting job creation and business

opportunities. A monitoring system led by the *Comissão para a Ação Climática* (CAC)⁴⁶ will ensure progress evaluation and a sustainable governance of the twin transition.

1.2 The aim

The aim of the Portuguese twin transition policy is to achieve carbon neutrality by 2050 while promoting a just and equitable transition. Those propositions are outlined in the National Energy and Climate Plan 2030 (PNEC 2030) and the Roadmap for Carbon Neutrality 2050 (RNC 2050). The approach consists in integrate climate and energy policies to drive economic growth and job creation while reducing greenhouse gas (GHG) emissions. The primary targets include a 45%-55% reduction in GHG emissions by 2030 compared to 2005 levels, a 17% reduction in non-ETS sectors, achieving a 47% share of renewable energy in gross final energy consumption, a 25% reduction in energy consumption through improved efficiency, and enhancing electricity grid interconnections to 15% by 2030. The policy encompasses measures such as phasing out coal, promoting renewable gases like hydrogen, prioritizing sustainable mobility and energy efficiency, supporting circular economical practices, investing in low-carbon technology research, and ensuring a fair transition for the benefit of citizens' life.

To finance this transition, Portugal relies on public and private investments, with over 60% of the National Investment Plan (PNI) 2030 directed toward decarbonization. European funds, such as the Multiannual Financial Framework (MFF) 2021-2027 and the LIFE program, are allocated for climate action, along with fiscal measures like carbon taxes are used to fund decarbonization and just transition initiatives. The impact of these efforts is expected to be transformative, driving economic growth through green jobs and industries, reducing air pollution, enhancing energy security, and improving resilience to climate change.

Governance is structured at multiple levels to ensure successful implementation and monitoring. At the political level, the Commission for Climate Action (CAC) oversees policy coordination, while the Monitoring and Projection System (SPeM) handles technical implementation and reporting. Key public authorities include the Ministry of Environment and Climate Action (MAAC)⁴⁷, the Directorate-General for Energy and Geology (DGEG)⁴⁸, the Portuguese Environment Agency (APA)⁴⁹, and the Energy Agency (ADENE)⁵⁰. This approach balances the environmental goals with economic and social progress, ensuring broad stakeholder participation and inclusive governance.

1.3 Roles and responsibilities

The Portuguese approach to the twin transition is based on a multi-layered governance structure ensuring that policy is carefully planned, drafted, implemented, and monitored. This structure involves various public authorities at different levels and promotes the participation of civil society and other organizations.

⁴⁶ <https://apambiente.pt/clima/comissao-para-acao-climatica>

⁴⁷ <https://www.portugal.gov.pt/pt/gc23/area-de-governo/ambiente-e-acao-climatica>

⁴⁸ <https://www.dgeg.gov.pt/>

⁴⁹ <https://apambiente.pt/>

⁵⁰ <https://www.adene.pt/>

Considering the political level, the Commission for Climate Action (CAC), formerly the Interministerial Commission for Air, Climate Change and Circular Economy (CIAAC), plays a crucial role. The CAC is responsible for policy coordination, promotion, supervision, and monitoring of the National Energy and Climate Plan (PNEC). It provides political guidance on climate change and air policies and ensures the integration of these policies across different sectors. The commission comprises representatives from various government departments, including energy, spatial planning, finance, agriculture, and transport, as well as regional governments. This broad representation is meant to ensure the coherence of the different components of the national plan. The CAC's responsibilities also include validating responsibilities and setting implementation deadlines and ensuring alignment with European and international commitments.

The Monitoring and Projection System (SPeM) is key for ensuring the implementation of the policies. This system is responsible for monitoring progress, tracking results, and reporting on the effectiveness of climate policies. It also contributes to the development of emission projections and ensures that the country meets its national and international obligations. The SPeM system works in collaboration with the CAC and provides the necessary data and analysis to inform policy decisions.

Specific public authorities such as the Ministry of Environment and Climate Action (MAAC), the Directorate-General for Energy and Geology (DGEG), the Portuguese Environment Agency (APA), and the Energy Agency (ADENE) also have distinct roles. The MAAC is a key entity, given its responsibility for environmental and climate policy, including the implementation of the National Roadmap for Carbon Neutrality 2050 (RNC 2050). The DGEG is responsible for the energy sector, including regulating energy markets and promoting energy efficiency and renewable energy. The APA focuses on environmental protection and sustainable development and the implementation of the national strategy for ecological public procurement. ADENE, the Energy Agency, supports the promotion of energy efficiency and the development of renewable energy technologies. These entities work both independently and in coordination to draft, implement and monitor specific policies related to their areas of expertise. The National government also sets the regulatory and fiscal framework for the energy transition. The regulatory authorities, such as the Energy Services Regulatory Authority (ERSE)⁵¹, oversee and enforce compliance with energy sector regulations. The Directorate-General for Natural Resources, Safety and Maritime Services also has responsibilities for marine spatial planning and for the maritime aspects of renewable energy projects.

Civil society organizations, social partners, and other bodies play a role throughout the policy cycle. These organizations contribute to the planning of the policy by participating in public consultations and providing input on the needs and priorities of different segments of society. They also contribute to drafting the policy, as the Council of Ministers Resolution specifies the need for participation of the various forces in the search for a consensus on energy policies. During implementation, they contribute through the operation of citizen energy communities, the promotion of best practices, and the engagement of individuals in behavioural change and sustainable consumption. In the monitoring and evaluation phase, these organizations may play a watchdog role and ensure that the policies are delivering the expected results and that they are

⁵¹ <https://www.erse.pt/inicio/>

benefiting vulnerable populations and other groups fairly. The creation of platforms for dialogue, debate, and the exchange of experiences is a means of engaging these actors in the transition.

The commitment to gender equality and support for vulnerable groups is integrated into the policy framework, although specific details on the mechanisms for achieving this are not always fully detailed in the sources. Policies are meant to ensure a just transition and address energy poverty through measures such as building rehabilitation, promotion of renewable energy and educational campaigns. There is also the understanding that the energy transition must promote social cohesion and protect vulnerable populations by providing access to affordable energy services, and that no one is left behind. Civil society organizations also contribute through awareness-raising and ensuring these groups are not overlooked by the policies. Furthermore, the regulatory authorities are tasked with monitoring the implementation of measures that protect vulnerable customers. The government also addresses energy poverty by including measures within the general social security system.

In summary, successful governance of Portugal's twin transition will be guided by a clear division of responsibilities across government bodies, and active collaboration with civil society and other stakeholders, all working toward a shared goal of a sustainable, equitable and carbon-neutral future.

1.4 Links between levels of governance

The successful implementation of Portugal's twin transition policy relies on strong links across different levels of governance, ensuring that planning, drafting, implementation, monitoring and evaluation are coherent and operative. These links are also crucial for confirming that shared equality standards and objectives, particularly concerning gender equality and vulnerable groups, are integrated throughout the policy chain. A tangible policy planning strategy benefits from the engagement of multiple levels of governance. At the highest level, is the Commission for Climate Action (CAC) that sets the overall direction, ensuring that climate and energy policies are aligned with national and international goals. The CAC works closely with various ministries, establishing a link between national level policy and the specific responsibilities of sectorial ministries. This is complemented by the technical expertise of the Monitoring and Projection System (SPeM), which provides data and analysis to support informed decision making at the policy planning stage. The public authorities, such as the Ministry of Environment and Climate Action (MAAC) and the Directorate-General for Energy and Geology (DGEG), are connected through the CAC, and contribute their respective areas of expertise. These connections ensure that all relevant factors are considered during the planning process. Local and regional authorities are also involved through representation in the CAC and by the results achieved with the public consultation processes.

Policy drafting is also a collaborative endeavour, as it results from different contributions from government to the representatives of different sectors of society. The CAC's role in providing political guidance ensures that draft policies are consistent with national objectives. Public authorities, such as the MAAC, the DGEG, the APA, and ADENE, take the lead in drafting specific policy measures, drawing on their expertise and experience, and with oversight from the CAC. The involvement of various stakeholders in consultations provides valuable input from diverse perspectives, including civil society organizations, social partners, and the public. This consultation

process is intended to ensure that policies are well-informed, feasible and address the needs of all sectors of society.

Policy implementation is facilitated by the links between different levels of governance. The national government sets the regulatory and fiscal framework, creating the necessary environment for the transition. Local and regional authorities are responsible for implementing policies at the local level, working within the national framework but also responding to local contexts. Public authorities like the DGEG and ADENE provide technical support and guidance to ensure operative and consistent implementation of the transition strategy at all levels. The regulatory authorities, such as the Energy Services Regulatory Authority (ERSE), play a critical role in enforcing regulations and promoting compliance. This is all supported by the SPeM's monitoring system, which tracks the implementation of policies and provides feedback to the different stakeholders.

Policy monitoring and evaluation is designed to assess the success of the implementation and to suggest all the necessary adjustments. The SPeM continuously monitors the progress of the twin transition, tracking indicators and identifying areas where policies are not achieving their intended outcomes. The CAC uses the information provided by the SPeM to make strategic adjustments, ensuring accountability. The analysis of such data also helps different levels of governance make informed decisions about the course of the twin transition. Public authorities use the data from the SPeM to assess the effectiveness of specific policy measures and to make further revisions. Civil society organizations can also play an important watchdog role in this process.

The commitment to shared equality standards and objectives is ensured through several means. The CAC, as the policy coordinator, has the responsibility to integrate objectives for social inclusion, gender equality, and support for vulnerable groups in the national strategy. Public authorities are meant to follow this guidance in their policy drafting and implementation activities. The consultation processes ensure civil society and social partners can advocate for equality issues to be incorporated at every stage. The government's commitment to a just transition means that policies are intended to address social issues such as energy poverty and the employment impacts of the energy transition and ensure that no one is left behind.

To ensure a common capacity among public authorities, there is a concerted effort to promote shared understanding and approaches. The involvement of all levels of government in policymaking and the technical implementation process means that they can each learn from the others and develop their skills and experience. Technical training and capacity-building efforts are focused on addressing both climate and social issues. The exchange of data and best practices also ensures that all levels of government can learn from each other and improve their work. The SPeM monitoring system also helps in identifying capacity gaps at different levels of governance and address those gaps with more training and resources.

Learning on equality is enabled through the continuous monitoring and evaluation processes, in which civil society organizations and social partners may also participate. The SPeM's tracking data includes disaggregated data, where possible, which can reveal which segments of the population are not benefitting from the policy as expected. Public authorities at all levels are expected to use the monitoring data to learn from their experience, adjust policies, and improve their effectiveness, especially in addressing equality issues. The ongoing dialogue and consultation processes ensure that lessons learned are shared and incorporated in policy adjustments. This

ensures a dynamic and adaptive approach to the transition, with a focus on continuous improvement.

In summary, the links across different levels of governance in Portugal's twin transition strategy facilitate a collaborative, informed, and adaptive approach to policymaking and implementation. This collaborative model also ensures that shared equality standards and objectives, particularly those related to gender equality and support for vulnerable groups, are integrated throughout the policy chain, with continuous monitoring and learning from experience to make sure that the goals of the twin transition are achieved.

1.5 *Civil society*

The consultation process for Portugal's energy and climate policies involved several levels of governance, including national, regional, and local entities. It included public sessions, thematic workshops, and a formal public consultation period. The process sought to gather input from diverse stakeholders, including civil society organizations (CSOs) and social partners. These consultations provided opportunity for CSOs to describe their concerns and suggest policy directions, there's no explicit evidence in the sources that they had in setting the policy focus on gender equality or support for vulnerable groups. However, the intention of the consultation process was to ensure policies are well-informed and address the needs of all sectors of society, including those related to fair transition and mitigate energy poverty. The process also aimed to capture diverse perspectives, suggesting that issues of gender and vulnerable groups could be raised. The multi-level dialogues were designed to include civil society, mostly for discussion of energy and climate policies. As an additional perspective, those discussions are designed to be inclusive and to promote the involvement of stakeholders' policy makers and society.

1.6 *Barriers*

Several barriers may impede the successful and transformative implementation of twin transition agenda while attending to gender equality and support for vulnerable groups policies within Portugal's energy and climate framework. One significant barrier is the complexity of the administrative and regulatory landscape. This complexity can hinder the smooth implementation of policies, including those related to equality and vulnerable groups, and can cause delays. Overlapping responsibilities and a lack of clarity about the roles of different authorities can also reduce the effectiveness of policy implementation. Another barrier is the potential for inconsistent implementation across different regions and municipalities. While national policies provide a framework, their specific application can vary based on local capacity and resources. This is especially evident in the provision of local services for consumers, as electronic platforms may not reach all consumers, particularly those who are more vulnerable. A lack of disaggregated and localized data is another hurdle. The sources highlight the importance of data for understanding issues like energy poverty, but without detailed data, public authorities may struggle to identify vulnerable groups and tailor policies to specific needs. Furthermore, the efficacy of public consultation and engagement is a potential barrier. Although the consultation process is designed to gather input from diverse stakeholders, including civil society organizations (CSOs), the sources do not guarantee that their voices will be influential in driving policy on gender equality or support for vulnerable groups. There may be a gap between the intent of inclusion and the actual influence of CSOs in setting priorities. The sources also suggest a lack of specific

mechanisms to ensure that equality considerations are integrated into policy planning. While the government is committed to a just transition, there is no explicit information on how equality issues are specifically mainstreamed or prioritized. Finally, a lack of resources at the local and regional level may impede policy implementation. Although there is a focus on supporting local strategies, the sources do not guarantee the availability of the necessary funding to do this. In summary, while Portugal has established a governance framework, these barriers, including complexity, inconsistent implementation, data gaps, engagement limitations, a lack of specific equality mechanisms and a possible lack of resources, must be addressed to ensure a fair and just energy and climate policy.

1.7 Factors for success

The existence of effective links between public authorities across different levels of governance for gender equality and vulnerable group support require several key factors. The first would be multi-level dialogue that includes diverse stakeholders is essential, creating platforms for discussion at national and local levels, along with regional cooperation, particularly on cross-border issues. A clear governance structure, such as the Commission for Climate Action (CAC) in Portugal, helps to translate policy into action by defining responsibilities and monitoring implementation. The involvement of stakeholders including social partners, civil society and local authorities, is essential for comprehensive policy development and implementation, and there is a need to engage with them in a way that promotes understanding of energy transition.

Furthermore, specific attention should be given to vulnerable customers, with measures in place to ensure necessary energy supplies and support for energy efficiency. A long-term strategy to combat energy poverty is necessary, including clear ways to measure and monitor this issue. Regular monitoring and assessment of progress, effectiveness of support schemes and impacts on various consumer groups are also crucial. Transparency and accessibility of information, along with clear public consultation processes, are vital, as is the establishment of secure systems for data management with strong cybersecurity and data protection standards. Finally, the removal of regulatory and administrative barriers that hinder the development of renewable energy and energy communities, alongside streamlined procedures for authorization, certification and licensing, would also improve outcomes.

1.8 Conclusion

To enhance links between different levels of governance that promotes gender equality and vulnerable group support policies, priority action areas include: establishing permanent multi-level platforms for dialogue that involve diverse stakeholders; ensuring that these platforms address both national and local concerns, particularly with respect to cross-border issues; creating clear governance structures with defined responsibilities and monitoring systems, such as the Commission for Climate Action; implementing a long-term strategy to combat energy poverty, including mechanisms to measure and monitor its impact on vulnerable groups; ensuring transparency and accessibility of information related to policies, procedures, and support mechanisms; and identifying and removing regulatory barriers that hinder the development of renewable energy and energy communities, in order to ensure fair access to resources and services. These actions would be strengthened by streamlined procedures for authorization, certification, and licensing

The Twin Transition in Housing and Construction

2.1 Introduction

The pathway of governance for a just twin transition in housing requires coordinated policies across all government levels, addressing local needs and engaging citizens—especially vulnerable groups. Adequate resources, data, and monitoring ensure accountability and compliance with national and EU regulations. Priorities include building rehabilitation, sustainable construction, waste reduction, and secondary material use. Energy efficiency is key by means of better insulation, electrification, and renewables like heat pumps and solar thermal. Integrated urban planning helps combat sprawl and create pedestrian-friendly spaces. Combating energy poverty is crucial for health, well-being, and inclusion. This means prioritizing vulnerable populations, focusing on local building upgrades, and adopting renewable energy. Monitoring energy consumption (e.g., via SGCEE and SPeM) informs policy and highlights the importance of efficient buildings.

2.2 The aim

The twin transition in housing and construction aims for a carbon-neutral economy by 2050, with major emission cuts by 2030. It focuses on integrating renewable energy, improving building efficiency, and promoting sustainable urban development while ensuring fairness for vulnerable groups. Key actions include rehabilitating buildings to extend their lifespan, reduce waste, and enhance energy efficiency through better insulation and nearly zero-energy standards, alongside electrification using heat pumps, solar thermal, and renewable gases like hydrogen. Urban planning helps curb sprawl, create pedestrian-friendly areas, and strengthen communities. Fairness is addressed through job creation, training, and measures to combat energy poverty. Progress is assessed through regular monitoring of emissions, energy efficiency, renewables, and social impacts, with funding from public and private investment, as well as national and European financing sources.

2.3 Roles and responsibilities

In good governance, key public authorities have distinct roles across the policy chain. During the planning phase, authorities at all levels should engage in multi-level dialogues, incorporating diverse stakeholder views including those of vulnerable groups. A central body, such as the Commission for Climate Action (CAC), is responsible for drafting policy, integrating various objectives, such as climate, energy, gender equality and support for vulnerable groups, aligning policies with national and international commitments. Implementation involves national, regional, and local authorities, putting policies into action, efficiently using resources, and delivering services. For instance, the ECO.AP program promotes energy efficiency in the public sector. Monitoring and evaluation require a structured approach, with bodies such as the CAC overseeing implementation and efficacy, checking compliance and impact. Continuous evaluation, data management, and accountability are essential components of this process.

Civil society, social partners, and other organizations play crucial roles across the policy chain. During the planning phase, these groups contribute by participating in multi-level dialogues and consultations, providing diverse perspectives, and representing the needs of various communities, including vulnerable groups. Their involvement helps to ensure that policies are well-informed

and responsive to societal needs. While the drafting of policy is primarily the responsibility of public authorities, civil society and social partners can offer input and feedback, ensuring that policies are fair and inclusive. During the implementation phase, these organizations can act as key partners, assisting in the delivery of programs and services at the local level, and ensuring that policies are implemented successfully and equitably. Civil society and other organizations can help to create awareness and ensure community engagement, as well as assisting those facing energy poverty. In the monitoring and evaluation phase, these groups play a vital role in assessing the impact of policies, providing independent oversight, and holding authorities accountable. They can contribute by reporting on the effectiveness of policies and identifying areas for improvement, acting as watchdogs and ensuring that policies achieve their intended outcomes. Their active involvement ensures transparency, responsiveness, and accountability throughout the policy process.

Public authorities at all levels demonstrate commitment to gender equality and support for vulnerable groups through their actions and policies, while civil society and other organizations contribute to enhancing this commitment and capacity. Public authorities show commitment through inclusive planning, integrating gender equality and support for vulnerable groups into sector policies, putting policies into action, ensuring efficient resource allocation, actual service delivery and safeguards for vulnerable customers and universal service. They should ensure that public service obligations are clear, transparent, non-discriminatory and verifiable. They also demonstrate commitment by monitoring the impact of policies on different groups and collecting data to provide targeted support to those affected.

Public authorities show capacity by creating instruments for change promoting actions, allocating necessary resources, establishing mechanisms for data management, data protection and non-discriminatory access to data. They show capacity by monitoring transparency and success of market opening and streamlined authorization procedures for small-scale generation. Civil society and other organizations contribute to the commitment and capacity of public authorities by providing input and diverse perspectives, acting as partners in delivering programs, creating awareness and community engagement with policies and services, and providing independent oversight. Their active involvement ensures transparency and responsiveness throughout the policy process.

2.4 Links between levels of governance

A multi-level approach, incorporating dialogue and consultation at all levels of governance, ensures that diverse perspectives, including those of vulnerable groups, are considered throughout the policy chain. The commitment to shared equality standards and objectives is further embedded through the work of central bodies, such as the Commission for Climate Action (CAC), which integrates these standards into policy.

The establishment of central coordinating bodies, such as the CAC, ensures that all levels of public authorities are aligned with common goals related to equality, and that policies are designed with these goals in mind. The commitment to promoting equality is further reinforced by the requirement for multi-level dialogues, involving various stakeholders including vulnerable groups, at all stages of the policy chain, thus building the capacity for all levels of government to respond effectually. A key link across different levels of governance that enables

learning on equality is the requirement for multi-level dialogues that include diverse stakeholders, ensuring that insights from various perspectives inform all policy steps. Furthermore, the monitoring and evaluation processes, conducted at different levels, provide essential feedback and learning, allowing public authorities to adapt and improve their practices related to equality throughout the policy chain.

2.5 *Civil society*

Portugal's National Energy and Climate Plan (PNEC) consultation process involved multiple levels of governance to include diverse stakeholders. At the national level, this included institutional and civic participation as well as involvement of the national Parliament. Local and regional authorities also participated, including specific meetings with delegations from the Autonomous Regions.

Thematic workshops were organized to address specific topics such as energy efficiency and poverty. A public consultation process received 59 contributions from diverse sectors. Additionally, cross-border consultations with other member states were conducted. A multi-level energy dialogue was held with local authorities, civil society organizations, businesses, investors, and other relevant stakeholders.

While the sources indicate that civil society organizations were encouraged to participate in this process, they do not explicitly detail the extent to which these groups drove the policy focus on gender equality or support for vulnerable groups¹. However, the focus on a "fair transition" in the consultation process suggests that the needs of vulnerable groups were considered, ensuring access to the benefits of the energy transition for all, particularly vulnerable consumers.

2.6 *Barriers*

Establishing effective links between key public authorities across different governance levels is often hindered by a lack of coherent coordination. Citizens' disengagement and poor alignment between national, regional, and local authorities, along with fragmented responsibilities, can obstruct policy implementation. This context often results from unclear divisions of responsibilities that may create undefined and vacuum of leadership in this area, further weakening governance structures. Another significant challenge is the insufficient consideration of local contexts. Policies designed at higher governance levels may not adequately reflect the specific needs of local communities, making them less relevant in practice. This issue is also constrained by limited resources at the local level, reducing the capacity of local authorities to implement and sustain valuable initiatives. The absence of public participation is also a critical limiting factor. The voices of women and vulnerable groups are often not sufficiently heard in the policymaking process, leading to policies that fail to also consider their needs. A lack of data and monitoring mechanisms further limits mitigation measures as it does not allow for evidence-based decision-making and the ability to track progress of such measures. Another potential problem is the siloed approach to policy implementation, where different departments or governance levels work independently thus preventing integrated and comprehensive solutions. To overcome these barriers, it is essential to improve coordination mechanisms, ensure clear definition of responsibilities, consider local needs, encourage inclusive public participation, allocate sufficient resources, and enhance data collection and monitoring processes.

2.7 *Factors for success*

The establishment of links between public authorities at different levels of governance require several factors to successfully implement gender equality and support for vulnerable groups policies. Clear and consistent policy frameworks are needed to ensure alignment of objectives and actions across all levels. Strong central coordinating bodies, such as the Commission for Climate Action (CAC), are essential for setting standards, monitoring progress, and ensuring coherence. It is also necessary to have interoperable data management systems for efficient data sharing and policy implementation. The establishment of multi-level dialogues and consultations is vital to facilitate the exchange of knowledge and best practices. Adequate funding and resources need to be allocated to support local initiatives. Additionally, streamlined administrative procedures and clear assignment of responsibilities are crucial to minimize complexity and ensure efficient implementation. The creation of a national system to monitor energy poverty, as well as the factors that influence it, is important to help direct policies that protect vulnerable consumers. Finally, transparency and public access to information are key to ensuring accountability.

2.8 *Conclusion*

To achieve successful governance in supporting vulnerable groups and promoting a fair twin transition requires stronger coordination across different levels of government. A multi-level governance approach, integrating national, regional, and local authorities, is essential to ensure that policies are aligned, inclusive, and responsive to the needs of all communities. Strengthening institutional frameworks, such as central coordinating bodies, can facilitate policy coherence and accountability while promoting shared objectives for social equity. One key priority is to enhance structured dialogue and cooperation among stakeholders, ensuring that vulnerable groups actively participate in decision-making processes. Inclusive governance mechanisms must be established to guarantee that policies are shaped by diverse perspectives and that marginalized voices are heard. Another critical area of action is ensuring adequate funding and capacity-building for local authorities and representative organizations, empowering them to implement targeted policies effectively. Financial and technical resources must be distributed equitably to enable local governments to address specific challenges, particularly in housing, employment, and social protection. Furthermore, improving data collection and monitoring systems is necessary to track the impact of policies and inform future decision-making. Transparent and interoperable data-sharing mechanisms will strengthen accountability and allow for evidence-based policymaking. Additionally, policies should be designed with an integrated approach, avoiding fragmented implementation and fostering cross-sector collaboration to maximize impact. Addressing these priorities will empower governance structures to become more inclusive, ensuring that gender equality and the protection of vulnerable groups are considered and embedded in public policies.

References

Decreto-Lei n.º 15/2022 | DR. (n.d.). Retrieved January 30, 2025, from <https://diariodarepublica.pt/dr/detalhe/decreto-lei/15-2022-177634016>

Directive - 2018/2001 - EN - EUR-Lex. (n.d.). Retrieved January 30, 2025, from <https://eur-lex.europa.eu/eli/dir/2018/2001/oj/eng>

Directive - 2019/944 - EN - EUR-Lex. (n.d.). Retrieved January 30, 2025, from <https://eur-lex.europa.eu/eli/dir/2019/944/oj/eng>

Portugal's National Energy and Climate Plan for 2021-2030 - Climate Change Laws of the World. (n.d.). Retrieved January 30, 2025, from https://climate-laws.org/document/portugal-s-national-energy-and-climate-plan-for-2021-2030_47f6

Regulation - 2018/1999 - EN - EUR-Lex. (n.d.). Retrieved January 30, 2025, from <https://eur-lex.europa.eu/eli/reg/2018/1999/oj/eng>

Resolução do Conselho de Ministros n.º 53/2020 | DR. (n.d.). Retrieved January 30, 2025, from <https://diariodarepublica.pt/dr/detalhe/resolucao-conselho-ministros/53-2020-137618093>

Resolução do Conselho de Ministros n.º 56/2015 | DR. (n.d.). Retrieved February 2, 2025, from <https://diariodarepublica.pt/dr/detalhe/resolucao-conselho-ministros/56-2015-69905665>

Resolução do Conselho de Ministros n.º 63/2020 | DR. (n.d.). Retrieved January 30, 2025, from <https://diariodarepublica.pt/dr/detalhe/resolucao-conselho-ministros/63-2020-140346286>

Resolução do Conselho de Ministros n.º 107/2019 | DR. (n.d.). Retrieved January 30, 2025, from <https://diariodarepublica.pt/dr/detalhe/resolucao-conselho-ministros/107-2019-122777644>

RNC2050 - Roteiro para a Neutralidade Carbónica. (n.d.). Retrieved January 31, 2025, from <https://descarbonizar2050.apambiente.pt/en/roadmap/>

MFF 2021-2027 - European Commission. (n.d.). Retrieved January 31, 2025, from https://commission.europa.eu/strategy-and-policy/eu-budget/long-term-eu-budget/2021-2027_en

PLANO NACIONAL ENERGIA E CLIMA 2021-2030. (2024, October 1). https://apambiente.pt/sites/default/files/Clima/Planeamento/20241030_pnec2030_maen.pdf

PNI 2030 | Infraestruturas de Portugal. (n.d.). Retrieved January 31, 2025, from

<https://www.infraestruturasdeportugal.pt/pt-pt/infraestruturas/investimentos/programas/planos-estrategicos/pni2030>